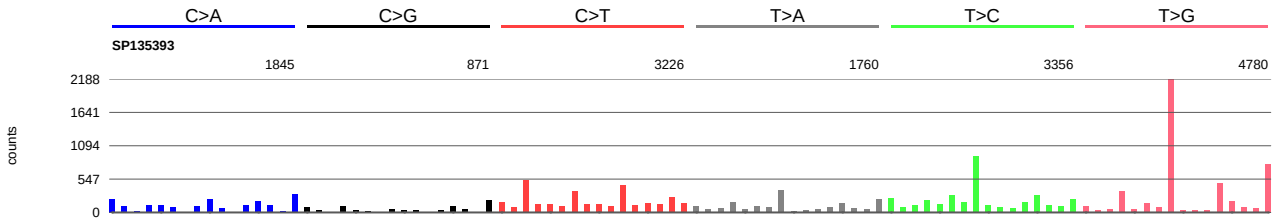
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SP135393
1845
871
3226
1760
3356
4780
counts
2188
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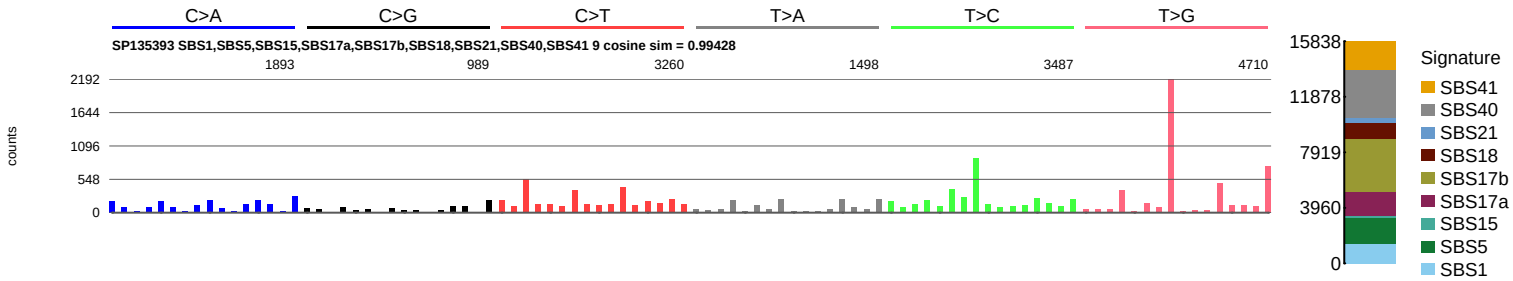
1094
547
0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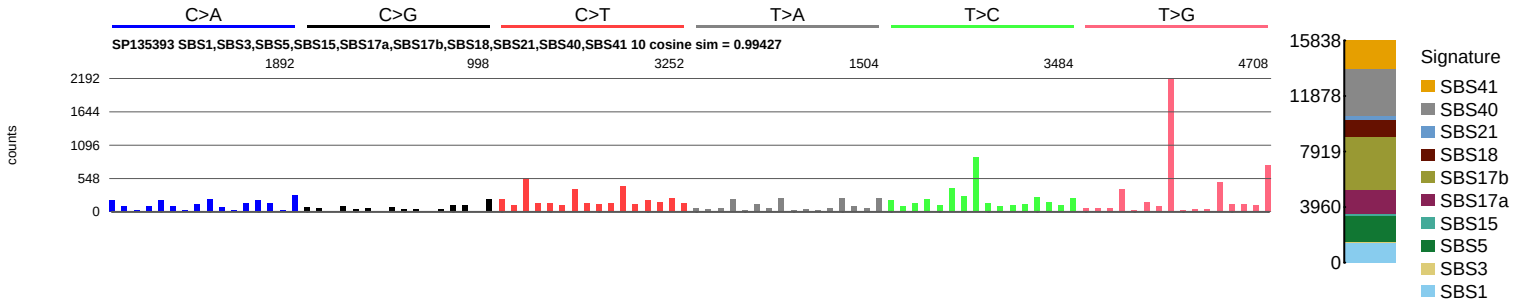
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99428
1893
989
3260
1498
3487
4710
counts
2192
1644
1096
548
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 10 cosine sim = 0.99427
1892
998
3252
1504
3484
4708
counts
2192
1644
1096
548
0
15838
11878
7919
3960
0
Signature
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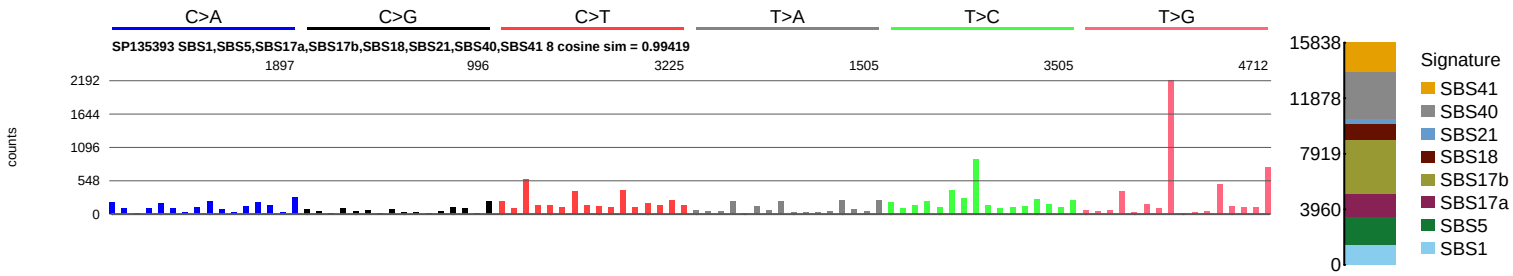
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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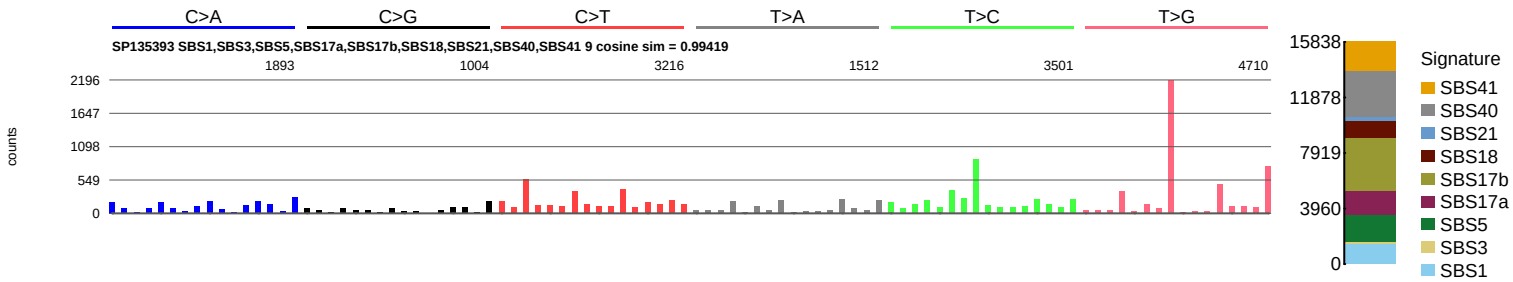
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99419
1897
996
3225
1505
3505
4712
counts
2192
1644
1096
548
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1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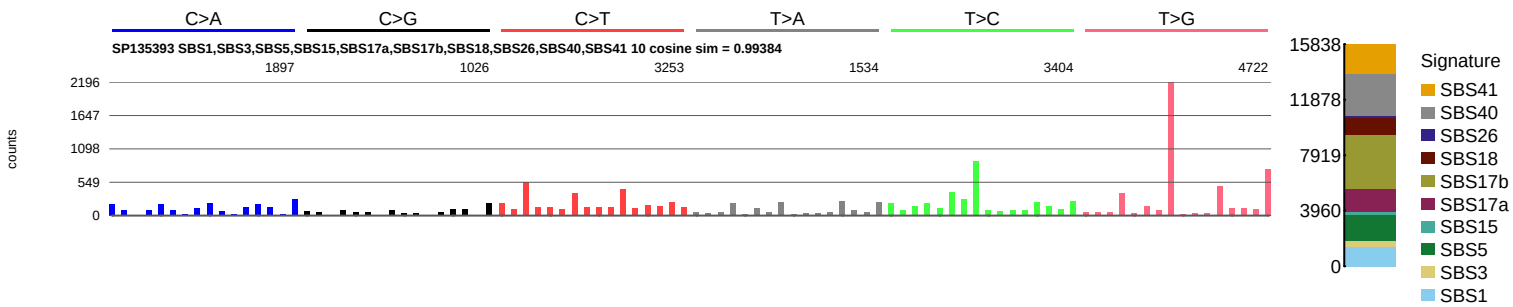
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99419
1893
1004
3216
1512
3501
4710
counts
2196
1647
1098
549
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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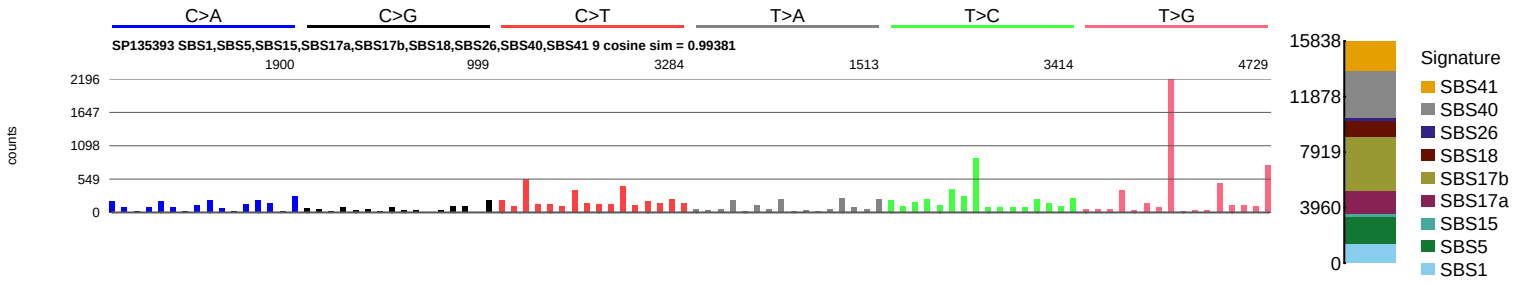
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99384
1897
1026
3253
1534
3404
4722
counts
2196
1647
1098
549
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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C>G
C>T
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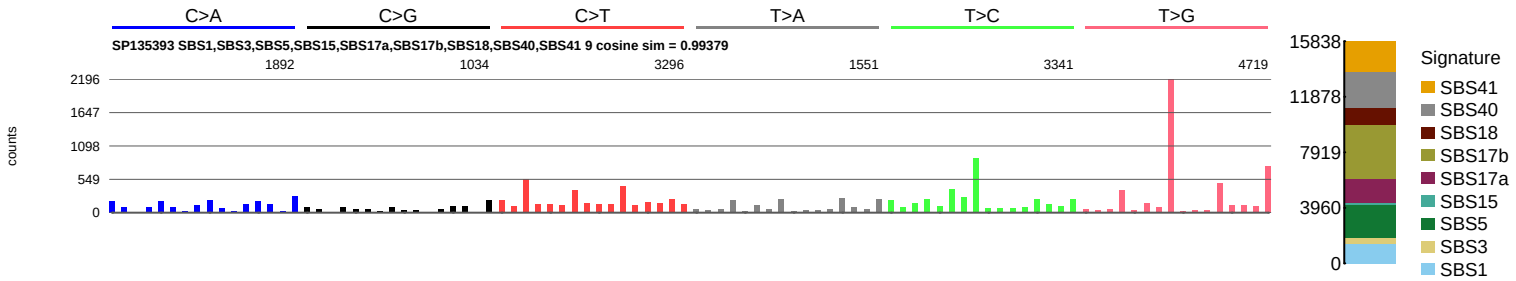
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99381
1900
999
3284
1513
3414
4729
counts
2196
1647
1098
549
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>T
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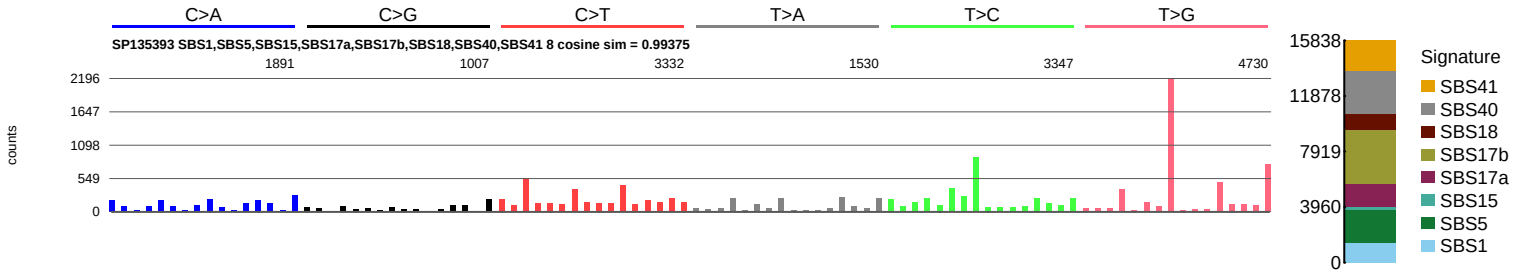
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99379
1892
1034
3296
1551
3341
4719
counts
2196
1647
1098
549
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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C>G
C>T
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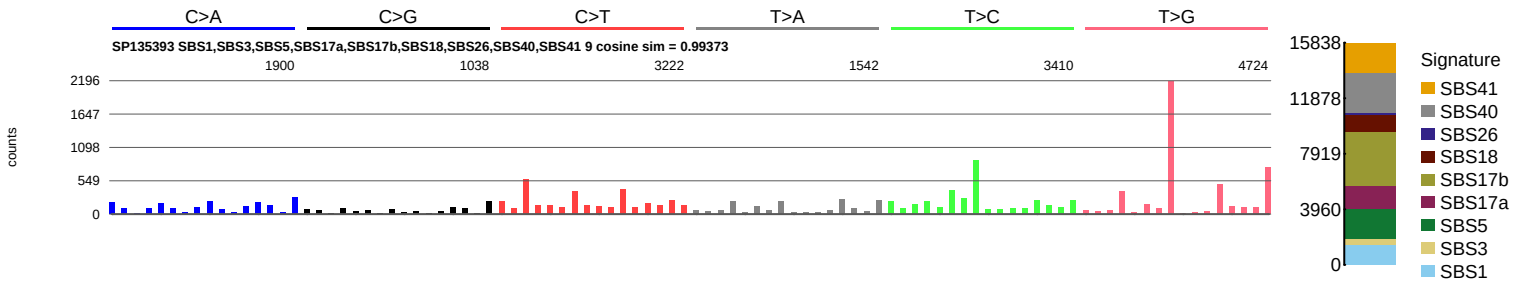
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99375
1891
1007
3332
1530
3347
4730
counts
2196
1647
1098
549
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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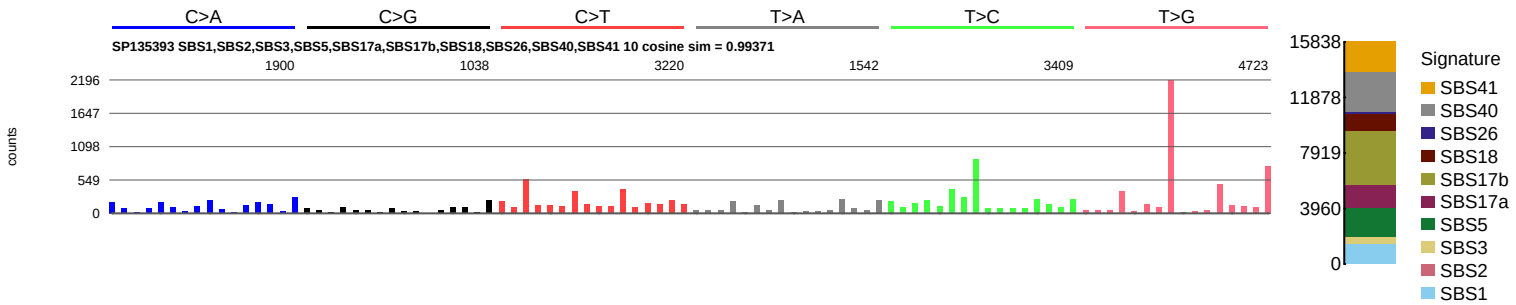
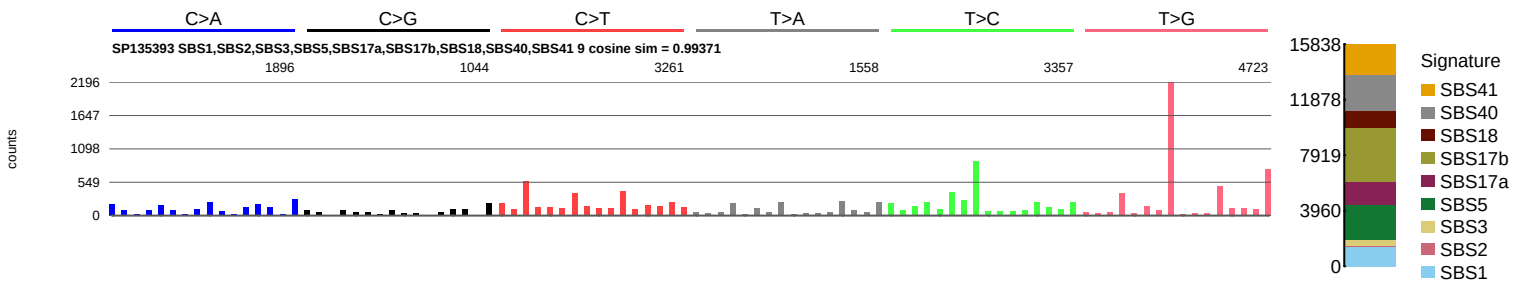
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99373
1900
1038
3222
1542
3410
4724
counts
2196
1647
1098
549
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 10 cosine sim = 0.99371
1900
1038
3220
1542
3409
4723
counts
2196
1647
1098
549
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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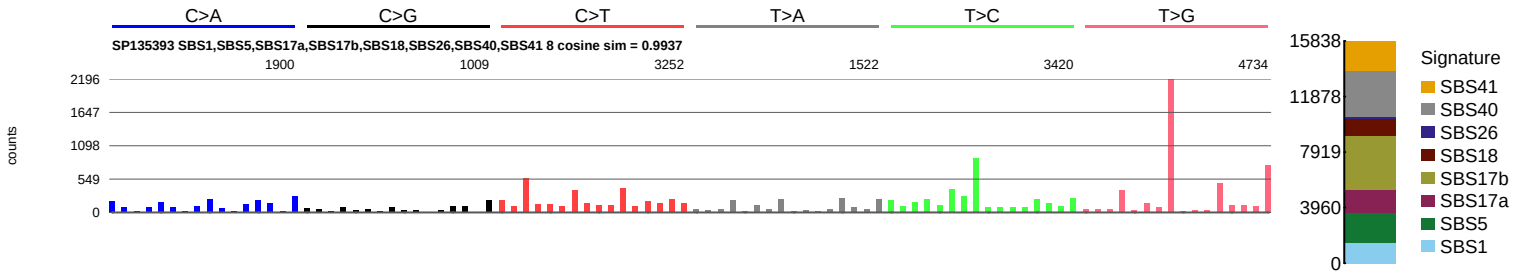
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 9 cosine sim = 0.99371
1896
1044
3261
1558
3357
4723
counts
2196
1647
1098
549
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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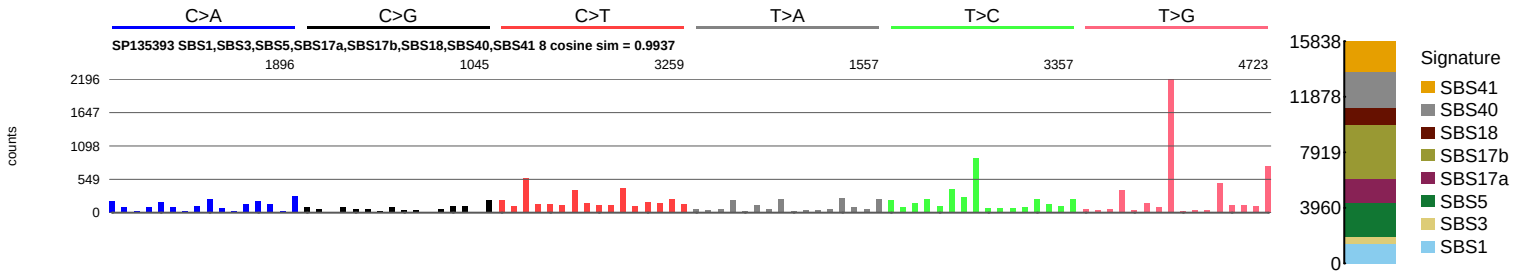
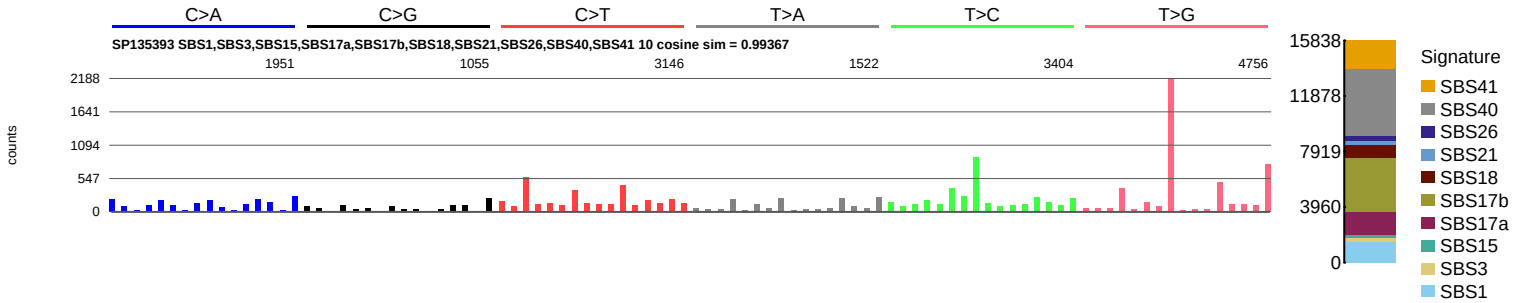
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.9937
1900
1009
3252
1522
3420
4734
counts
2196
1647
1098
549
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.9937
1896
1045
3259
1557
3357
4723
counts
2196
1647
1098
549
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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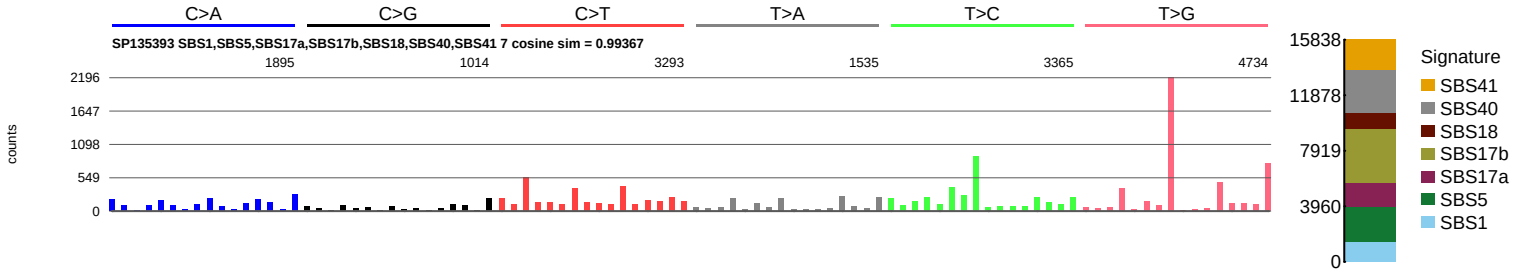
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 10 cosine sim = 0.99367
1951
1055
3146
1522
3404
4756
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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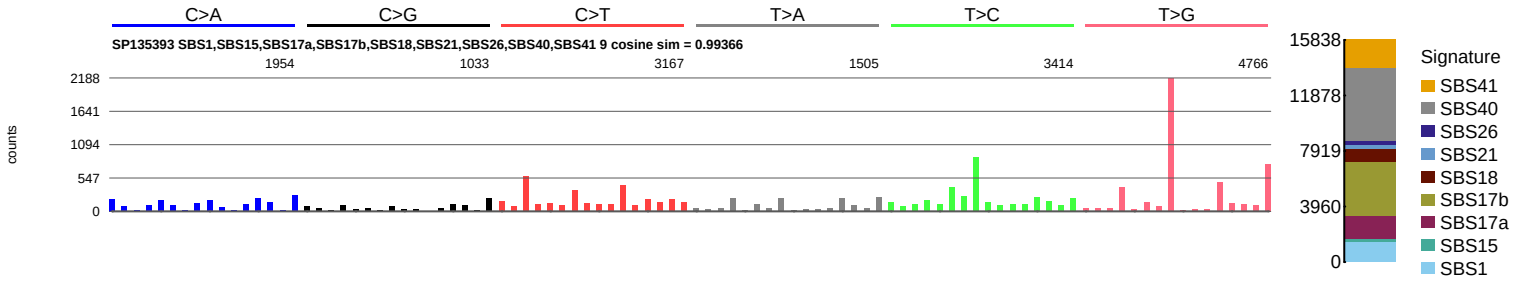
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99367
1895
1014
3293
1535
3365
4734
counts
2196
1647
1098
549
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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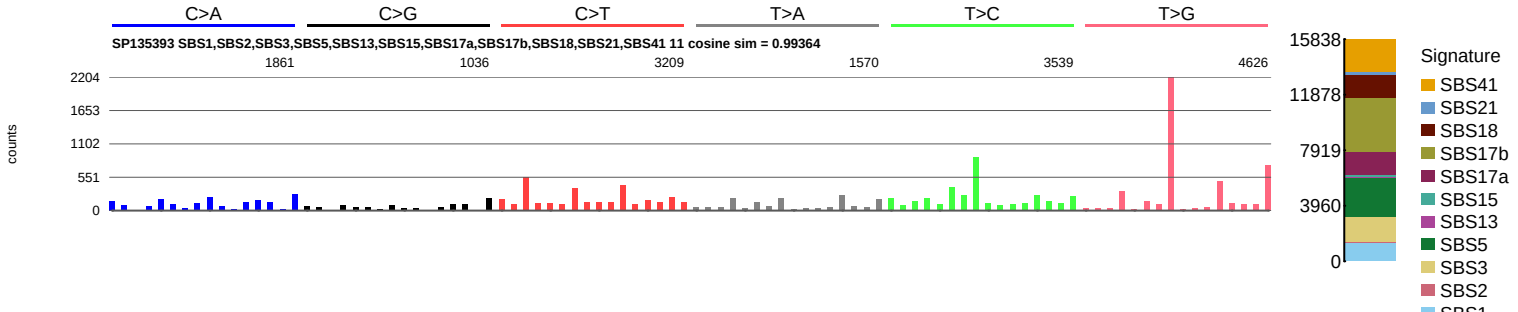
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.99366
1954
1033
3167
1505
3414
4766
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
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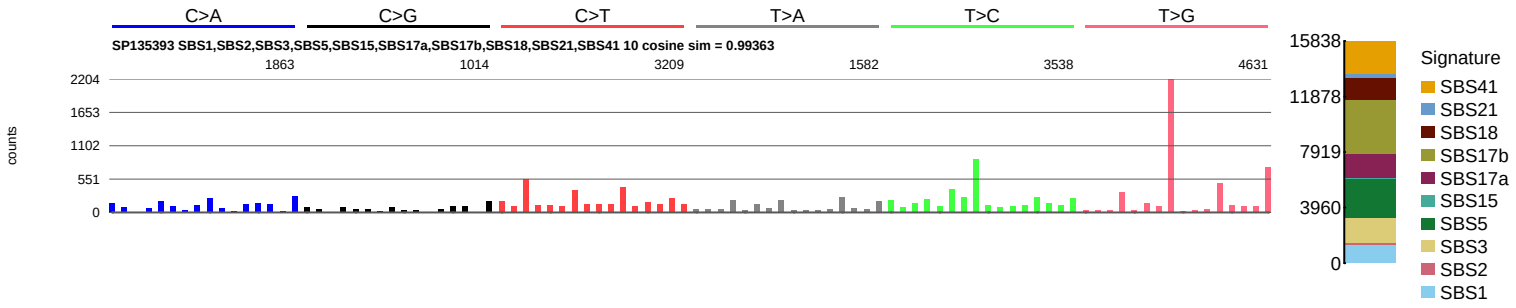
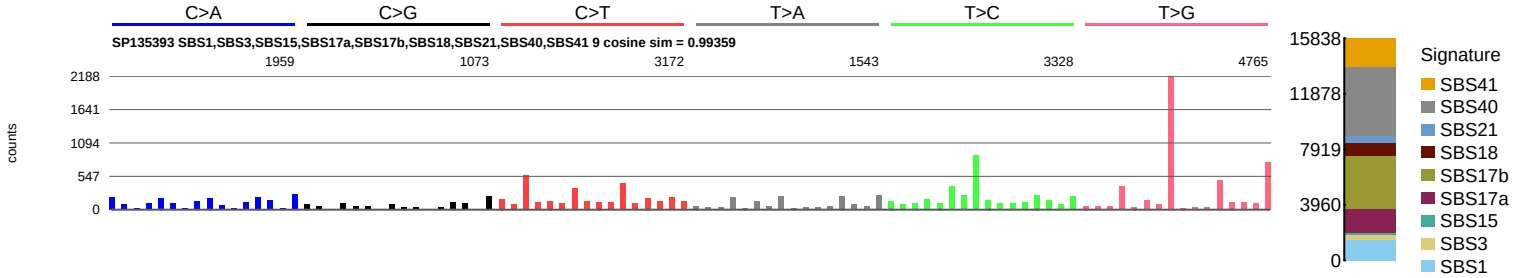
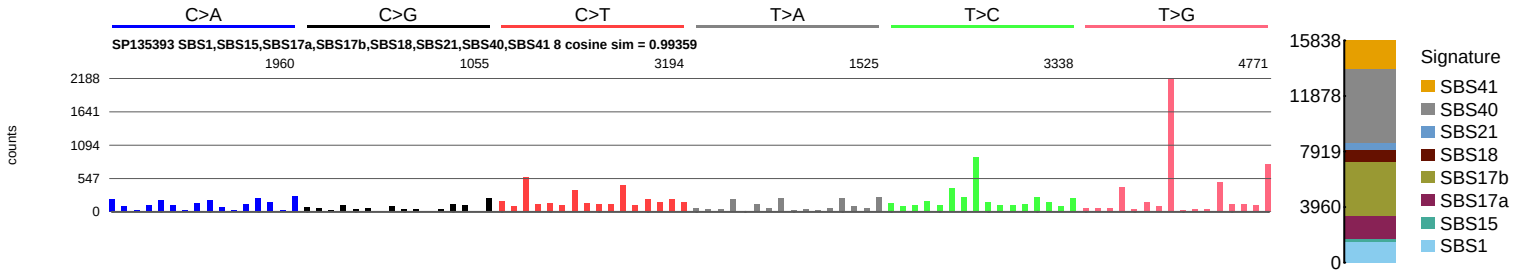
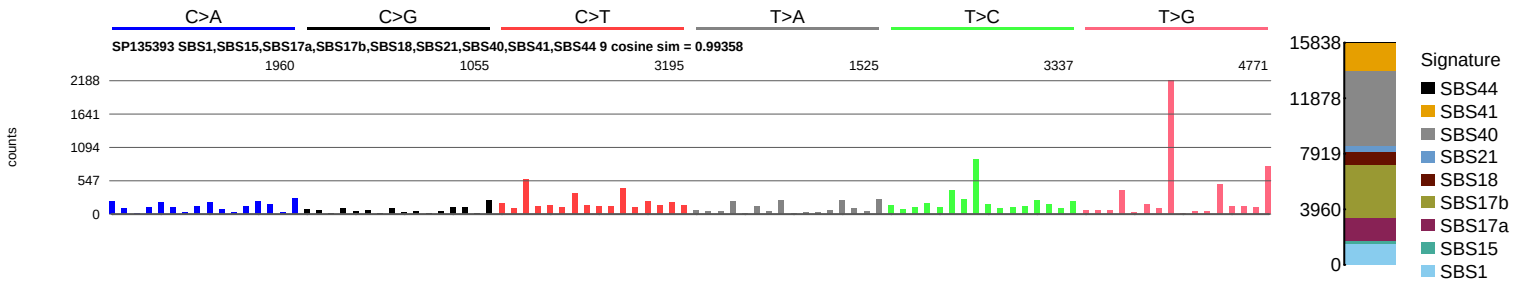
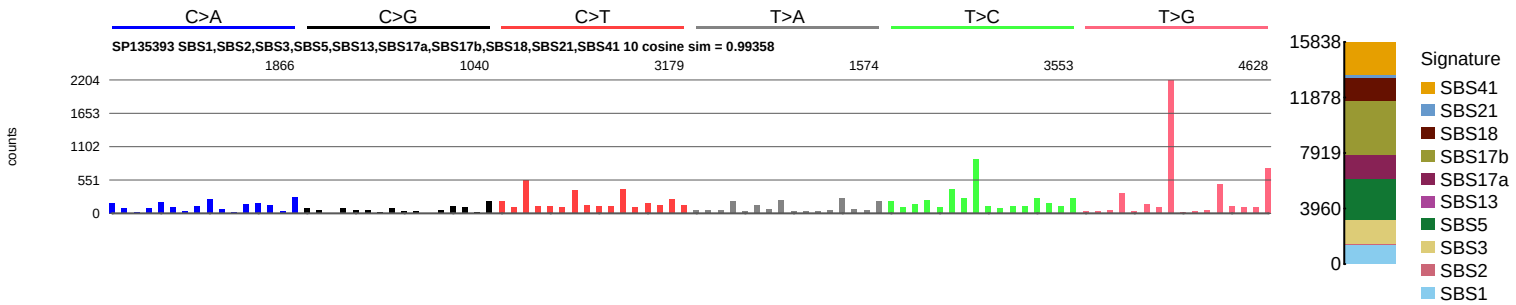
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 11 cosine sim = 0.99364
1861
1036
3209
1570
3539
4626
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2

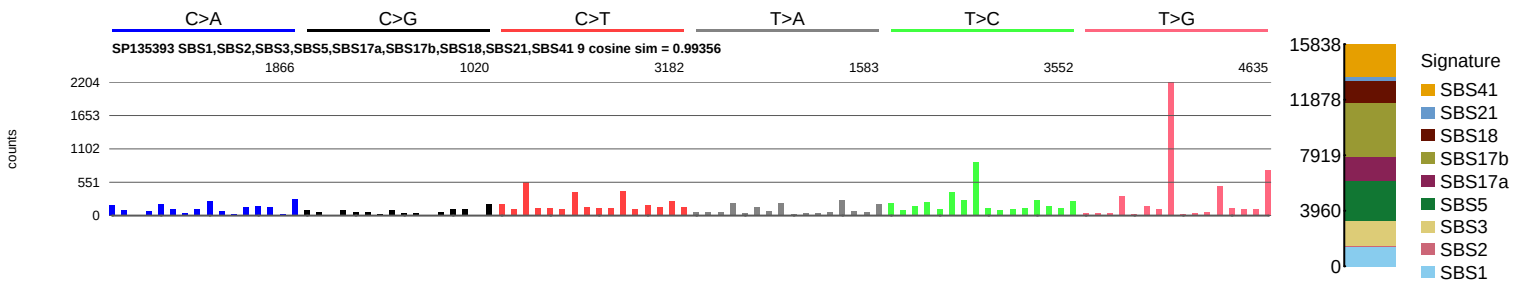

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99363
1863
1014
3209
1582
3538
4631
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

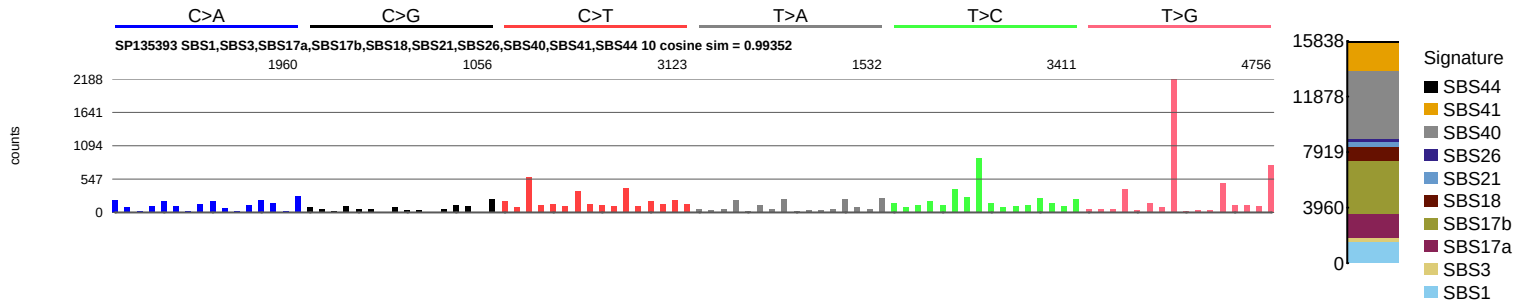

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 9 cosine sim = 0.99359
1959
1073
3172
1543
3328
4765
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

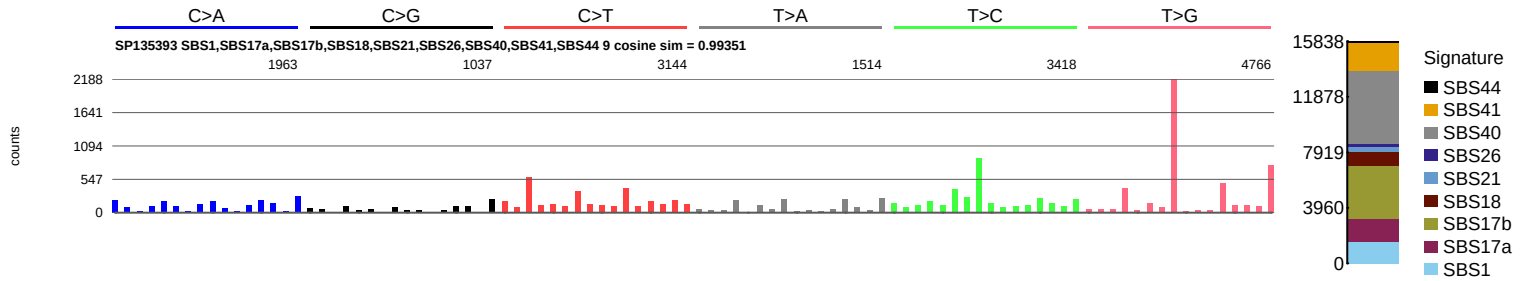

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99359
1960
1055
3194
1525
3338
4771
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

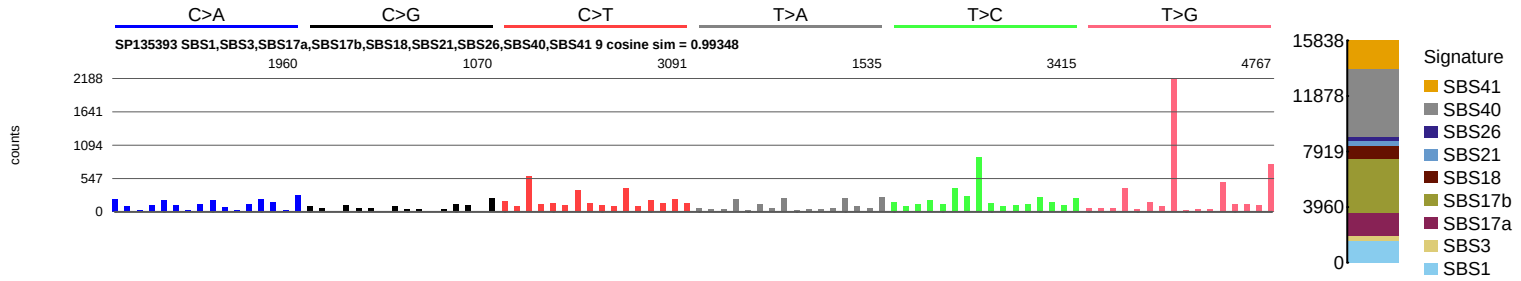

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99358
1960
1055
3195
1525
3337
4771
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1

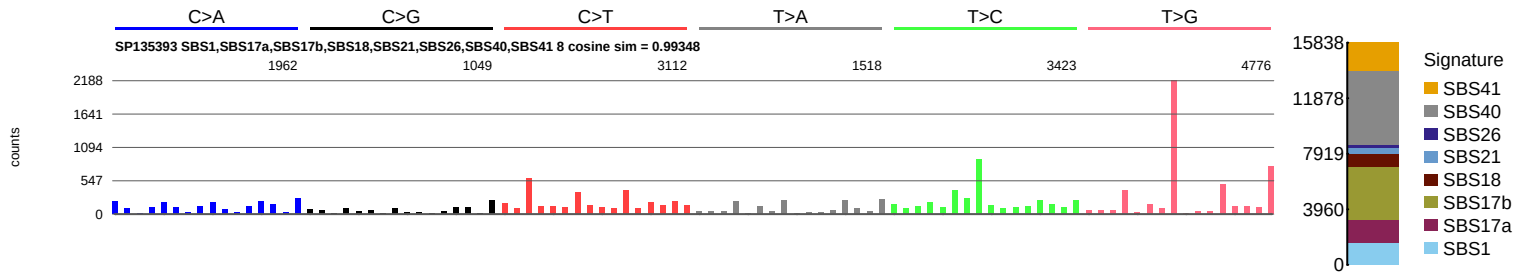

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99358
1866
1040
3179
1574
3553
4628
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

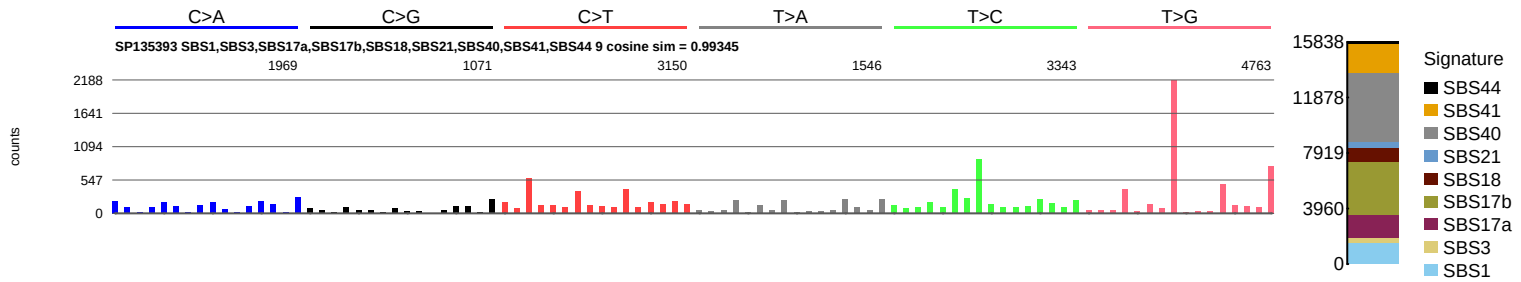

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99356
1866
1020
3182
1583
3552
4635
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

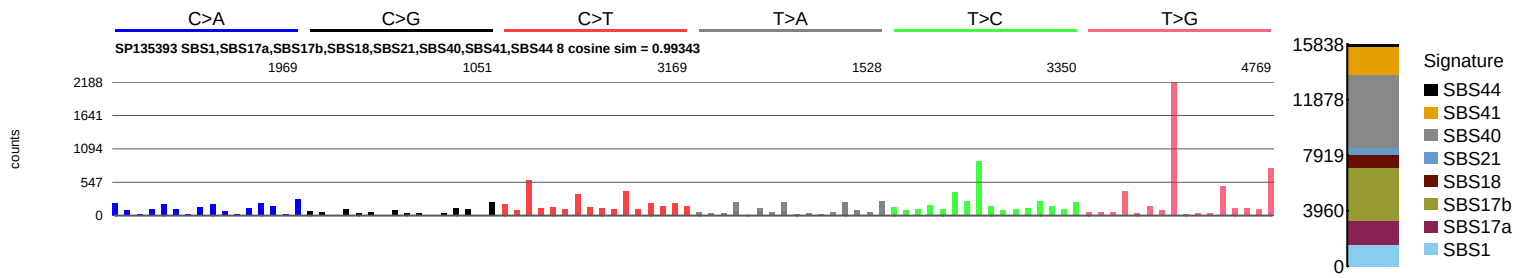

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99352
1960
1056
3123
1532
3411
4756
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

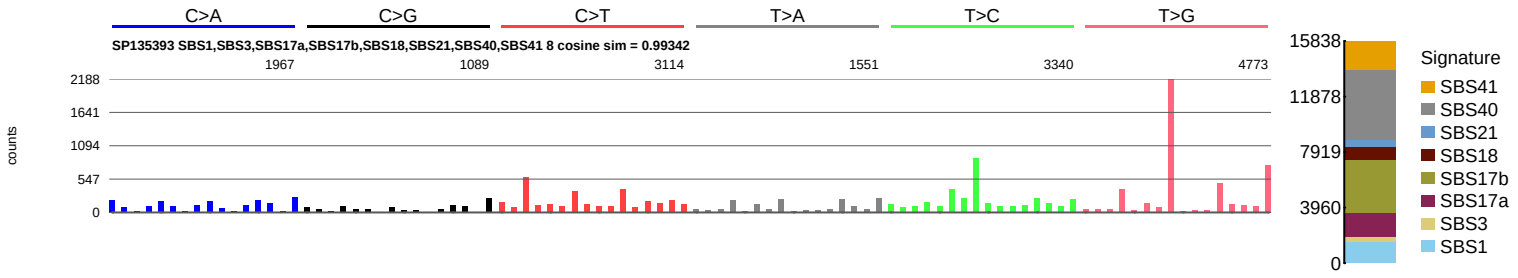

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99351
1963
1037
3144
1514
3418
4766
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1

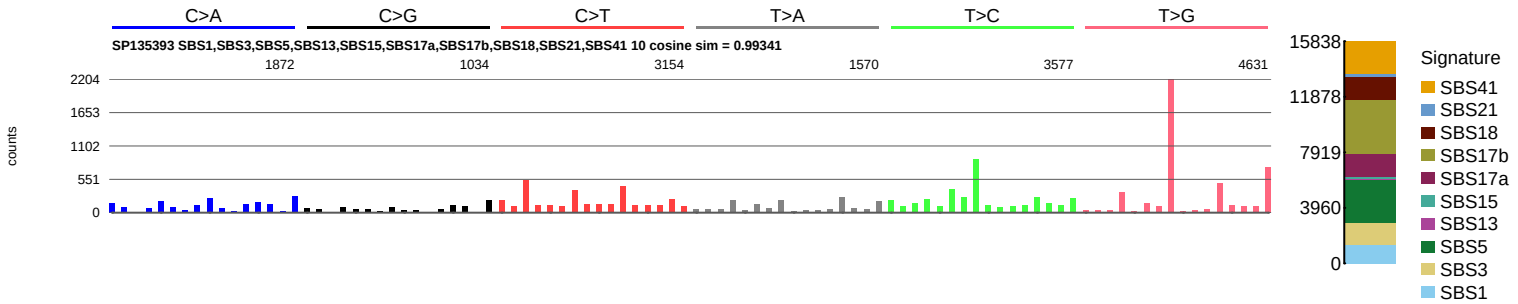

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 9 cosine sim = 0.99348
1960
1070
3091
1535
3415
4767
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

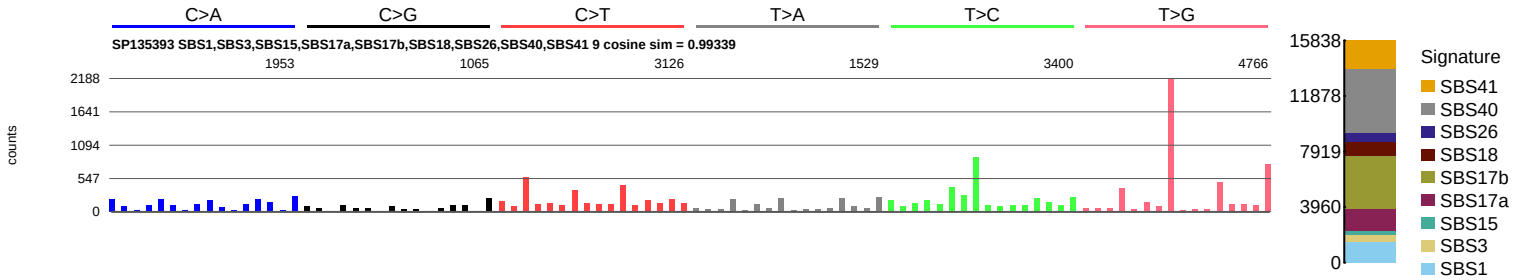

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.99348
1962
1049
3112
1518
3423
4776
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1

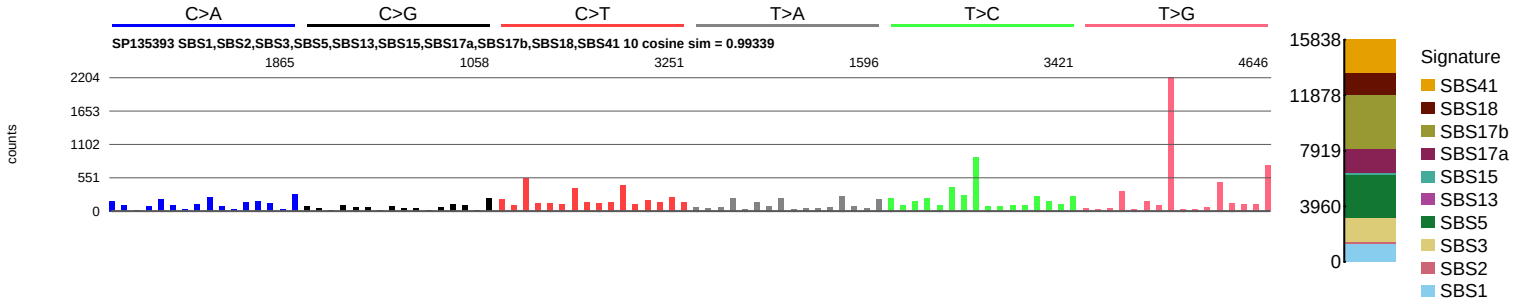

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 9 cosine sim = 0.99345
1969
1071
3150
1546
3343
4763
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

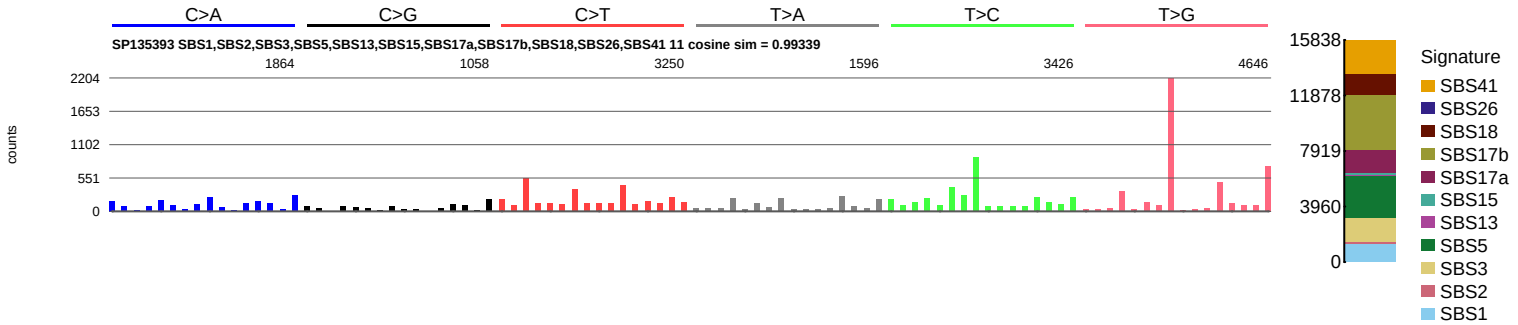

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.99343
1969
1051
3169
1528
3350
4769
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

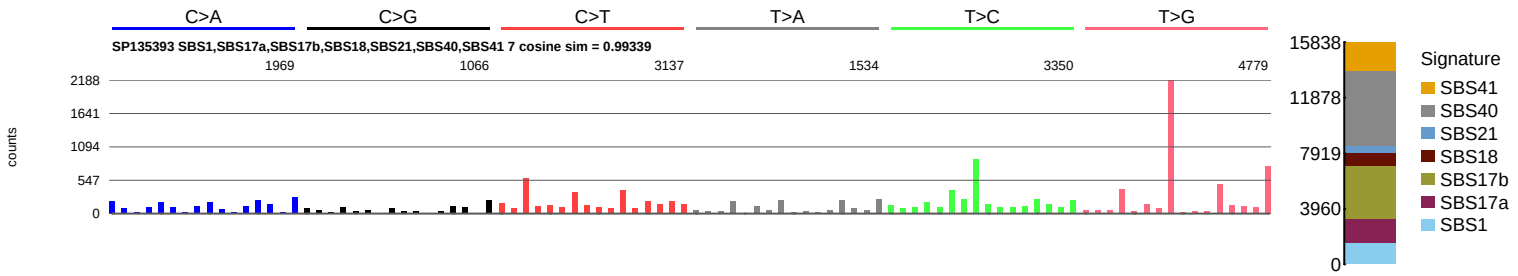

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 8 cosine sim = 0.99342
1967
1089
3114
1551
3340
4773
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

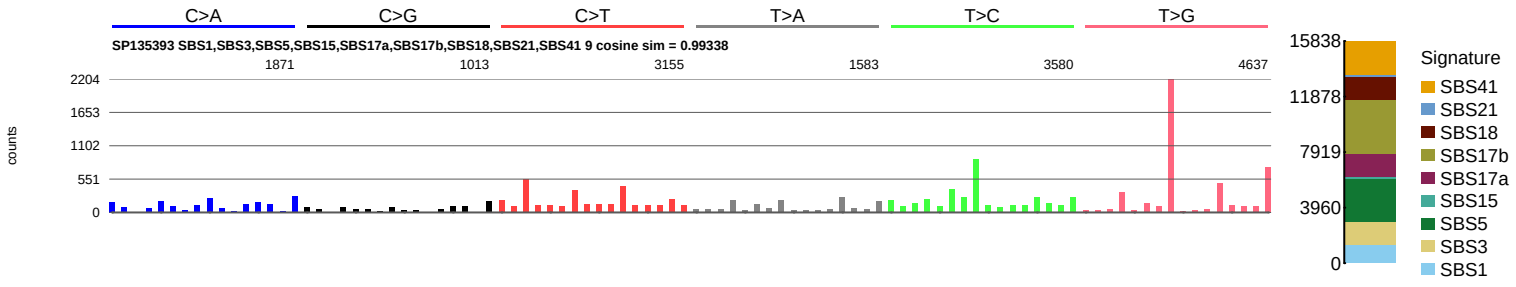

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99341
1872
1034
3154
1570
3577
4631
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

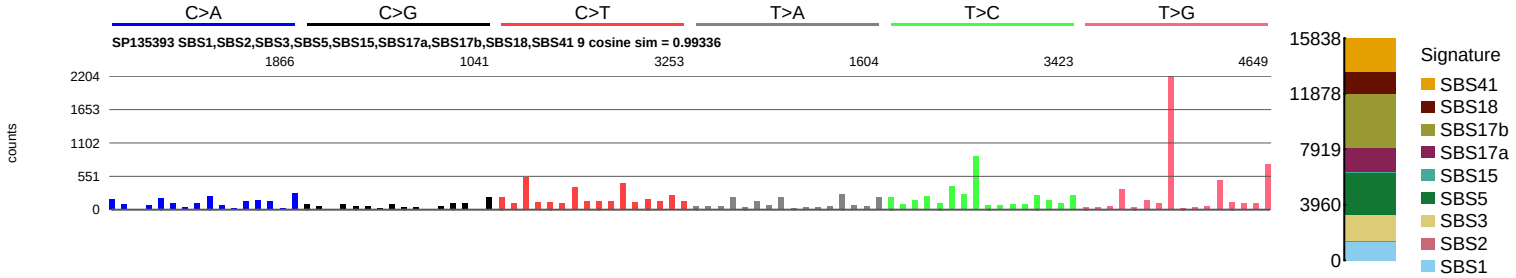

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99339
1953
1065
3126
1529
3400
4766
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

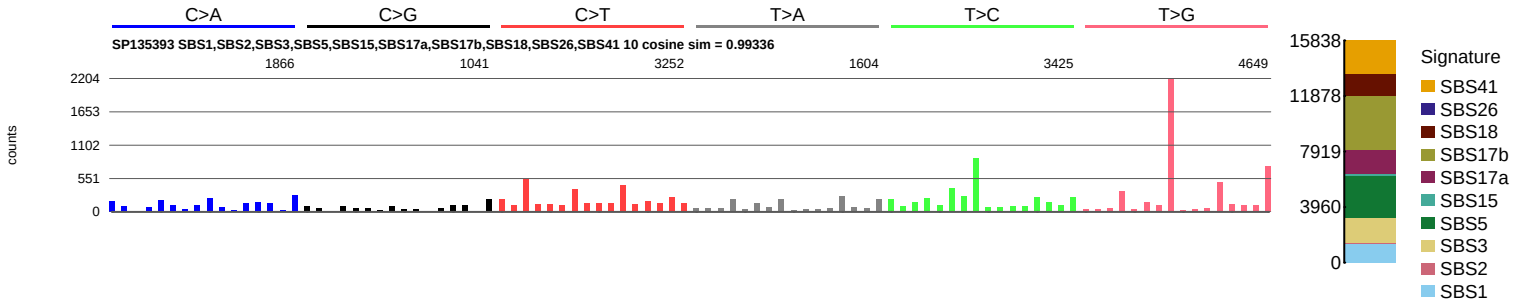

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 10 cosine sim = 0.99339
1865
1058
3251
1596
3421
4646
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

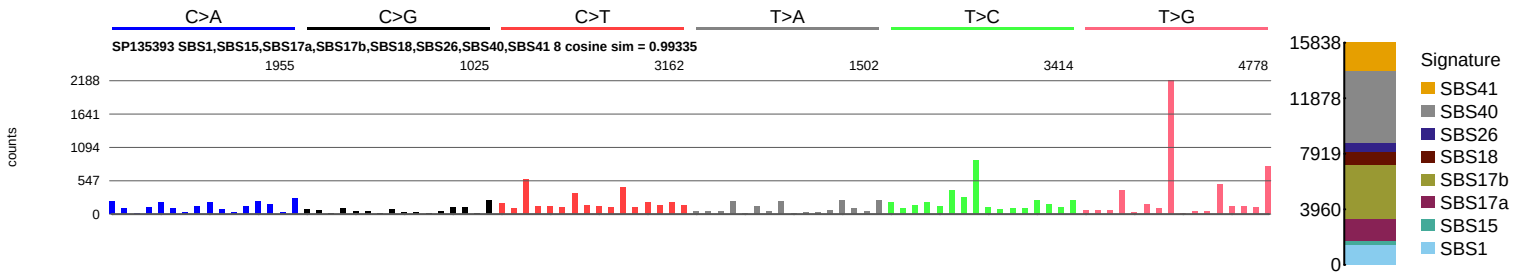

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 11 cosine sim = 0.99339
1864
1058
3250
1596
3426
4646
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

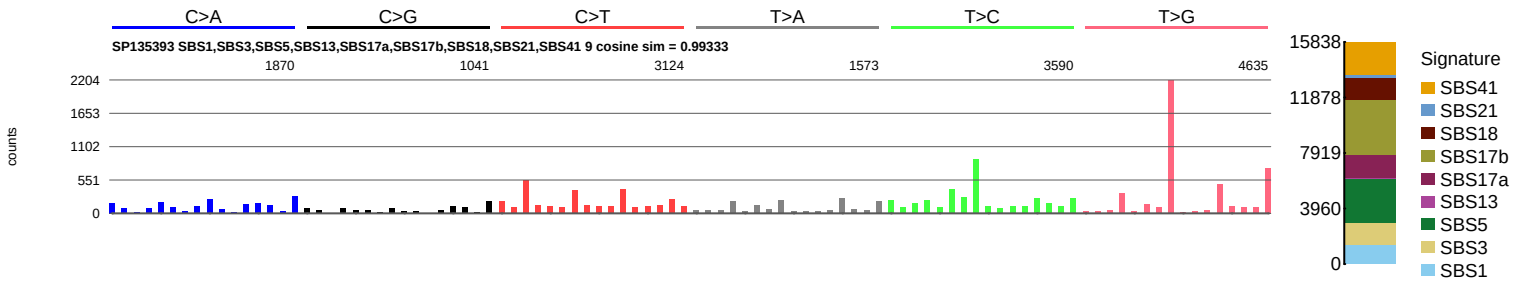

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS41 7 cosine sim = 0.99339
1969
1066
3137
1534
3350
4779
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1

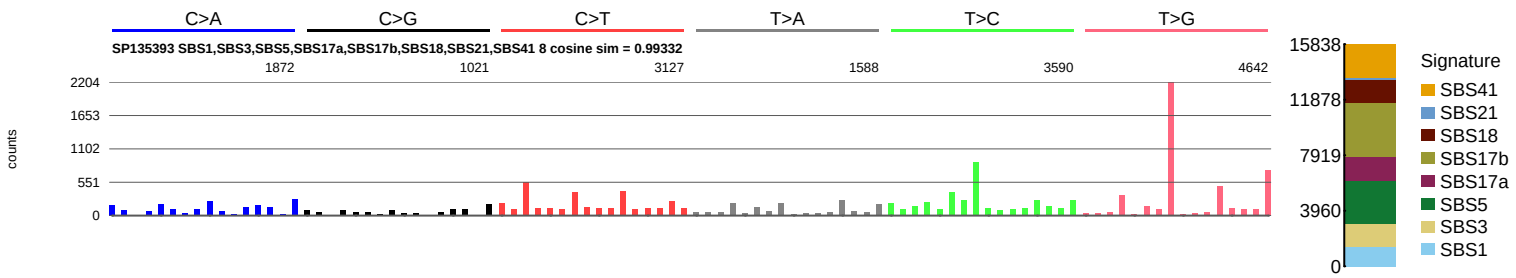

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99338
1871
1013
3155
1583
3580
4637
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1

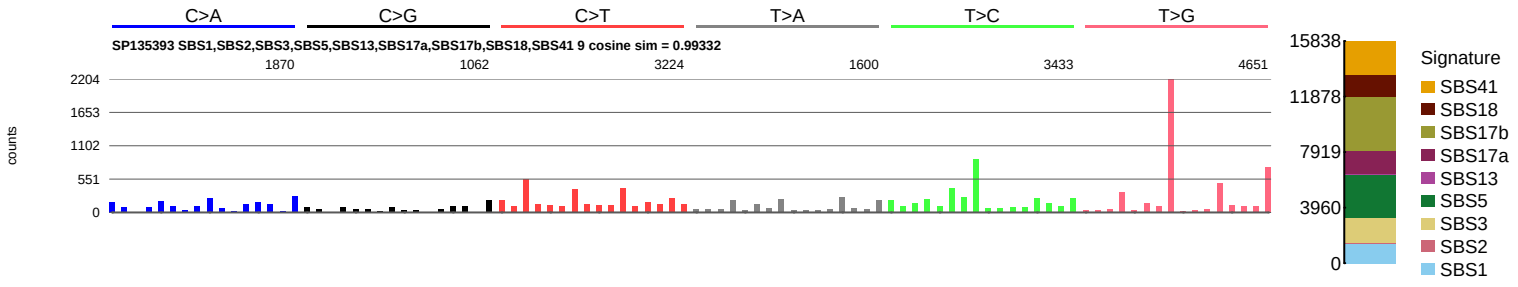

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99336
1866
1041
3253
1604
3423
4649
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

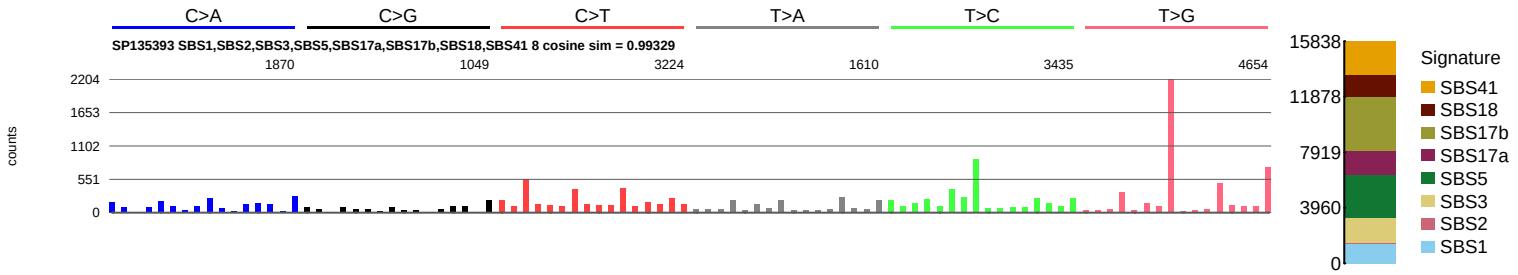

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.99336
1866
1041
3252
1604
3425
4649
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

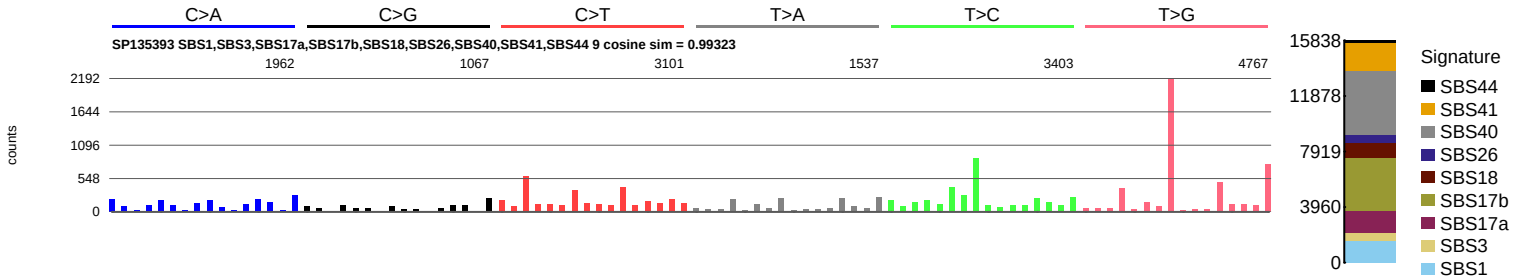

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.99335
1955
1025
3162
1502
3414
4778
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1

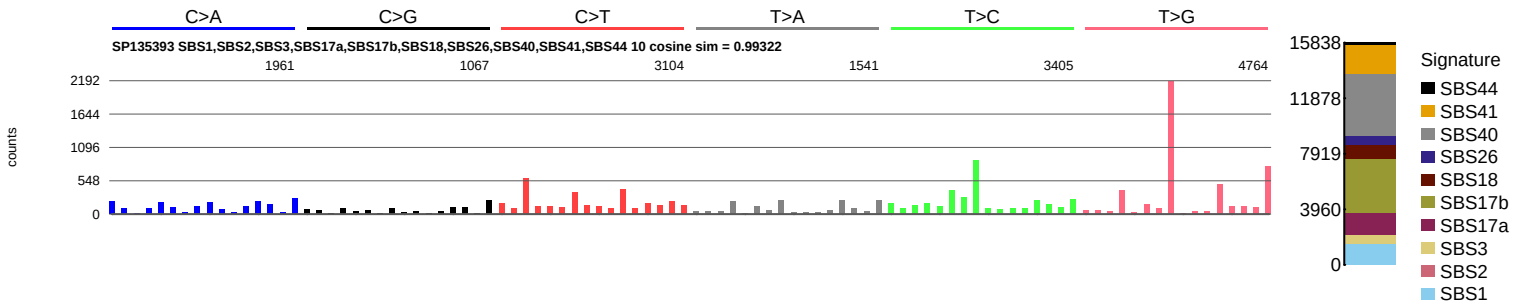

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99333
1870
1041
3124
1573
3590
4635
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

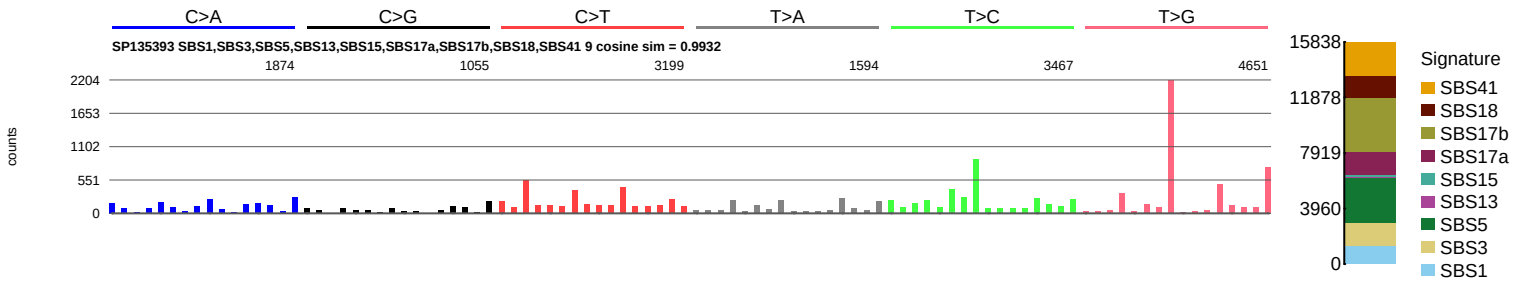

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99332
1872
1021
3127
1588
3590
4642
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

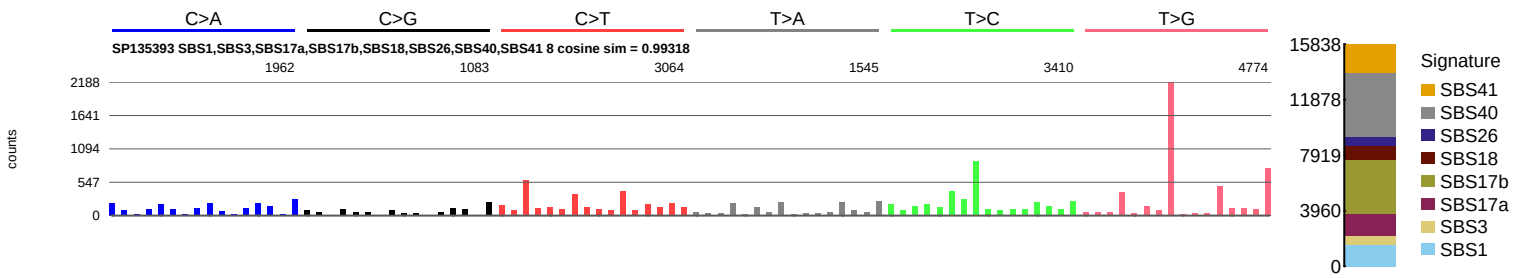

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99332
1870
1062
3224
1600
3433
4651
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

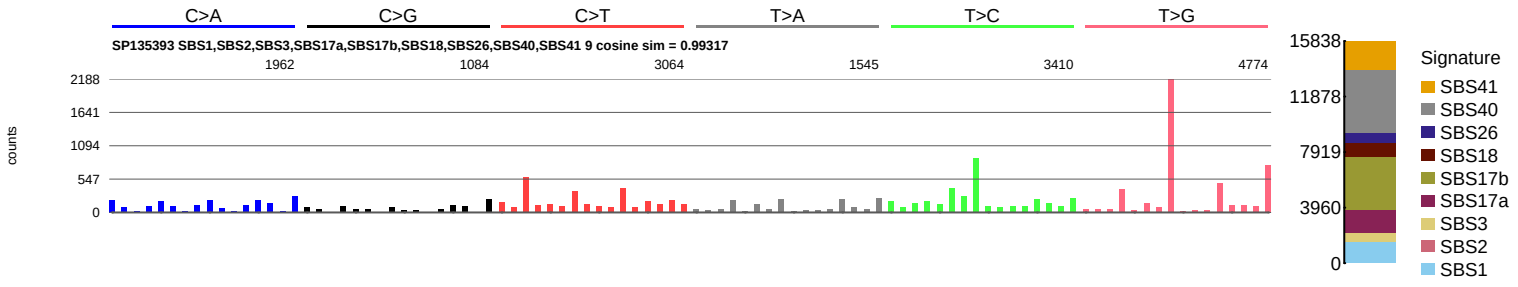

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99329
1870
1049
3224
1610
3435
4654
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

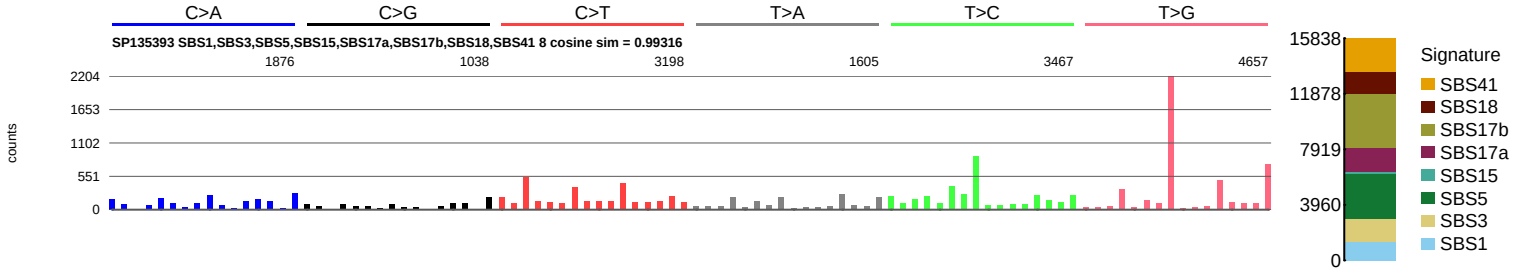

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 9 cosine sim = 0.99323
1962
1067
3101
1537
3403
4767
counts
2192
1644
1096
548
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 10 cosine sim = 0.99322
1961
1067
3104
1541
3405
4764
counts
2192
1644
1096
548
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.9932
1874
1055
3199
1594
3467
4651
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 8 cosine sim = 0.99318
1962
1083
3064
1545
3410
4774
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 9 cosine sim = 0.99317
1962
1084
3064
1545
3410
4774
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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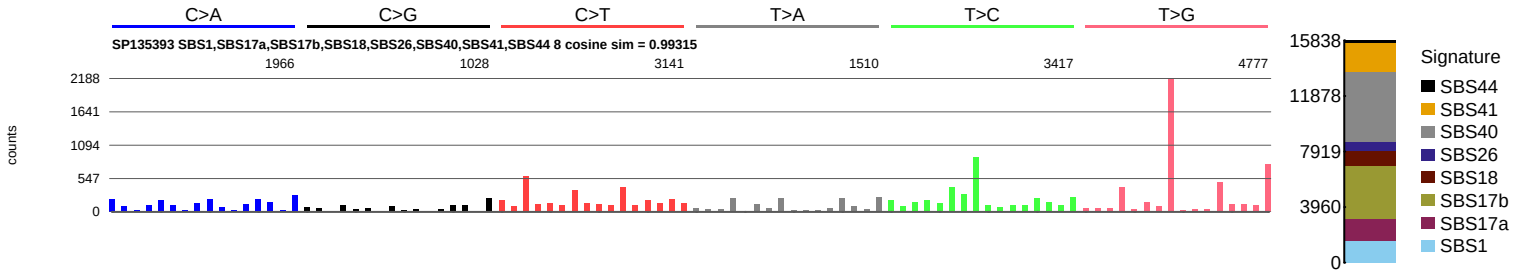
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99316
1876
1038
3198
1605
3467
4657
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
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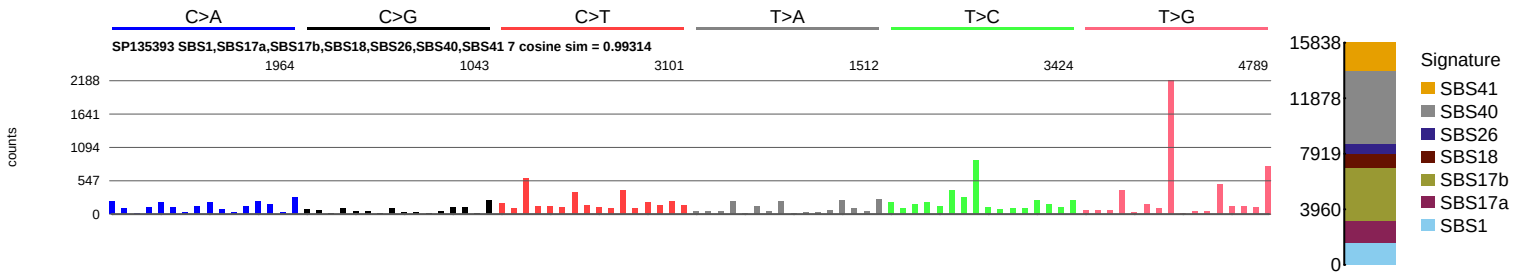
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99315
1966
1028
3141
1510
3417
4777
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
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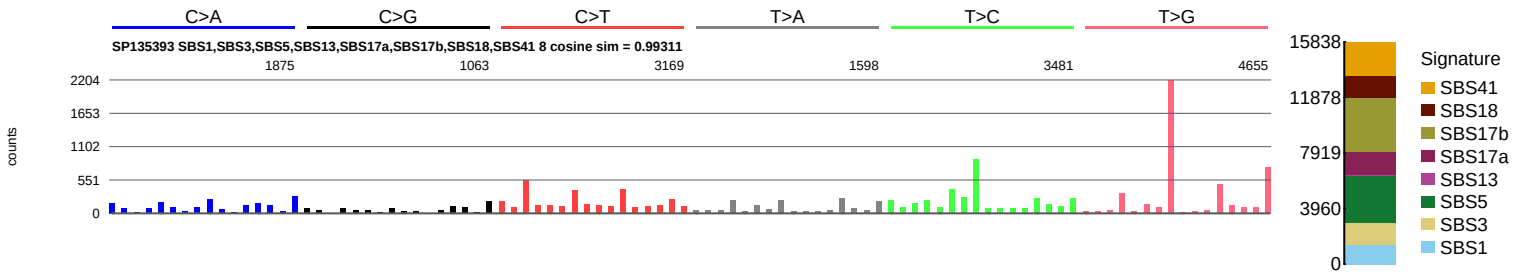
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS41 7 cosine sim = 0.99314
1964
1043
3101
1512
3424
4789
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
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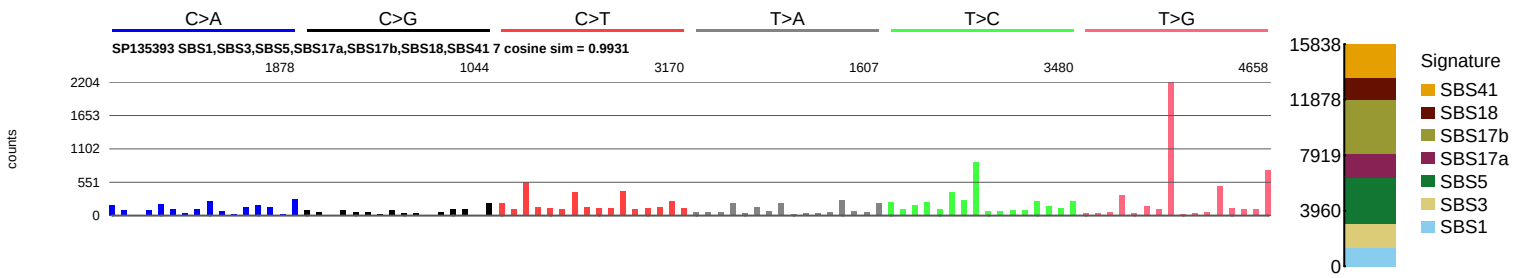
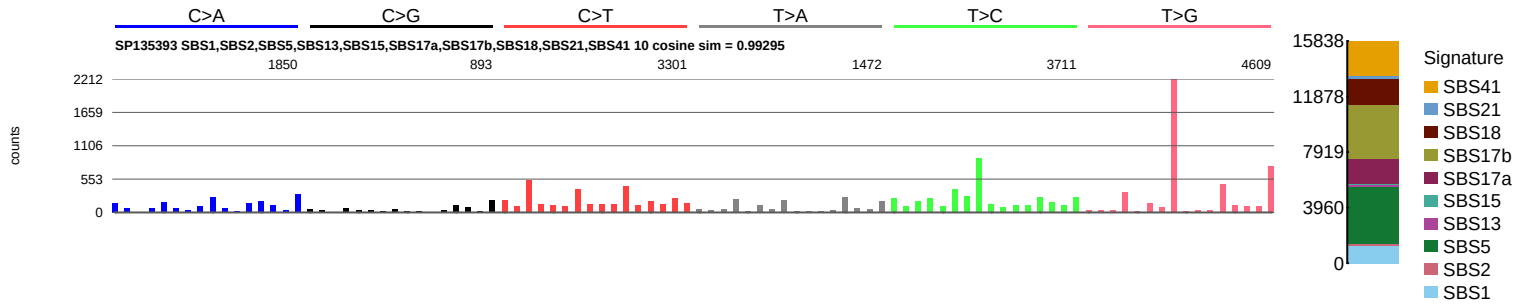
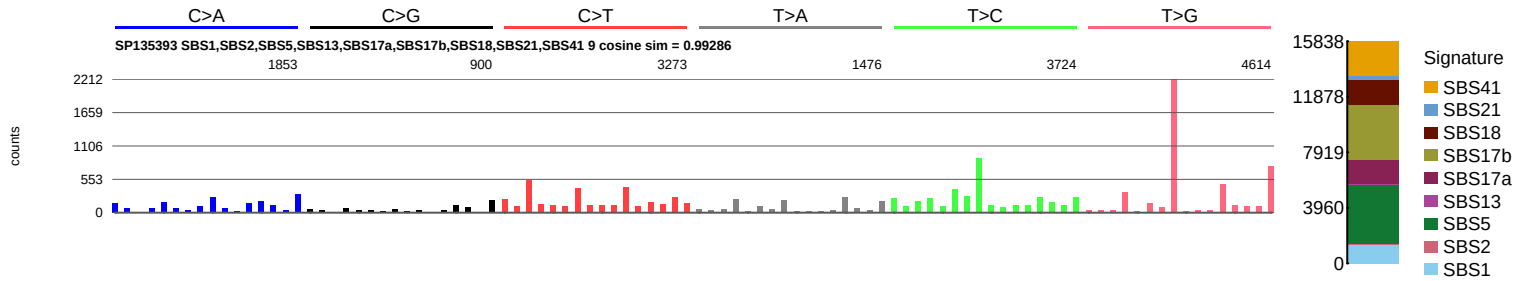
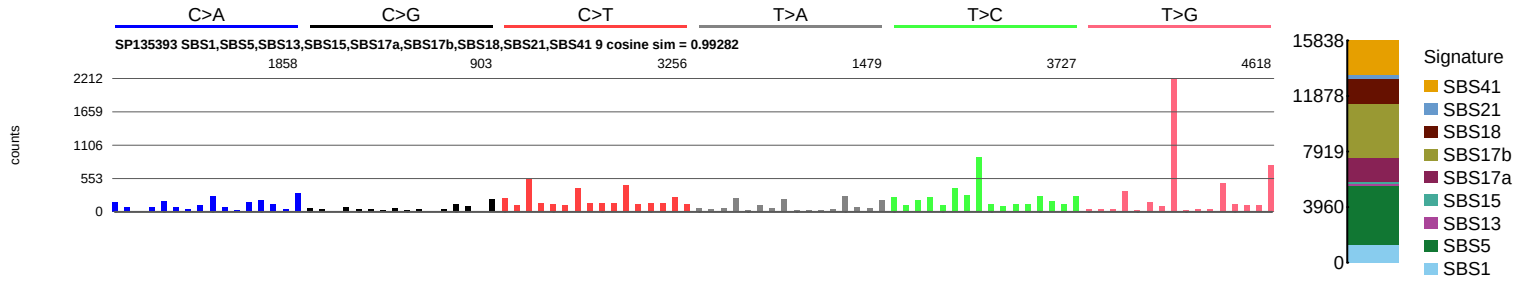
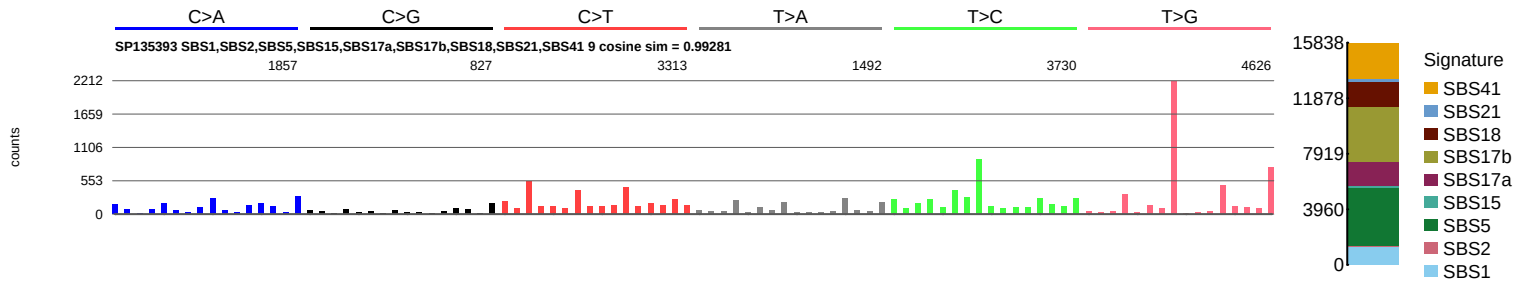
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99311
1875
1063
3169
1598
3481
4655
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

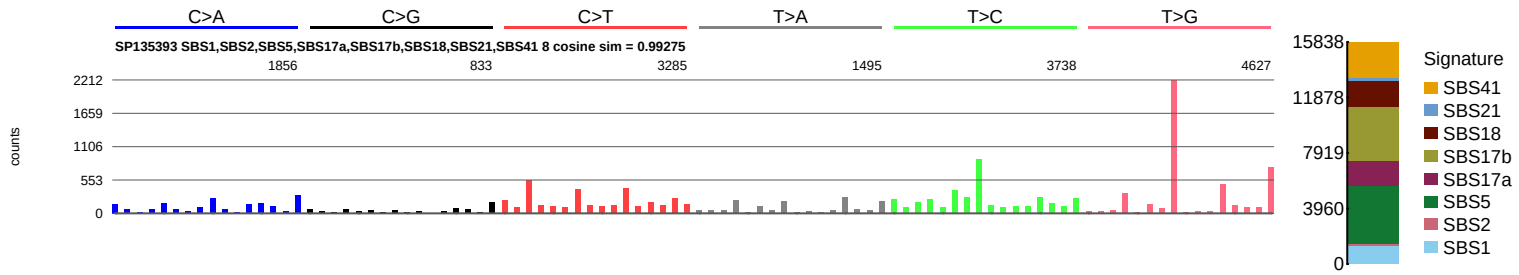

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9931
1878
1044
3170
1607
3480
4658
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1

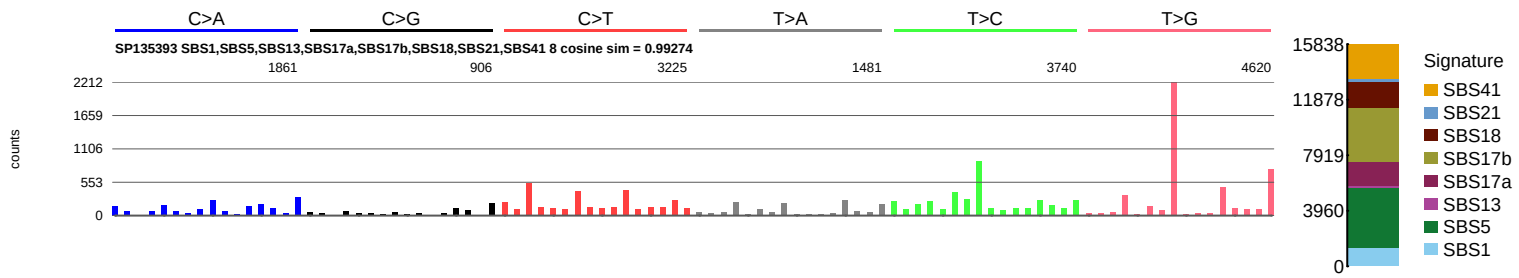

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 10 cosine sim = 0.99295
1850
893
3301
1472
3711
4609
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

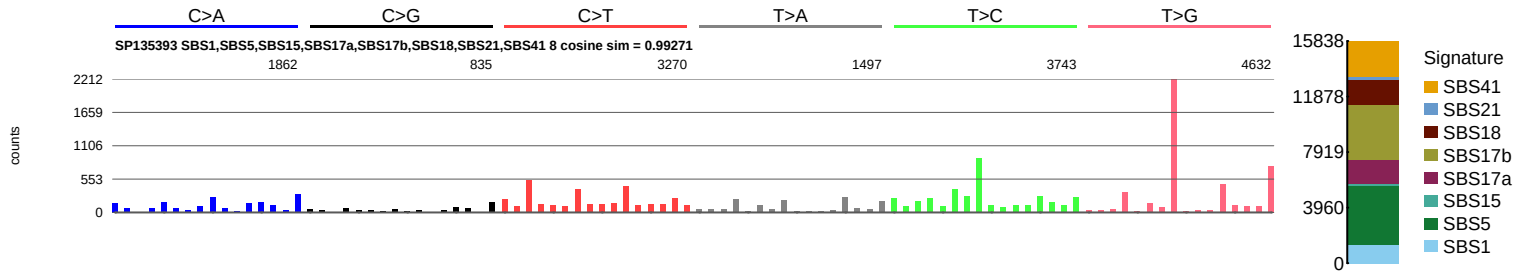

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99286
1853
900
3273
1476
3724
4614
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

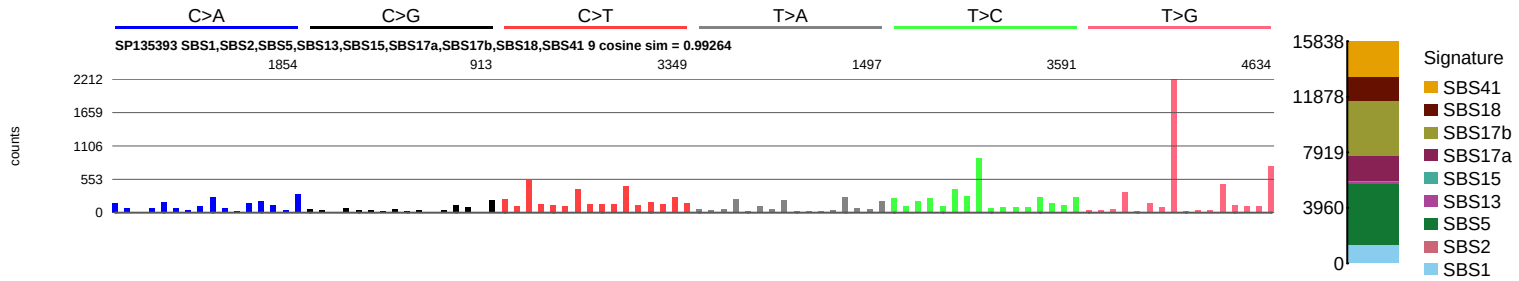

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99282
1858
903
3256
1479
3727
4618
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

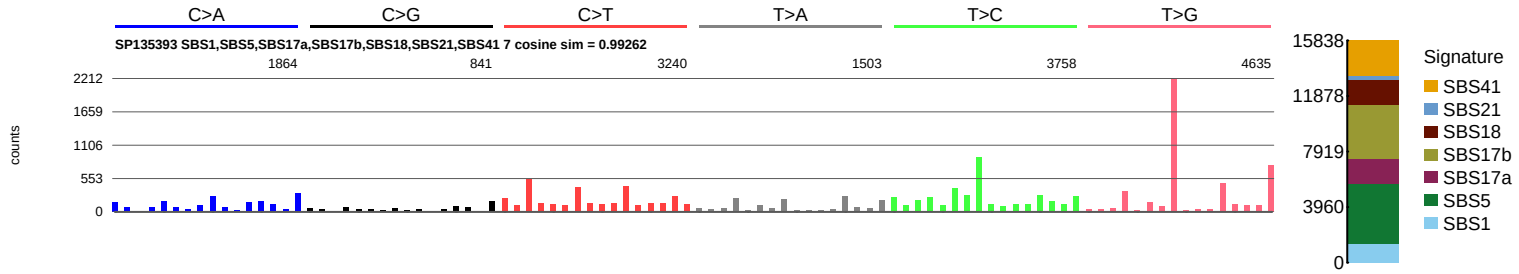

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99281
1857
827
3313
1492
3730
4626
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

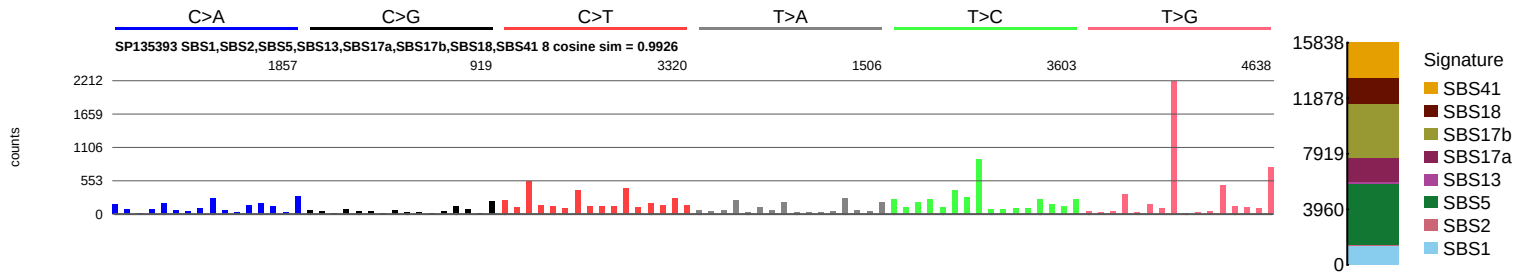

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99275
1856
833
3285
1495
3738
4627
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

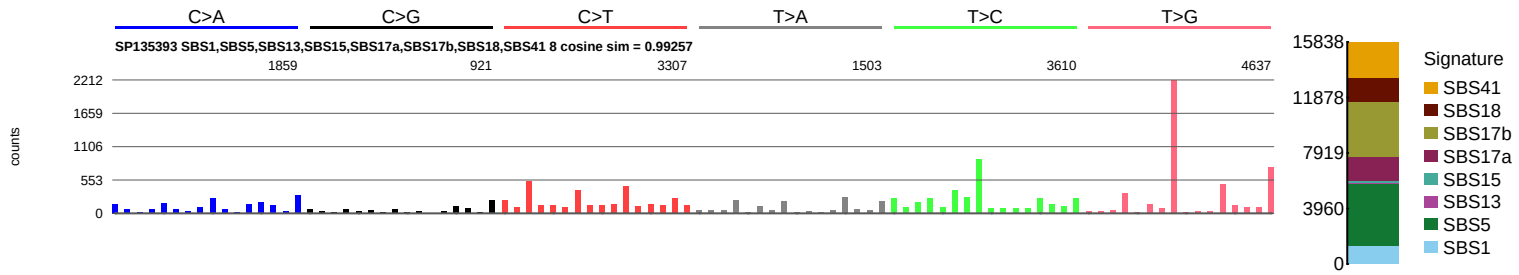

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99274
1861
906
3225
1481
3740
4620
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1

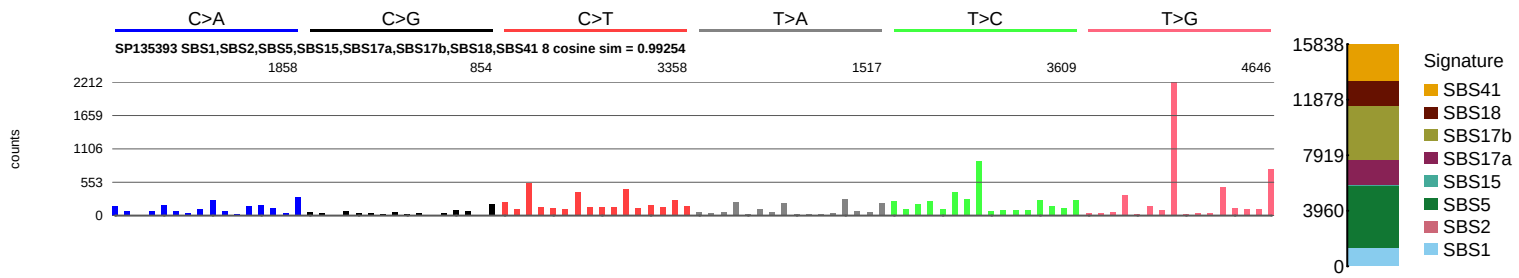

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99271
1862
835
3270
1497
3743
4632
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1

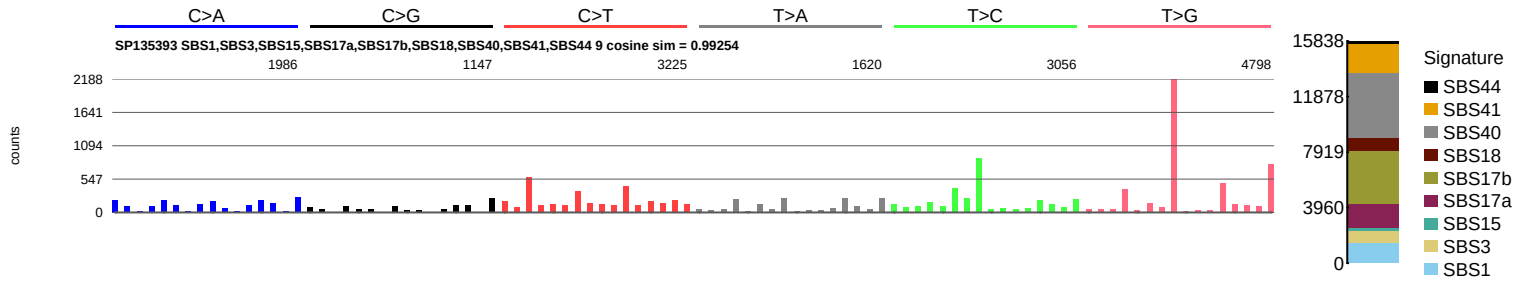

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 9 cosine sim = 0.99264
1854
913
3349
1497
3591
4634
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1

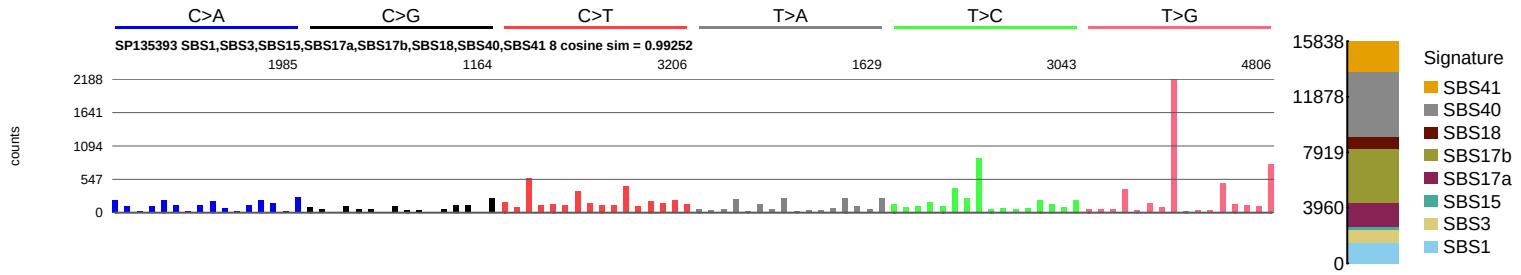

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.99262
1864
841
3240
1503
3758
4635
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

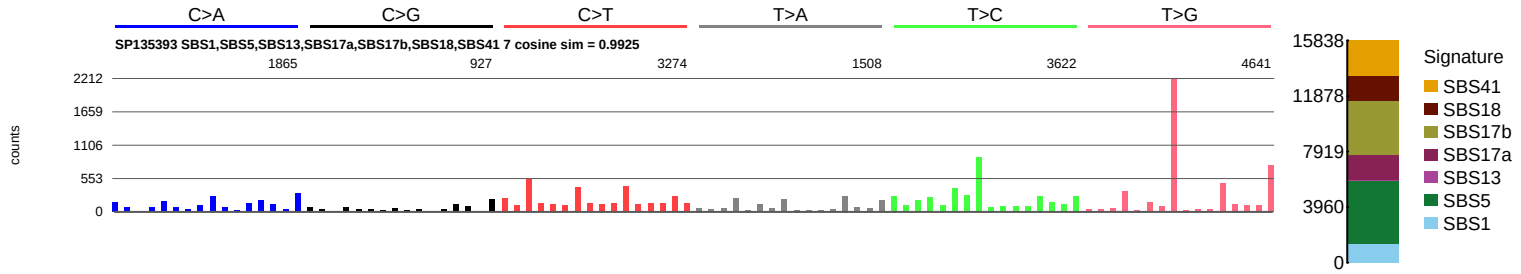

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.9926
1857
919
3320
1506
3603
4638
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1

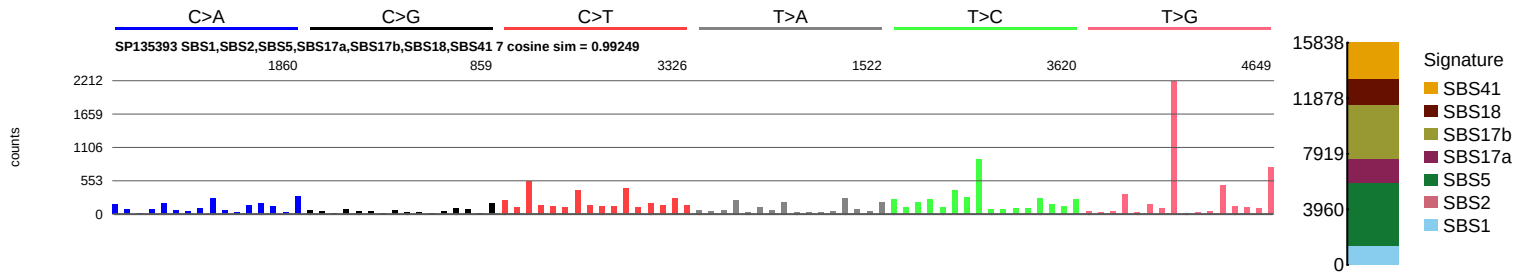

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99257
1859
921
3307
1503
3610
4637
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1

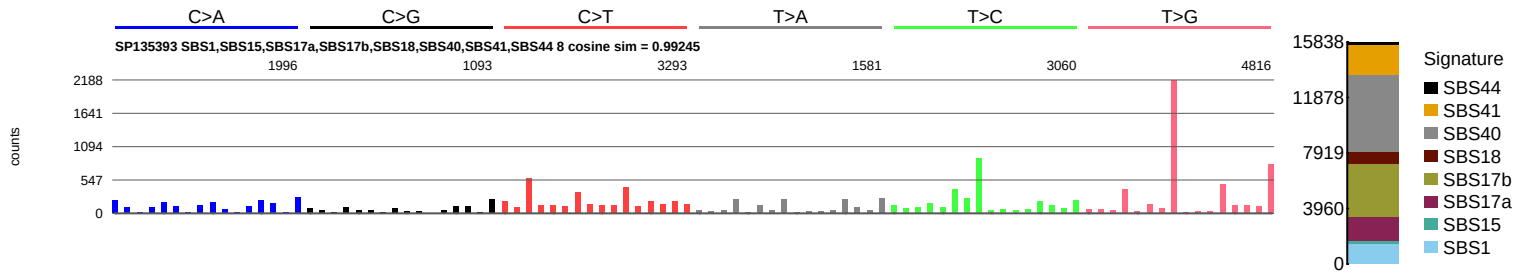

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99254
1858
854
3358
1517
3609
4646
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1

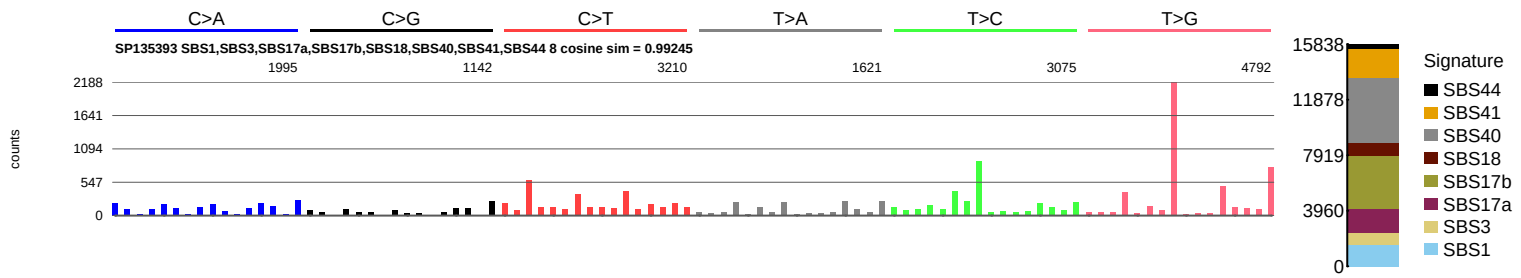

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 9 cosine sim = 0.99254
1986
1147
3225
1620
3056
4798
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

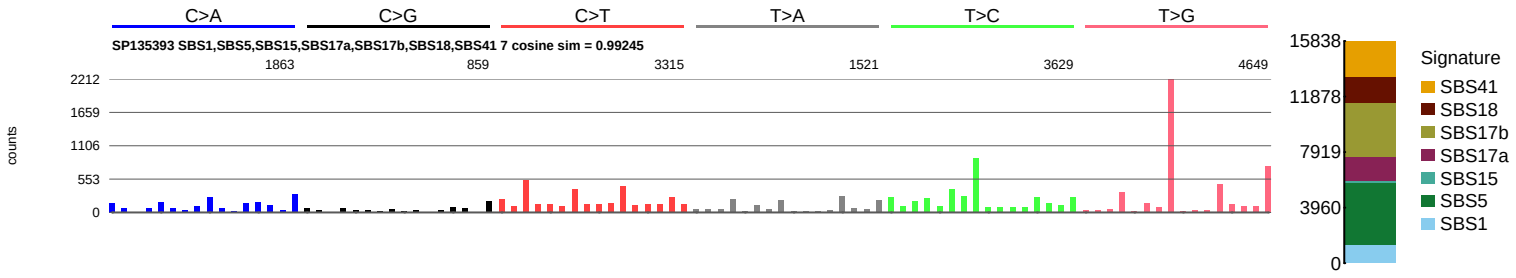

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 8 cosine sim = 0.99252
1985
1164
3206
1629
3043
4806
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

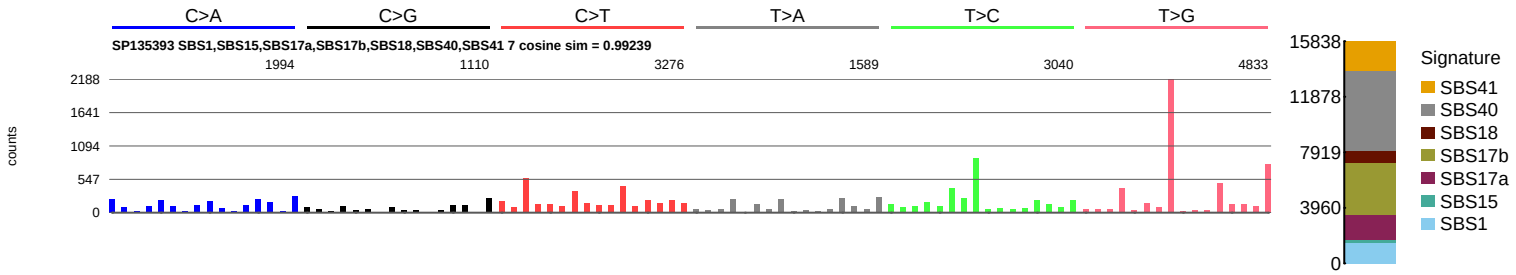

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.9925
1865
927
3274
1508
3622
4641
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99249
1860
859
3326
1522
3620
4649
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.99245
1996
1093
3293
1581
3060
4816
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 8 cosine sim = 0.99245
1995
1142
3210
1621
3075
4792
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99245
1863
859
3315
1521
3629
4649
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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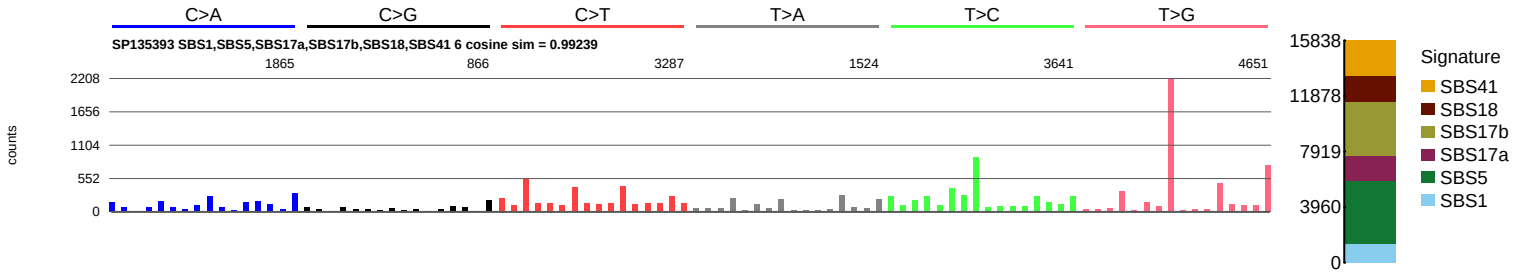
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99239
1994
1110
3276
1589
3040
4833
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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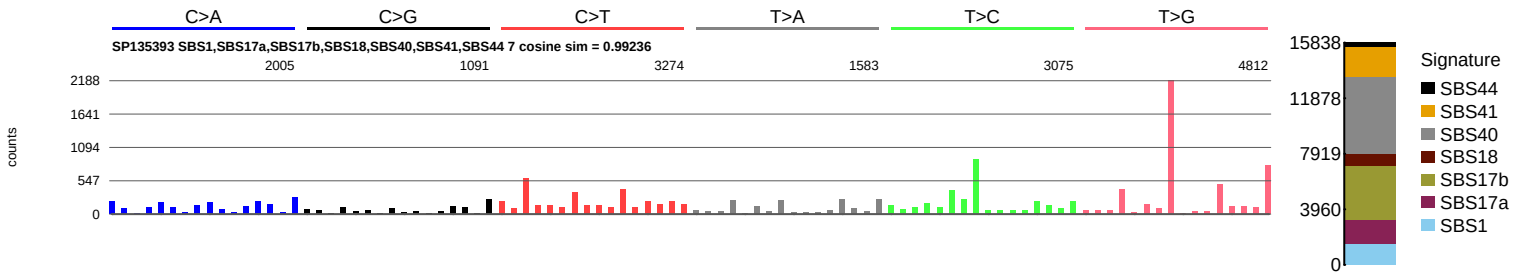
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.99239
1865
866
3287
1524
3641
4651
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS5
SBS1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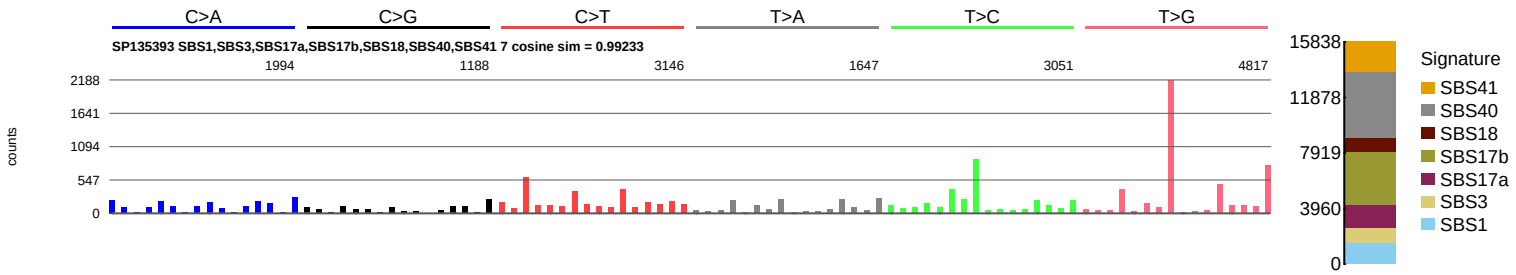
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41,SBS44 7 cosine sim = 0.99236
2005
1091
3274
1583
3075
4812
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1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C>G
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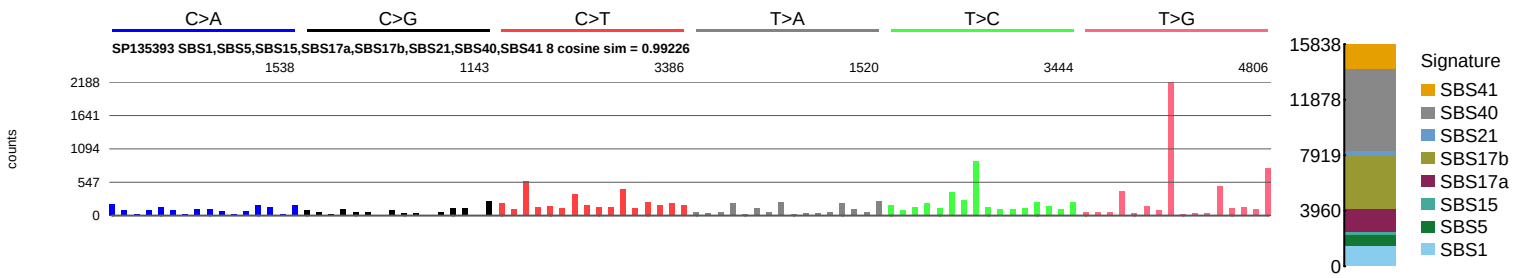
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS41 7 cosine sim = 0.99233
1994
1188
3146
1647
3051
4817
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS1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C>G
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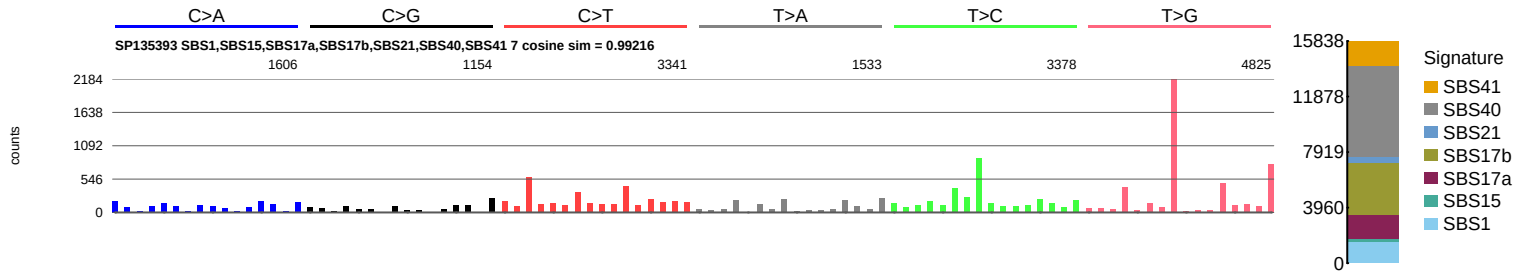
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 8 cosine sim = 0.99226
1538
1143
3386
1520
3444
4806
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
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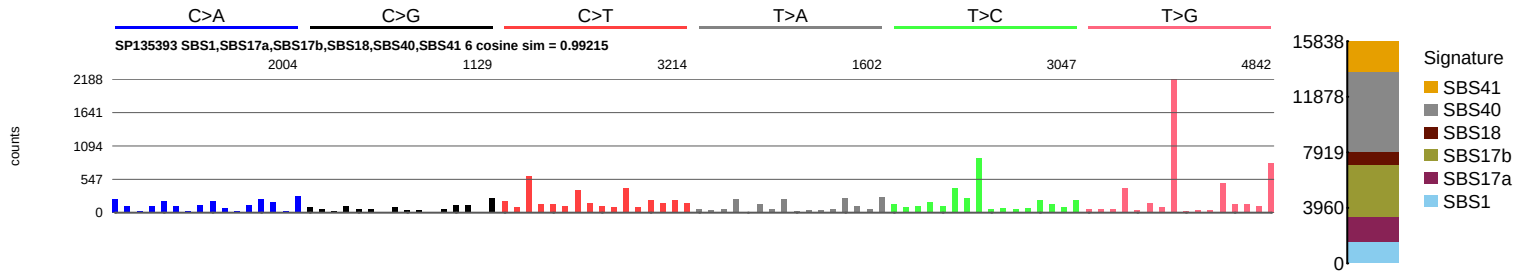
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99216
1606
1154
3341
1533
3378
4825
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1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C>G
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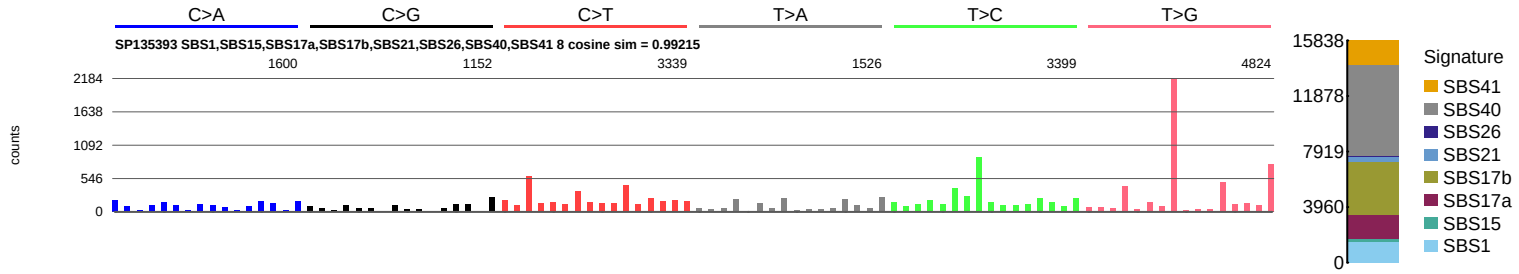
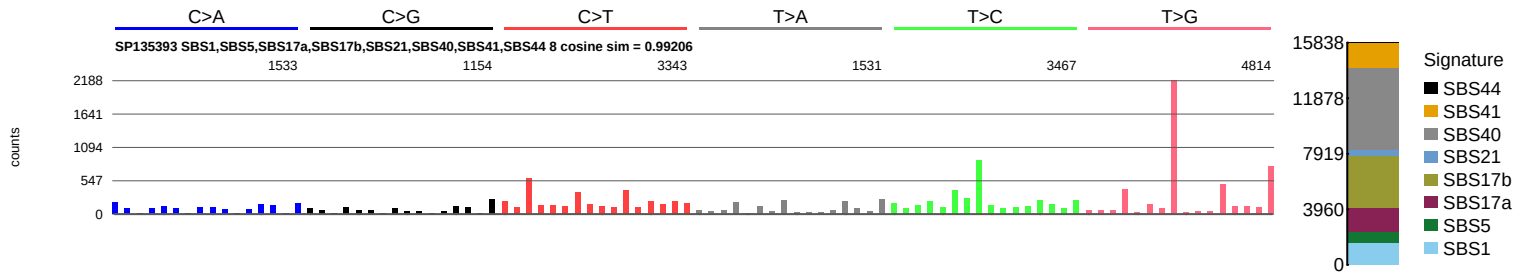
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS41 6 cosine sim = 0.99215
2004
1129
3214
1602
3047
4842
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 8 cosine sim = 0.99215
1600
1152
3339
1526
3399
4824
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
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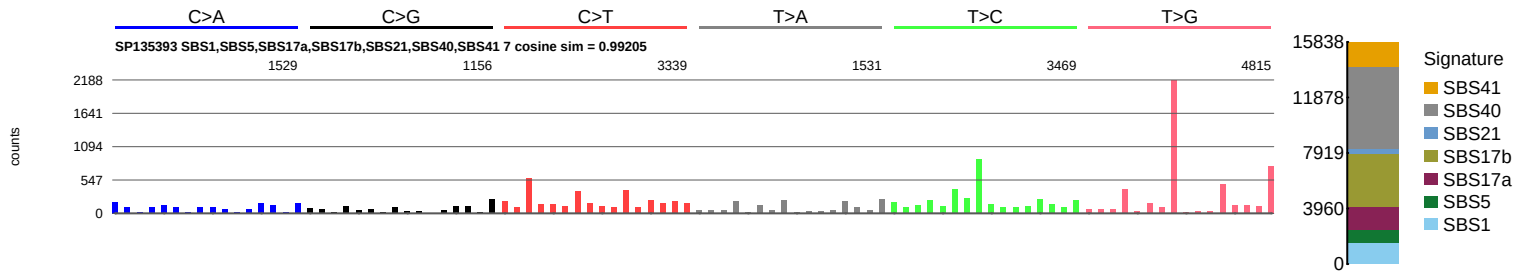
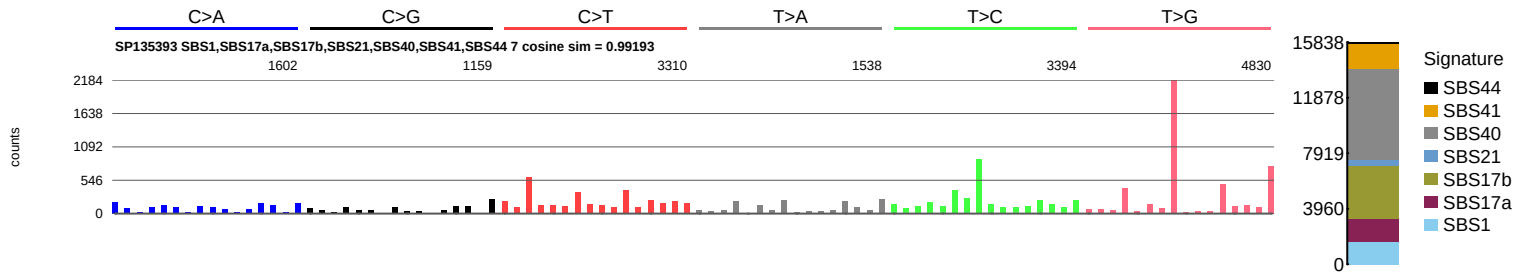
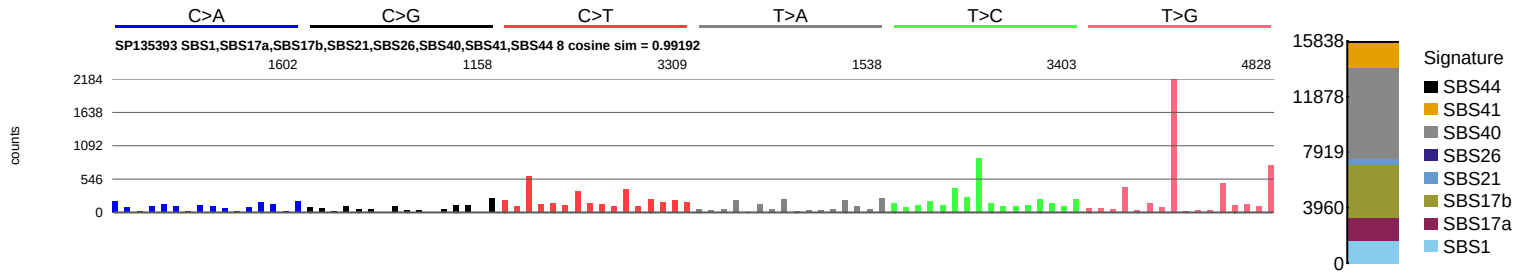
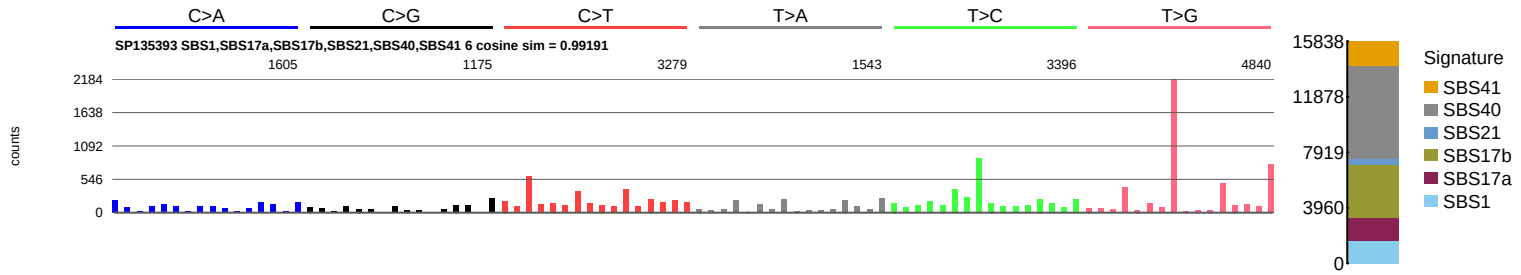
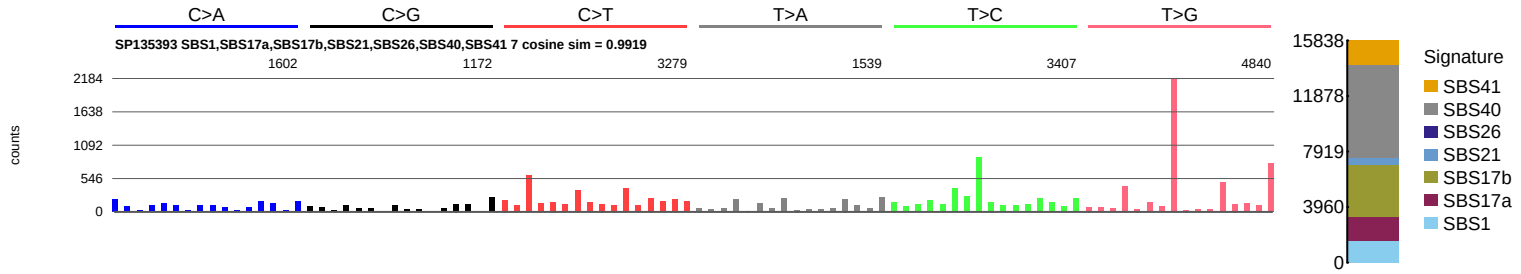
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 8 cosine sim = 0.99206
1533
1154
3343
1531
3467
4814
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1

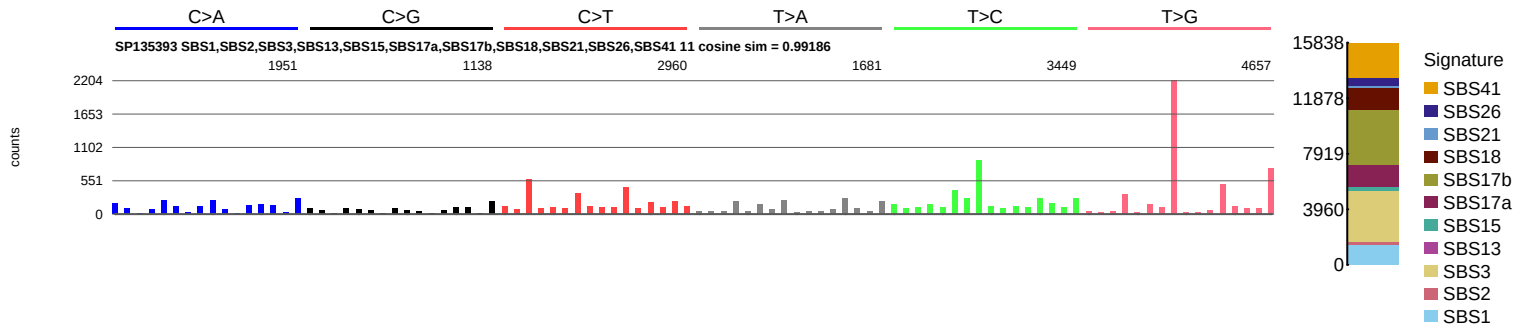

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS41 7 cosine sim = 0.99205
1529
1156
3339
1531
3469
4815
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41,SBS44 7 cosine sim = 0.99193
1602
1159
3310
1538
3394
4830
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99192
1602
1158
3309
1538
3403
4828
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS41 6 cosine sim = 0.99191
1605
1175
3279
1543
3396
4840
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS41 7 cosine sim = 0.9919
1602
1172
3279
1539
3407
4840
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1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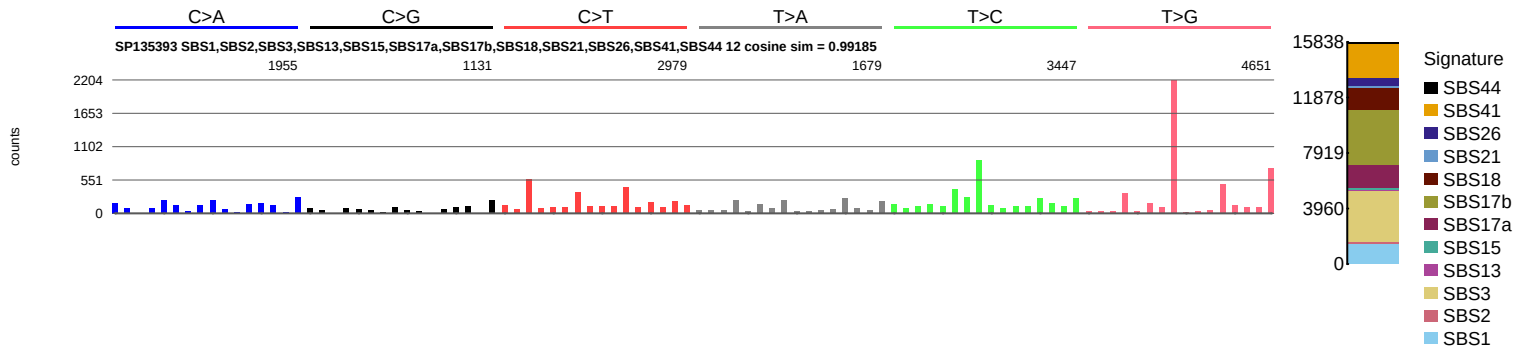
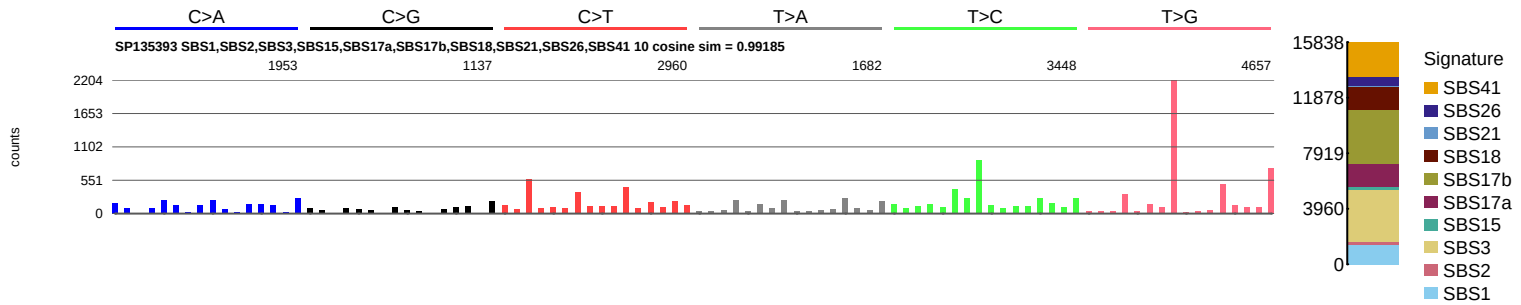
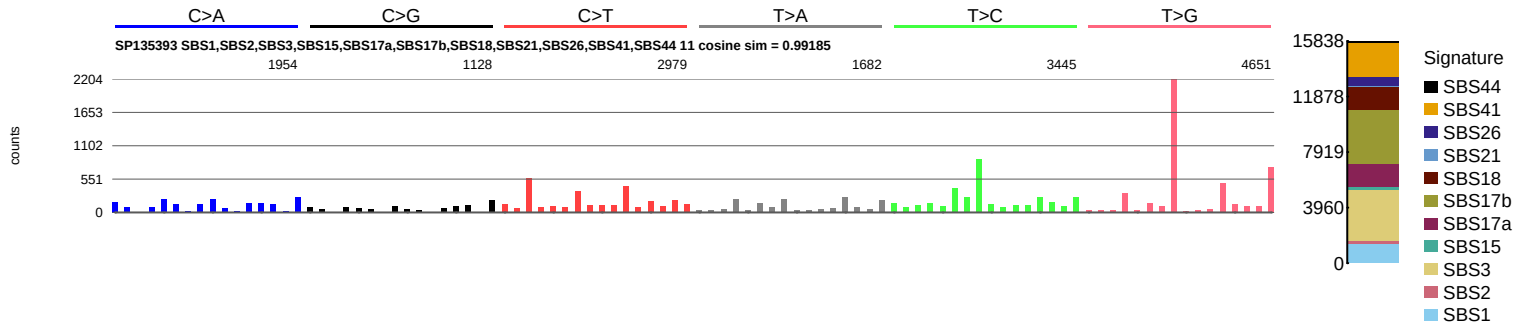
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 11 cosine sim = 0.99186
1951
1138
2960
1681
3449
4657
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 12 cosine sim = 0.99185
1955
1131
2979
1679
3447
4651
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.99185
1953
1137
2960
1682
3448
4657
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
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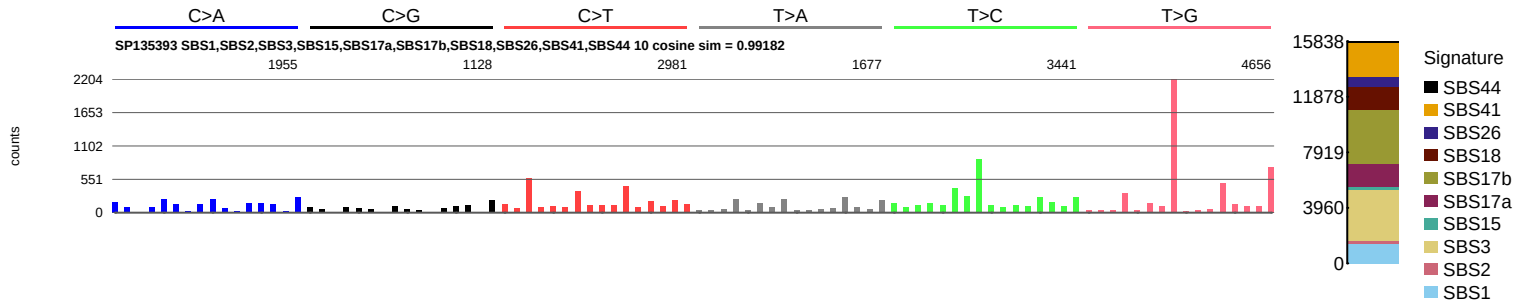
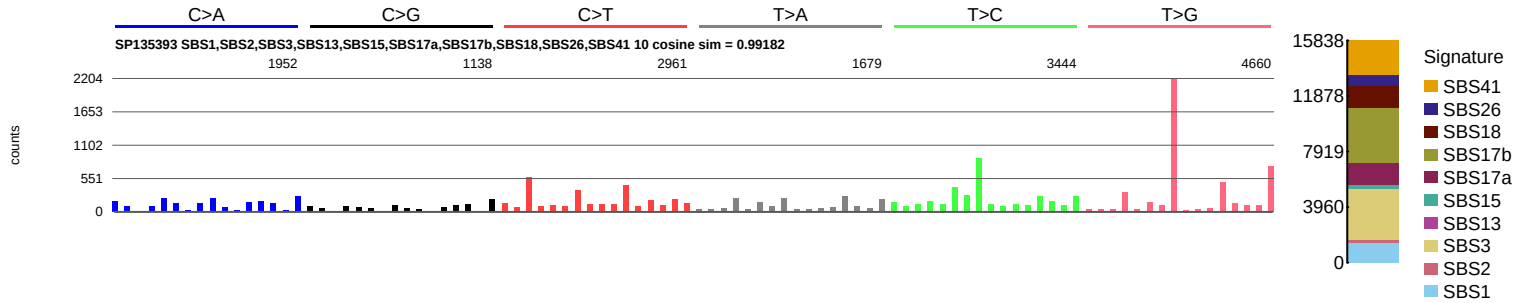
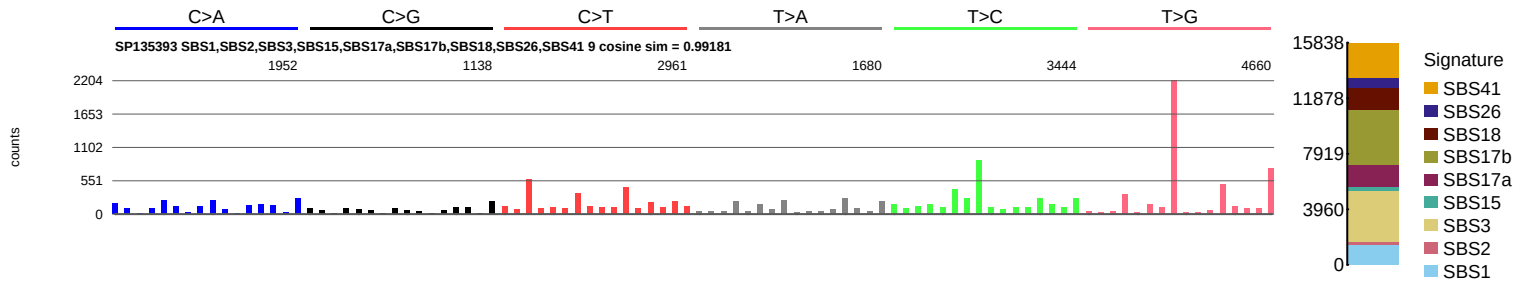
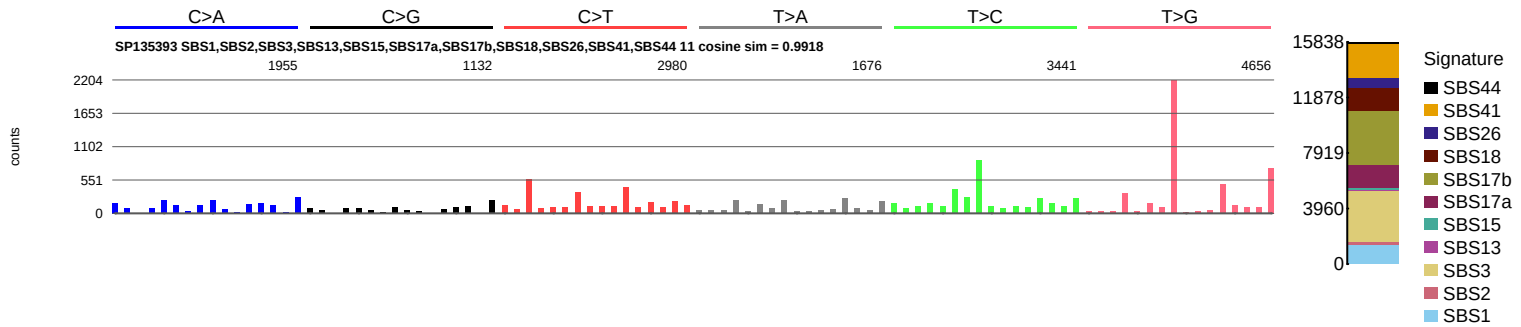
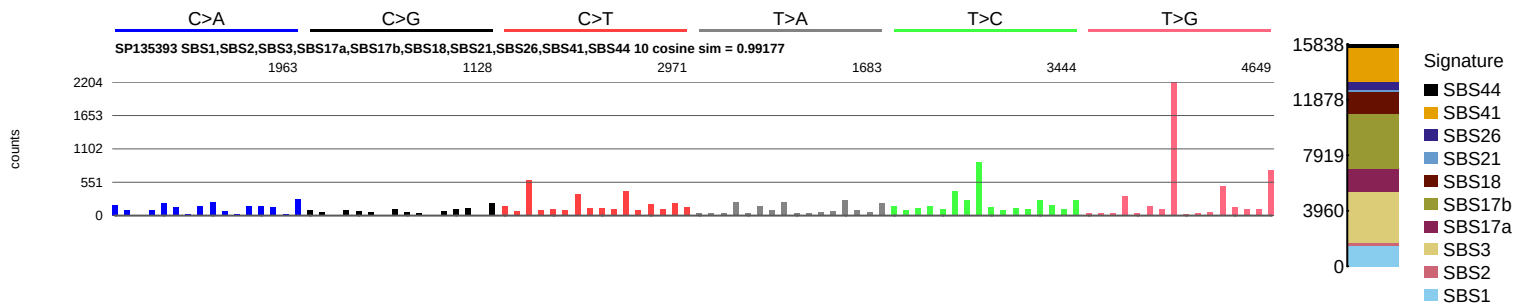
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.99185
1954
1128
2979
1682
3445
4651
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

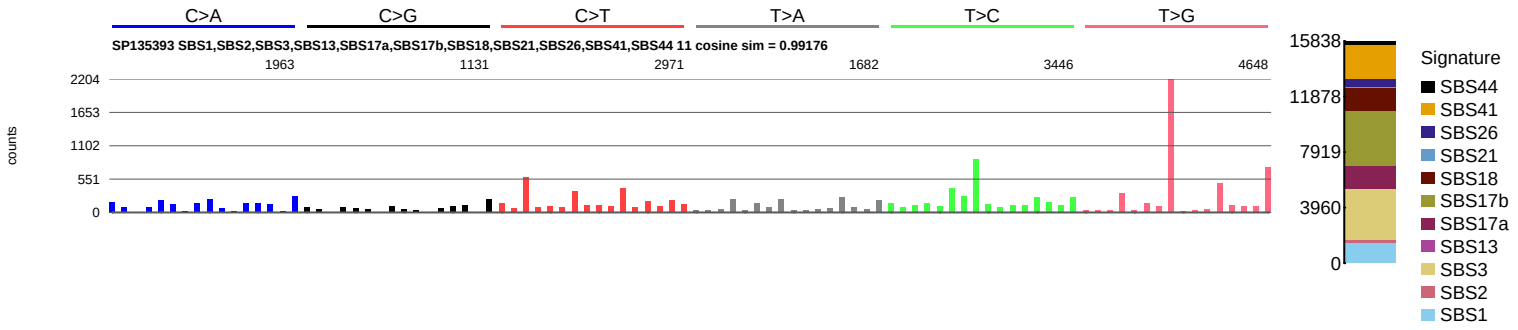

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.99182
1955
1128
2981
1677
3441
4656
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

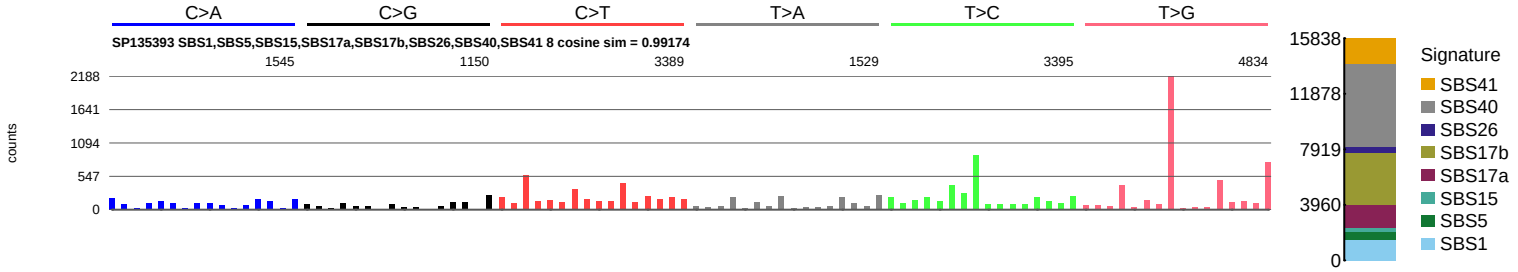

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 10 cosine sim = 0.99182
1952
1138
2961
1679
3444
4660
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.99181
1952
1138
2961
1680
3444
4660
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 11 cosine sim = 0.9918
1955
1132
2980
1676
3441
4656
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.99177
1963
1128
2971
1683
3444
4649
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.99176
1963
1131
2971
1682
3446
4648
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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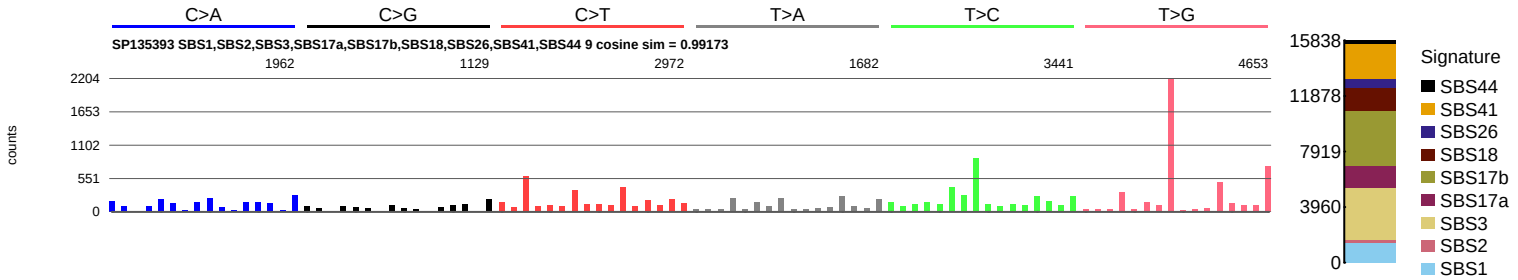
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.99174
1545
1150
3389
1529
3395
4834
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS1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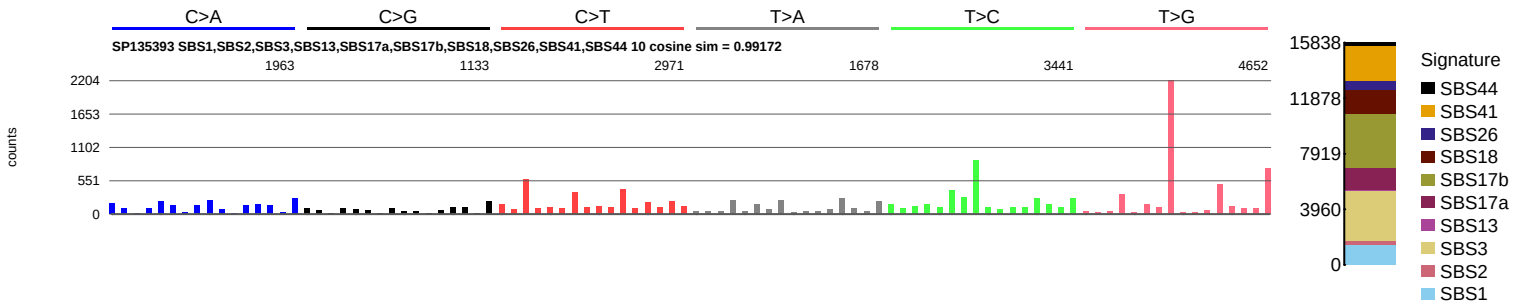
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99173
1962
1129
2972
1682
3441
4653
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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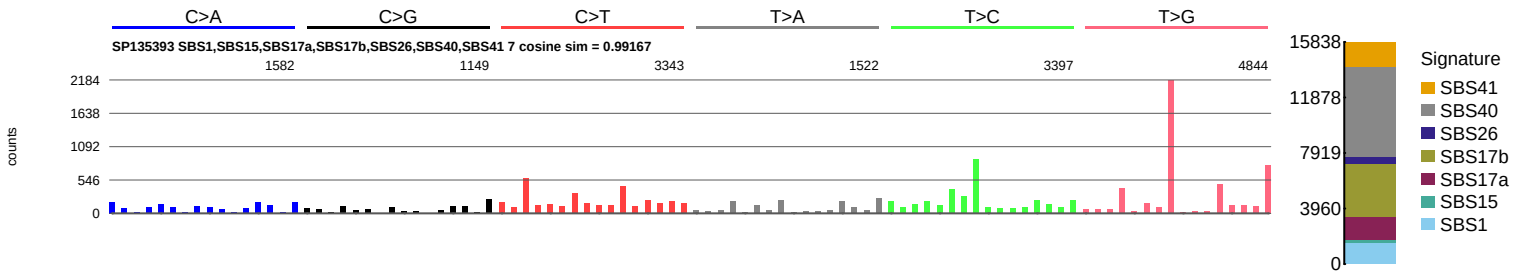
SP135393 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.99172
1963
1133
2971
1678
3441
4652
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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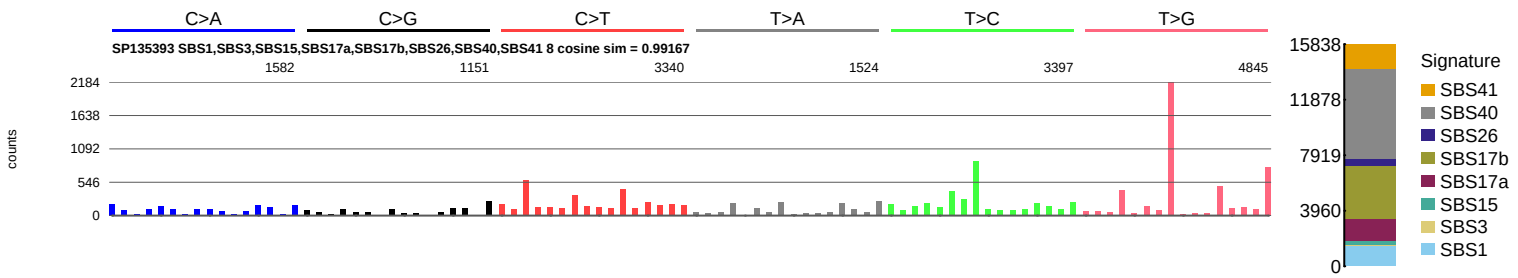
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99167
1582
1149
3343
1522
3397
4844
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS1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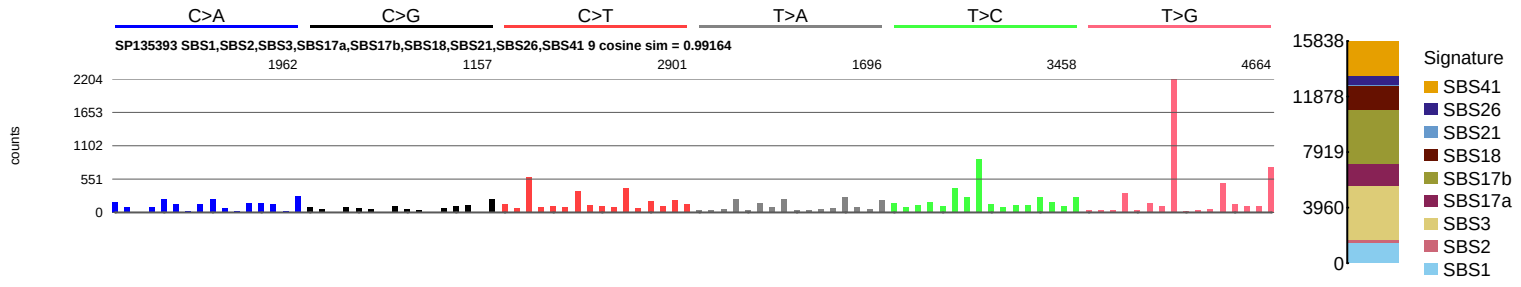
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS41 8 cosine sim = 0.99167
1582
1151
3340
1524
3397
4845
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1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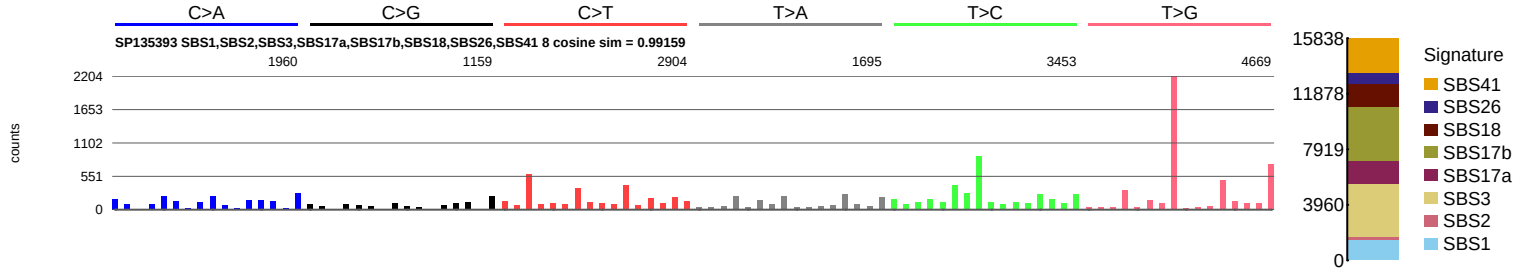
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99164
1962
1157
2901
1696
3458
4664
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
C>T
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T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99159
1960
1159
2904
1695
3453
4669
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
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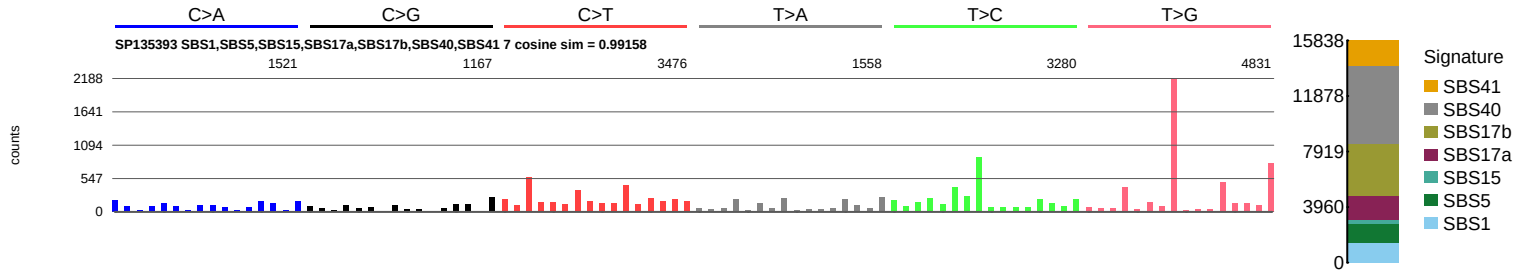
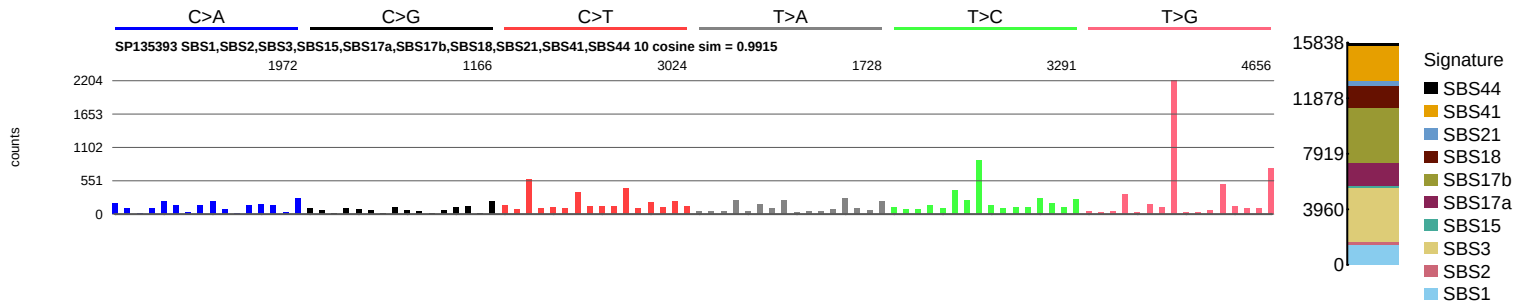
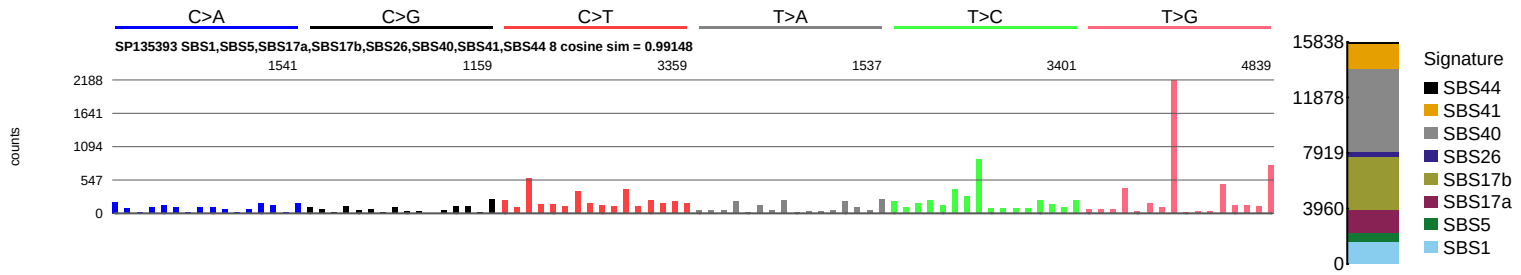
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99158
1521
1167
3476
1558
3280
4831
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.9915
1972
1166
3024
1728
3291
4656
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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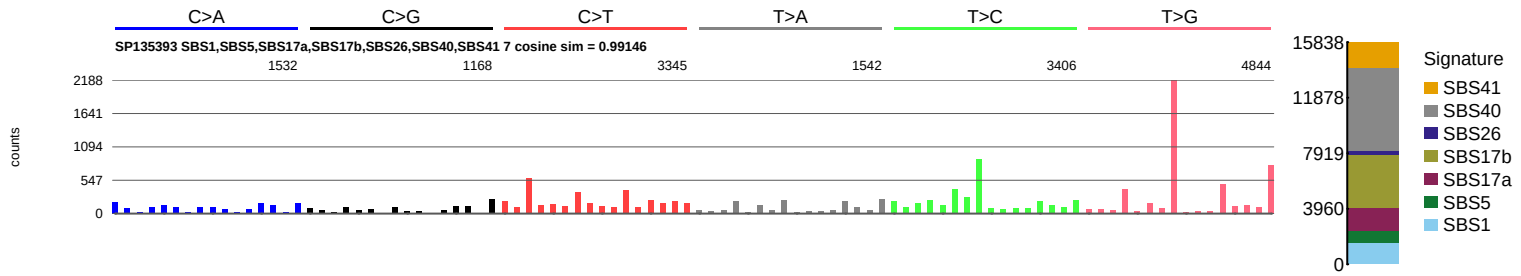
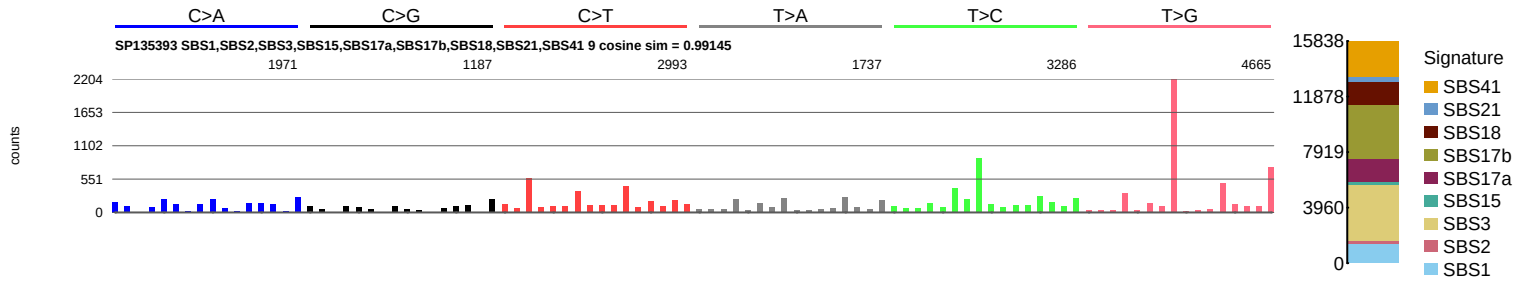
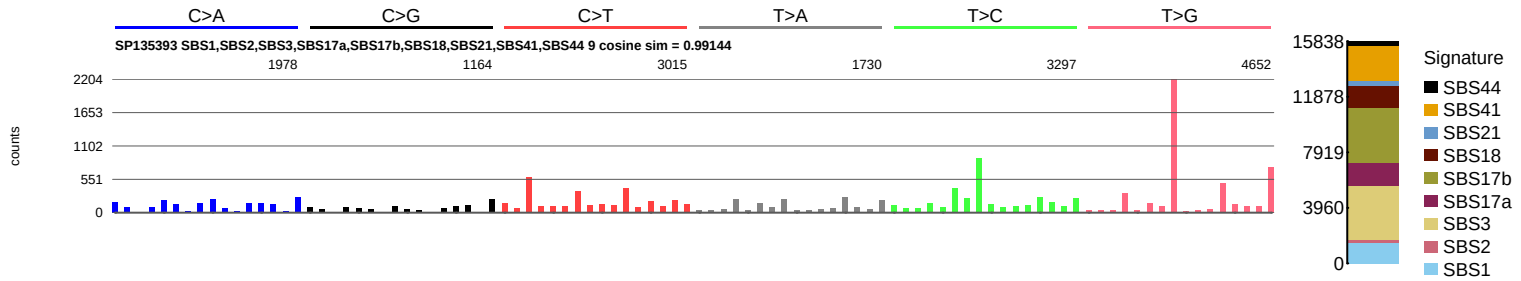
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99148
1541
1159
3359
1537
3401
4839
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99146
1532
1168
3345
1542
3406
4844
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.99145
1971
1187
2993
1737
3286
4665
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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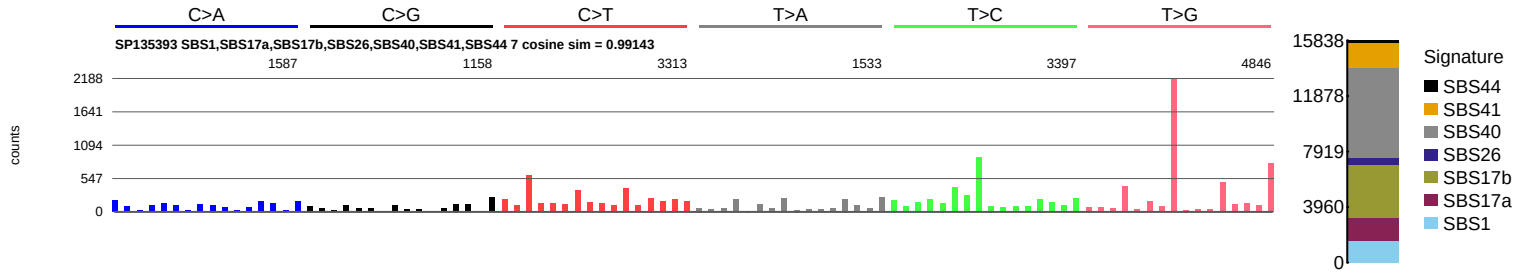
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.99144
1978
1164
3015
1730
3297
4652
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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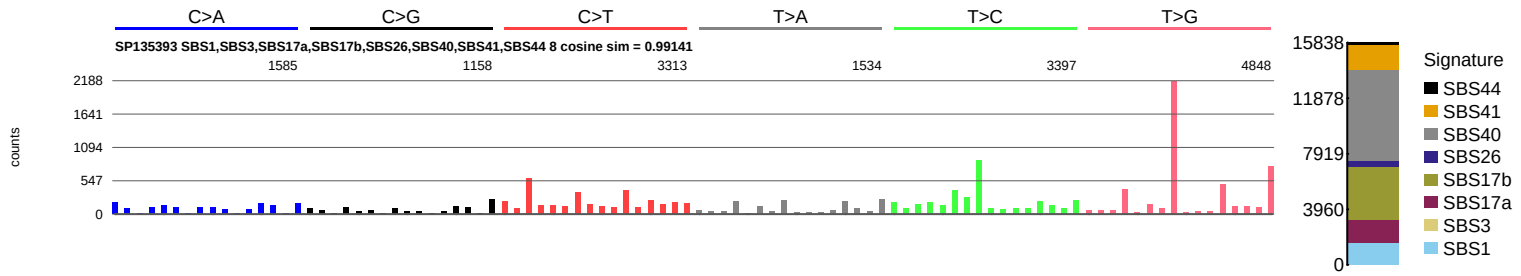
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 7 cosine sim = 0.99143
1587
1158
3313
1533
3397
4846
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1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C>G
C>T
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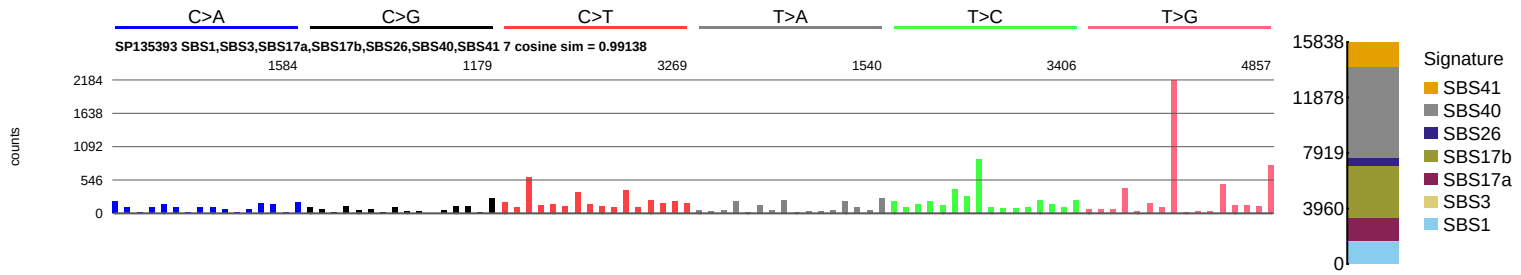
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS41,SBS44 8 cosine sim = 0.99141
1585
1158
3313
1534
3397
4848
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS1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T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS40,SBS41 7 cosine sim = 0.99138
1584
1179
3269
1540
3406
4857
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
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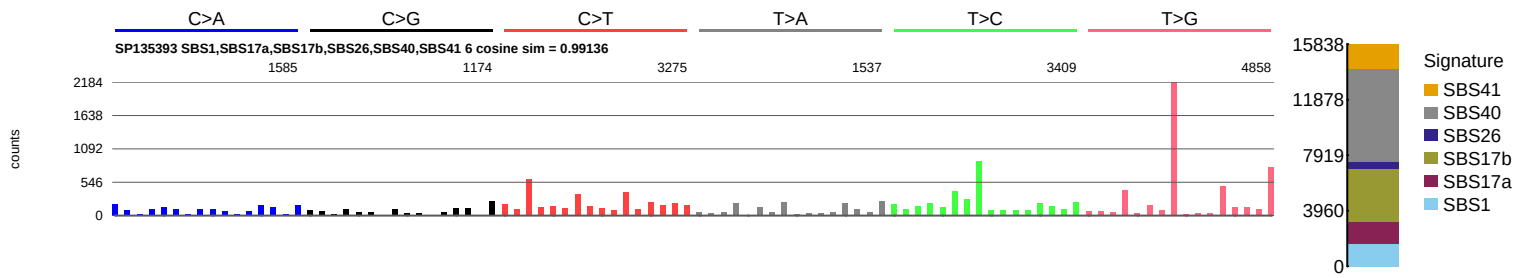
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS1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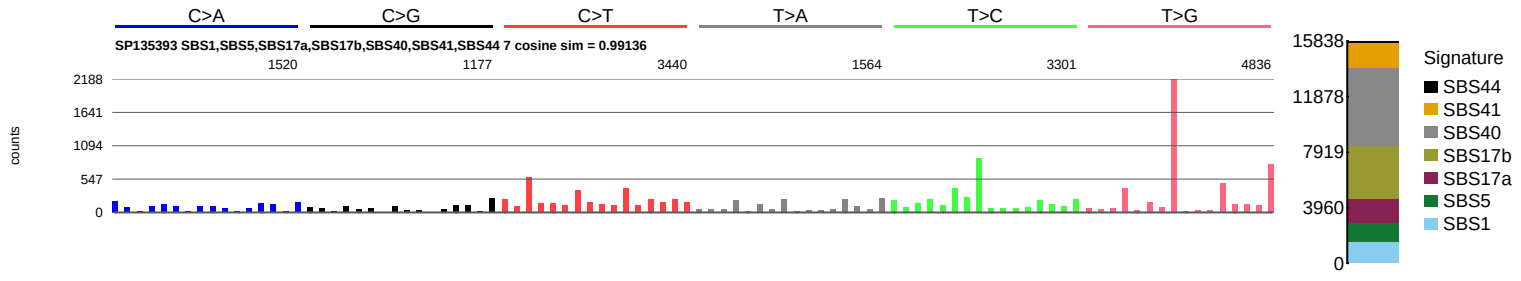
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS41 6 cosine sim = 0.99136
1585
1174
3275
1537
3409
4858
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS26
SBS17b
SBS17a
SBS1


C>A
C>G
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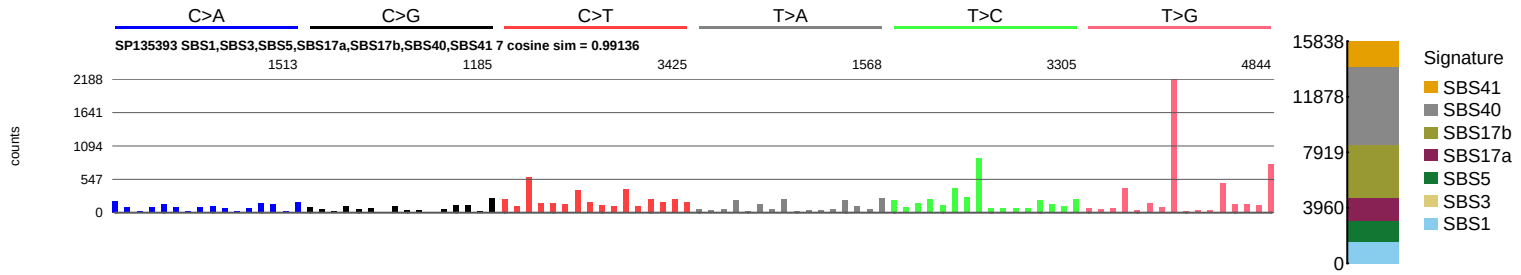
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.99136
1520
1177
3440
1564
3301
4836
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99136
1513
1185
3425
1568
3305
4844
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
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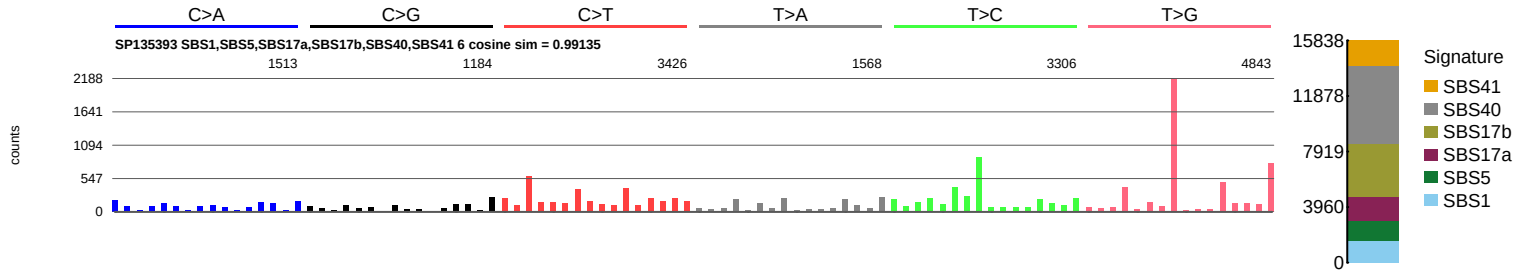
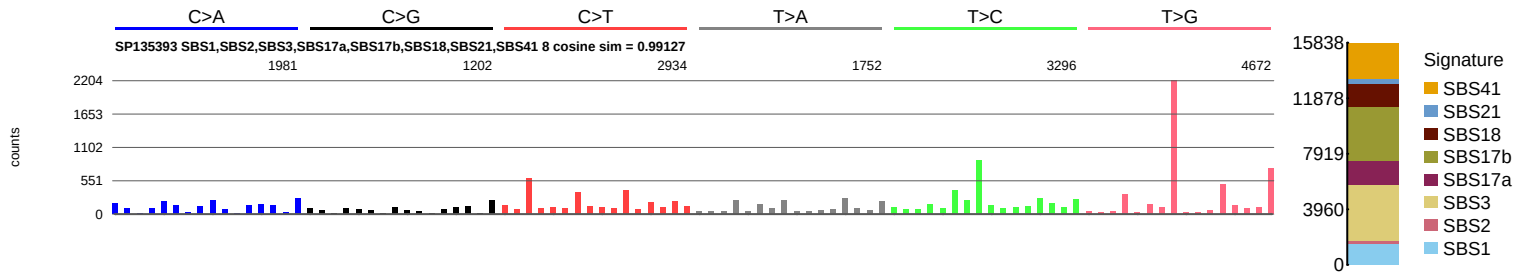
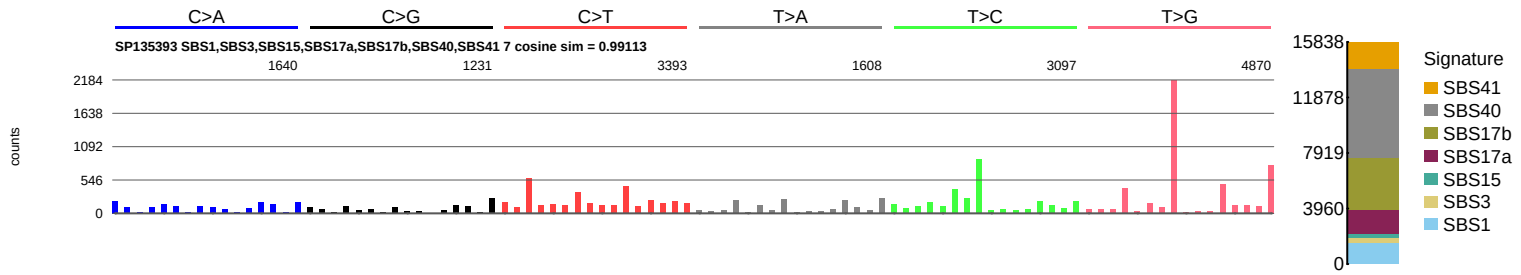
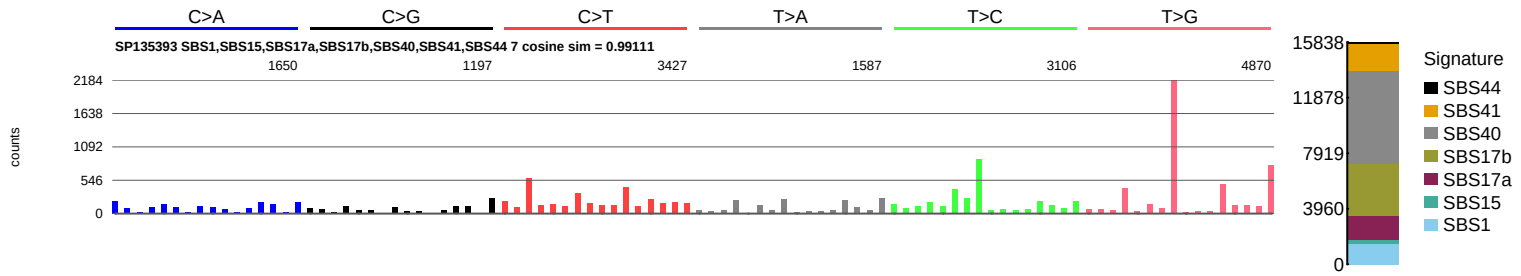
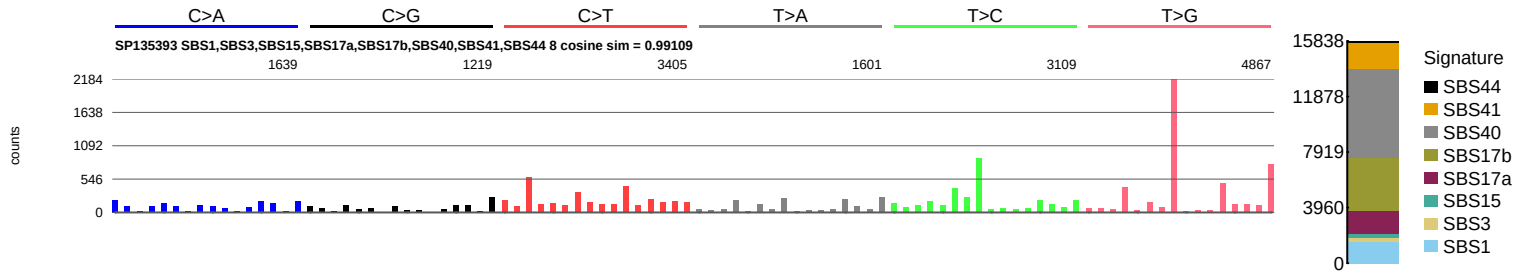
SBS40
SBS17b
SBS17a
SBS5
SBS3
SBS1

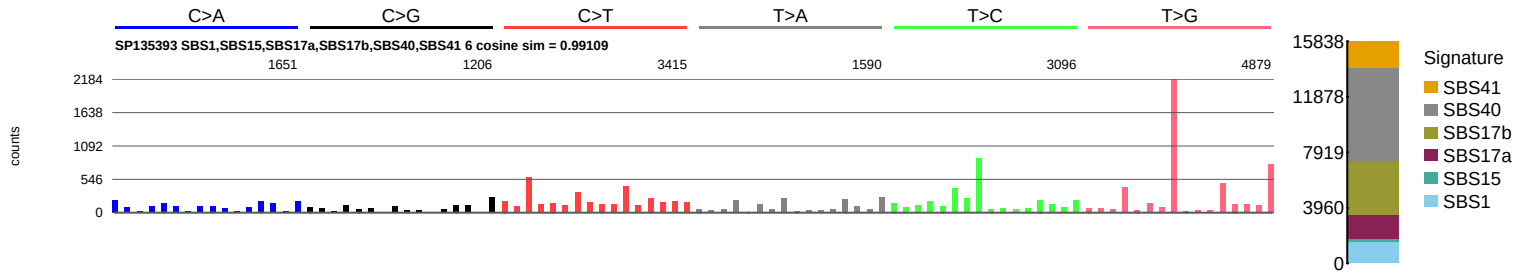

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99135
1513
1184
3426
1568
3306
4843
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS5
SBS1

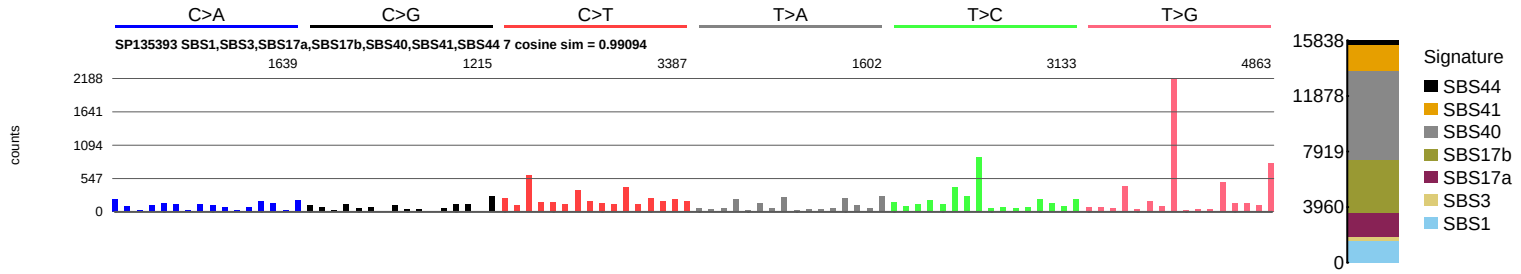

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99127
1981
1202
2934
1752
3296
4672
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41 7 cosine sim = 0.99113
1640
1231
3393
1608
3097
4870
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.99111
1650
1197
3427
1587
3106
4870
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS41,SBS44 8 cosine sim = 0.99109
1639
1219
3405
1601
3109
4867
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.99109
1651
1206
3415
1590
3096
4879
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS15
SBS1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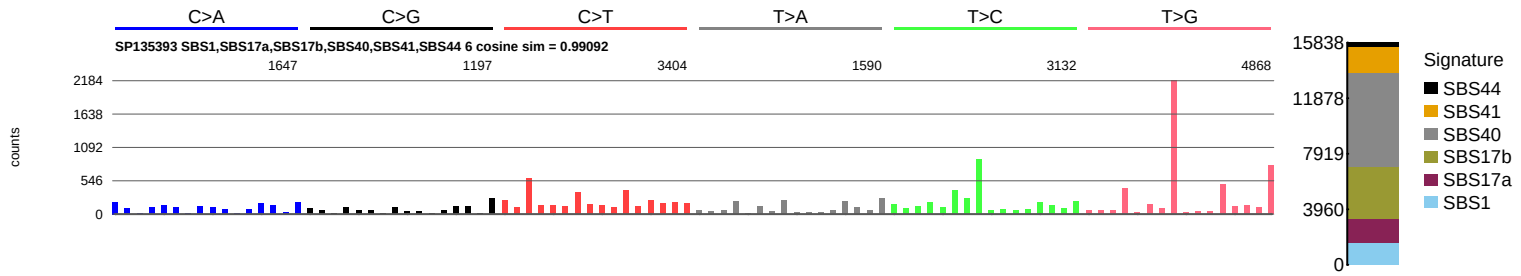
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS40,SBS41,SBS44 7 cosine sim = 0.99094
1639
1215
3387
1602
3133
4863
counts
2188
1641
1094
547
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS3
SBS1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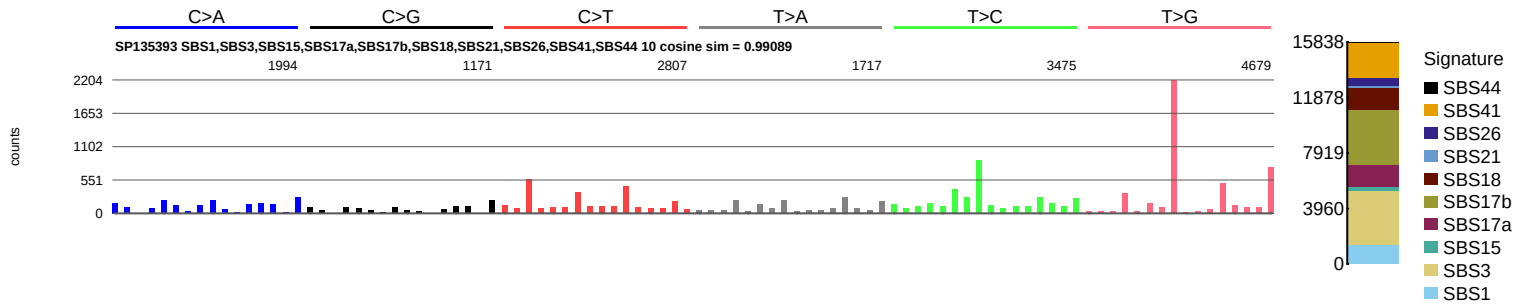
SP135393 SBS1,SBS17a,SBS17b,SBS40,SBS41,SBS44 6 cosine sim = 0.99092
1647
1197
3404
1590
3132
4868
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS40
SBS17b
SBS17a
SBS1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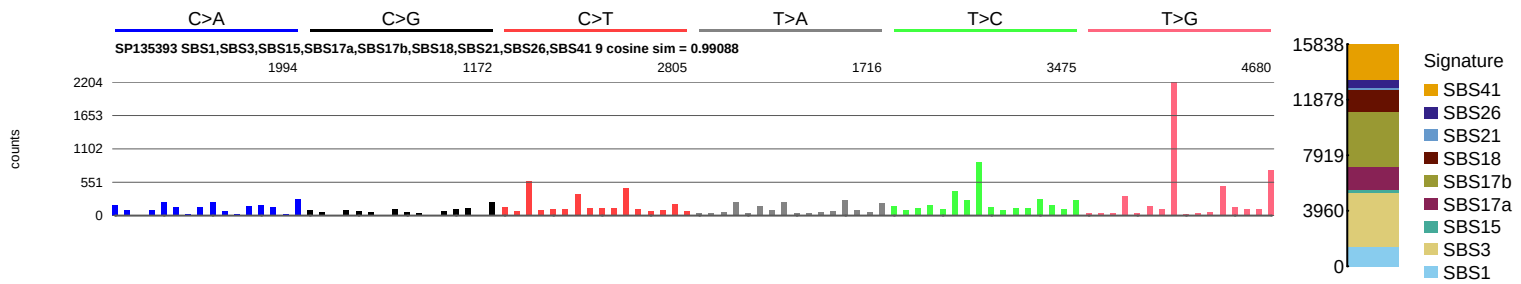
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.99089
1994
1171
2807
1717
3475
4679
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>T
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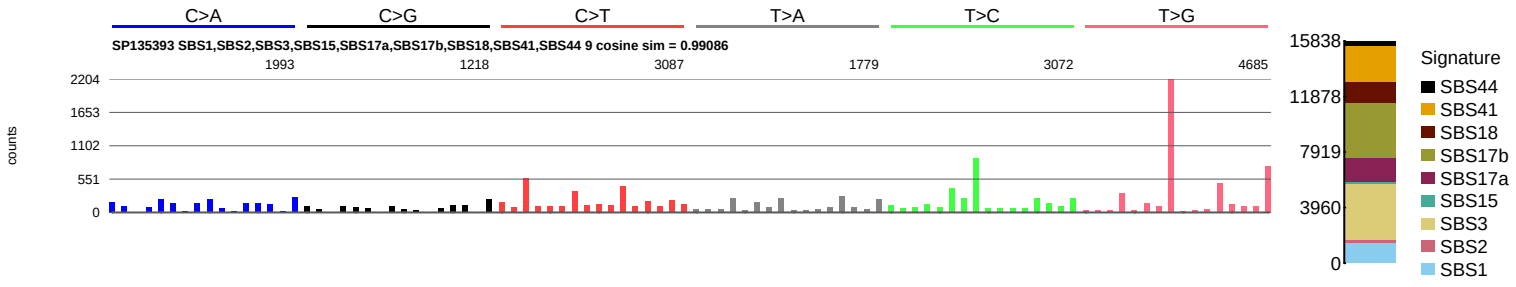
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.99088
1994
1172
2805
1716
3475
4680
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>G
C>T
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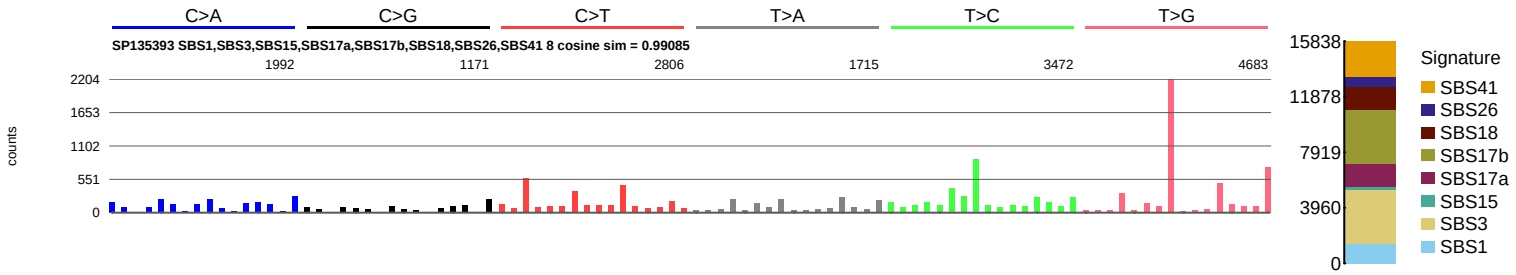
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 9 cosine sim = 0.99086
1993
1218
3087
1779
3072
4685
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
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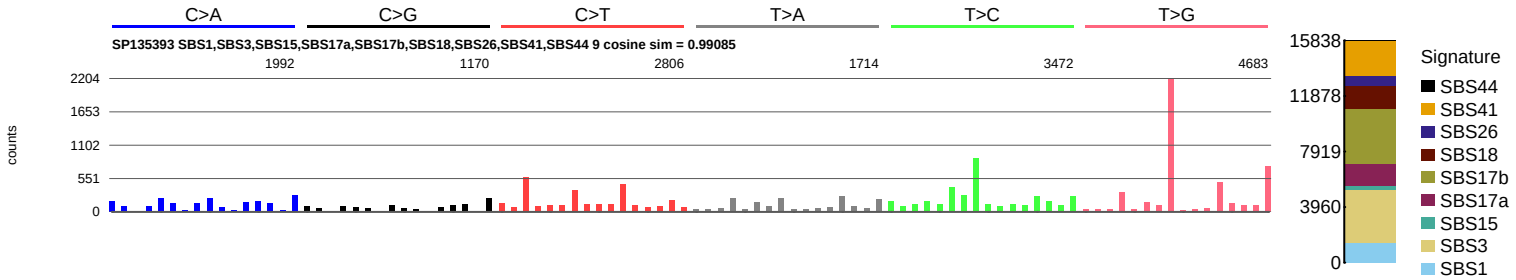
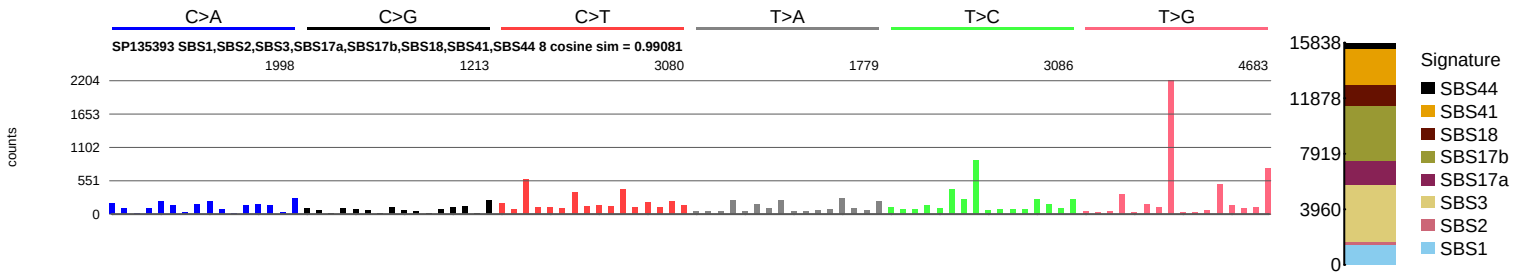
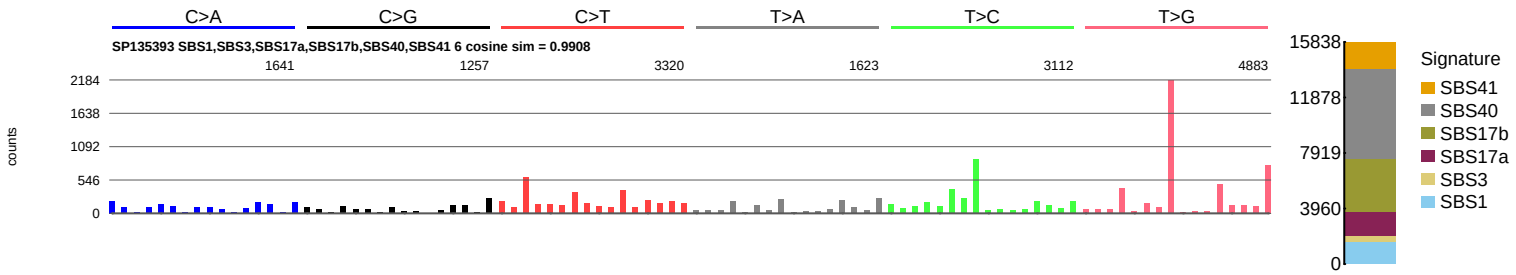
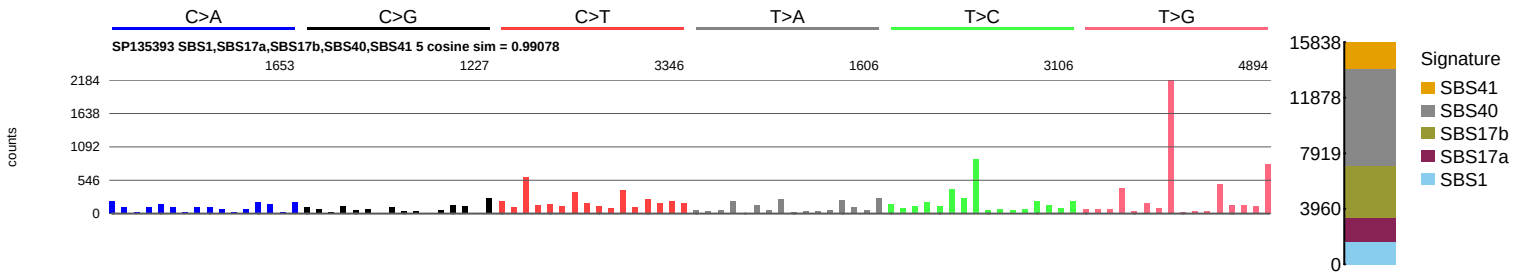
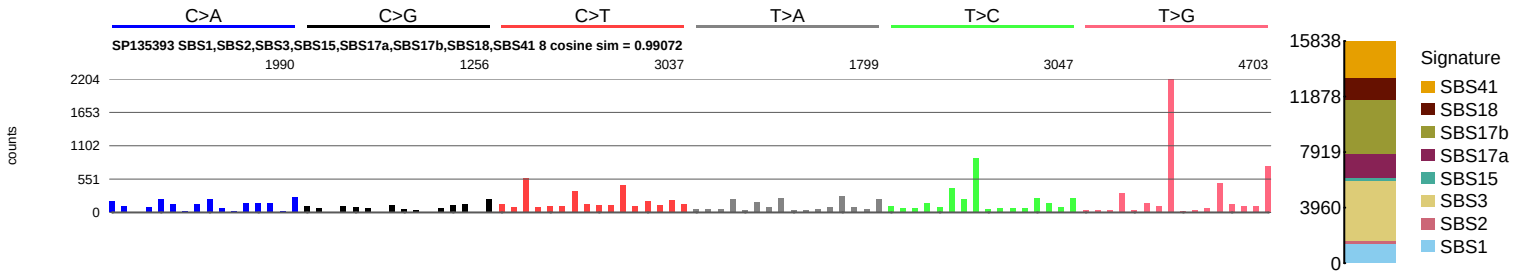
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.99085
1992
1171
2806
1715
3472
4683
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

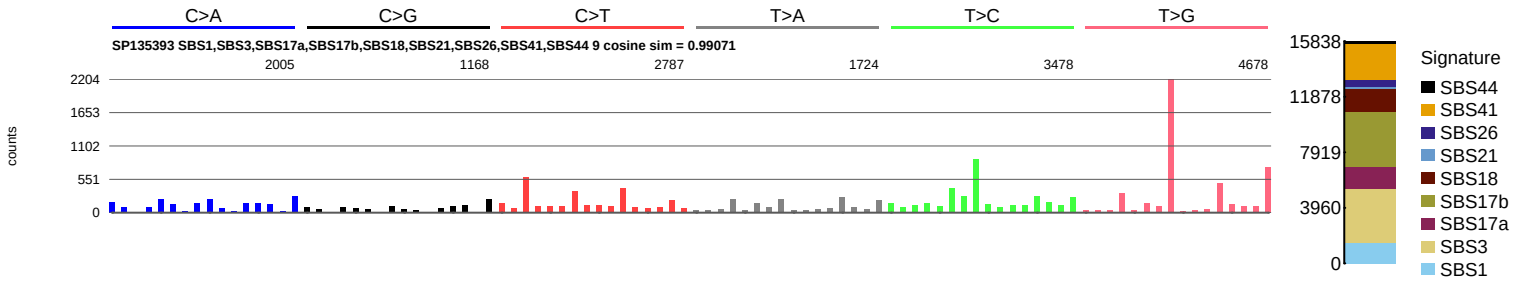

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.99085
1992
1170
2806
1714
3472
4683
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

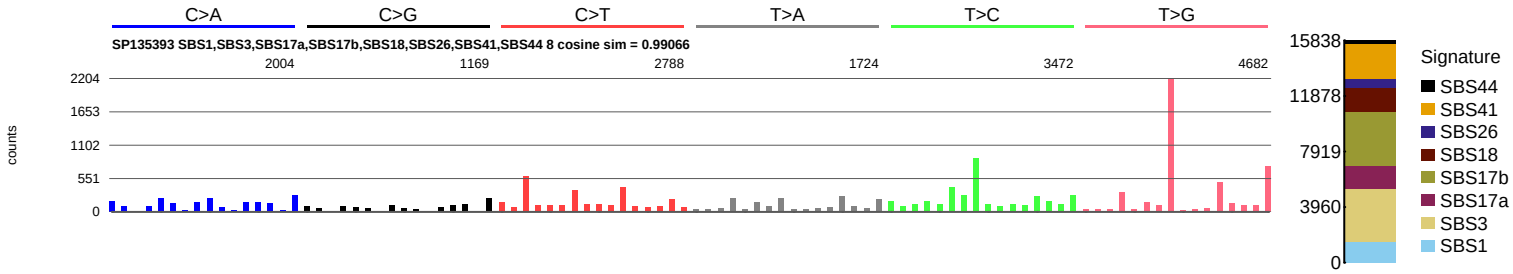

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.99081
1998
1213
3080
1779
3086
4683
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

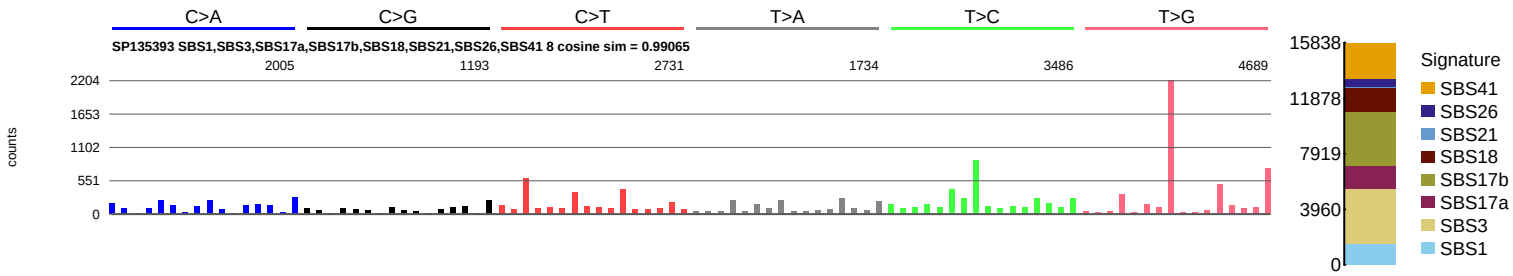

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS40,SBS41 6 cosine sim = 0.9908
1641
1257
3320
1623
3112
4883
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS3
SBS1

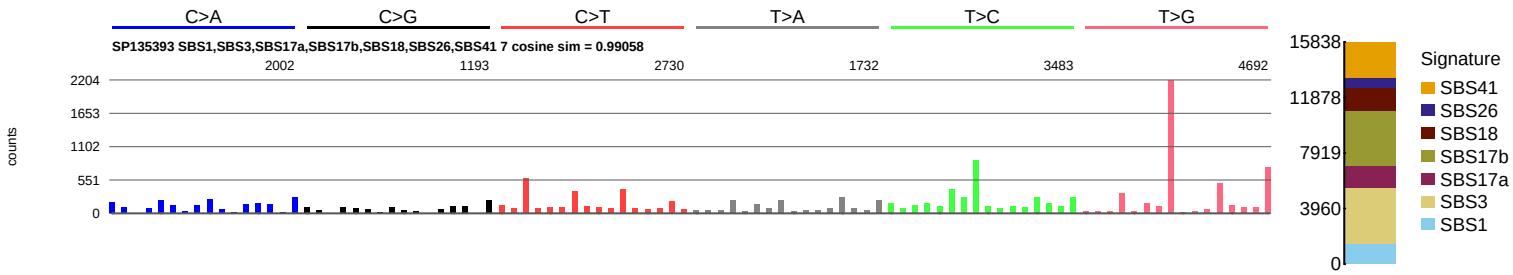

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS40,SBS41 5 cosine sim = 0.99078
1653
1227
3346
1606
3106
4894
counts
2184
1638
1092
546
0
15838
11878
7919
3960
0
Signature
SBS41
SBS40
SBS17b
SBS17a
SBS1

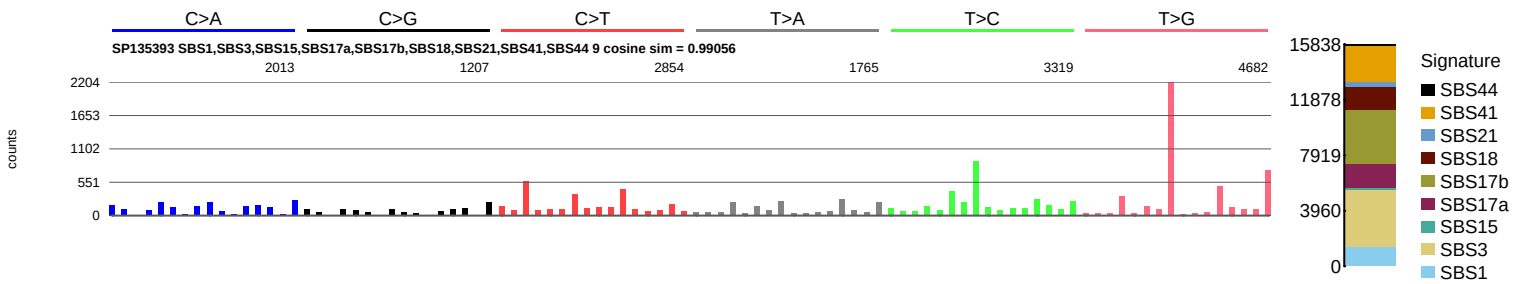

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.99072
1990
1256
3037
1799
3047
4703
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.99071
2005
1168
2787
1724
3478
4678
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.99066
2004
1169
2788
1724
3472
4682
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.99065
2005
1193
2731
1734
3486
4689
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.99058
2002
1193
2730
1732
3483
4692
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1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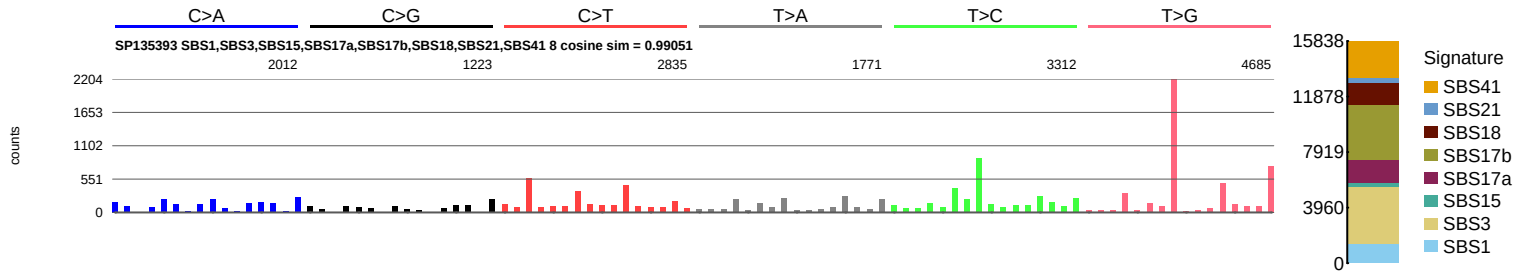
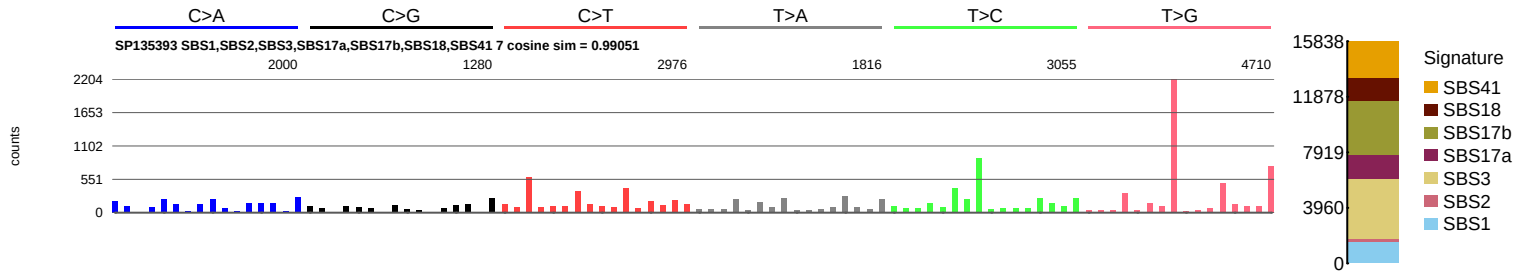
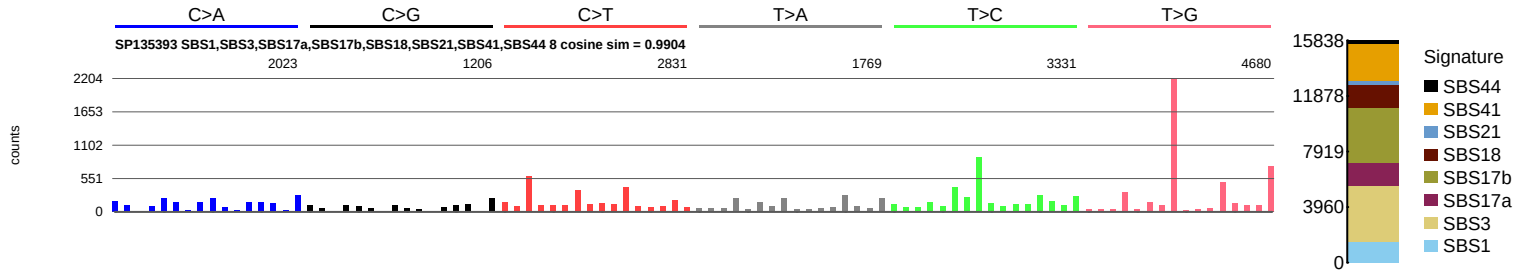
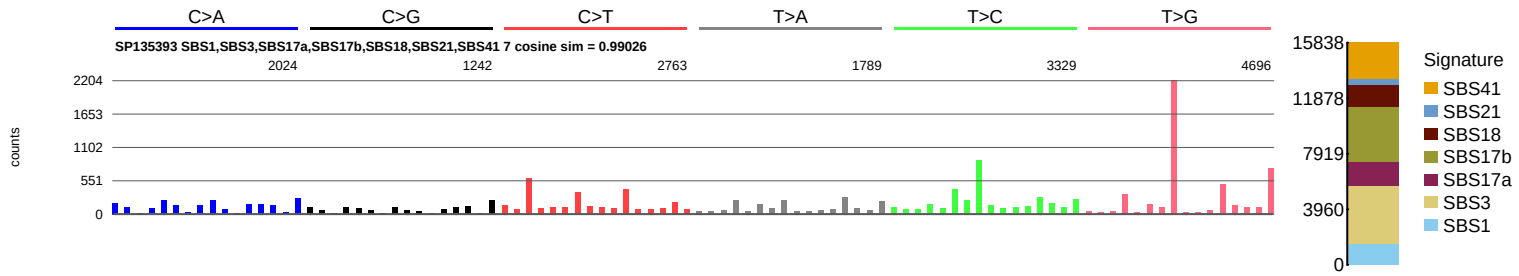
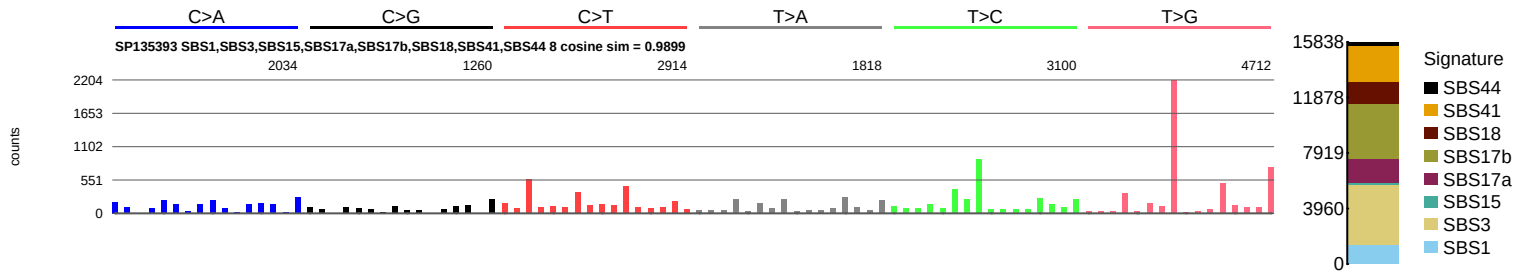
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.99056
2013
1207
2854
1765
3319
4682
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

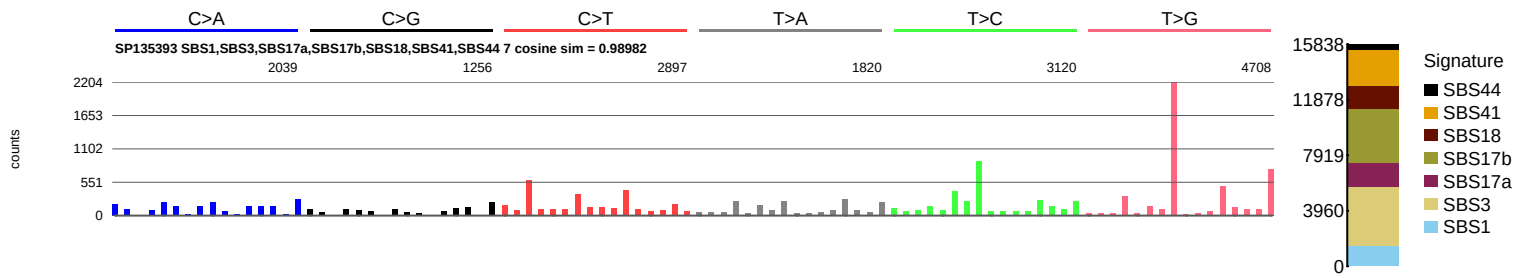

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.99051
2012
1223
2835
1771
3312
4685
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1

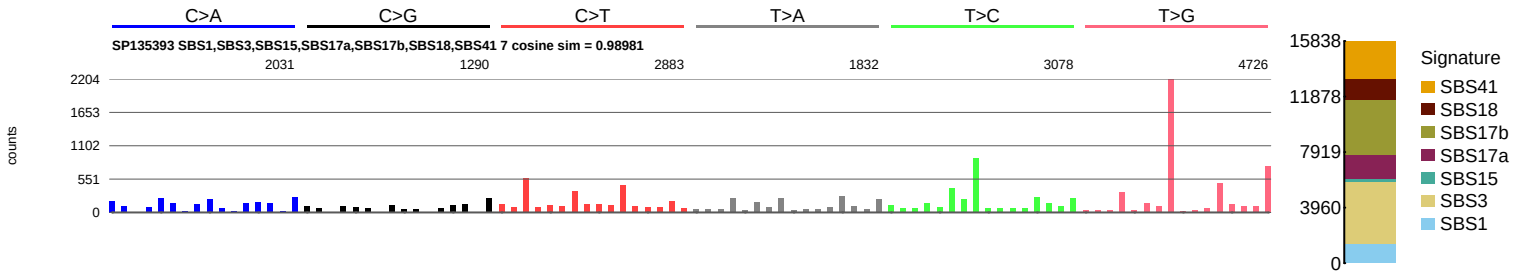

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.99051
2000
1280
2976
1816
3055
4710
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

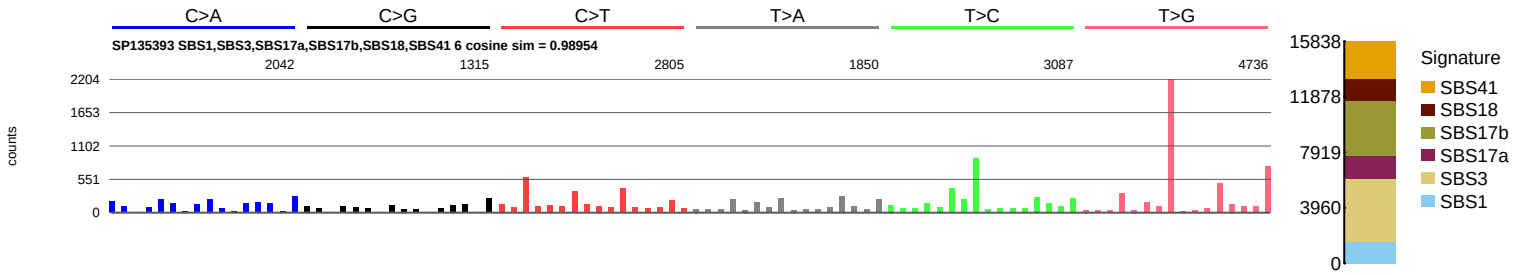

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.9904
2023
1206
2831
1769
3331
4680
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1

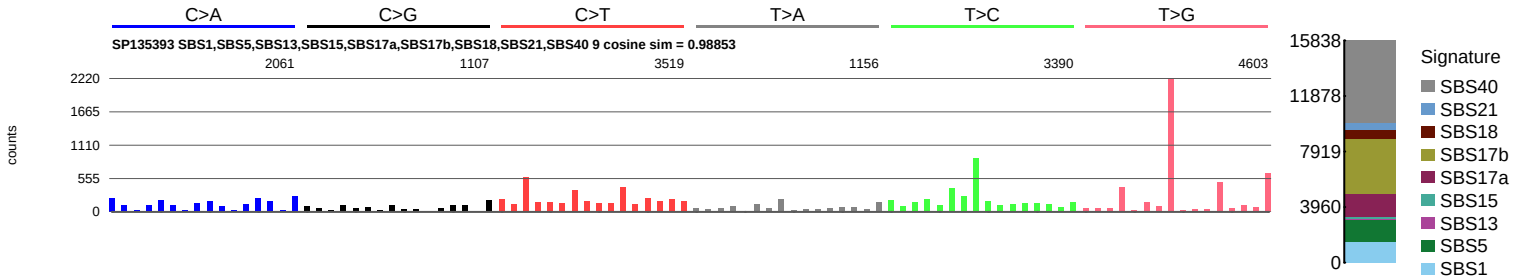

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.99026
2024
1242
2763
1789
3329
4696
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.9899
2034
1260
2914
1818
3100
4712
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98982
2039
1256
2897
1820
3120
4708
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98981
2031
1290
2883
1832
3078
4726
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS41 6 cosine sim = 0.98954
2042
1315
2805
1850
3087
4736
counts
2204
1653
1102
551
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98853
2061
1107
3519
1156
3390
4603
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
Signature
SBS40
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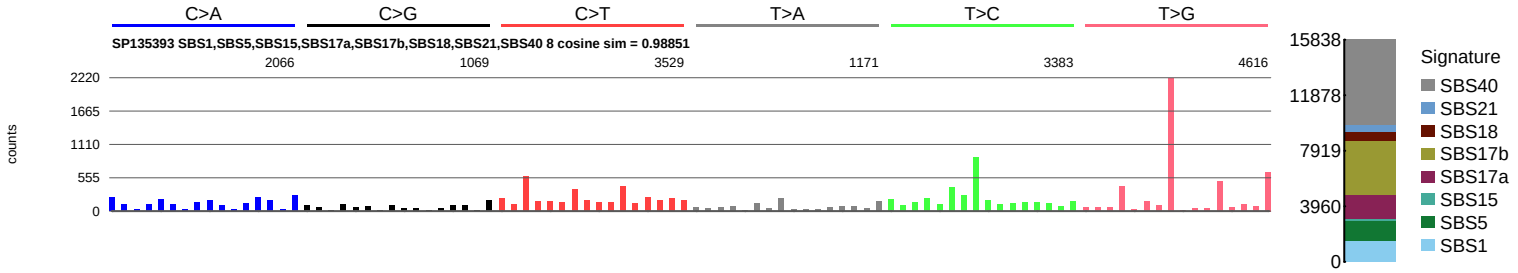
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2066
1069
3529
1171
3383
4616
counts
2220
1665
1110
555
0
15838
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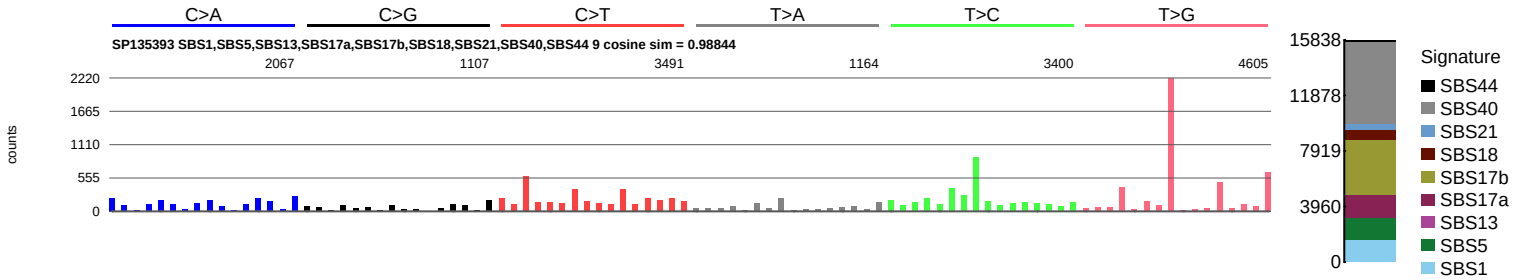
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2067
1107
3491
1164
3400
4605
counts
2220
1665
1110
555
0
15838
11878
7919
3960
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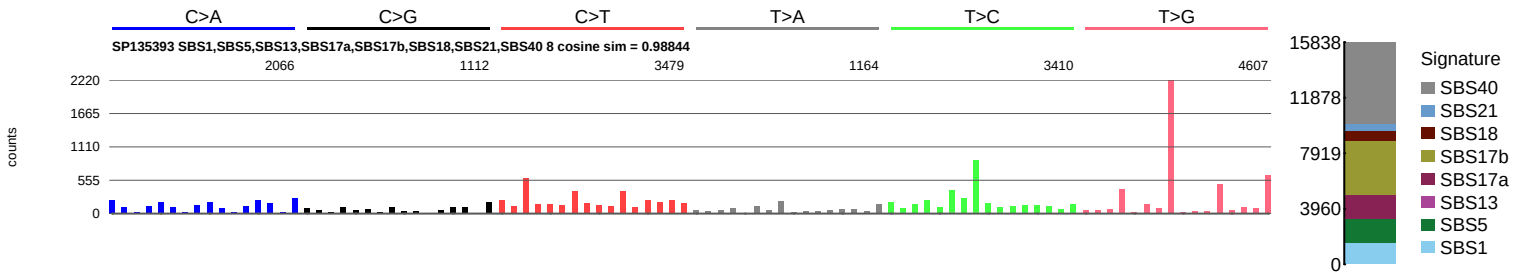
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SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98844
2066
1112
3479
1164
3410
4607
counts
2220
1665
1110
555
0
15838
11878
7919
3960
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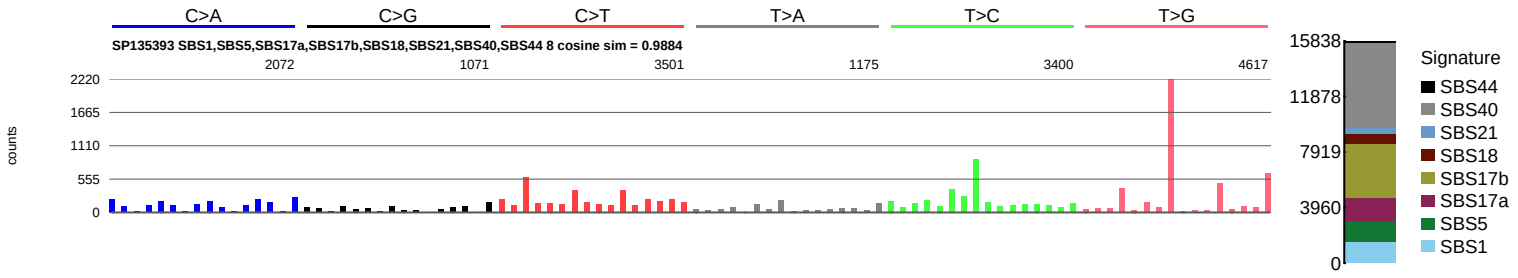
SBS5
SBS1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SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.9884
2072
1071
3501
1175
3400
4617
counts
2220
1665
1110
555
0
15838
11878
7919
3960
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Signature
SBS44
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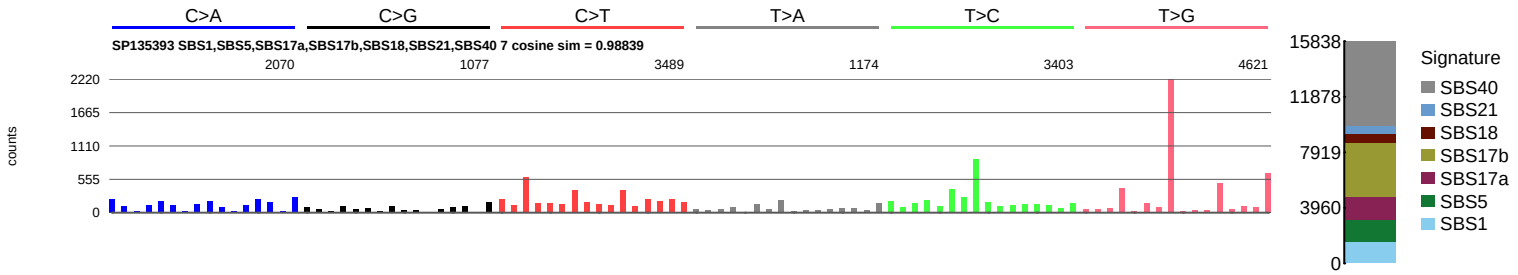
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C>G
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SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98839
2070
1077
3489
1174
3403
4621
counts
2220
1665
1110
555
0
15838
11878
7919
3960
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SBS21
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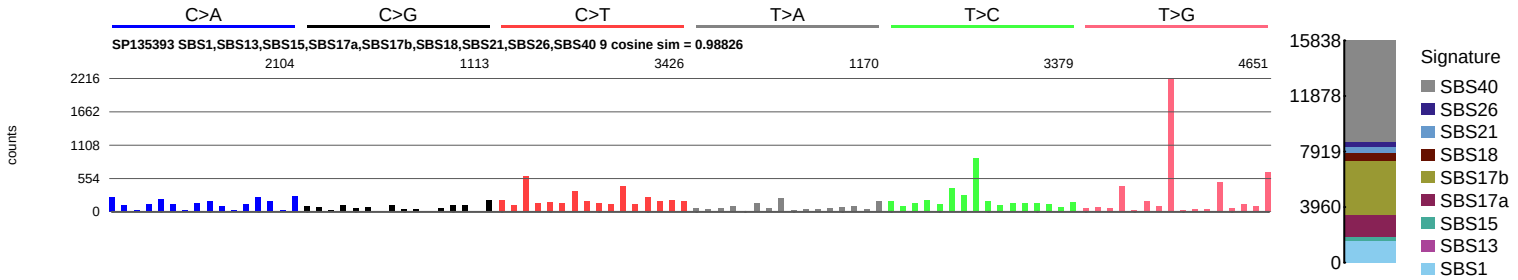
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2104
1113
3426
1170
3379
4651
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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SBS21
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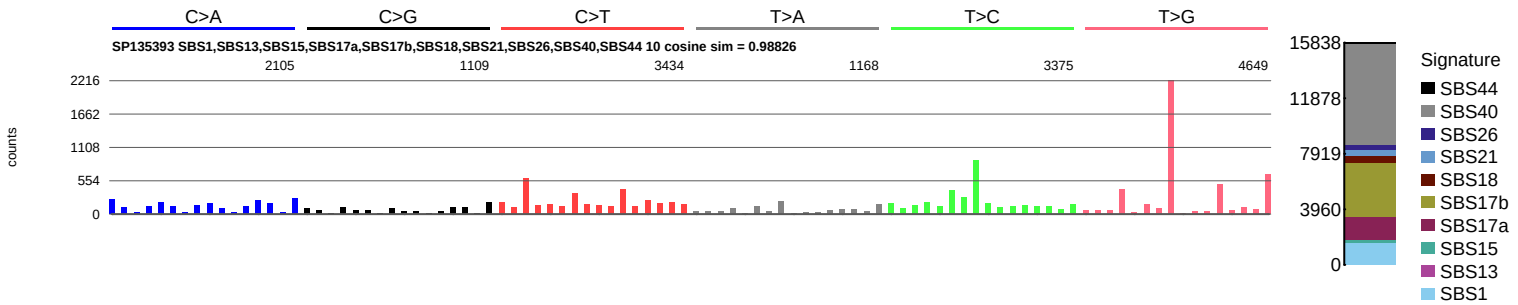
SBS13
SBS1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T>C
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SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98826
2105
1109
3434
1168
3375
4649
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
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SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
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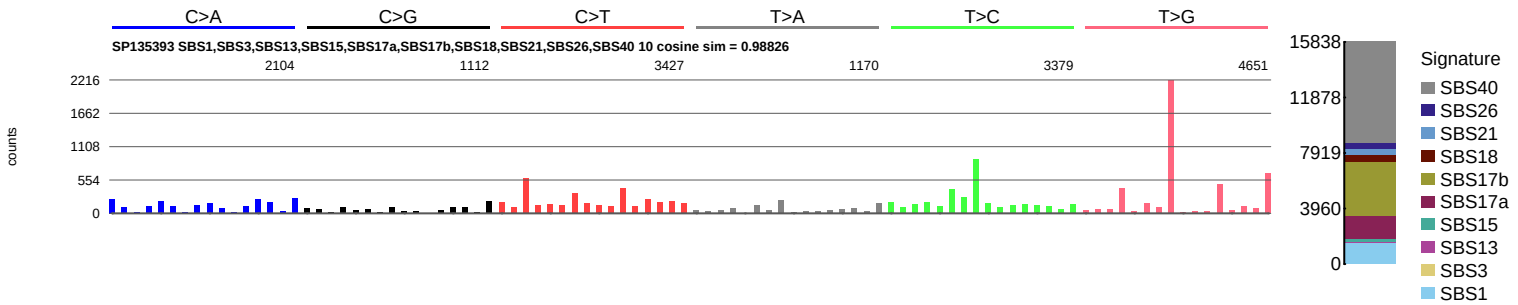
SBS13
SBS1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T>G
SP135393 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 10 cosine sim = 0.98826
2104
1112
3427
1170
3379
4651
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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Signature
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SBS18
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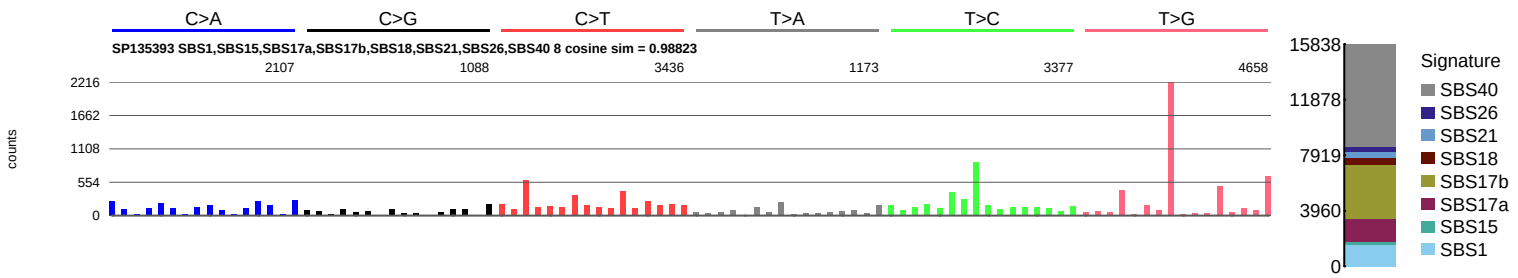
SBS13
SBS3
SBS1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T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98823
2107
1088
3436
1173
3377
4658
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
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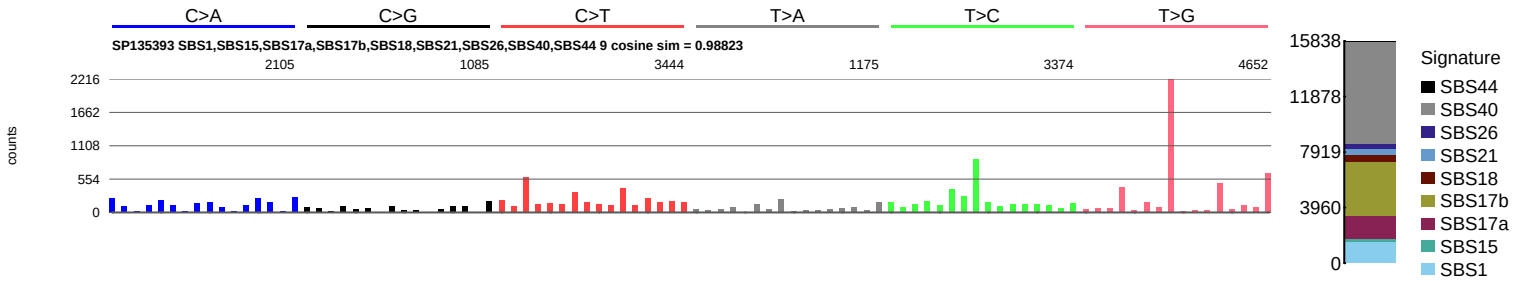
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98823
2105
1085
3444
1175
3374
4652
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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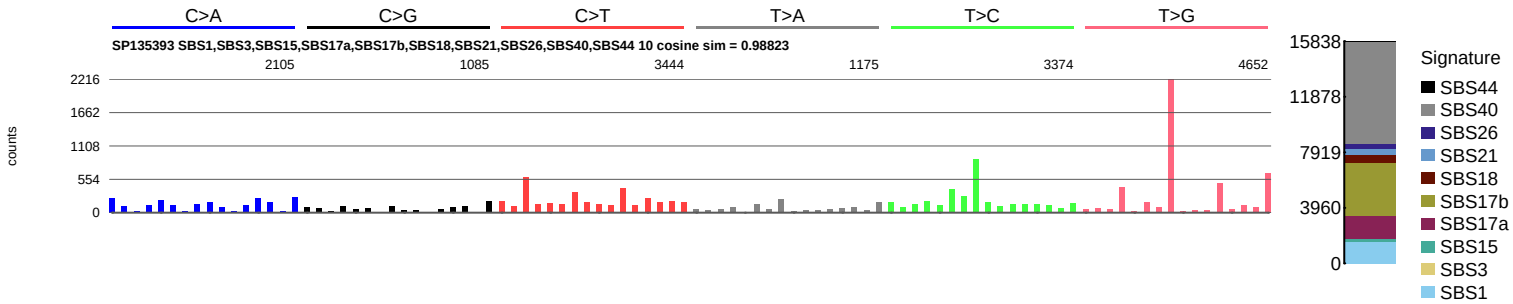
Signature
SBS44
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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T>C
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SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 10 cosine sim = 0.98823
2105
1085
3444
1175
3374
4652
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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Signature
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SBS40
SBS26
SBS21
SBS18
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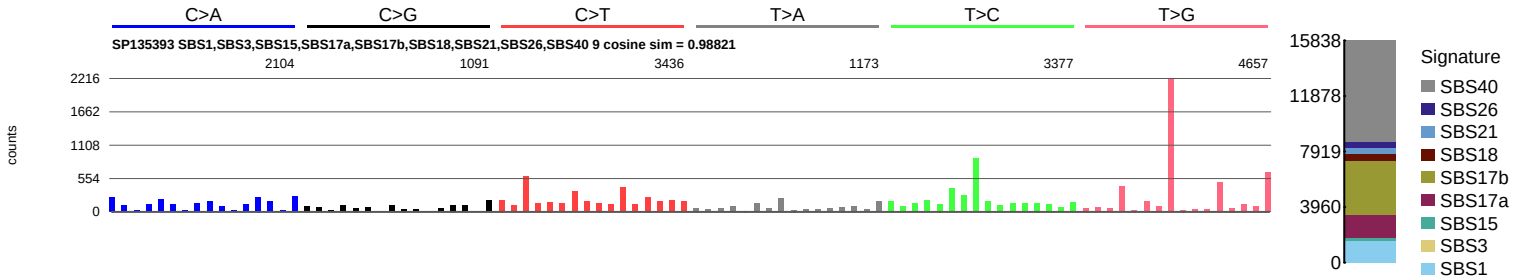
SBS17a
SBS15
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98821
2104
1091
3436
1173
3377
4657
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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SBS21
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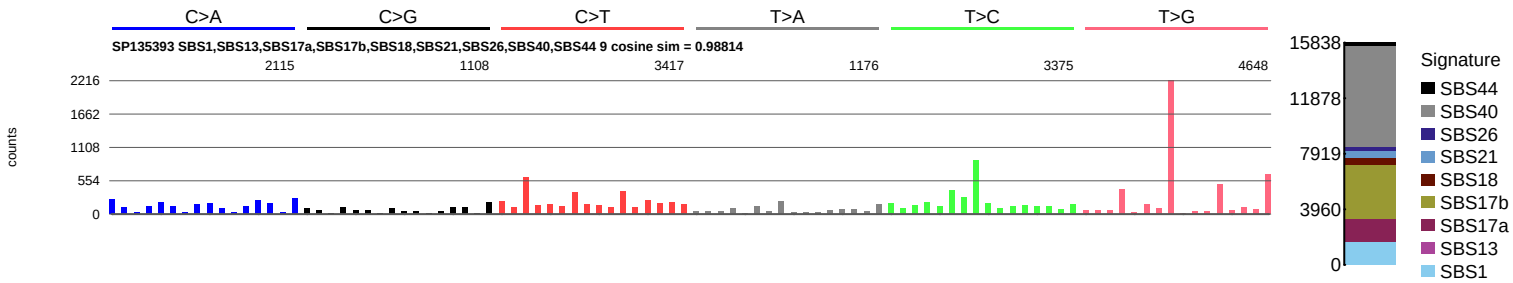
SBS15
SBS3
SBS1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SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98814
2115
1108
3417
1176
3375
4648
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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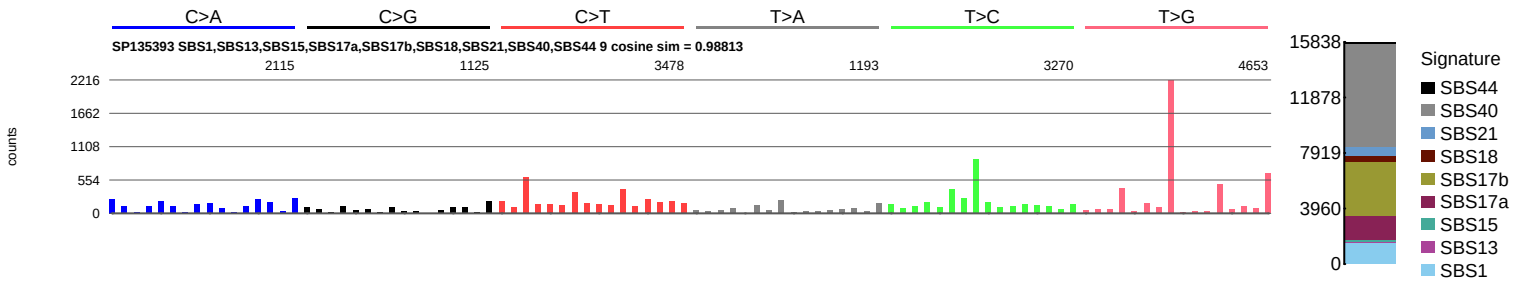
SBS17b
SBS17a
SBS13
SBS1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SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 9 cosine sim = 0.98813
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1125
3478
1193
3270
4653
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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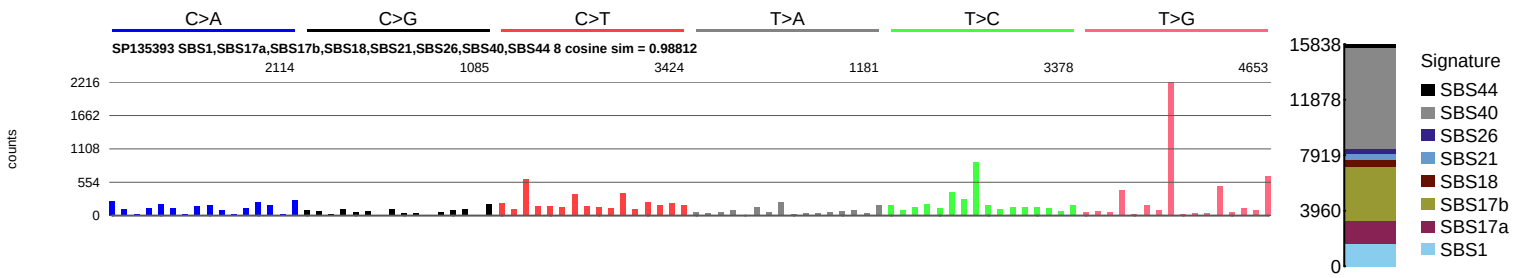
SBS15
SBS13
SBS1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SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98812
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1085
3424
1181
3378
4653
counts
2216
1662
1108
554
0
15838
11878
7919
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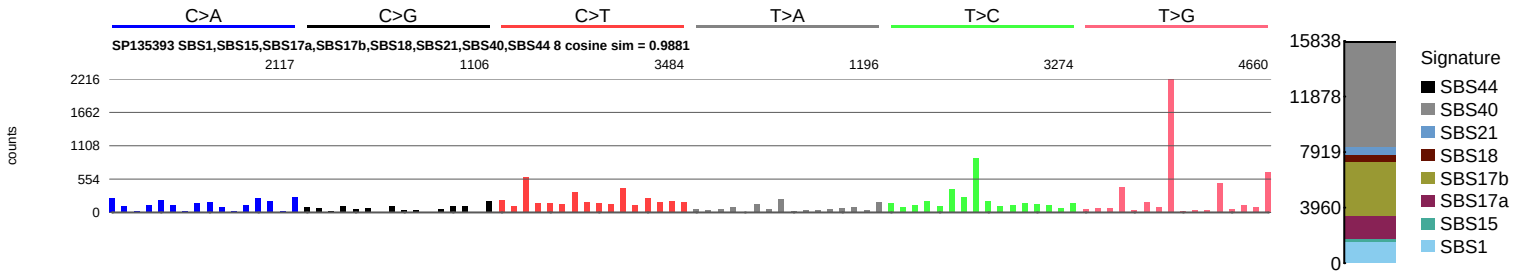
SBS17a
SBS1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SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 8 cosine sim = 0.9881
2117
1106
3484
1196
3274
4660
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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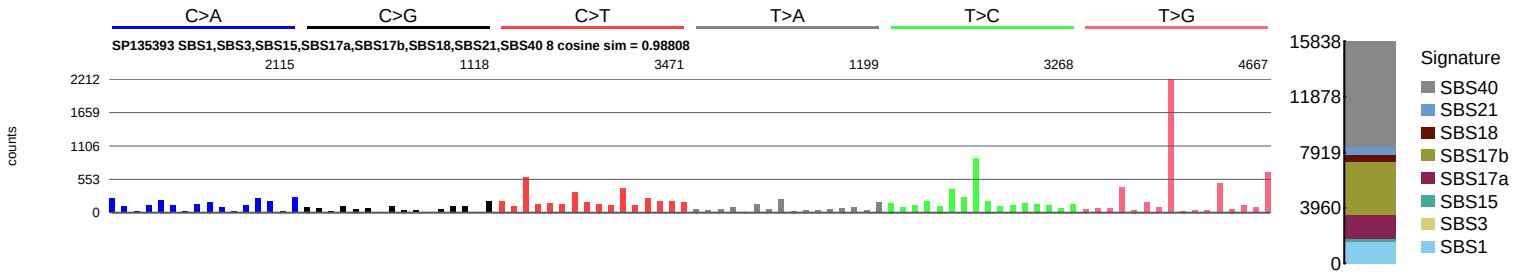
SBS17b
SBS17a
SBS15
SBS1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1118
3471
1199
3268
4667
counts
2212
1659
1106
553
0
15838
11878
7919
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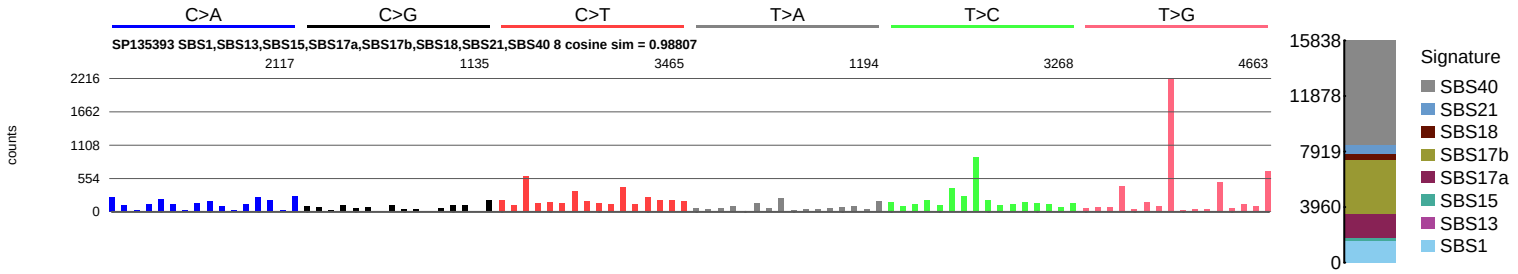
SBS17a
SBS15
SBS3
SBS1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1135
3465
1194
3268
4663
counts
2216
1662
1108
554
0
15838
11878
7919
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SBS21
SBS18
SBS17b
SBS17a
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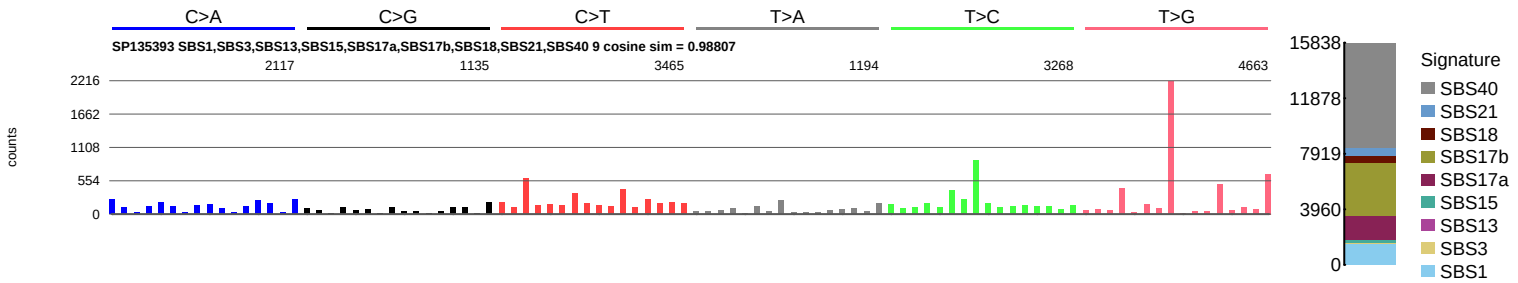
SBS13
SBS1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T>C
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SP135393 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 9 cosine sim = 0.98807
2117
1135
3465
1194
3268
4663
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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SBS21
SBS18
SBS17b
SBS17a
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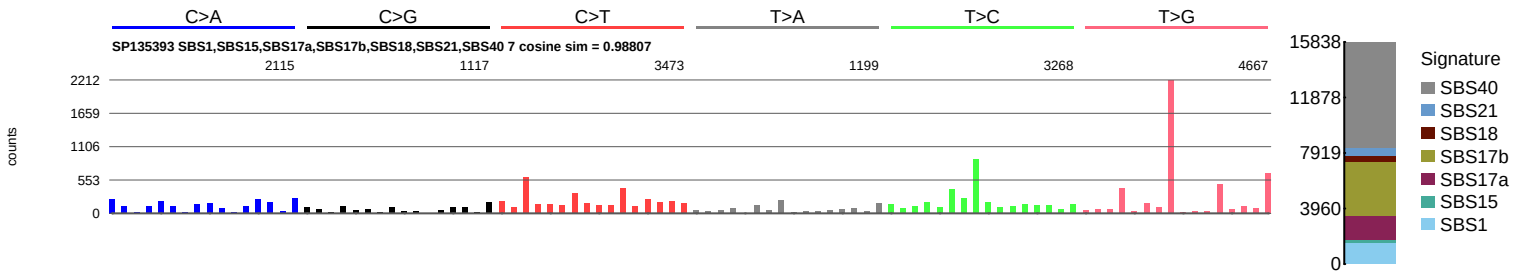
SBS13
SBS3
SBS1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C>T
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T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98807
2115
1117
3473
1199
3268
4667
counts
2212
1659
1106
553
0
15838
11878
7919
3960
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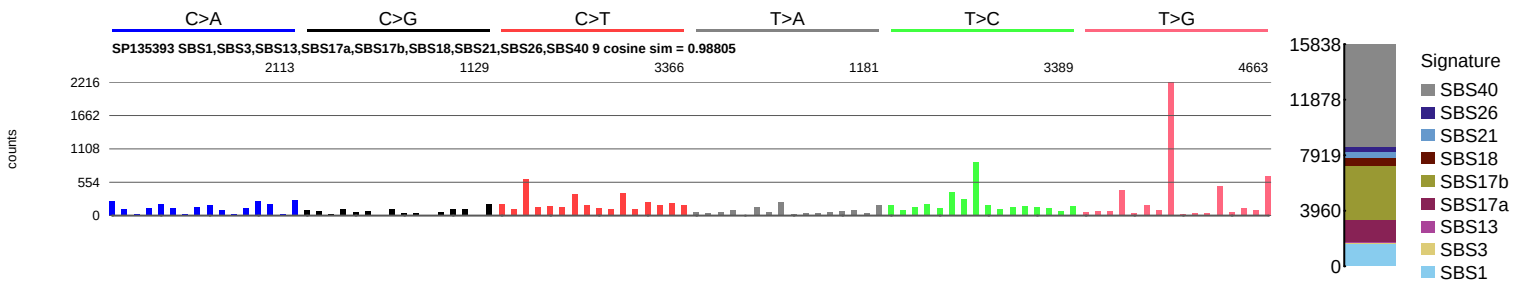
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>T
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T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 9 cosine sim = 0.98805
2113
1129
3366
1181
3389
4663
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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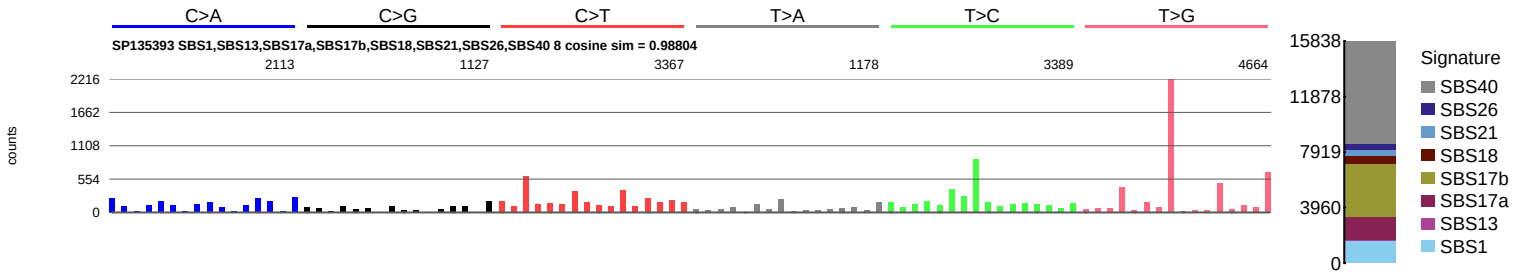
SBS3
SBS1


C>A
C>G
C>T
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T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98804
2113
1127
3367
1178
3389
4664
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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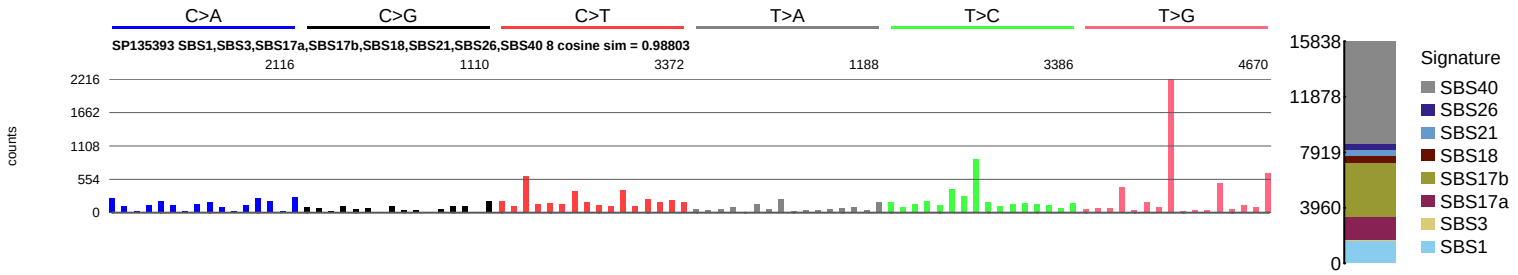
SBS13
SBS1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C>G
C>T
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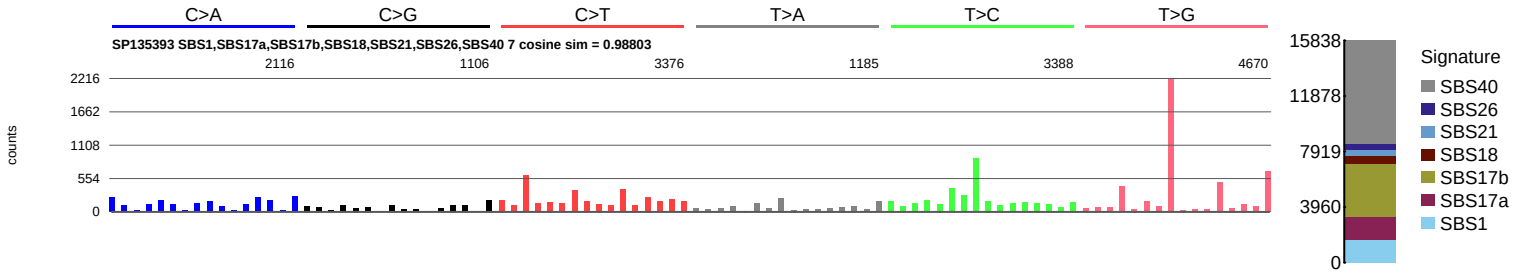
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 8 cosine sim = 0.98803
2116
1110
3372
1188
3386
4670
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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C>G
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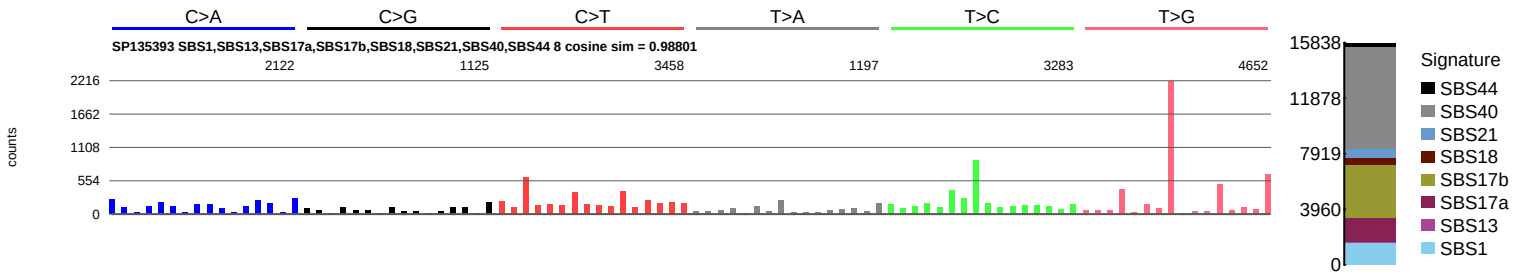
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS40 7 cosine sim = 0.98803
2116
1106
3376
1185
3388
4670
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1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2122
1125
3458
1197
3283
4652
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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Signature
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SBS17b
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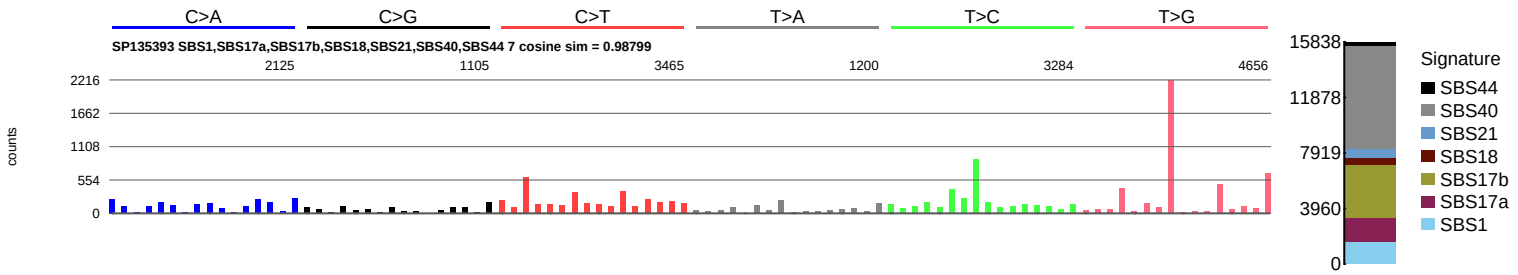
SBS13
SBS1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SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40,SBS44 7 cosine sim = 0.98799
2125
1105
3465
1200
3284
4656
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
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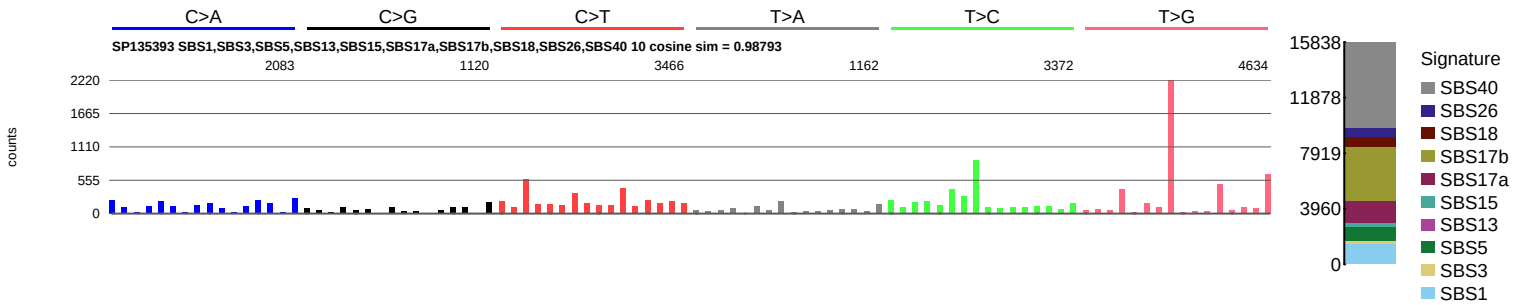
SBS21
SBS18
SBS17b
SBS17a
SBS1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SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 10 cosine sim = 0.98793
2083
1120
3466
1162
3372
4634
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
Signature
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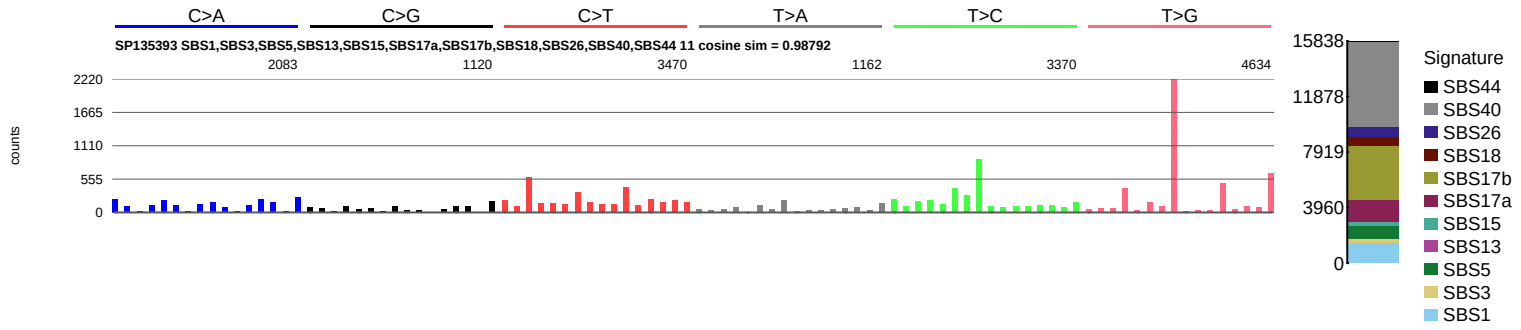
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 11 cosine sim = 0.98792
2083
1120
3470
1162
3370
4634
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
Signature
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SBS26
SBS18
SBS17b
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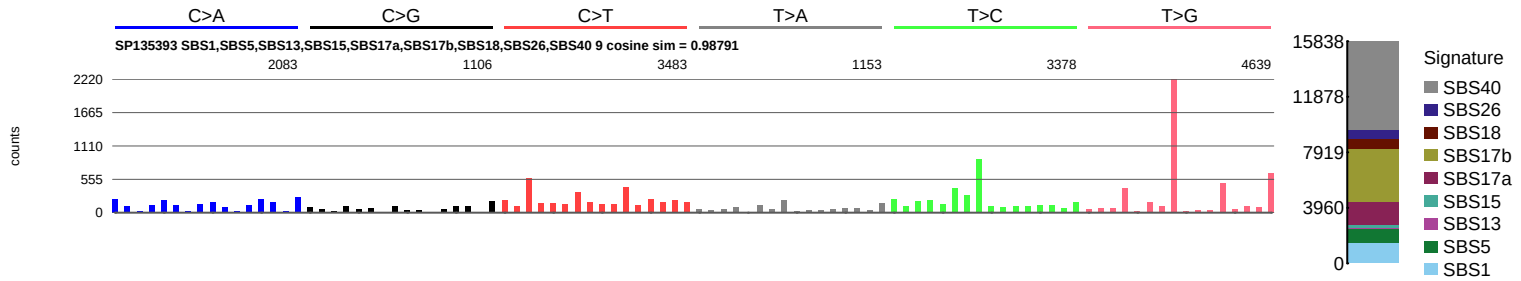
SBS15
SBS13
SBS5
SBS3
SBS1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T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98791
2083
1106
3483
1153
3378
4639
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
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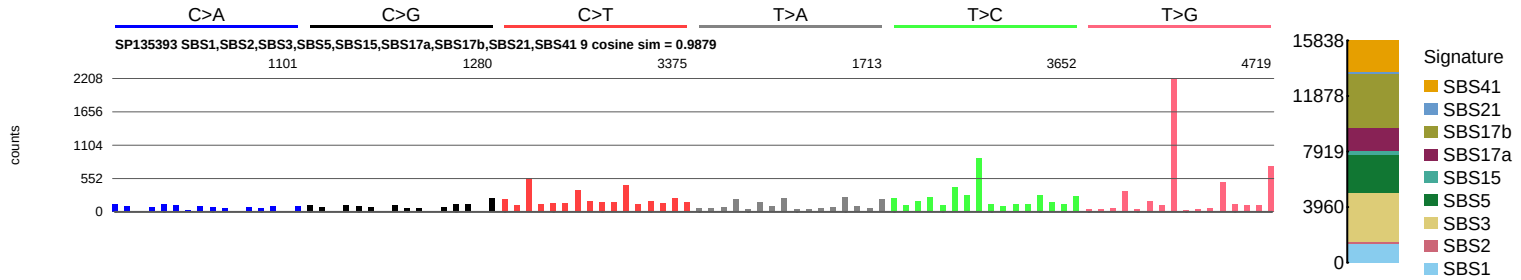
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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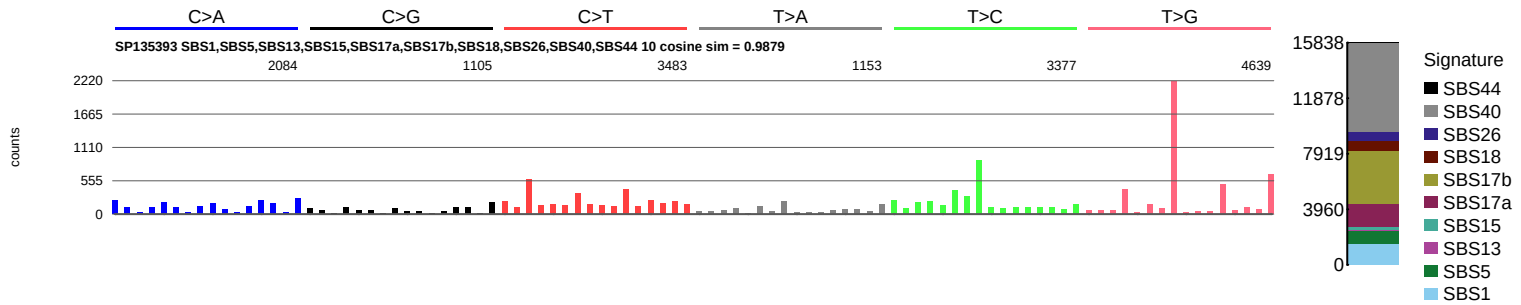
SP135393 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.9879
1101
1280
3375
1713
3652
4719
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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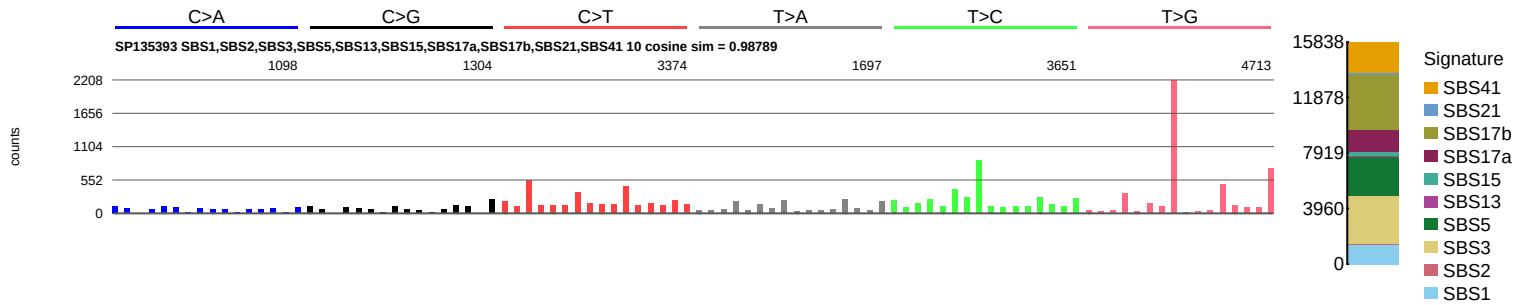
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.9879
2084
1105
3483
1153
3377
4639
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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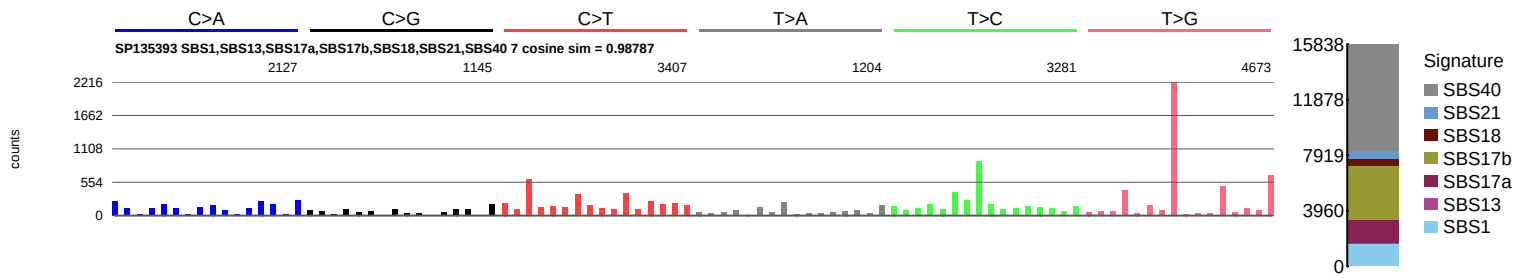
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 10 cosine sim = 0.98789
1098
1304
3374
1697
3651
4713
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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C>G
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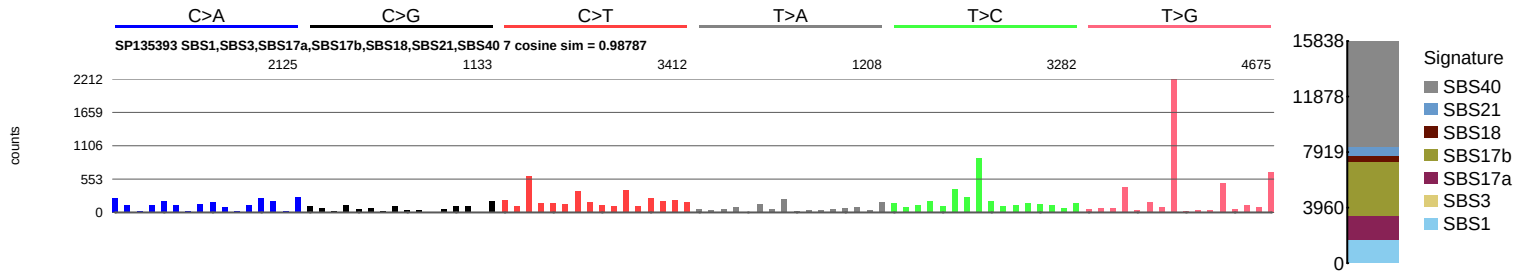
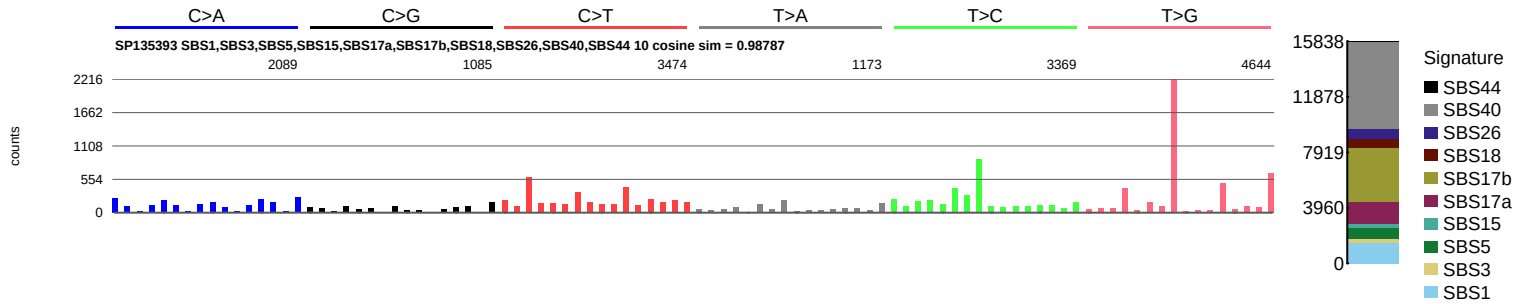
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98787
2127
1145
3407
1204
3281
4673
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS40 7 cosine sim = 0.98787
2125
1133
3412
1208
3282
4675
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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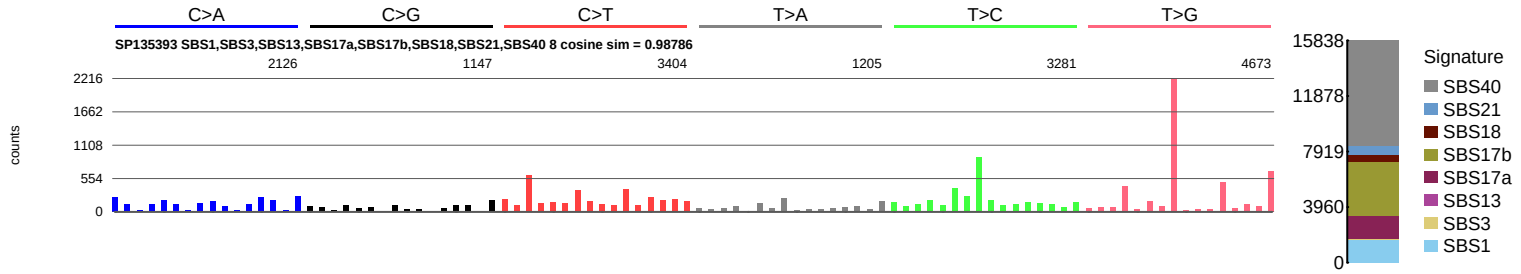
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98787
2089
1085
3474
1173
3369
4644
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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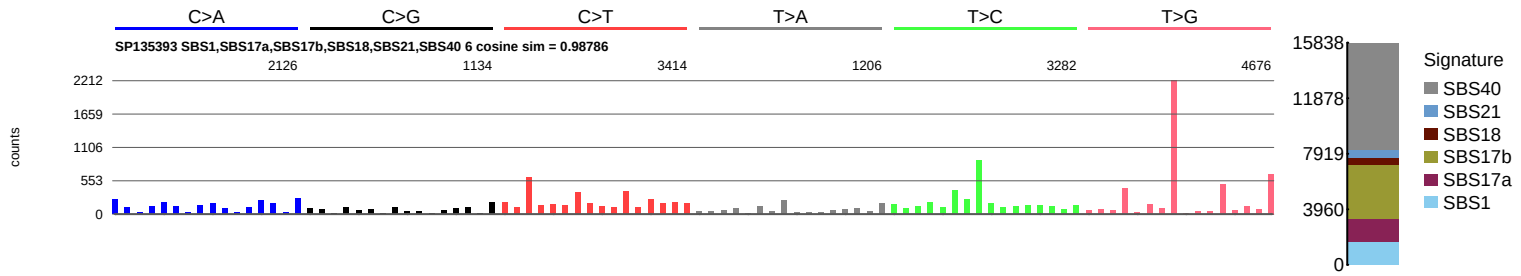
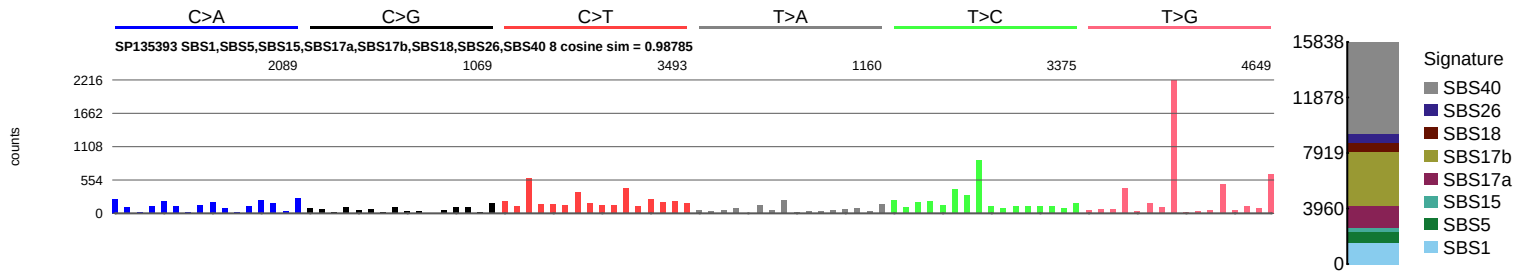
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS40 8 cosine sim = 0.98786
2126
1147
3404
1205
3281
4673
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS40 6 cosine sim = 0.98786
2126
1134
3414
1206
3282
4676
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS21
SBS18
SBS17b
SBS17a
SBS1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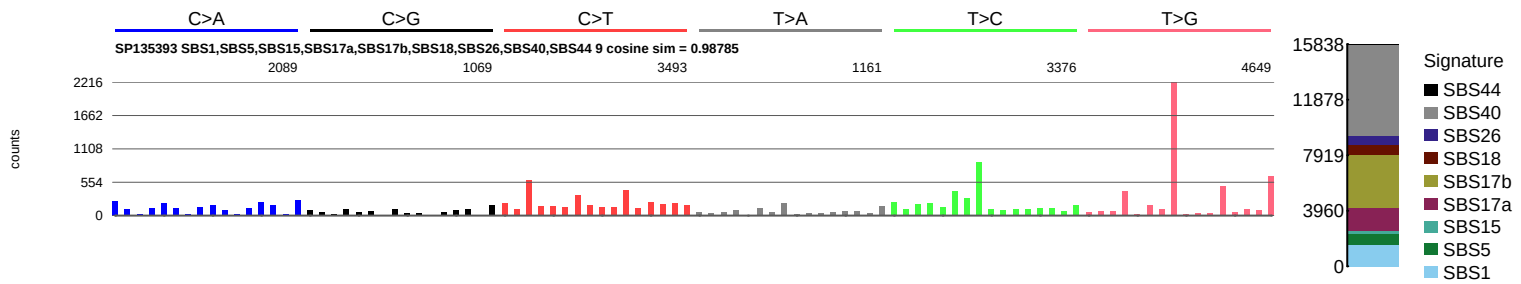
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98785
2089
1069
3493
1160
3375
4649
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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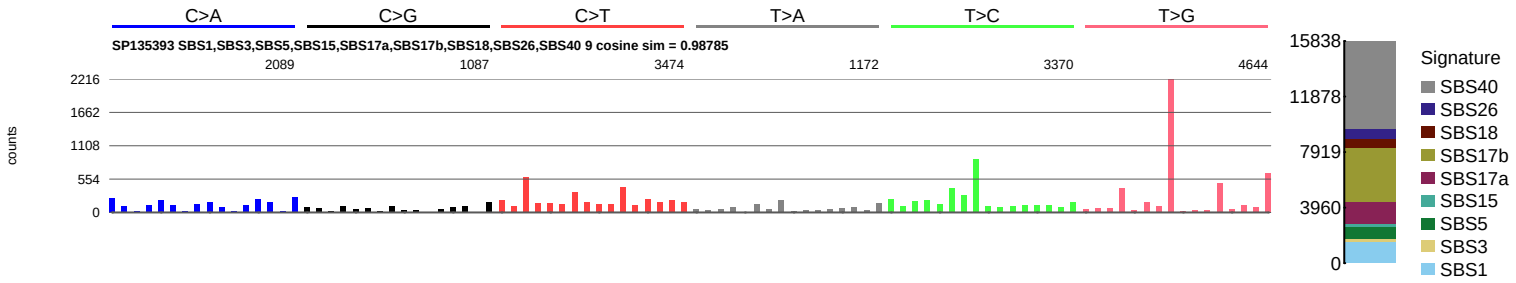
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98785
2089
1069
3493
1161
3376
4649
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
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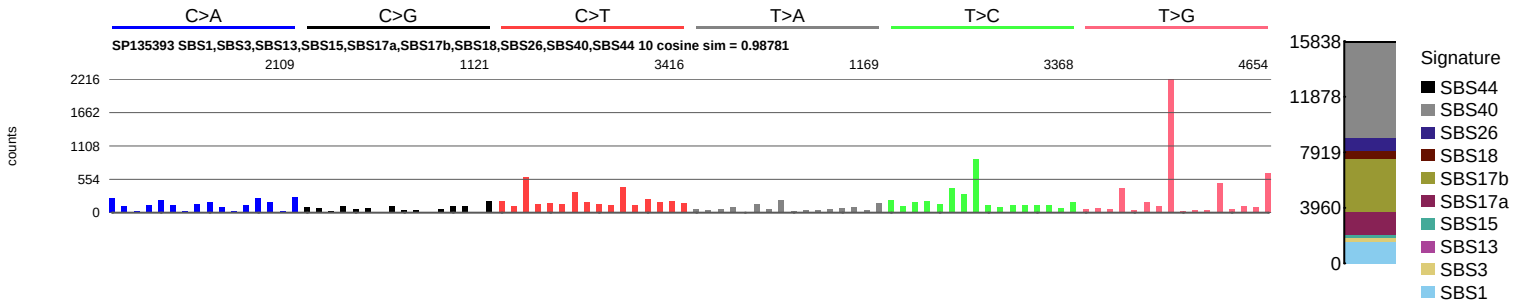
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.98785
2089
1087
3474
1172
3370
4644
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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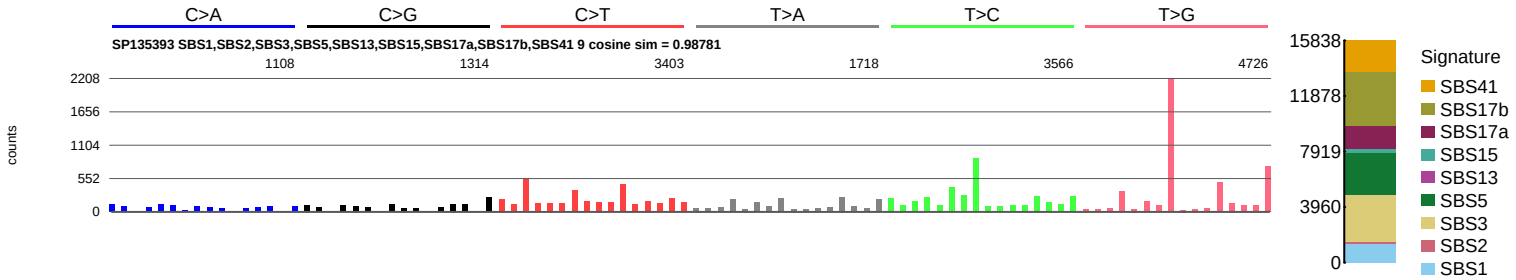
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98781
2109
1121
3416
1169
3368
4654
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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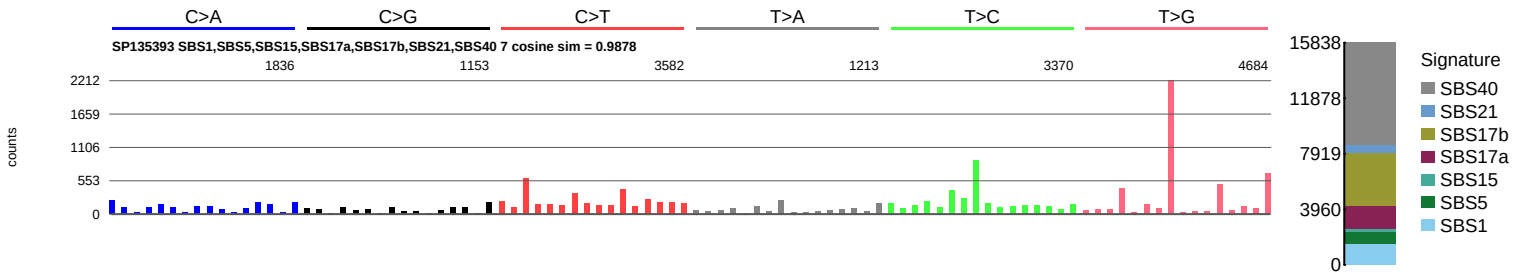
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 9 cosine sim = 0.98781
1108
1314
3403
1718
3566
4726
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1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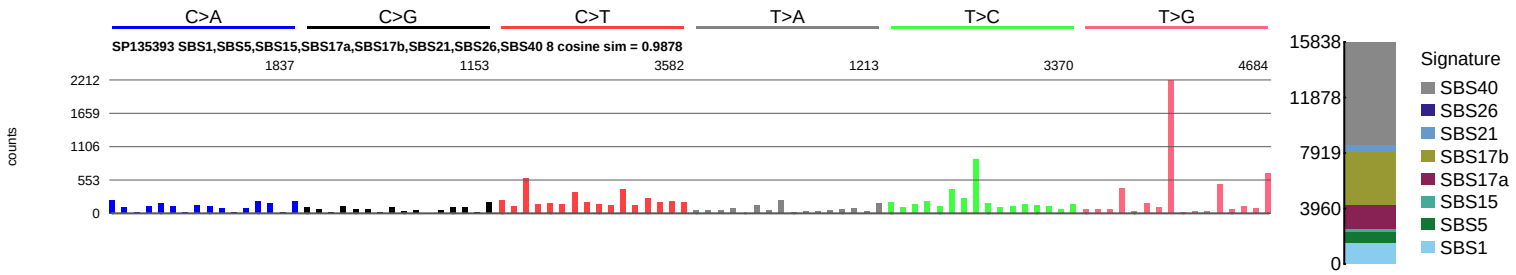
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.9878
1836
1153
3582
1213
3370
4684
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1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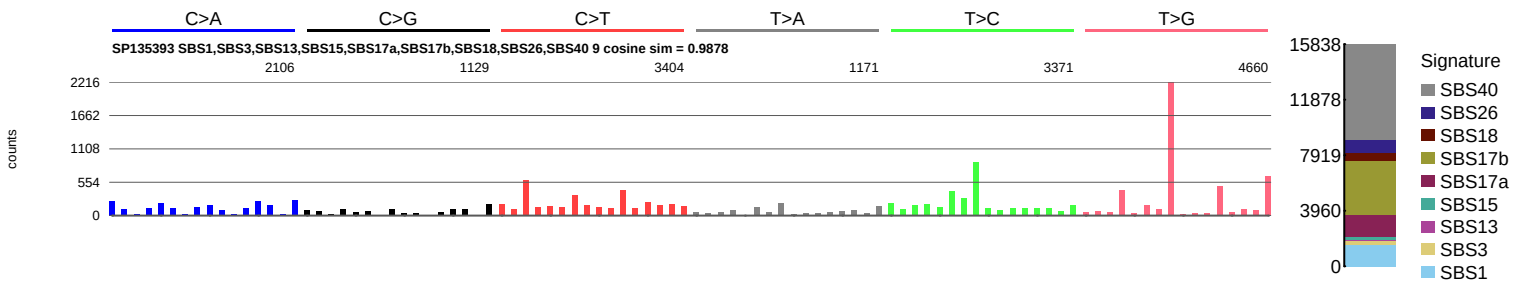
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.9878
1837
1153
3582
1213
3370
4684
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1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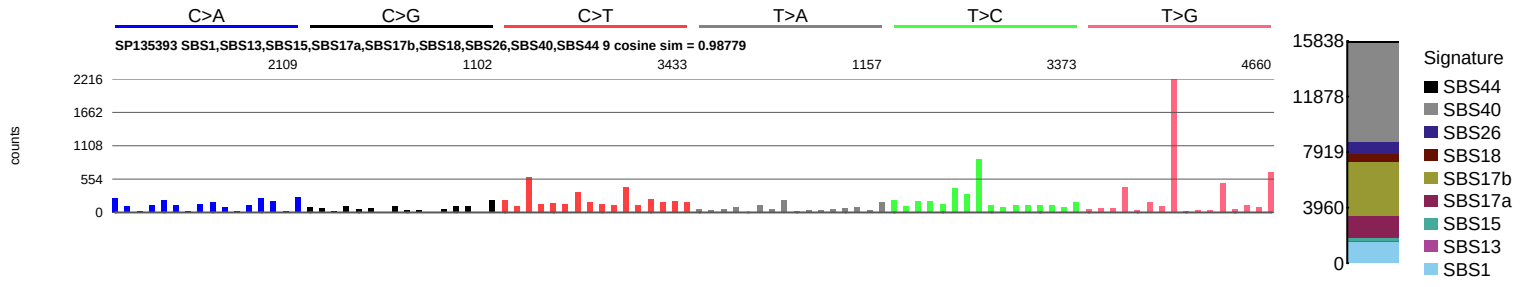
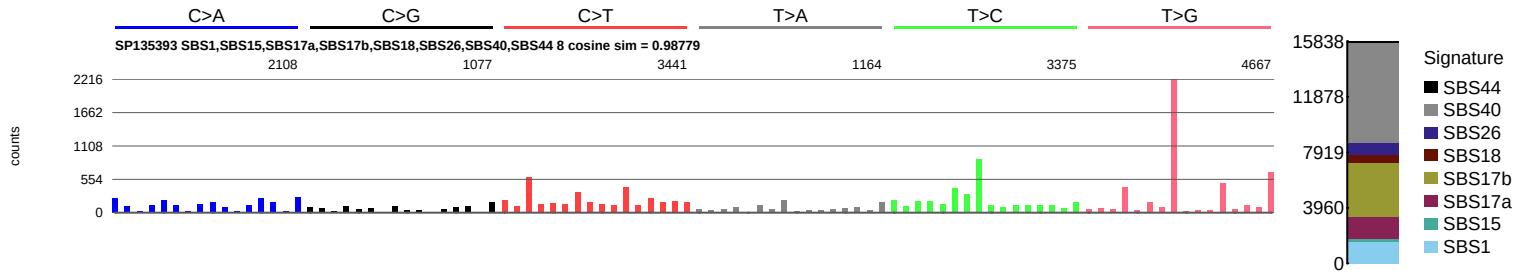
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.9878
2106
1129
3404
1171
3371
4660
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98779
2109
1102
3433
1157
3373
4660
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98779
2108
1077
3441
1164
3375
4667
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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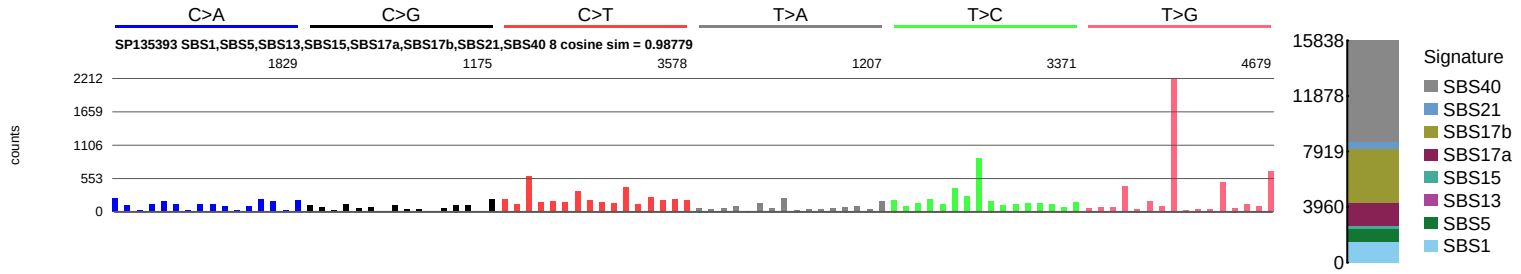
SBS15
SBS1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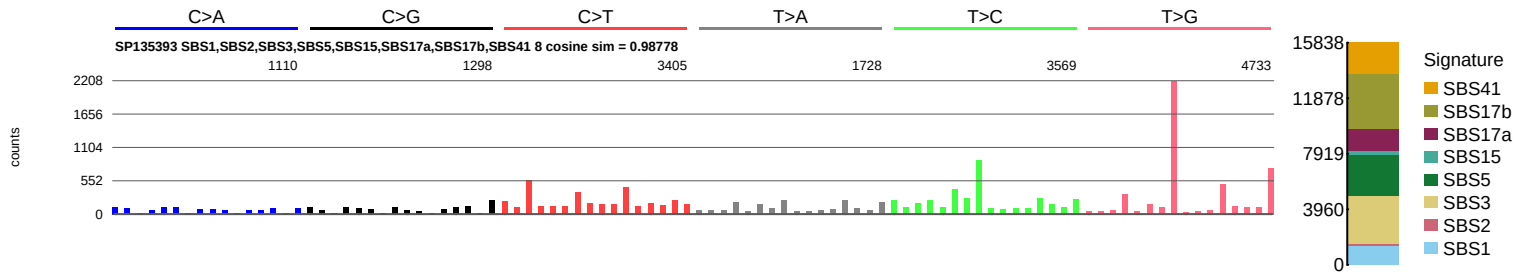
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 8 cosine sim = 0.98779
1829
1175
3578
1207
3371
4679
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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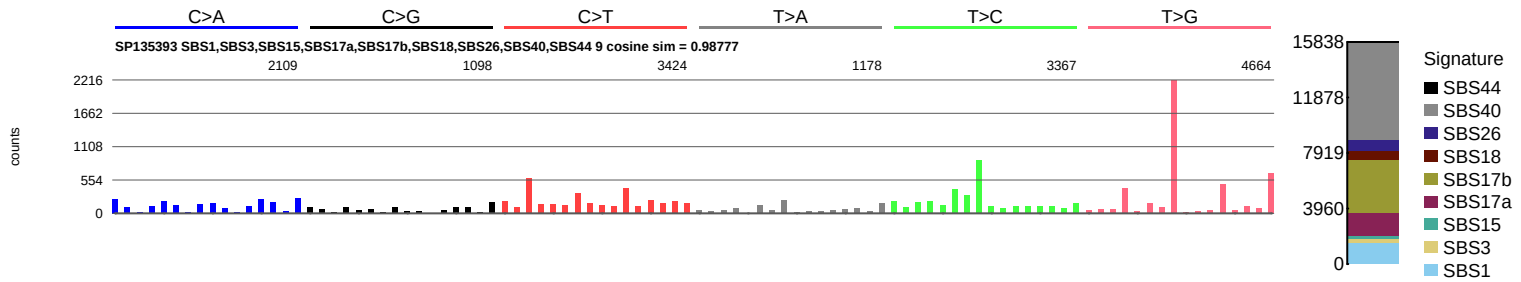
SP135393 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98778
1110
1298
3405
1728
3569
4733
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1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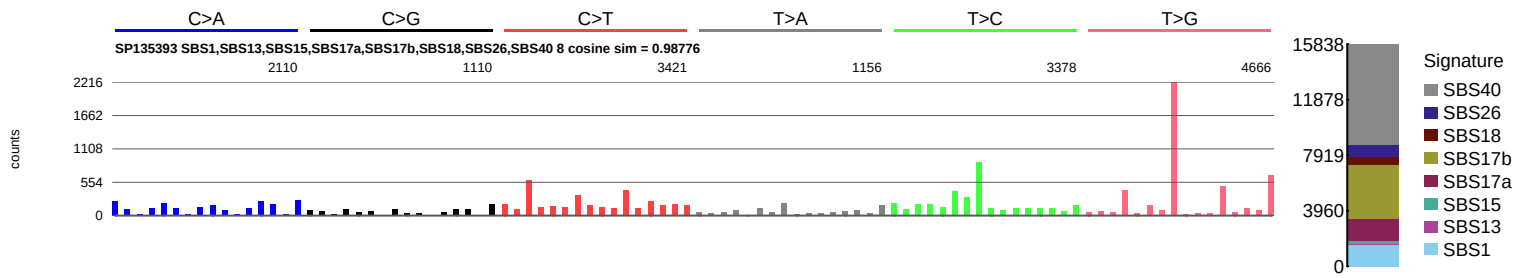
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98777
2109
1098
3424
1178
3367
4664
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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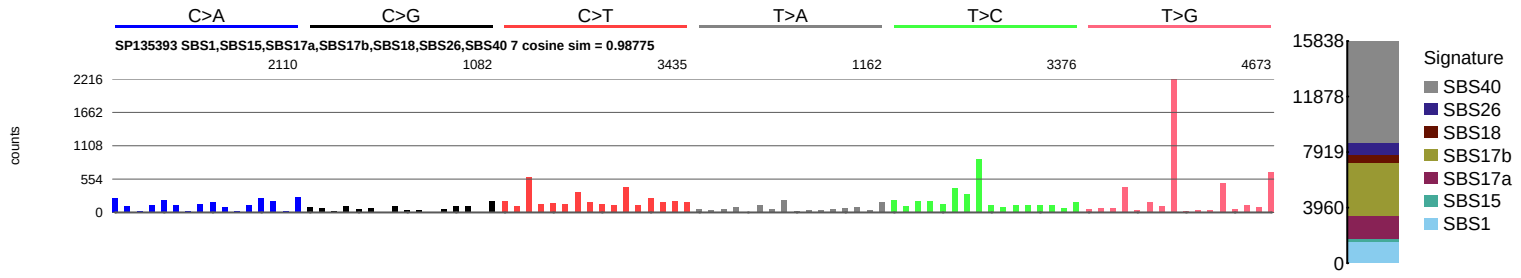
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98776
2110
1110
3421
1156
3378
4666
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
C>T
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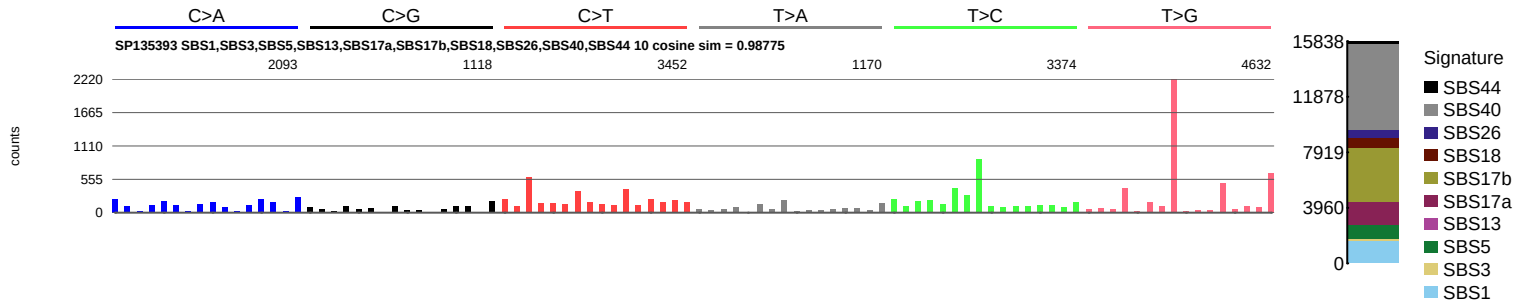
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98775
2110
1082
3435
1162
3376
4673
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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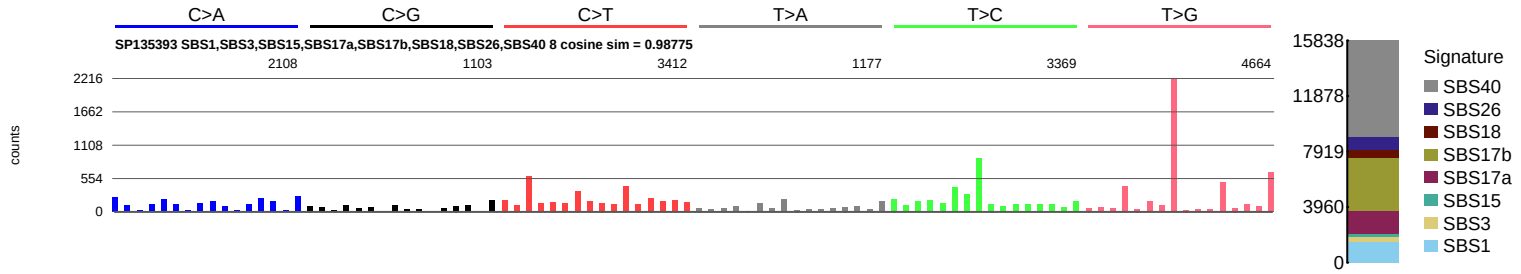
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 10 cosine sim = 0.98775
2093
1118
3452
1170
3374
4632
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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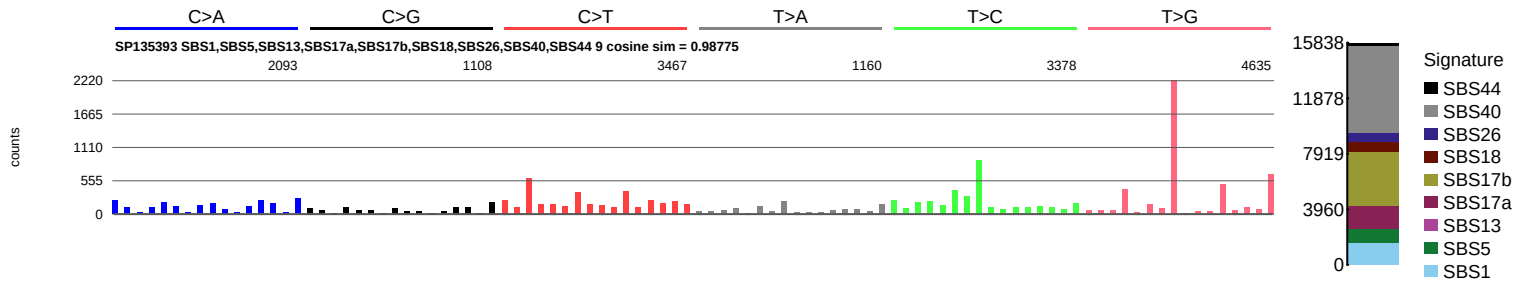
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98775
2108
1103
3412
1177
3369
4664
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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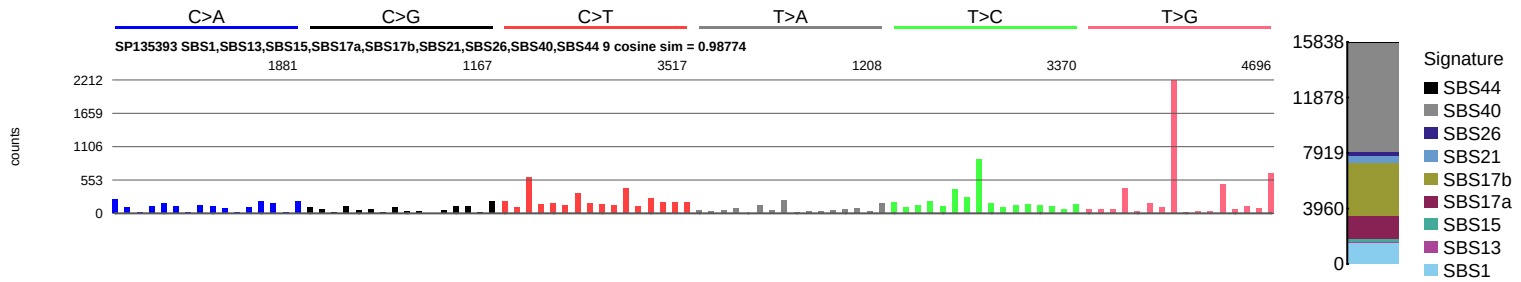
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98775
2093
1108
3467
1160
3378
4635
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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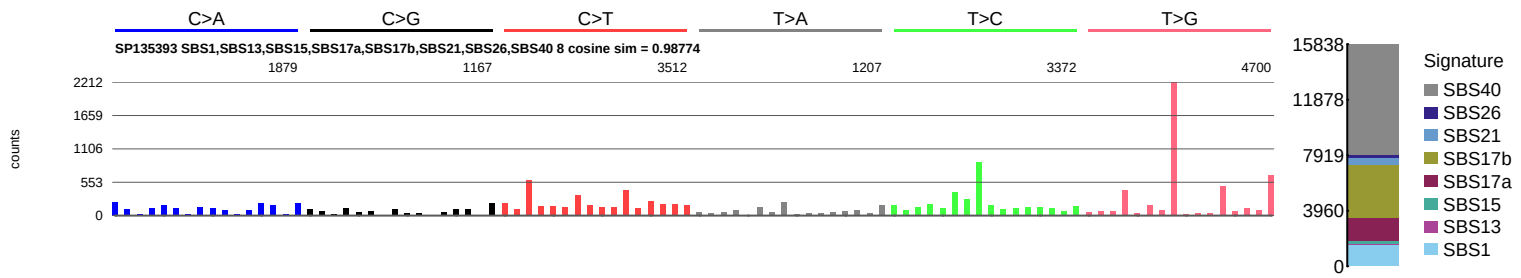
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 9 cosine sim = 0.98774
1881
1167
3517
1208
3370
4696
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1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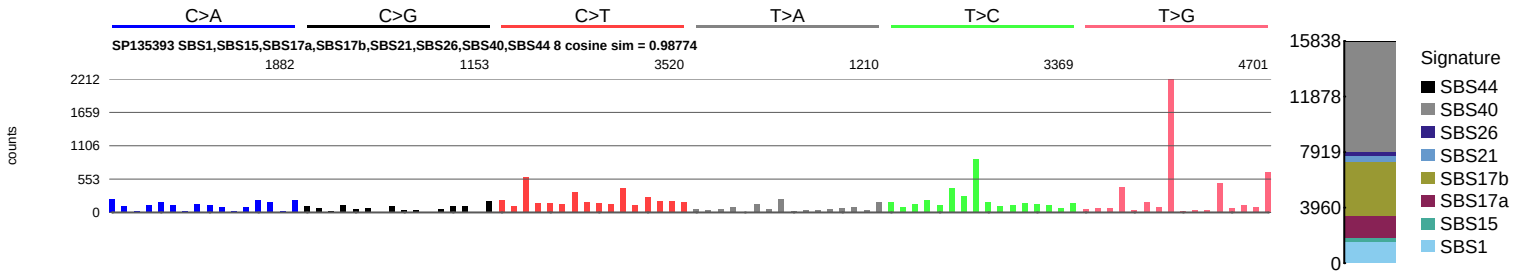
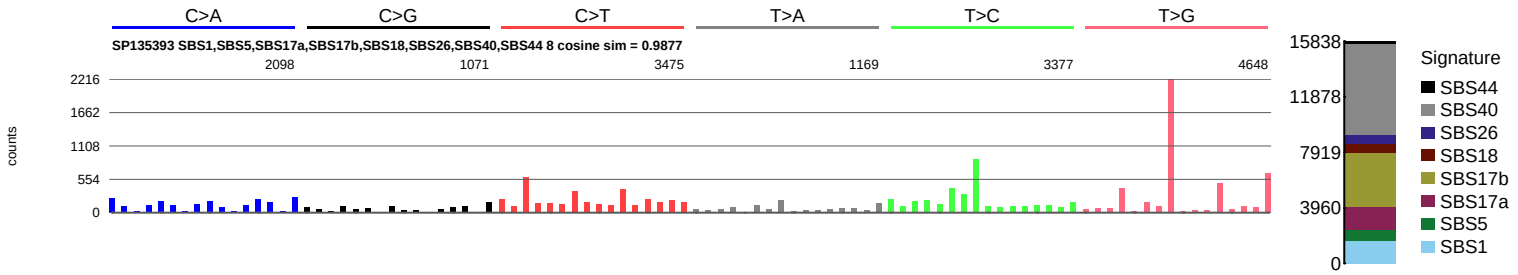
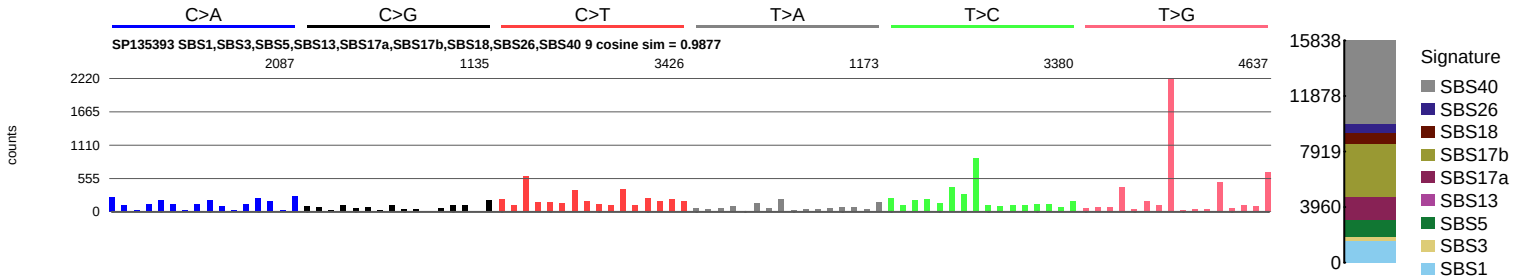
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 8 cosine sim = 0.98774
1879
1167
3512
1207
3372
4700
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98774
1882
1153
3520
1210
3369
4701
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.9877
2098
1071
3475
1169
3377
4648
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1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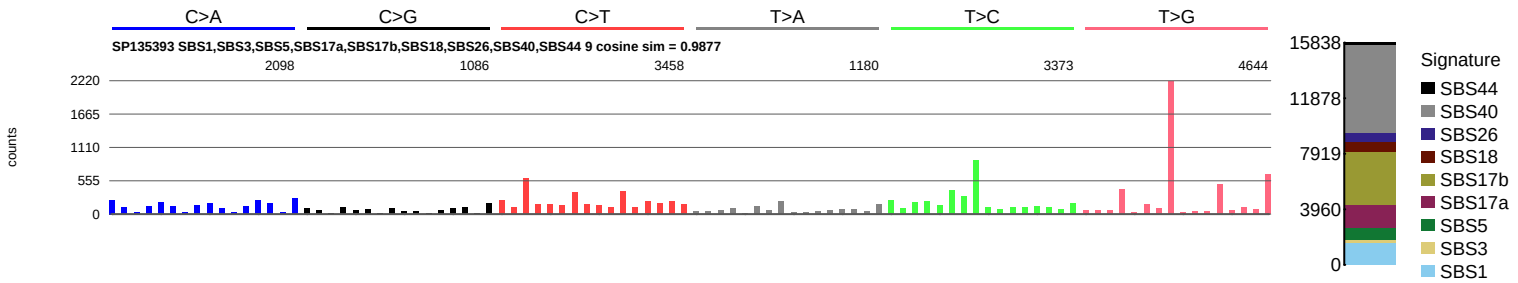
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 9 cosine sim = 0.9877
2087
1135
3426
1173
3380
4637
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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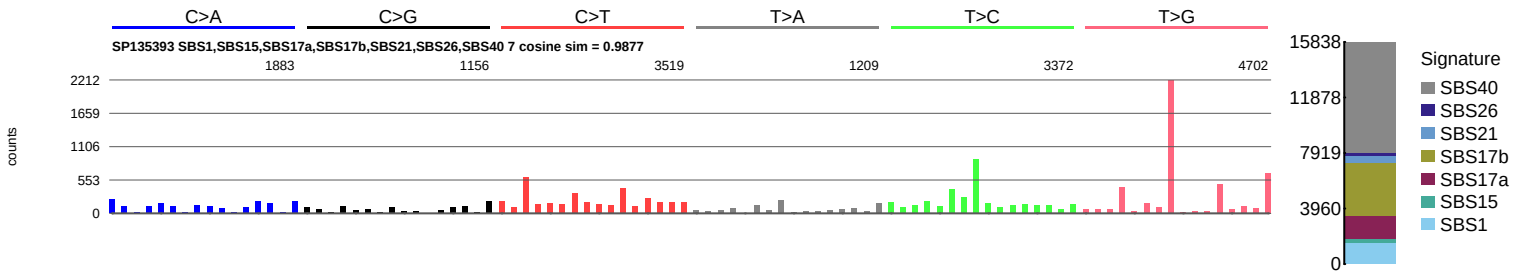
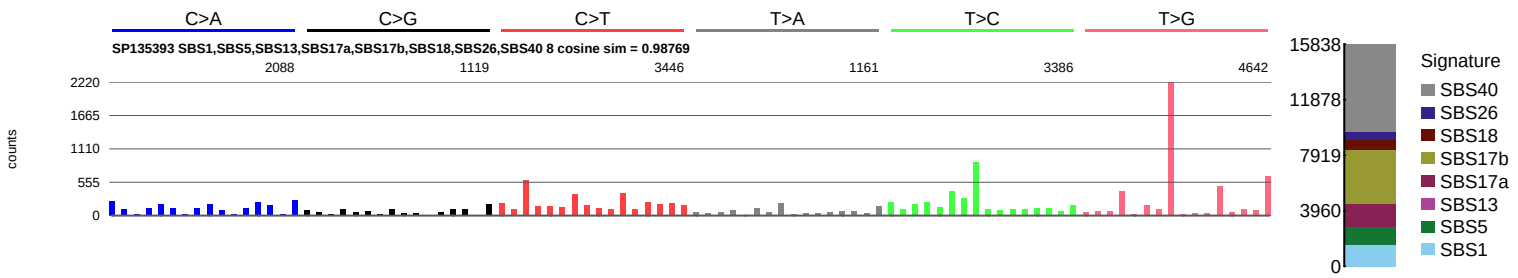
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.9877
2098
1086
3458
1180
3373
4644
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.9877
1883
1156
3519
1209
3372
4702
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS1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C>G
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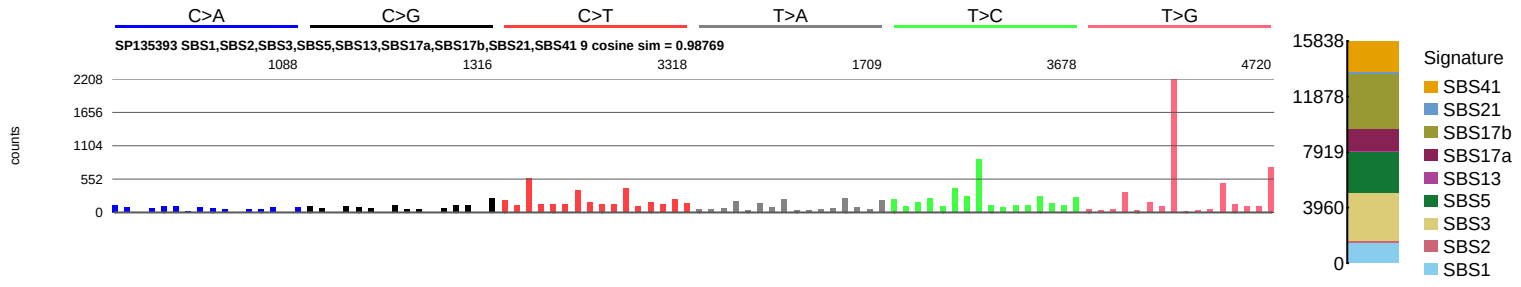
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98769
2088
1119
3446
1161
3386
4642
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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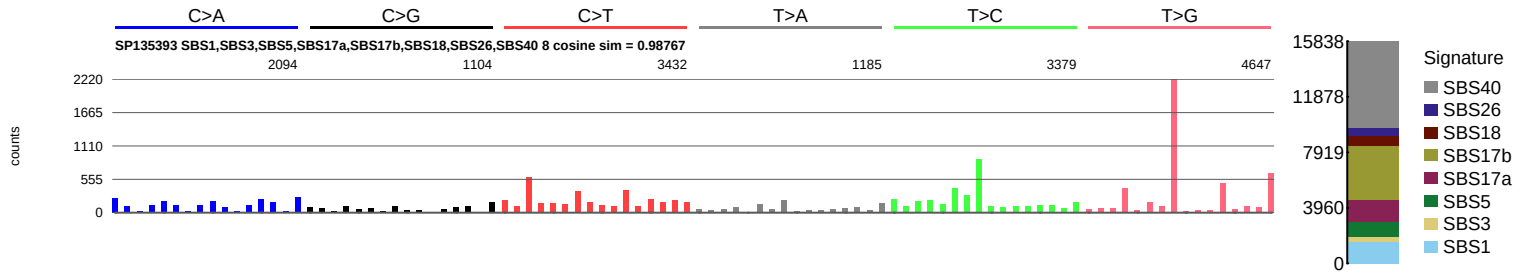
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98769
1088
1316
3318
1709
3678
4720
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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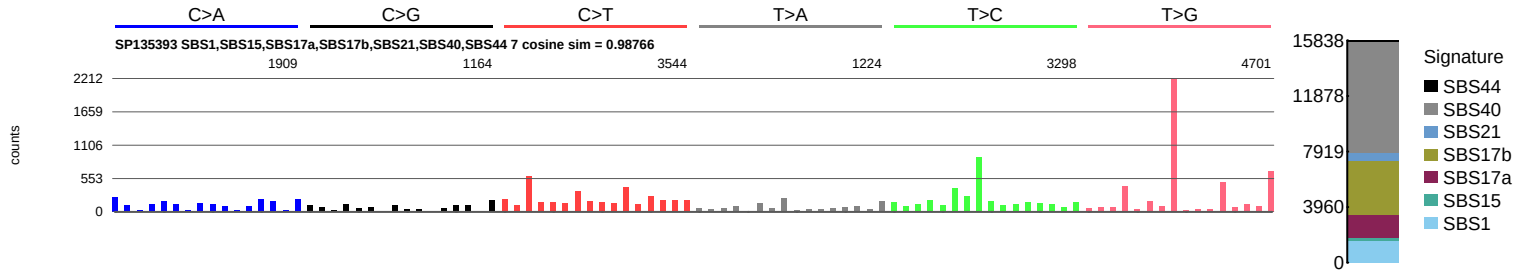
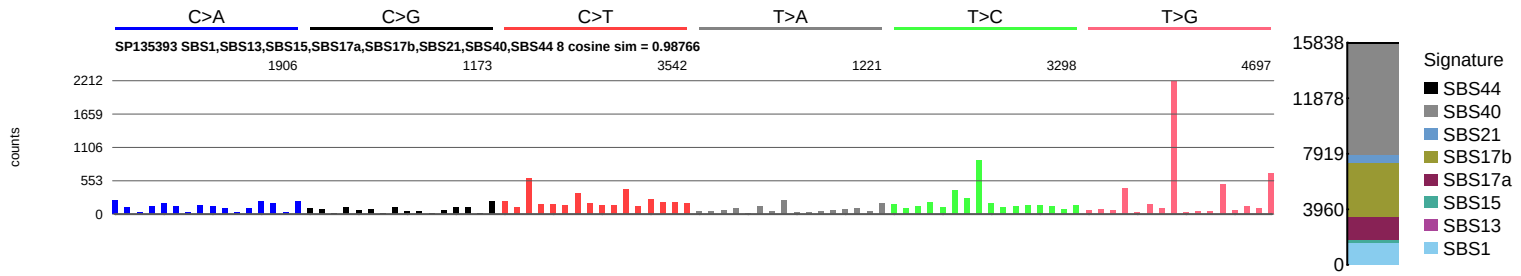
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98767
2094
1104
3432
1185
3379
4647
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98766
1909
1164
3544
1224
3298
4701
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1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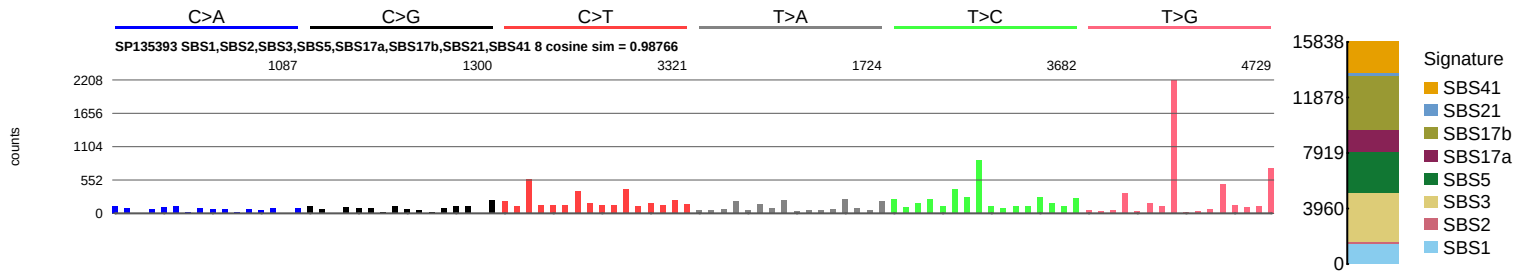
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98766
1906
1173
3542
1221
3298
4697
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98766
1087
1300
3321
1724
3682
4729
counts
2208
1656
1104
552
0
15838
11878
7919
3960
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Signature
SBS41
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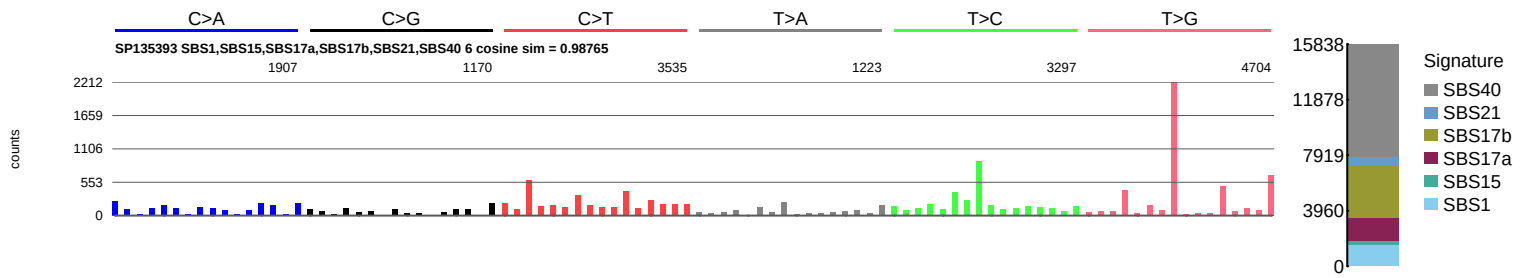
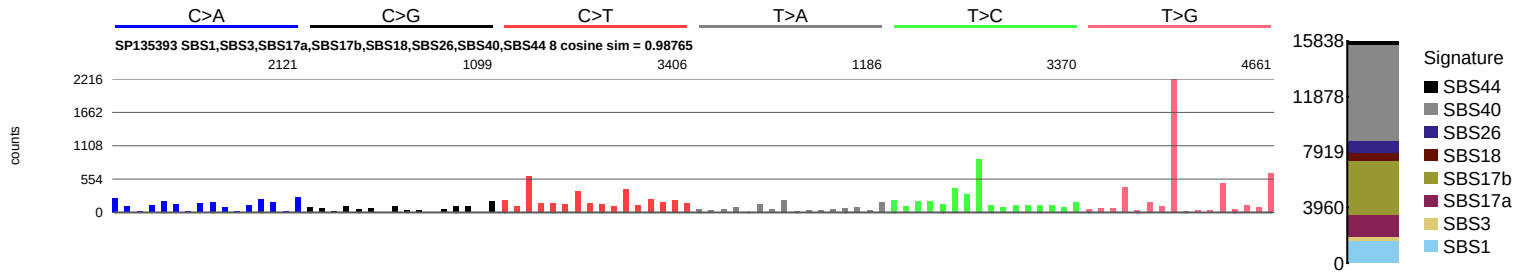
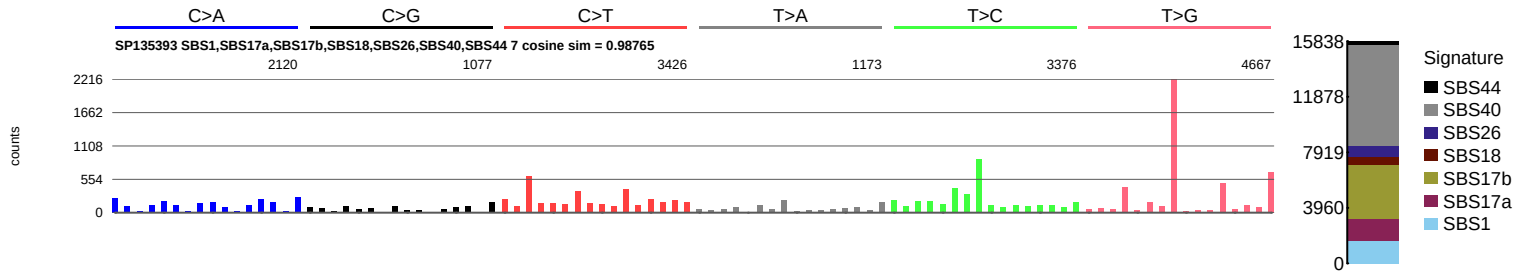
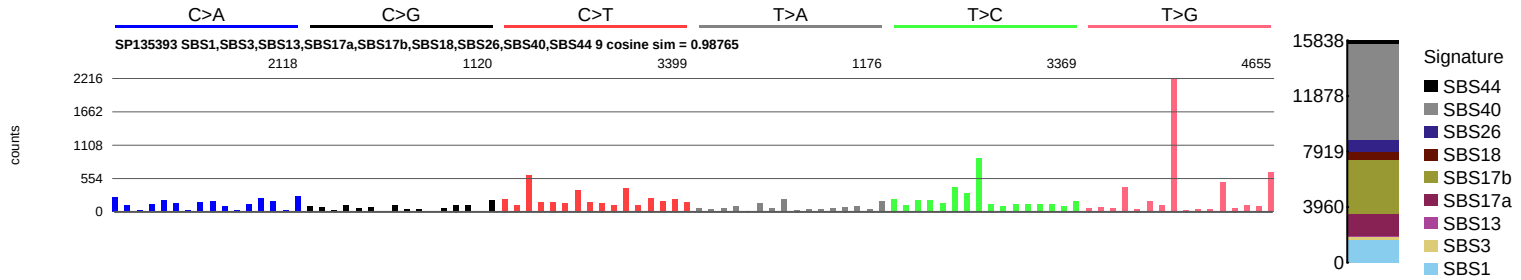
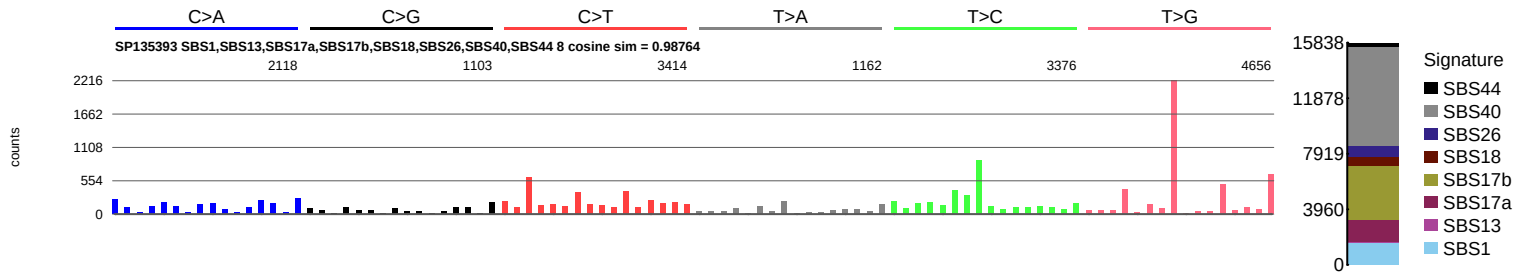
SBS2
SBS1

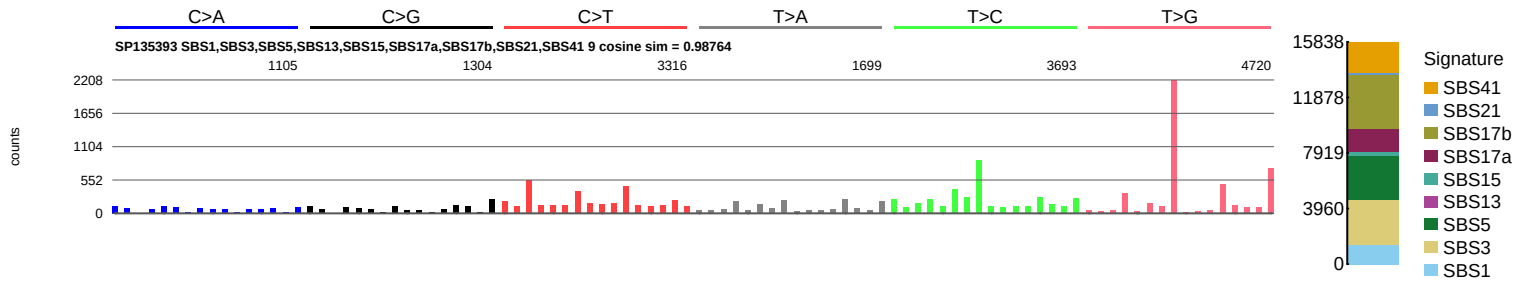

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98765
1907
1170
3535
1223
3297
4704
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98765
2121
1099
3406
1186
3370
4661
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 7 cosine sim = 0.98765
2120
1077
3426
1173
3376
4667
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 9 cosine sim = 0.98765
2118
1120
3399
1176
3369
4655
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40,SBS44 8 cosine sim = 0.98764
2118
1103
3414
1162
3376
4656
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1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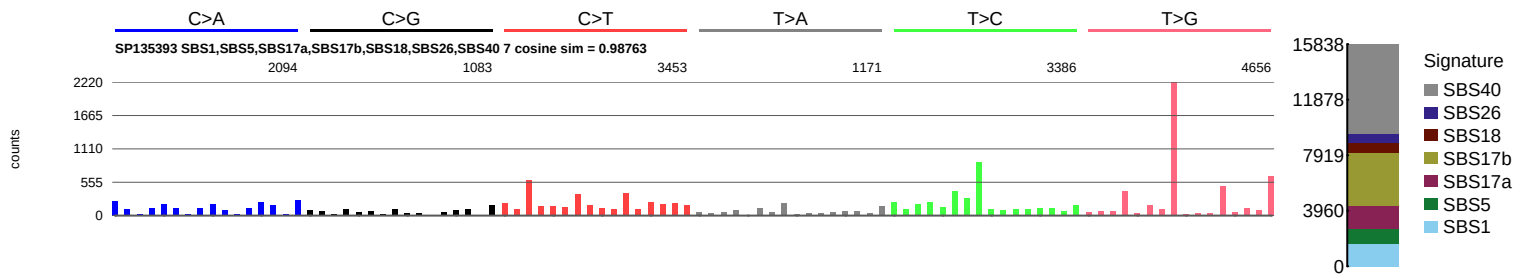
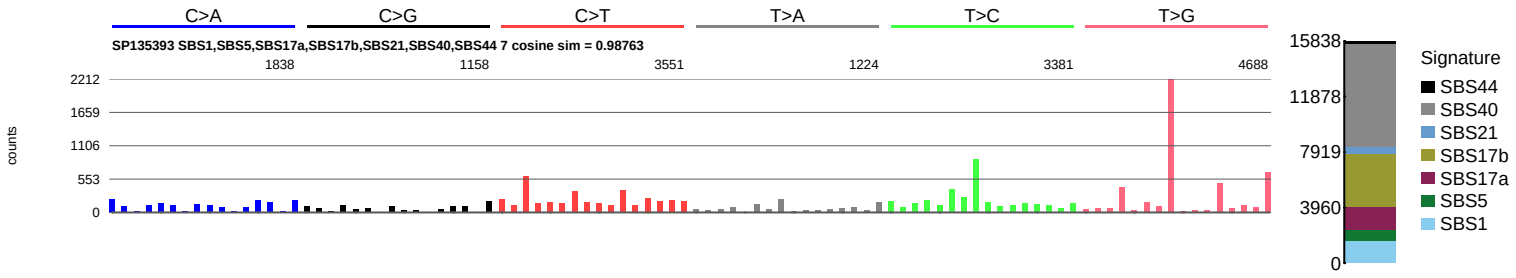
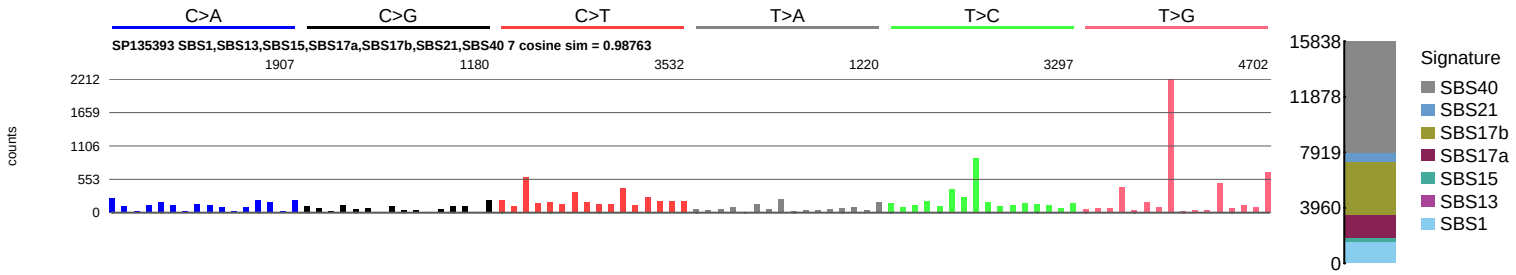
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98764
1105
1304
3316
1699
3693
4720
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98763
2094
1083
3453
1171
3386
4656
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98763
1838
1158
3551
1224
3381
4688
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1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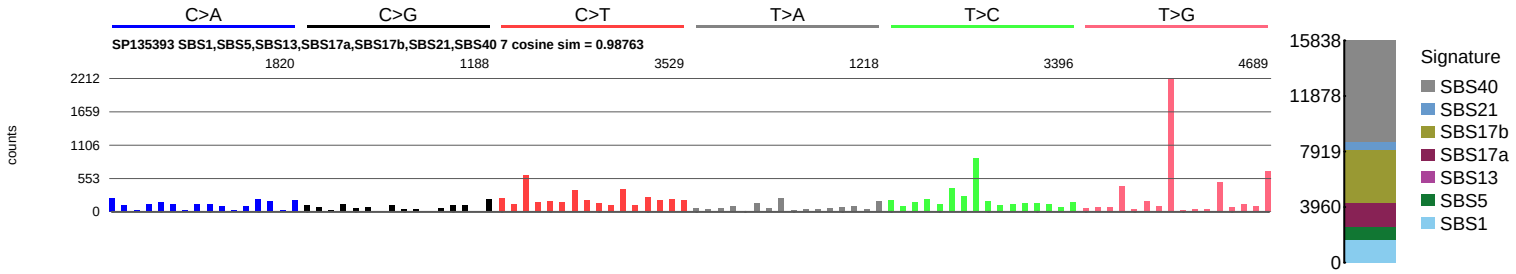
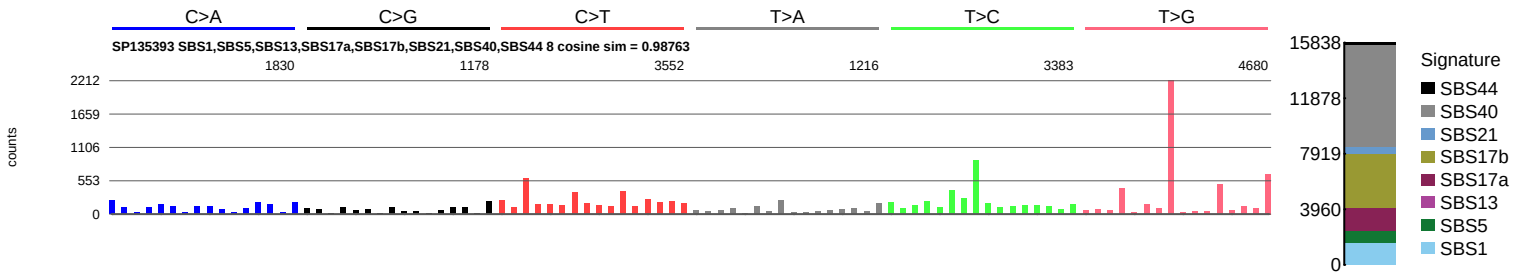
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98763
1907
1180
3532
1220
3297
4702
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40 7 cosine sim = 0.98763
1820
1188
3529
1218
3396
4689
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1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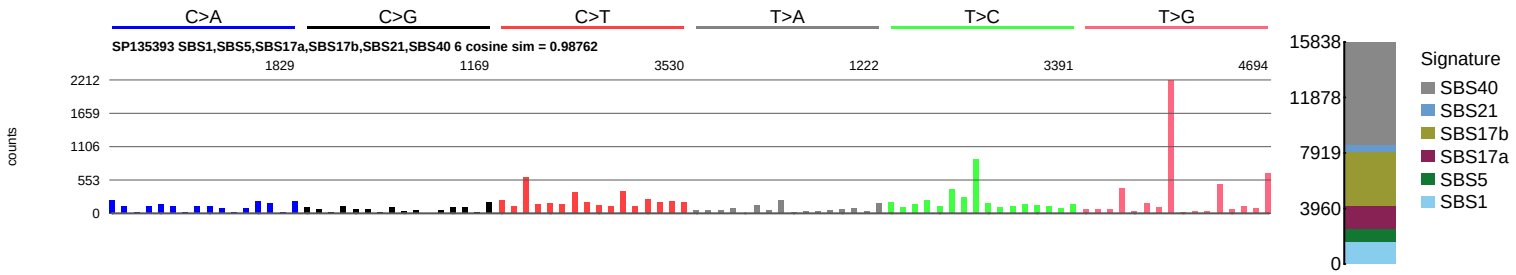
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 8 cosine sim = 0.98763
1830
1178
3552
1216
3383
4680
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1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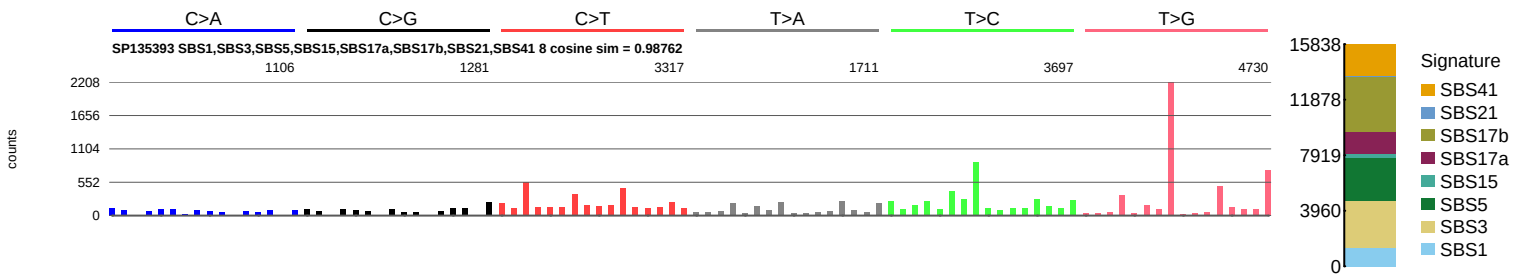
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98762
1829
1169
3530
1222
3391
4694
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS5
SBS1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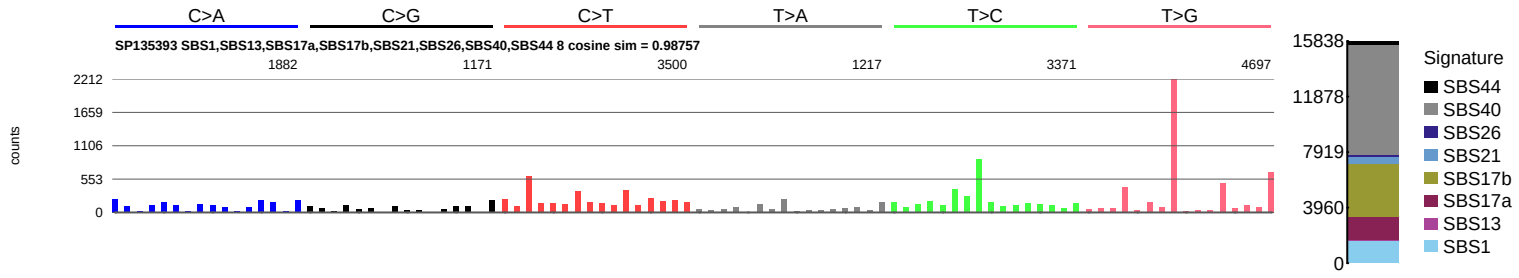
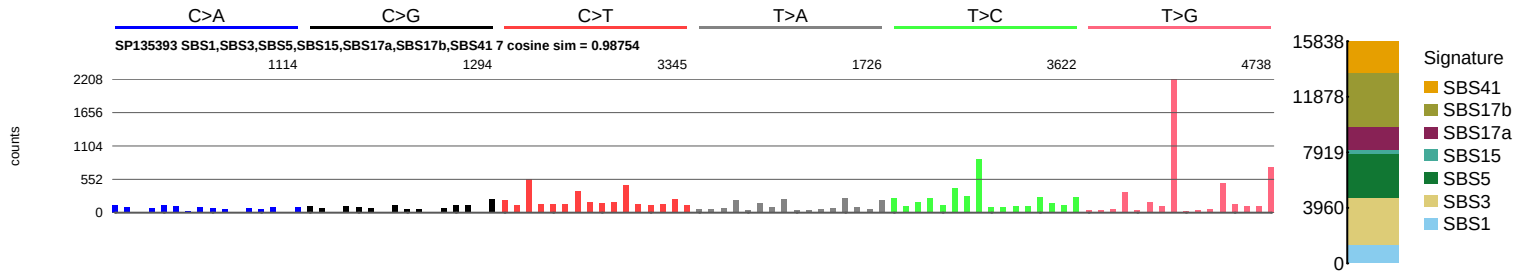
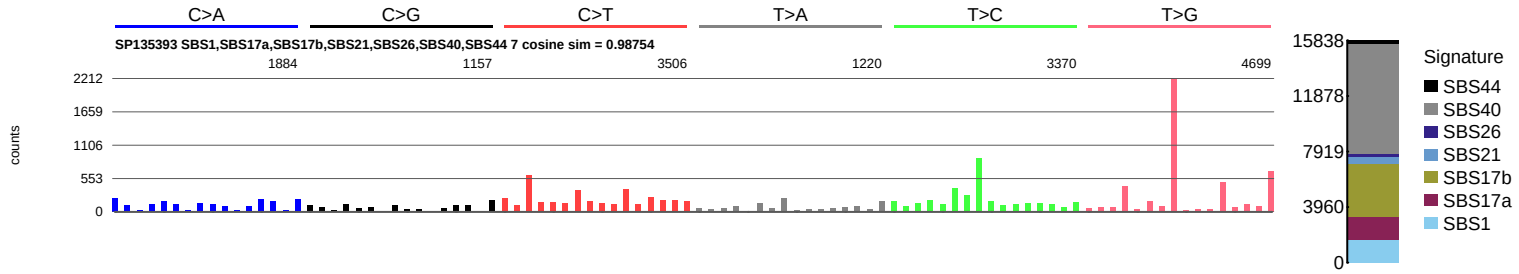
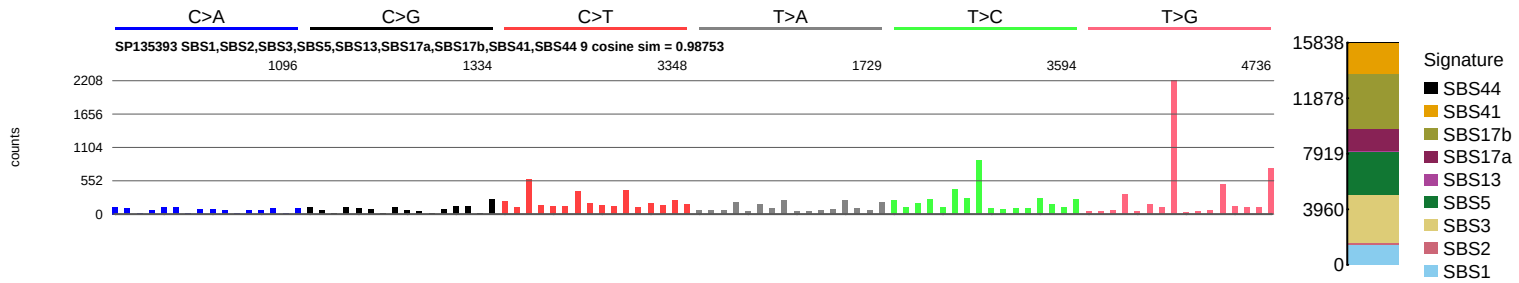
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98762
1106
1281
3317
1711
3697
4730
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 8 cosine sim = 0.98757
1882
1171
3500
1217
3371
4697
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98754
1114
1294
3345
1726
3622
4738
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40,SBS44 7 cosine sim = 0.98754
1884
1157
3506
1220
3370
4699
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1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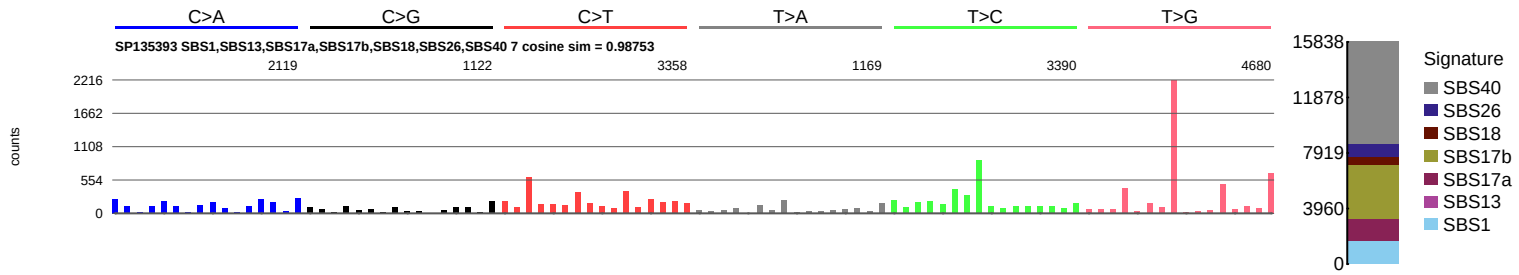
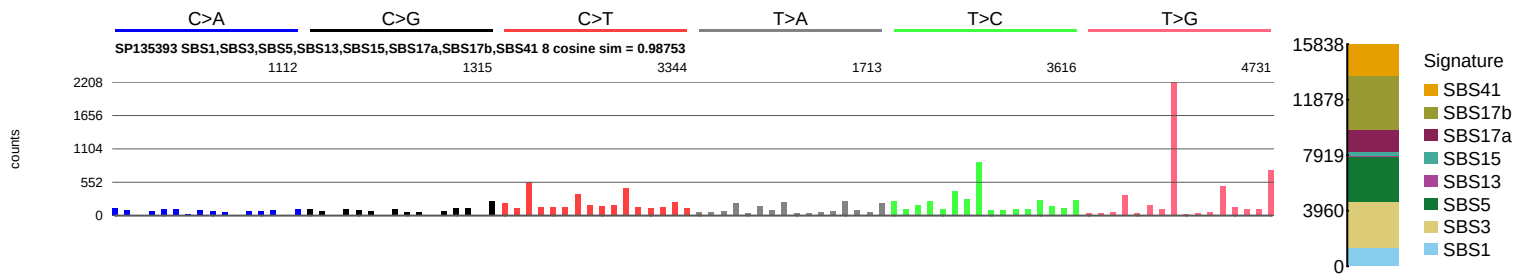
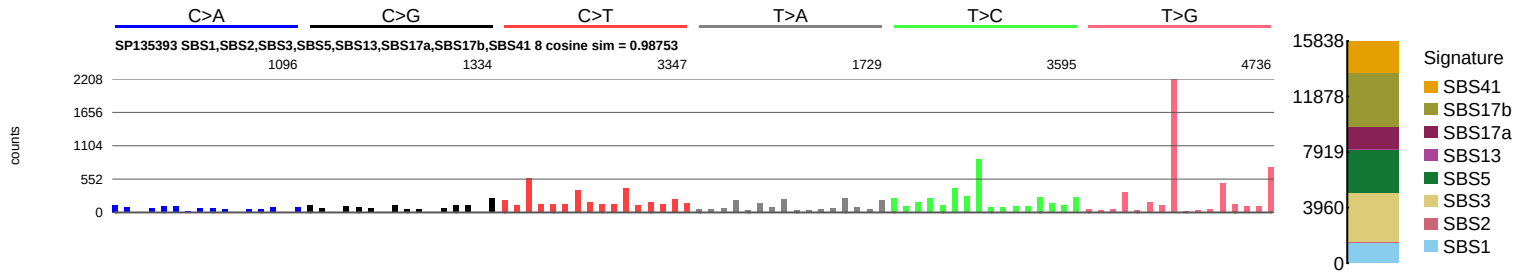
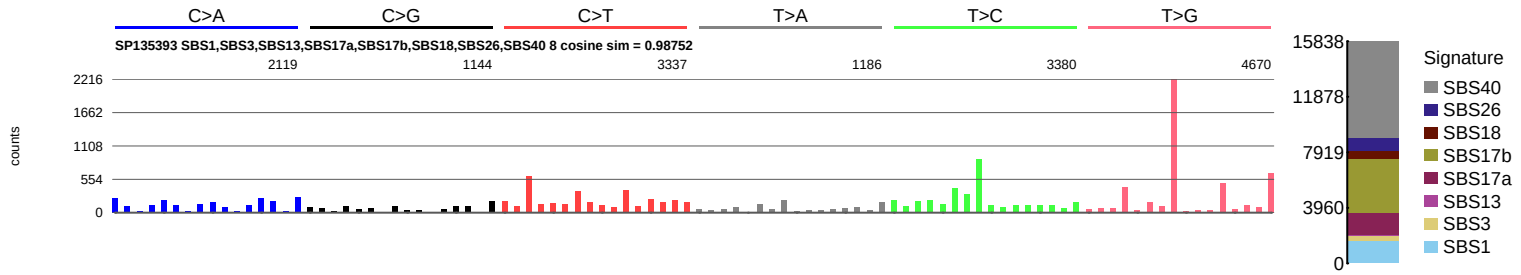
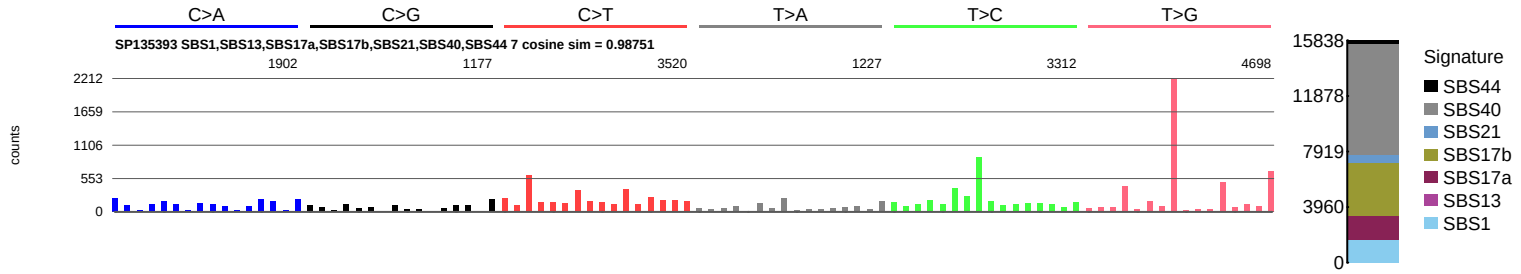
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41,SBS44 9 cosine sim = 0.98753
1096
1334
3348
1729
3594
4736
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

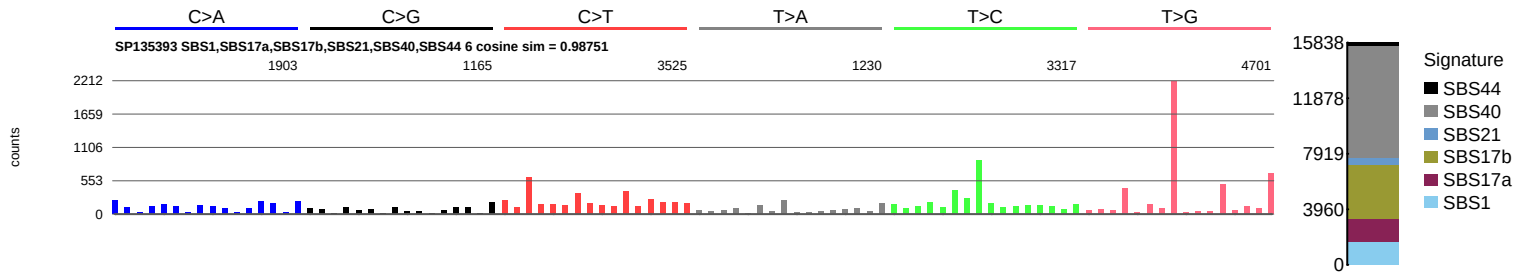

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98753
2119
1122
3358
1169
3390
4680
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1

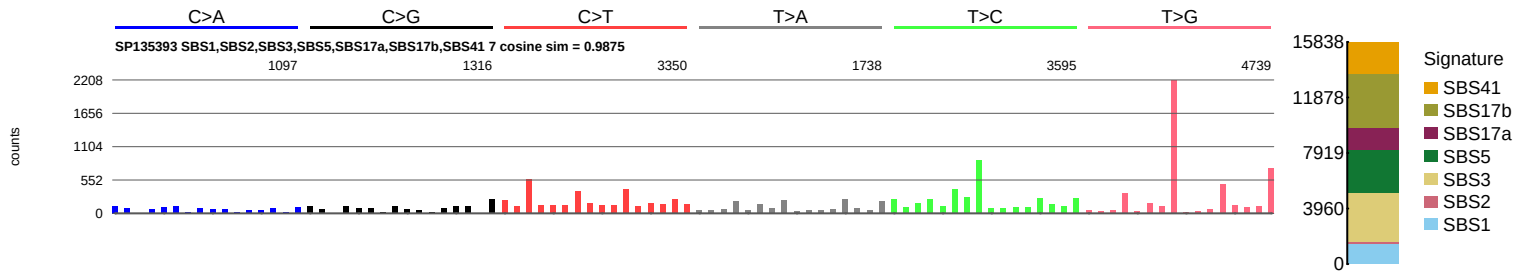

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98753
1112
1315
3344
1713
3616
4731
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

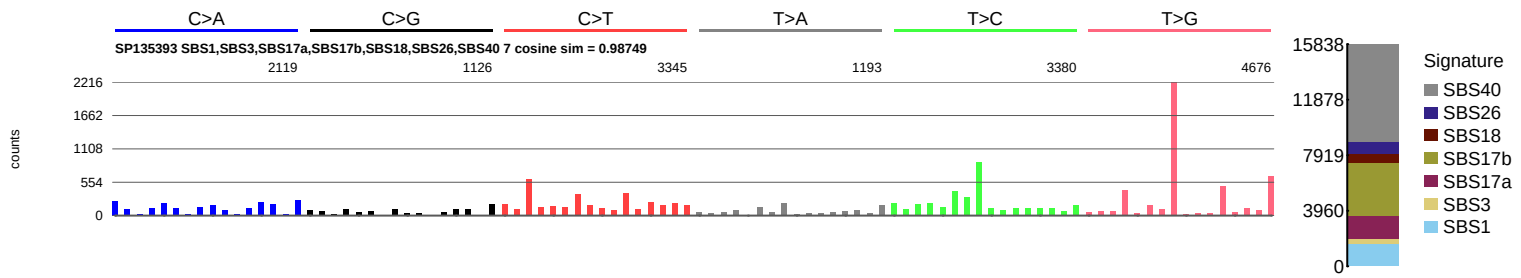

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98753
1096
1334
3347
1729
3595
4736
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1

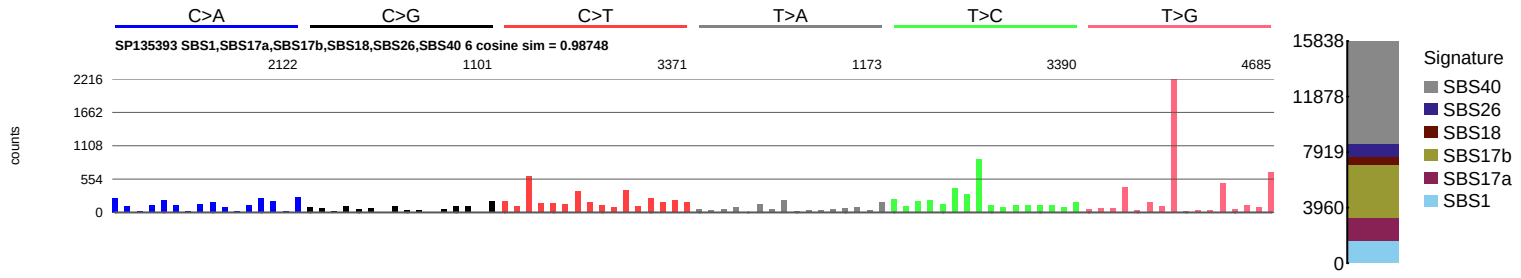

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS40 8 cosine sim = 0.98752
2119
1144
3337
1186
3380
4670
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1

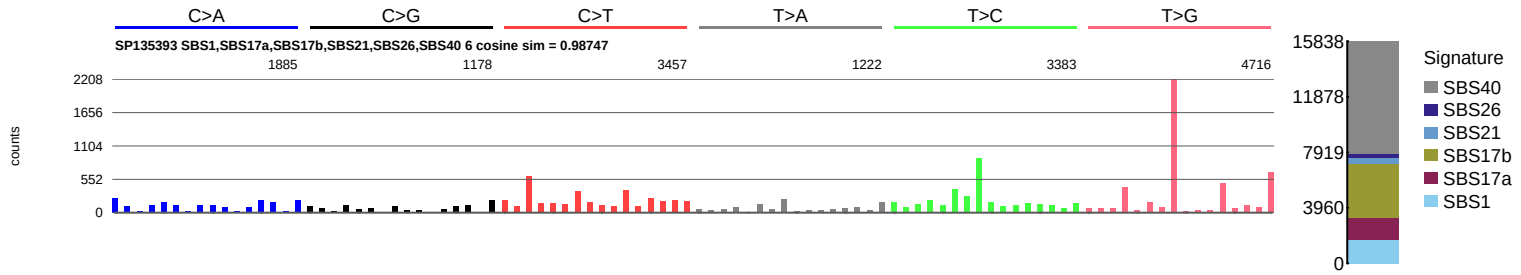

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS40,SBS44 7 cosine sim = 0.98751
1902
1177
3520
1227
3312
4698
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS1

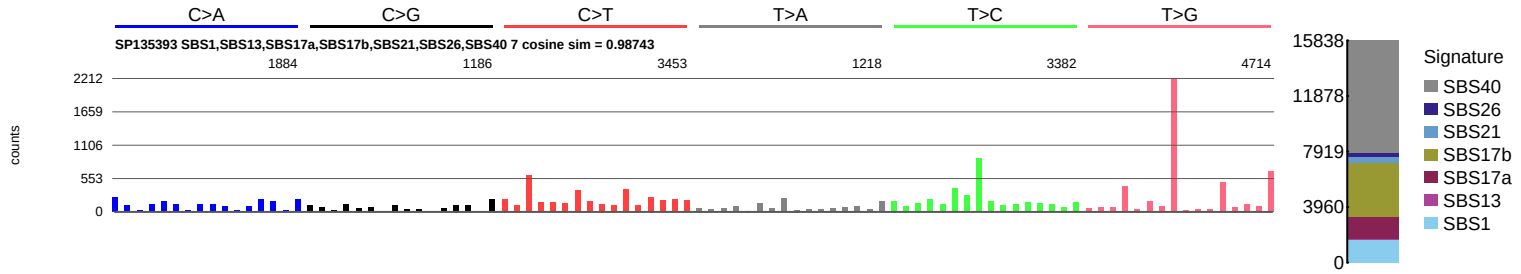

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS21,SBS40,SBS44 6 cosine sim = 0.98751
1903
1165
3525
1230
3317
4701
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS21
SBS17b
SBS17a
SBS1

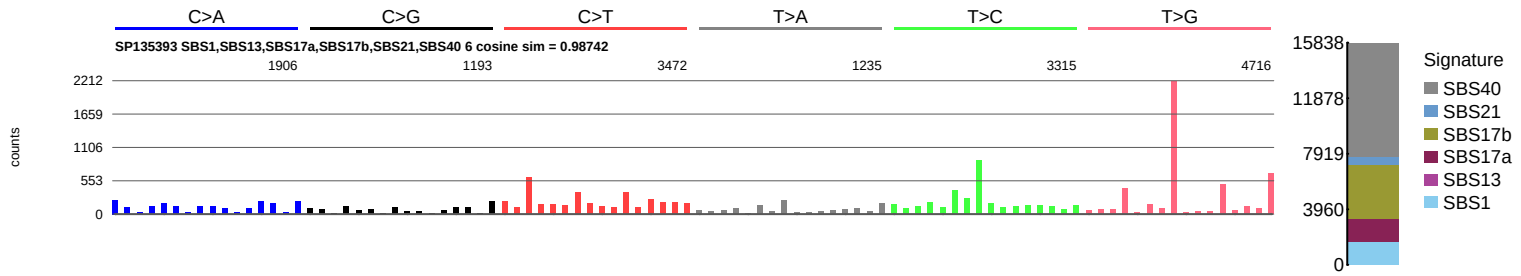

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS41 7 cosine sim = 0.9875
1097
1316
3350
1738
3595
4739
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

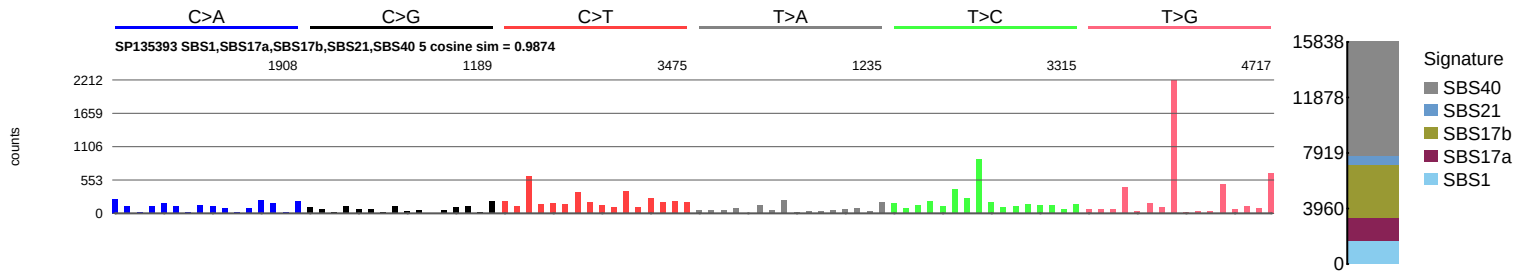

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS40 7 cosine sim = 0.98749
2119
1126
3345
1193
3380
4676
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS40 6 cosine sim = 0.98748
2122
1101
3371
1173
3390
4685
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS21,SBS26,SBS40 6 cosine sim = 0.98747
1885
1178
3457
1222
3383
4716
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS40 7 cosine sim = 0.98743
1884
1186
3453
1218
3382
4714
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS21,SBS40 6 cosine sim = 0.98742
1906
1193
3472
1235
3315
4716
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS21
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS21,SBS40 5 cosine sim = 0.9874
1908
1189
3475
1235
3315
4717
counts
2212
1659
1106
553
0
15838
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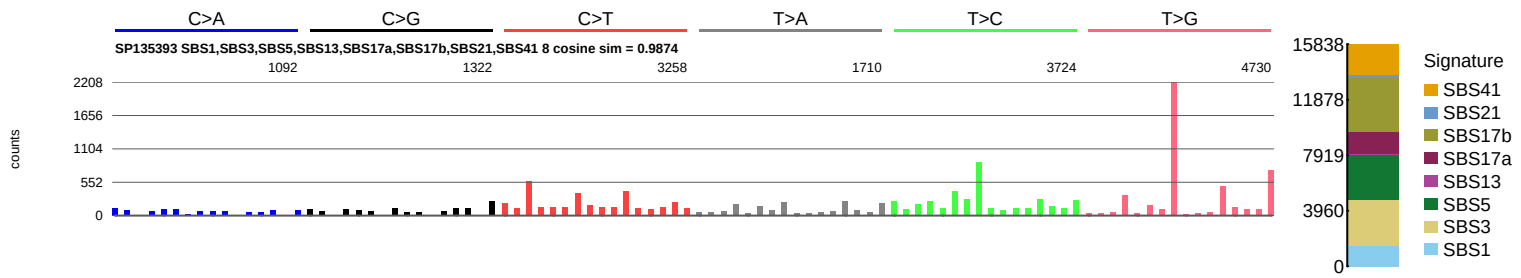
SBS40
SBS21
SBS17b
SBS17a
SBS1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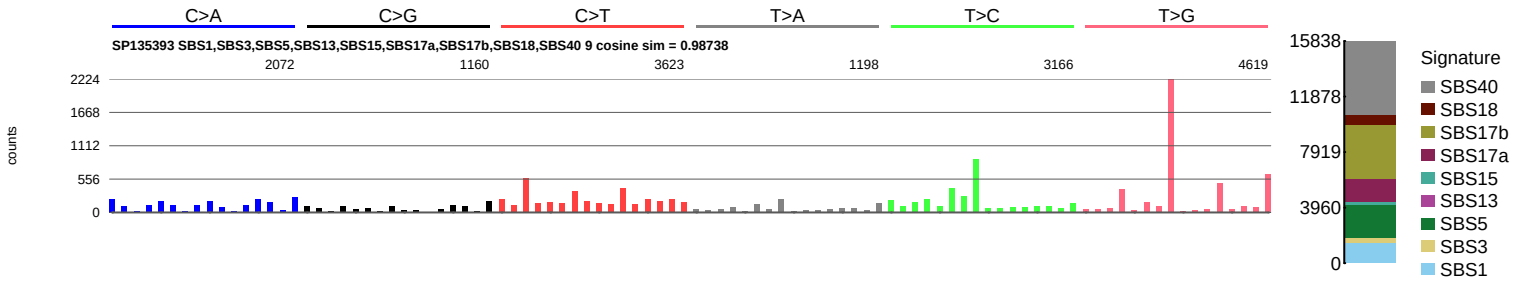
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.9874
1092
1322
3258
1710
3724
4730
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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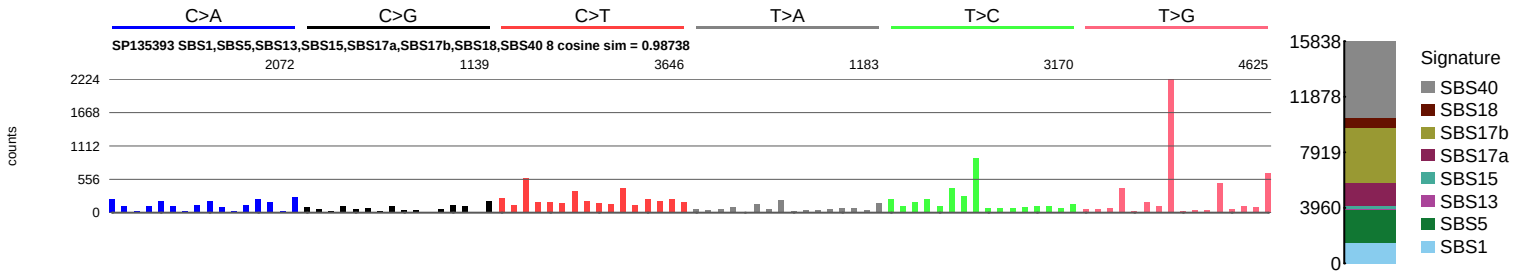
SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 9 cosine sim = 0.98738
2072
1160
3623
1198
3166
4619
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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C>G
C>T
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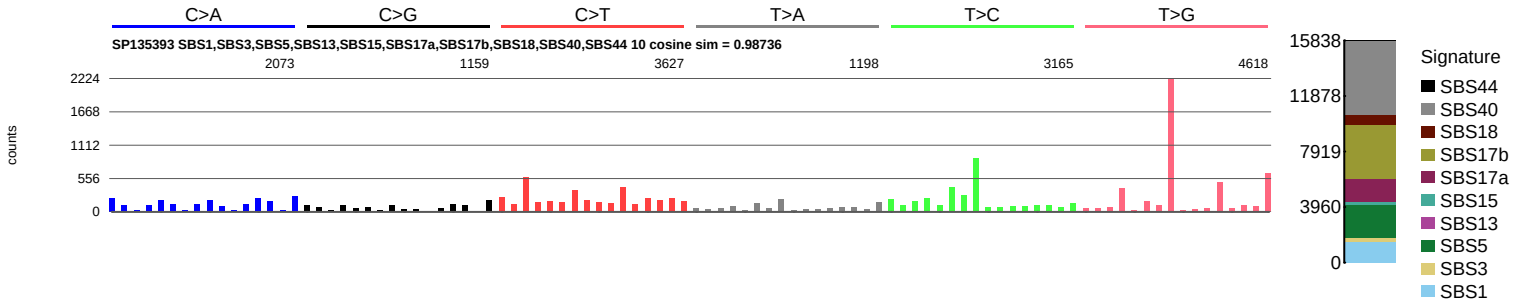
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98738
2072
1139
3646
1183
3170
4625
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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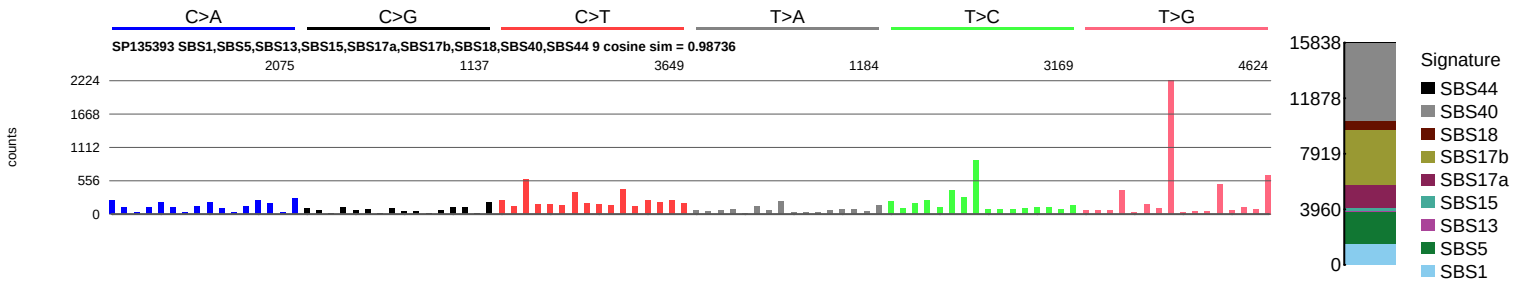
SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 10 cosine sim = 0.98736
2073
1159
3627
1198
3165
4618
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1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C>G
C>T
T>A
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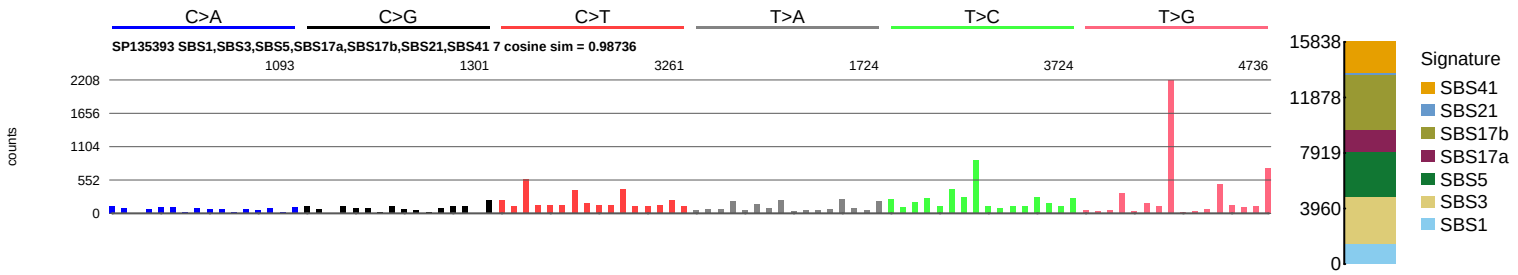
T>G
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98736
2075
1137
3649
1184
3169
4624
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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C>G
C>T
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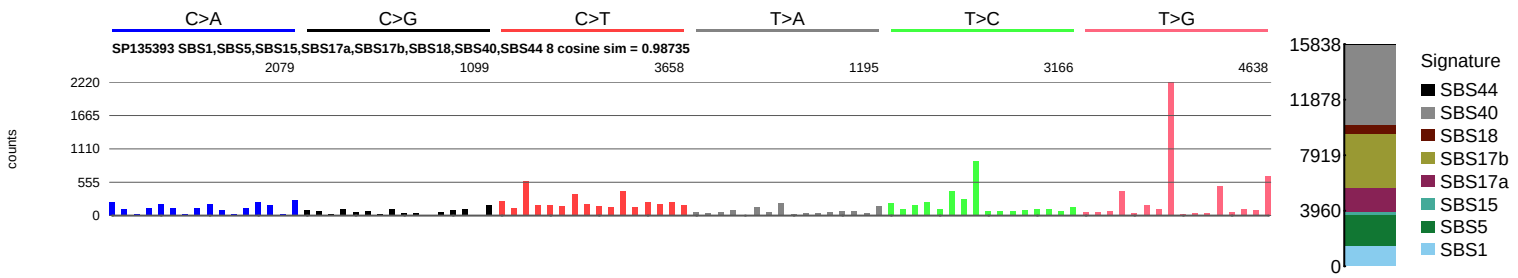
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98736
1093
1301
3261
1724
3724
4736
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS1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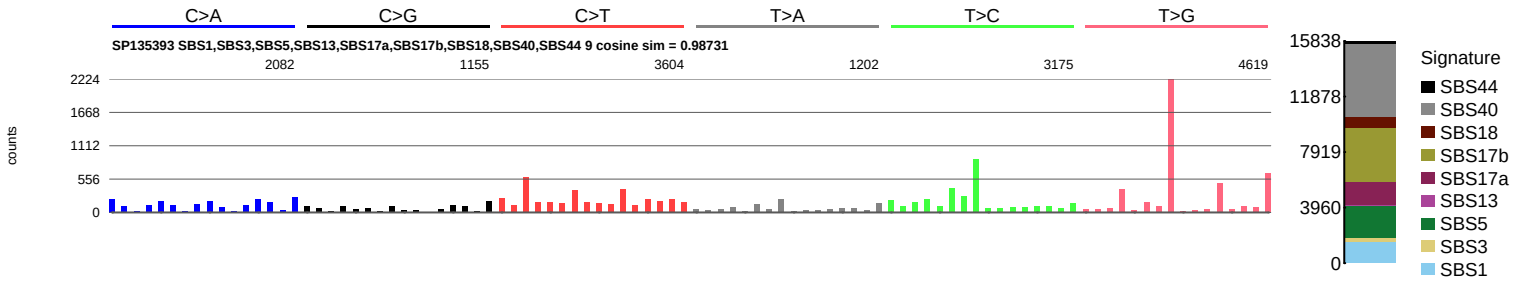
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98735
2079
1099
3658
1195
3166
4638
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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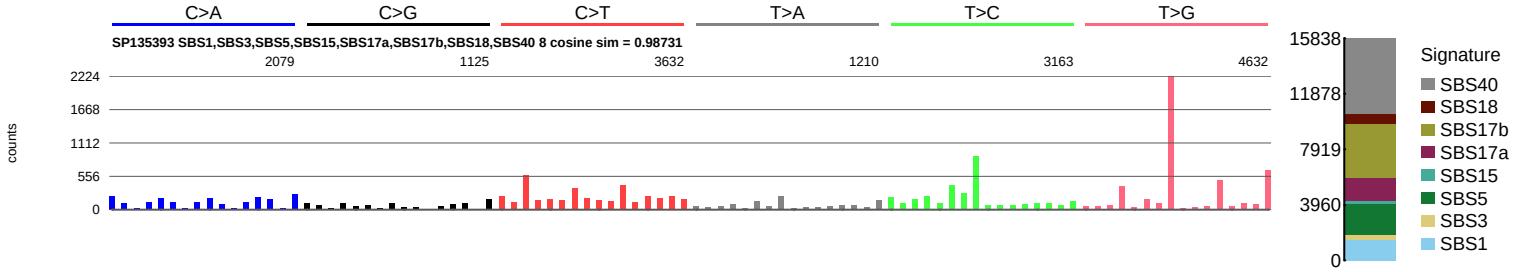
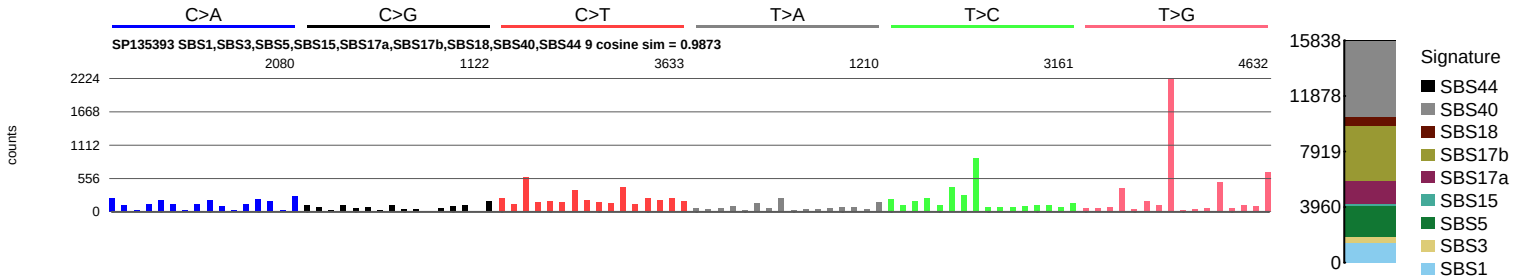
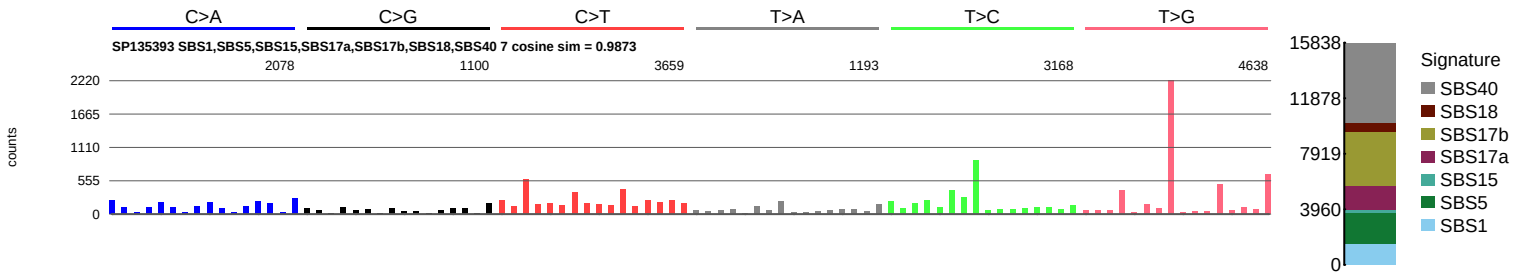
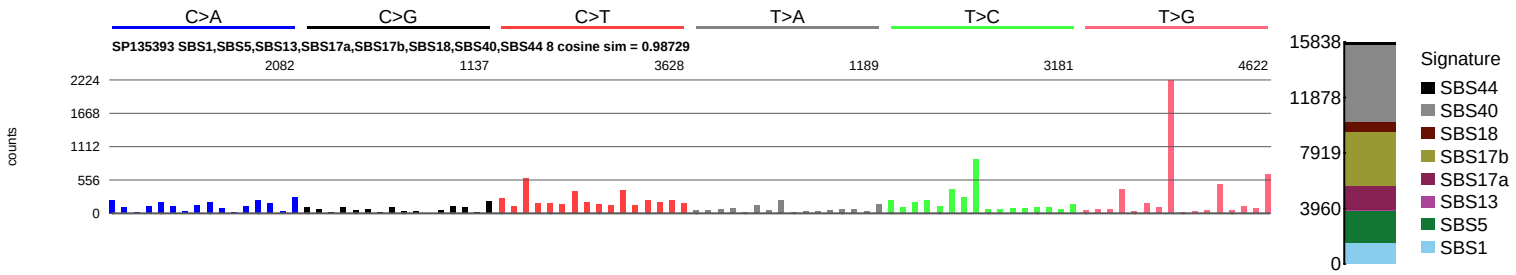
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98731
2082
1155
3604
1202
3175
4619
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98731
2079
1125
3632
1210
3163
4632
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.9873
2080
1122
3633
1210
3161
4632
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.9873
2078
1100
3659
1193
3168
4638
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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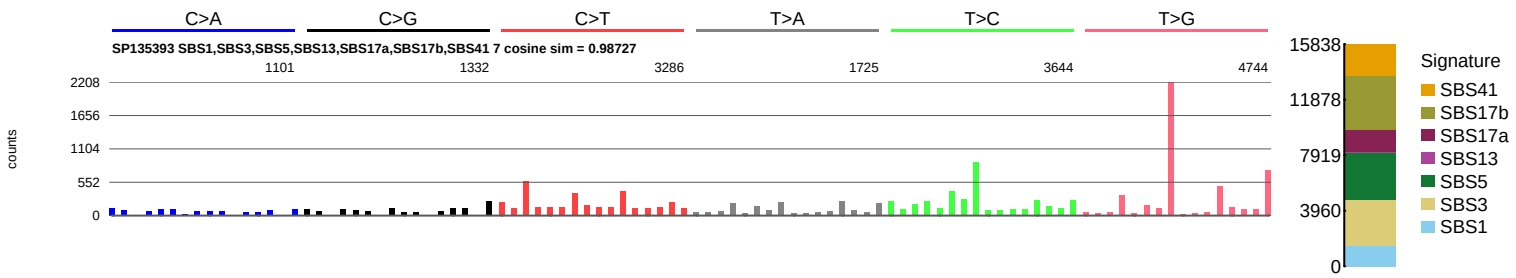
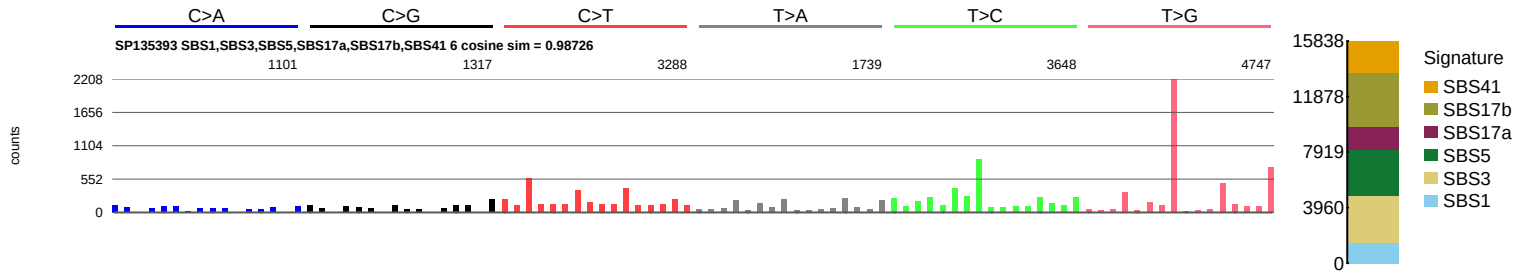
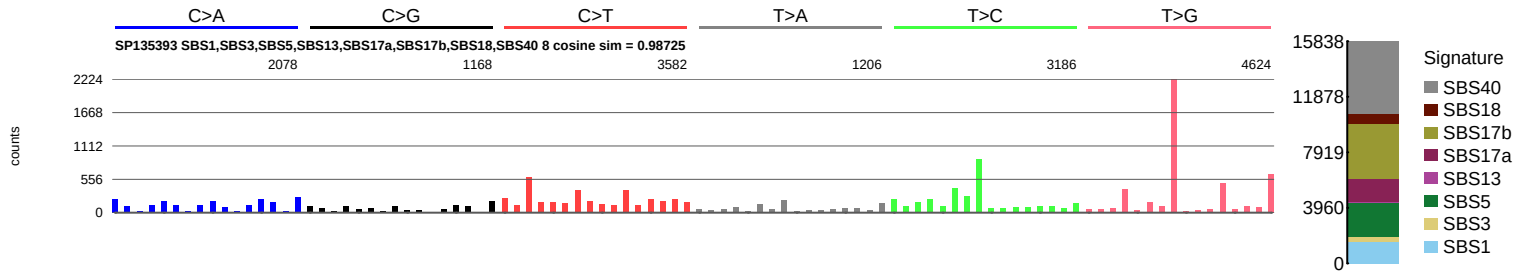
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98729
2082
1137
3628
1189
3181
4622
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98727
1101
1332
3286
1725
3644
4744
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98726
1101
1317
3288
1739
3648
4747
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS3
SBS1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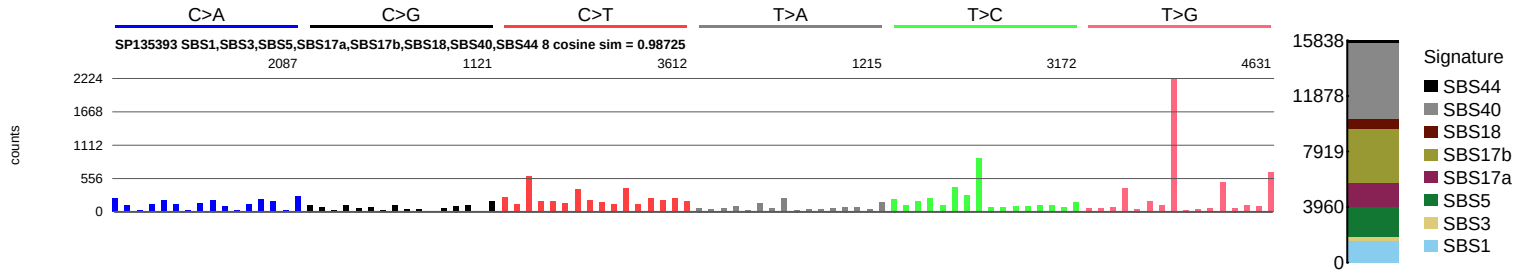
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 8 cosine sim = 0.98725
2078
1168
3582
1206
3186
4624
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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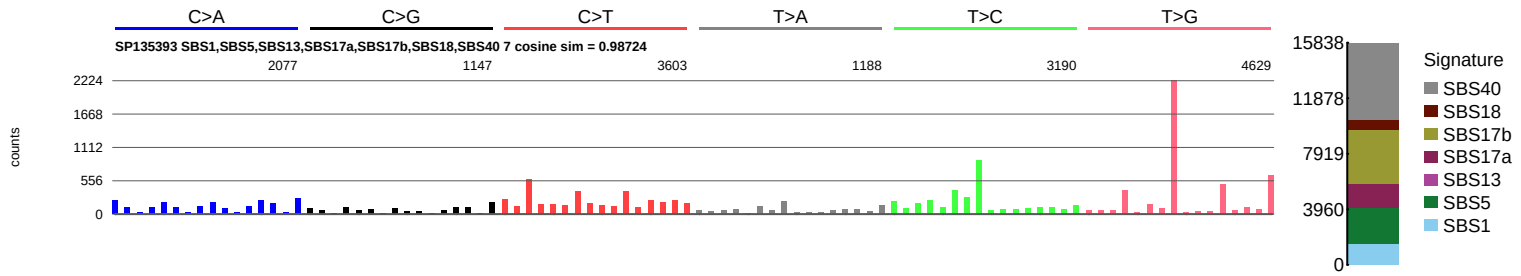
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98725
2087
1121
3612
1215
3172
4631
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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C>G
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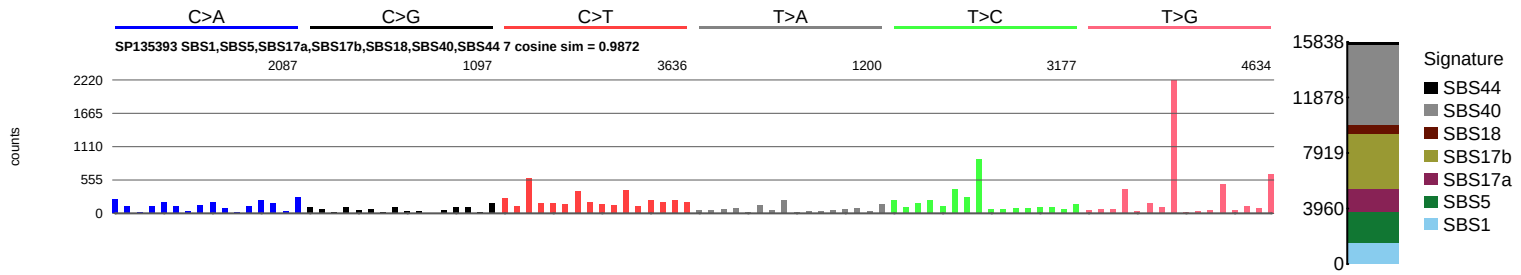
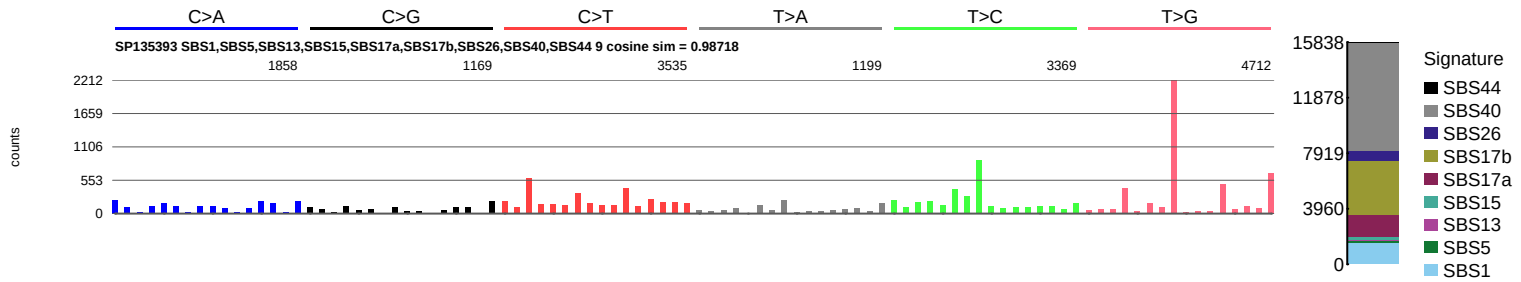
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98724
2077
1147
3603
1188
3190
4629
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.9872
2087
1097
3636
1200
3177
4634
counts
2220
1665
1110
555
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS1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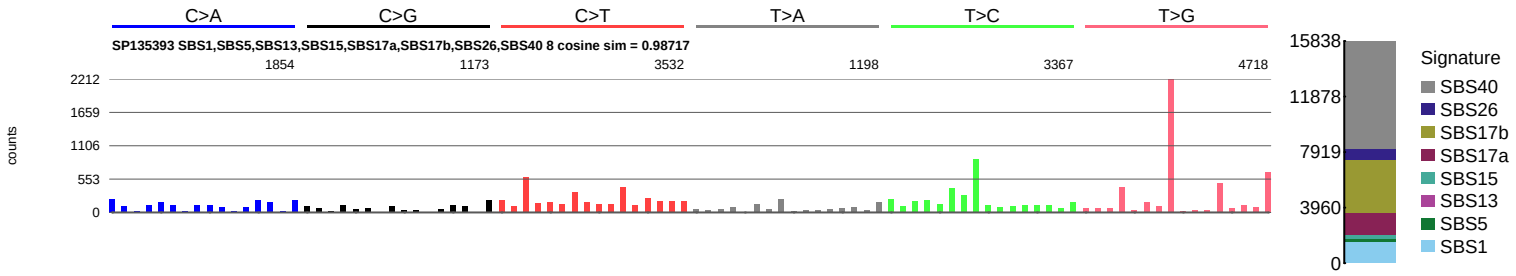
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 9 cosine sim = 0.98718
1858
1169
3535
1199
3369
4712
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 8 cosine sim = 0.98717
1854
1173
3532
1198
3367
4718
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
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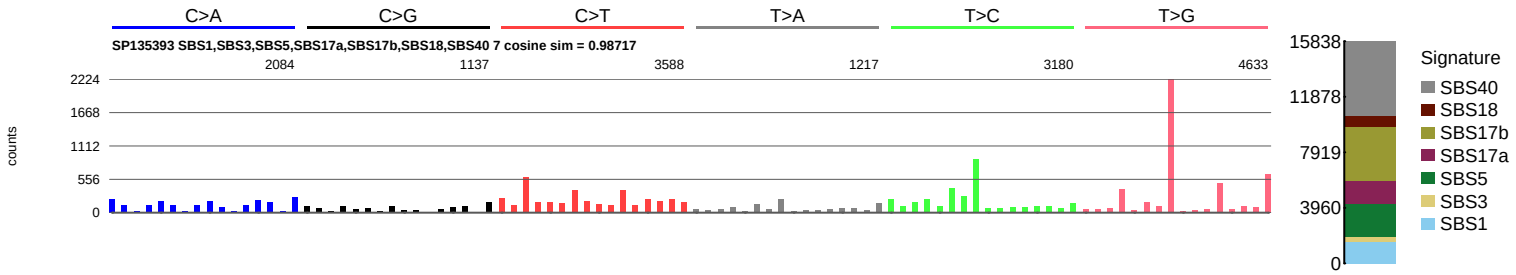
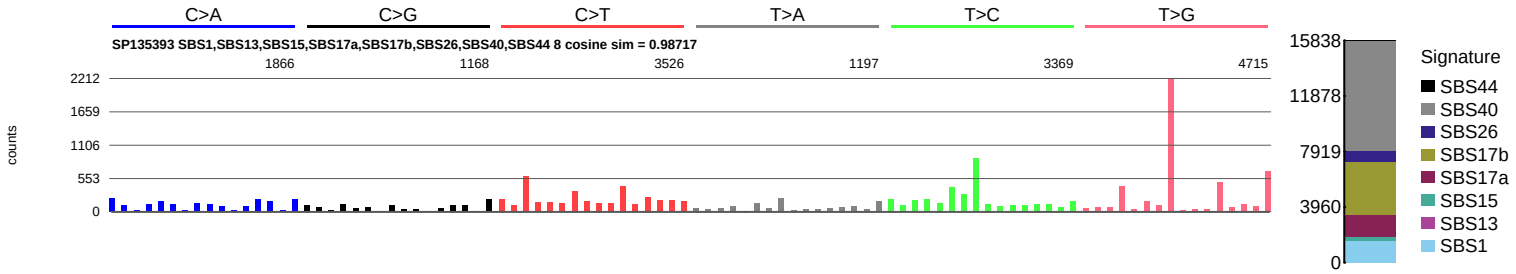
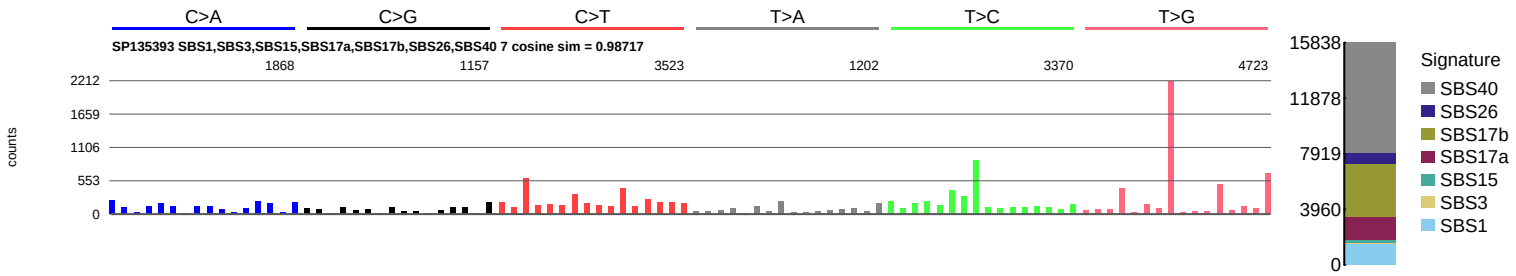
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98717
2084
1137
3588
1217
3180
4633
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98717
1866
1168
3526
1197
3369
4715
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98717
1868
1157
3523
1202
3370
4723
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS17b
SBS17a
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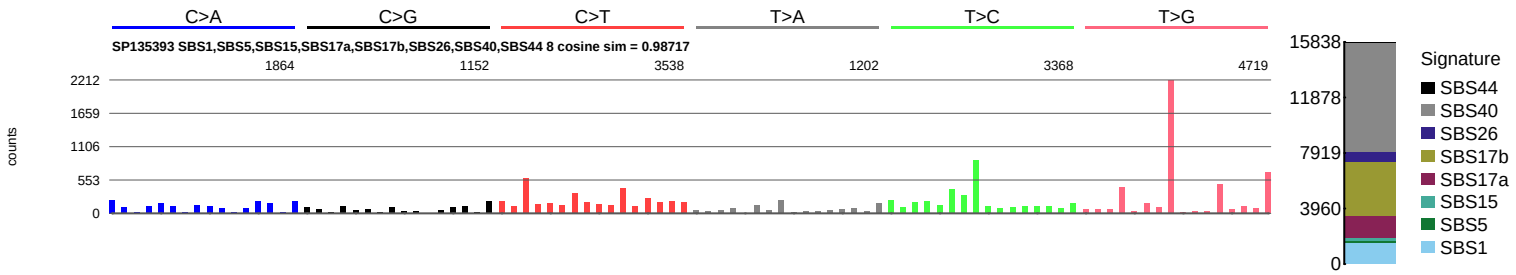
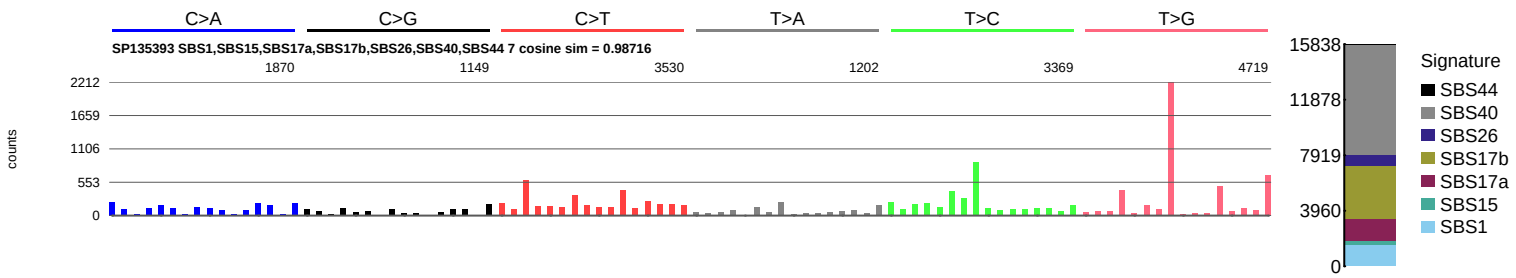
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98717
1864
1152
3538
1202
3368
4719
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98716
1870
1149
3530
1202
3369
4719
counts
2212
1659
1106
553
0
15838
11878
7919
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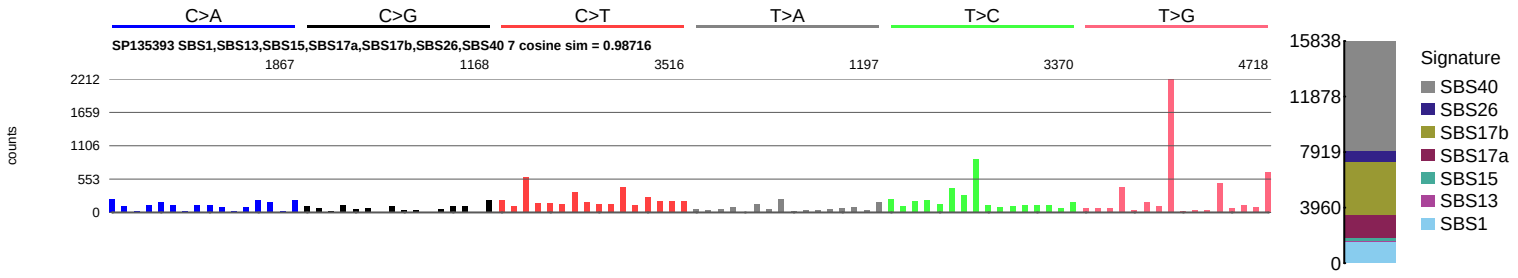
SBS15
SBS1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SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98716
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1168
3516
1197
3370
4718
counts
2212
1659
1106
553
0
15838
11878
7919
3960
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SBS13
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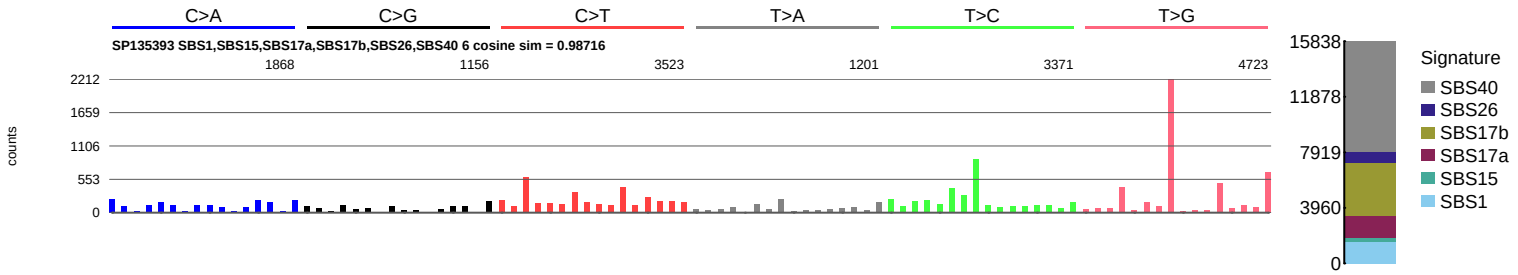
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SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98716
1868
1156
3523
1201
3371
4723
counts
2212
1659
1106
553
0
15838
11878
7919
3960
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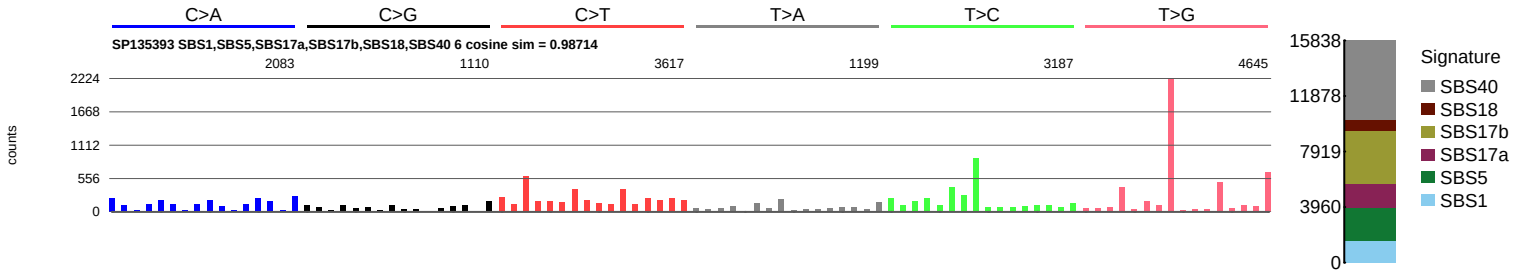
SBS17a
SBS15
SBS1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SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98714
2083
1110
3617
1199
3187
4645
counts
2224
1668
1112
556
0
15838
11878
7919
3960
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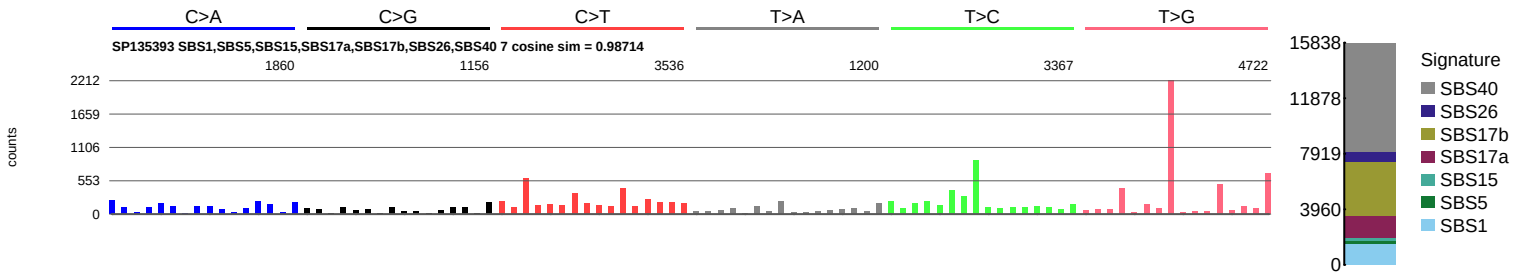
SBS17b
SBS17a
SBS5
SBS1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SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98714
1860
1156
3536
1200
3367
4722
counts
2212
1659
1106
553
0
15838
11878
7919
3960
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SBS26
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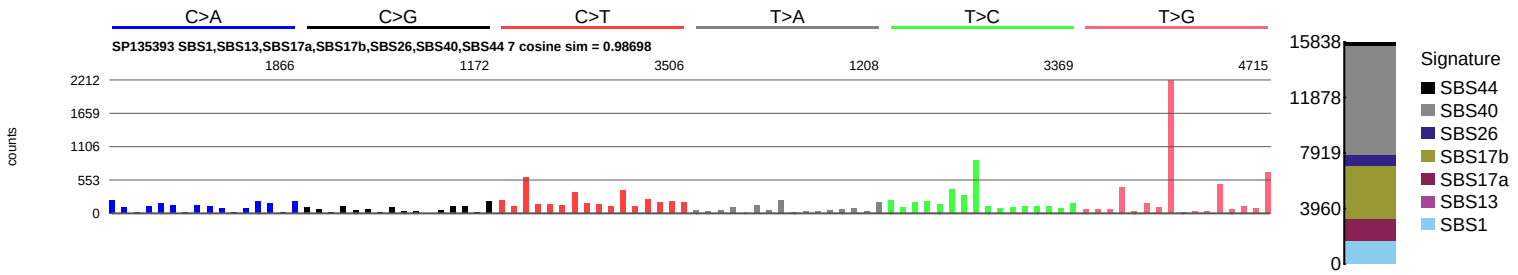
SBS17a
SBS15
SBS5
SBS1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SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98698
1866
1172
3506
1208
3369
4715
counts
2212
1659
1106
553
0
15838
11878
7919
3960
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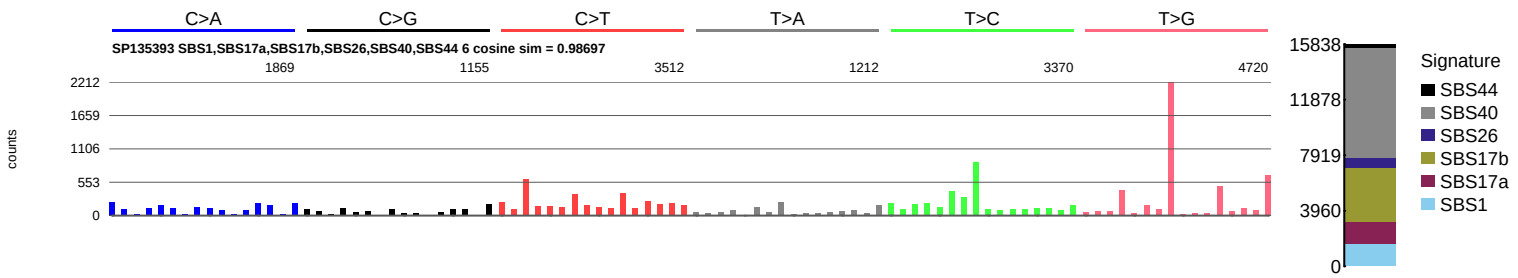
SBS17b
SBS17a
SBS13
SBS1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SP135393 SBS1,SBS17a,SBS17b,SBS26,SBS40,SBS44 6 cosine sim = 0.98697
1869
1155
3512
1212
3370
4720
counts
2212
1659
1106
553
0
15838
11878
7919
3960
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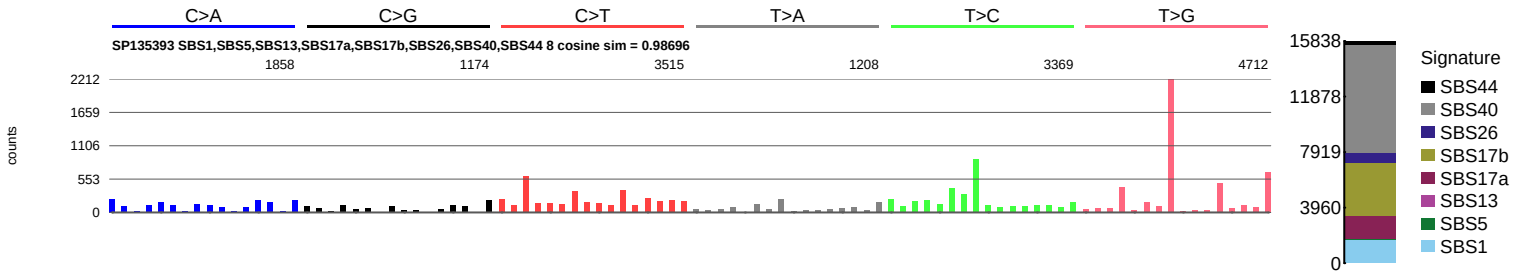
SBS17a
SBS1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SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40,SBS44 8 cosine sim = 0.98696
1858
1174
3515
1208
3369
4712
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
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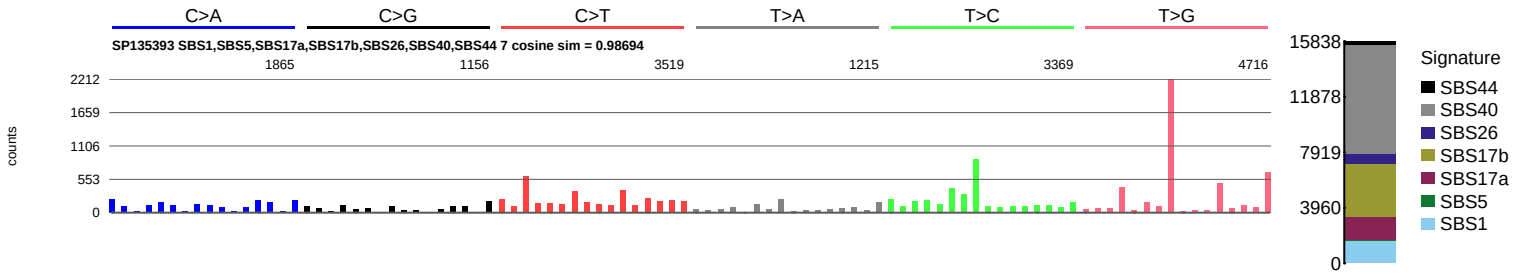
SBS44
SBS40
SBS26
SBS17b
SBS17a
SBS13
SBS5
SBS1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C>G
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SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40,SBS44 7 cosine sim = 0.98694
1865
1156
3519
1215
3369
4716
counts
2212
1659
1106
553
0
15838
11878
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Signature
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SBS40
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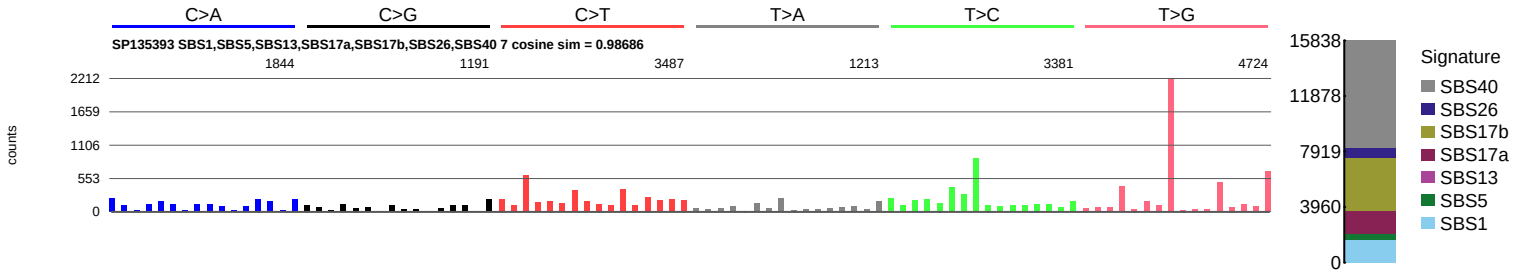
SBS17b
SBS17a
SBS5
SBS1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SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98686
1844
1191
3487
1213
3381
4724
counts
2212
1659
1106
553
0
15838
11878
7919
3960
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SBS26
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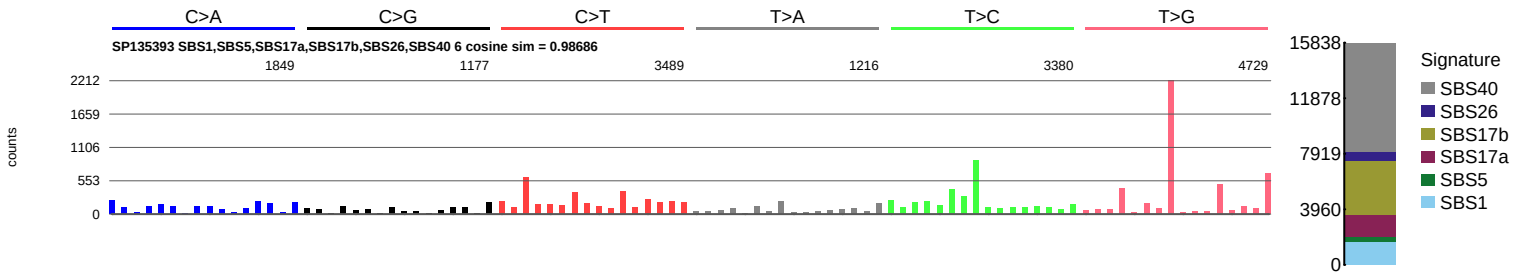
SBS17a
SBS13
SBS5
SBS1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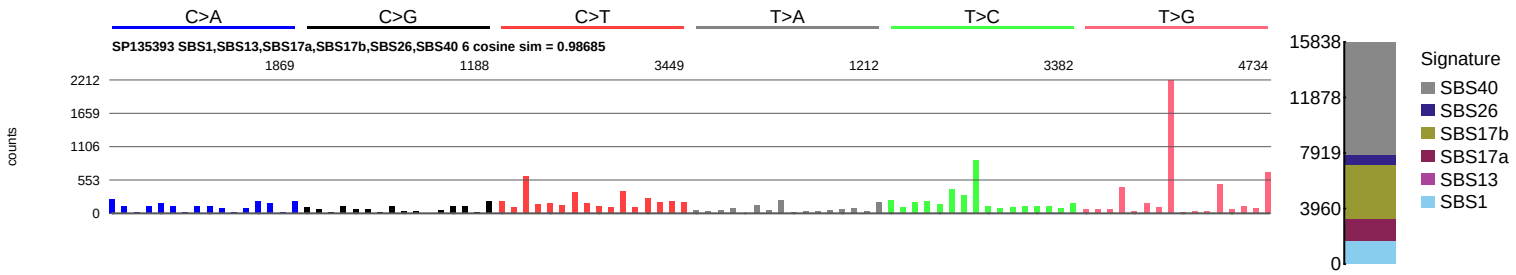
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98686
1849
1177
3489
1216
3380
4729
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS5
SBS1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C>G
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T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98685
1869
1188
3449
1212
3382
4734
counts
2212
1659
1106
553
0
15838
11878
7919
3960
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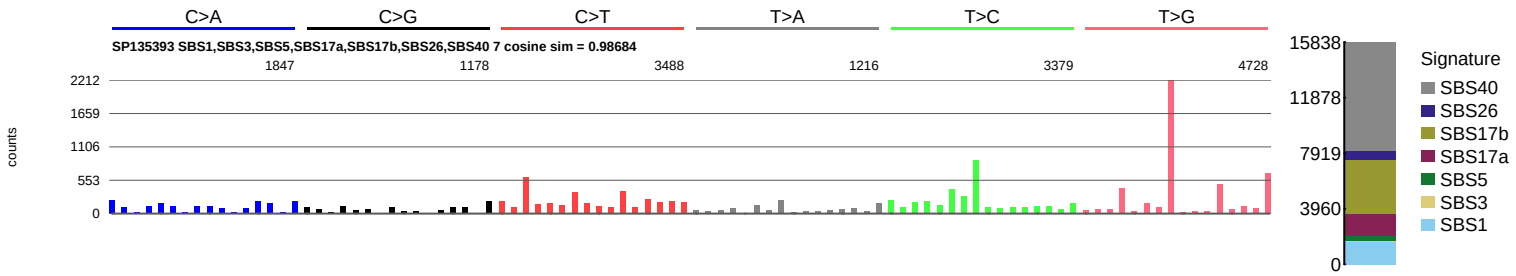
SBS26
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98684
1847
1178
3488
1216
3379
4728
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
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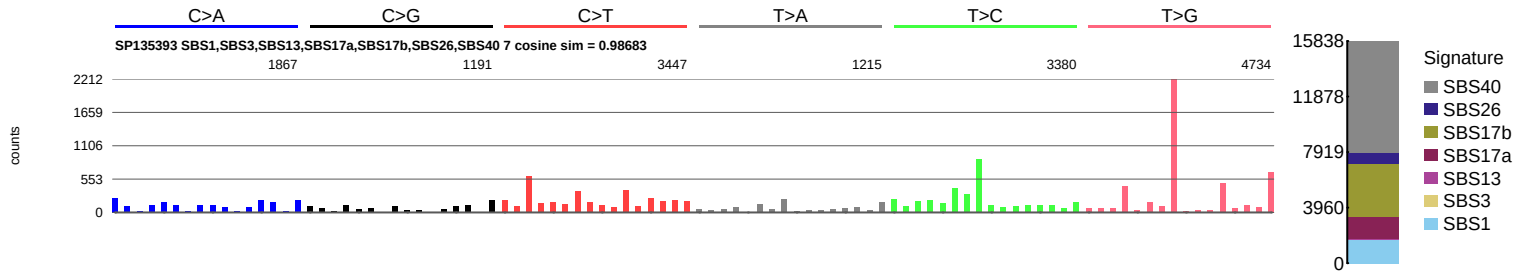
SBS26
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
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T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS40 7 cosine sim = 0.98683
1867
1191
3447
1215
3380
4734
counts
2212
1659
1106
553
0
15838
11878
7919
3960
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SBS26
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SBS17a
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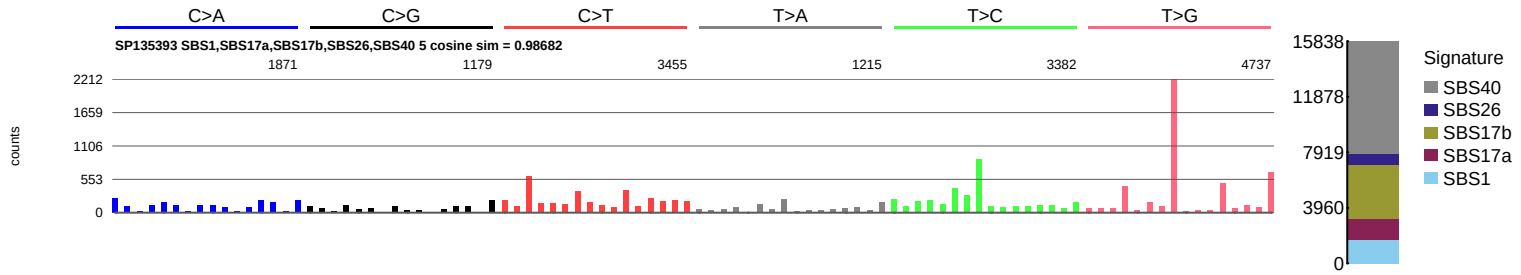
SBS3
SBS1


C>A
C>G
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T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS26,SBS40 5 cosine sim = 0.98682
1871
1179
3455
1215
3382
4737
counts
2212
1659
1106
553
0
15838
11878
7919
3960
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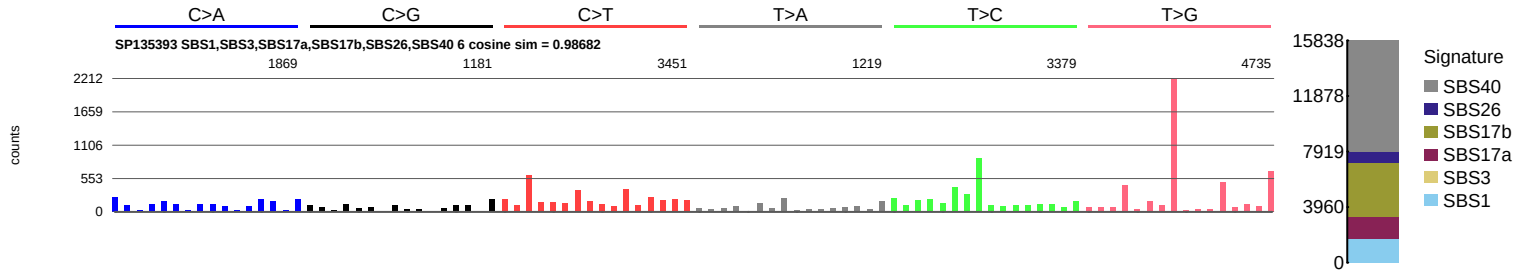
Signature
SBS40
SBS26
SBS17b
SBS17a
SBS1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C>G
C>T
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T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS40 6 cosine sim = 0.98682
1869
1181
3451
1219
3379
4735
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
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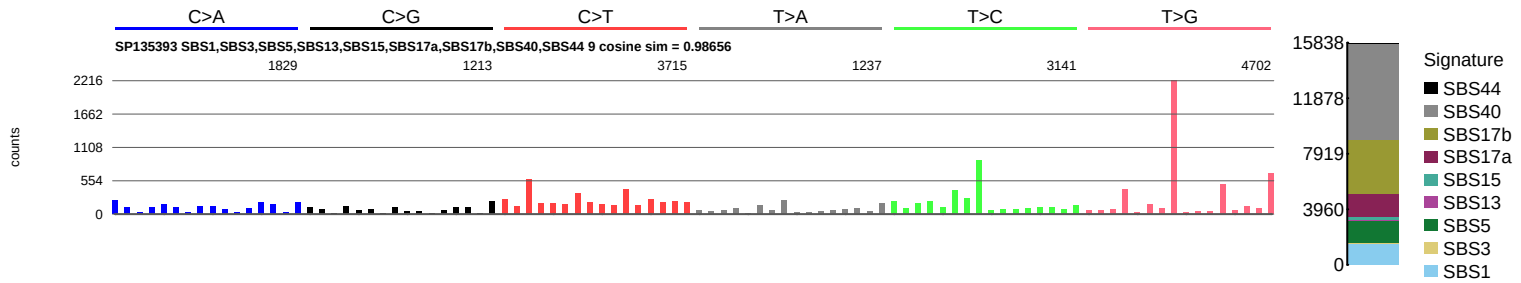
SBS40
SBS26
SBS17b
SBS17a
SBS3
SBS1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C>T
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T>C
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SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 9 cosine sim = 0.98656
1829
1213
3715
1237
3141
4702
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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SBS40
SBS17b
SBS17a
SBS15
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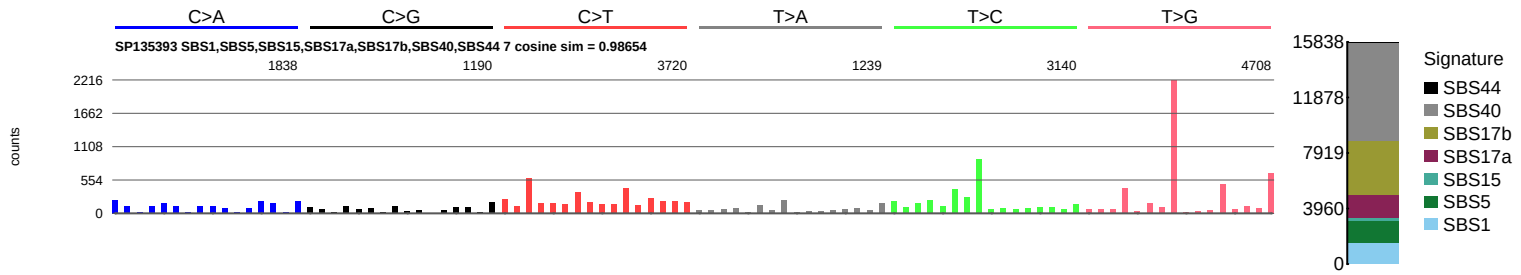
SBS5
SBS3
SBS1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T>C
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SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98654
1838
1190
3720
1239
3140
4708
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
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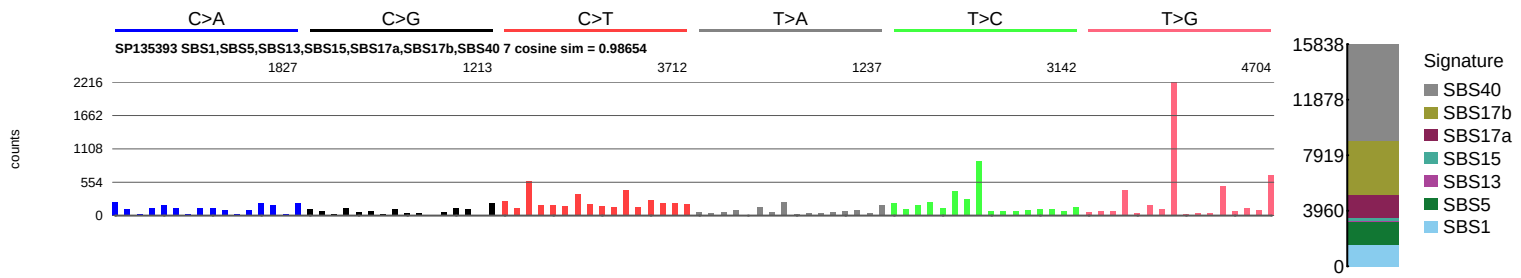
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1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C>G
C>T
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T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98653
1827
1213
3712
1237
3142
4704
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
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SBS17b
SBS17a
SBS15
SBS13
SBS5
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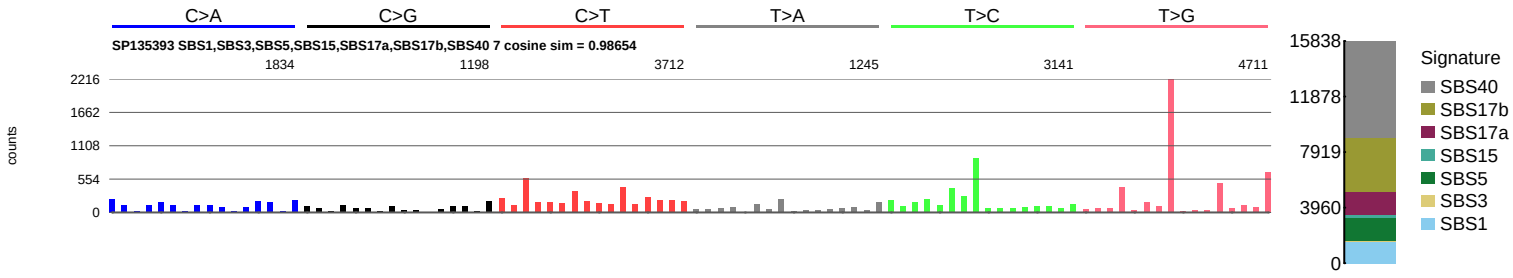
C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98654
1834
1198
3712
1245
3141
4711
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
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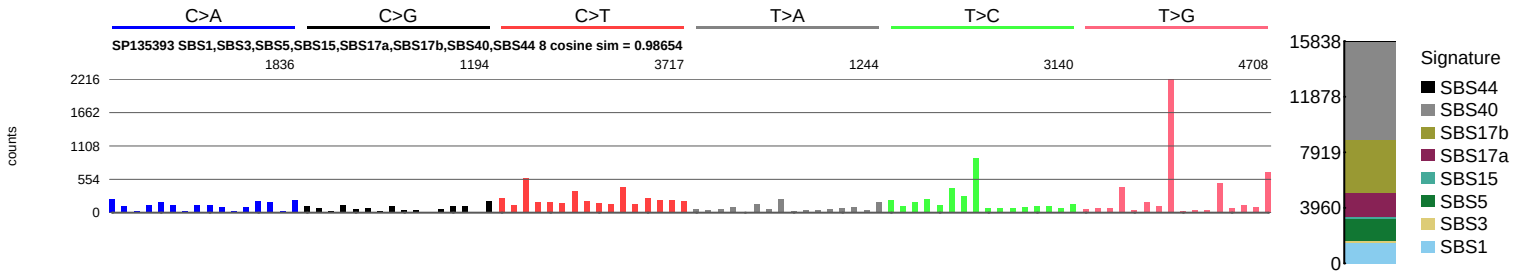
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98654
1836
1194
3717
1244
3140
4708
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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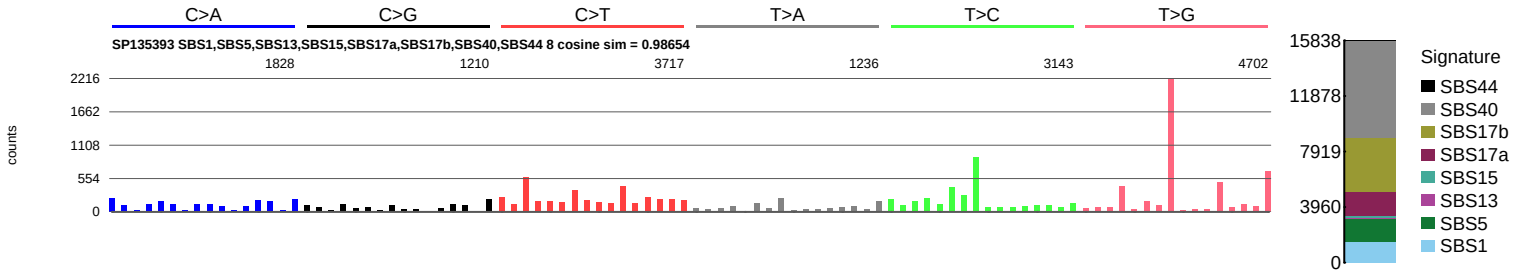
SBS3
SBS1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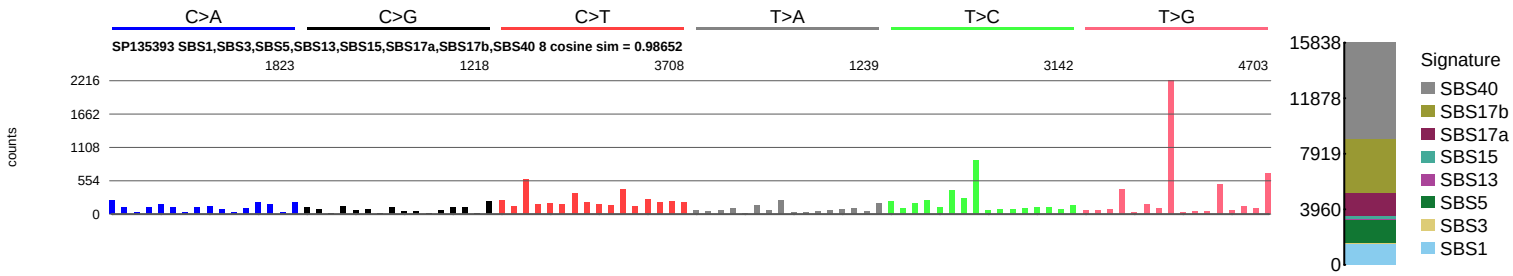
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98654
1828
1210
3717
1236
3143
4702
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS40 8 cosine sim = 0.98652
1823
1218
3708
1239
3142
4703
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
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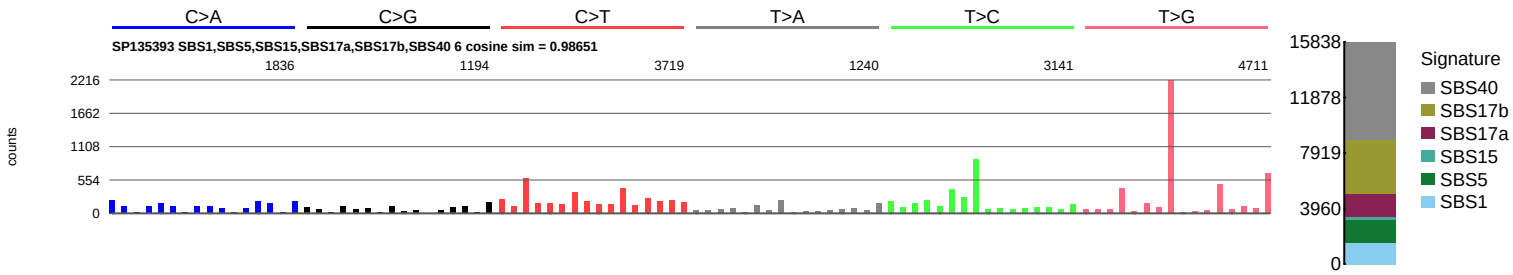
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98651
1836
1194
3719
1240
3141
4711
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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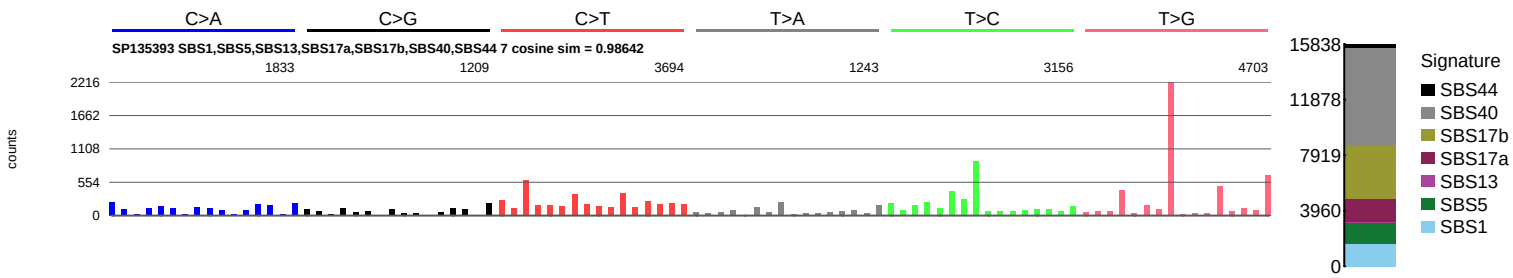
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98642
1833
1209
3694
1243
3156
4703
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
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SBS44
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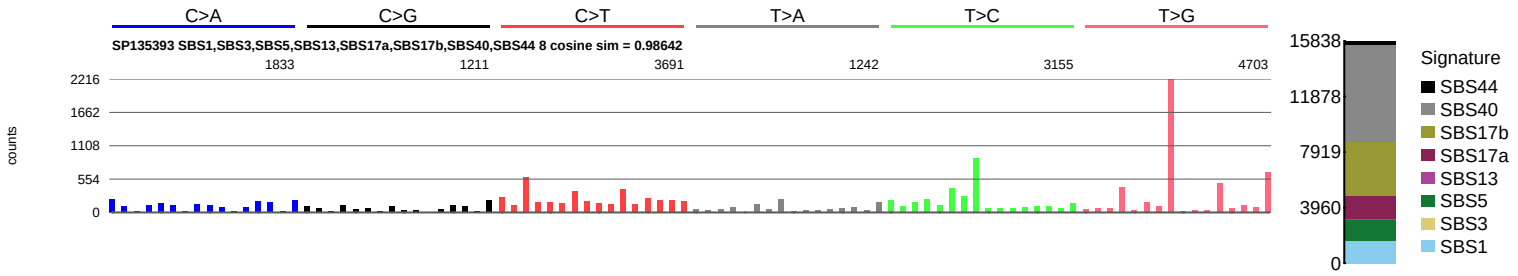
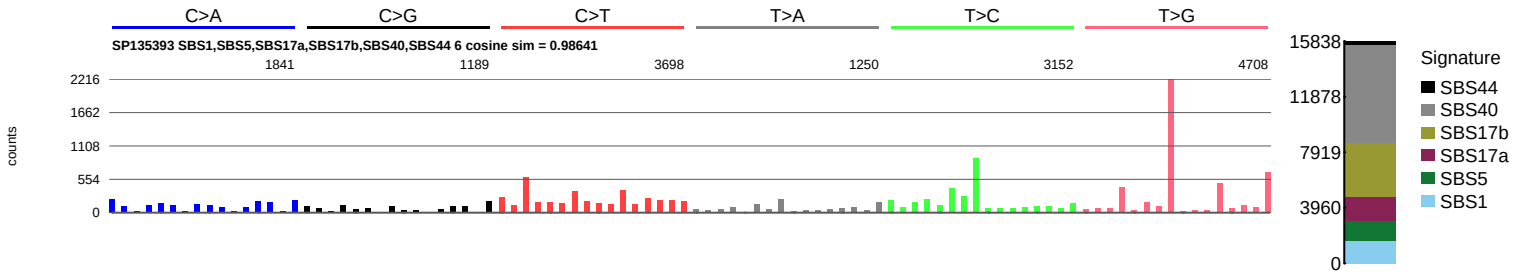
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98642
1833
1211
3691
1242
3155
4703
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98641
1841
1189
3698
1250
3152
4708
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
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SBS40
SBS17b
SBS17a
SBS5
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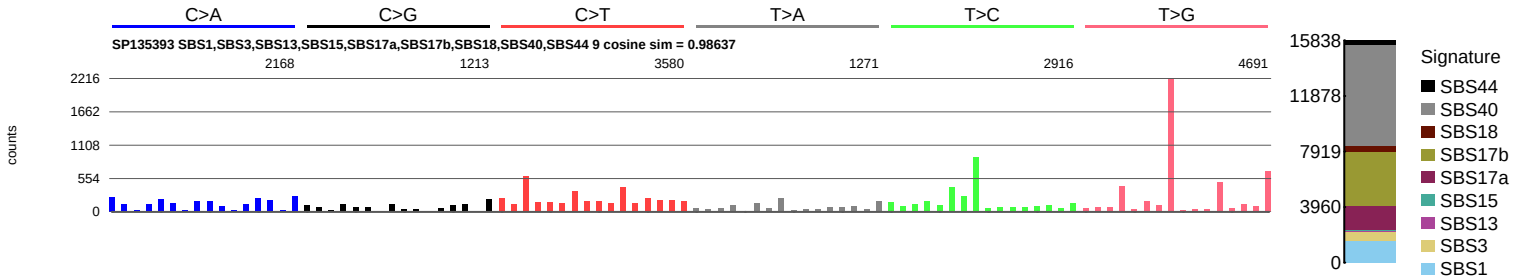
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SP135393 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 9 cosine sim = 0.98637
2168
1213
3580
1271
2916
4691
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
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SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
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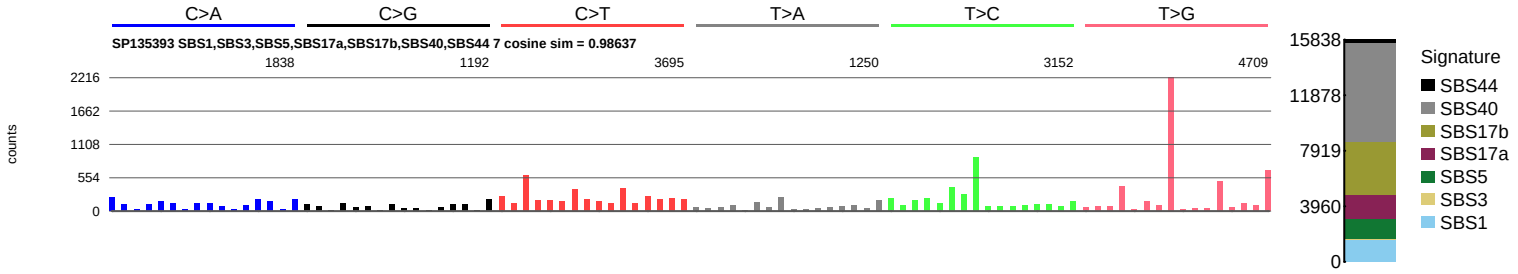
SBS3
SBS1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SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98637
1838
1192
3695
1250
3152
4709
counts
2216
1662
1108
554
0
15838
11878
7919
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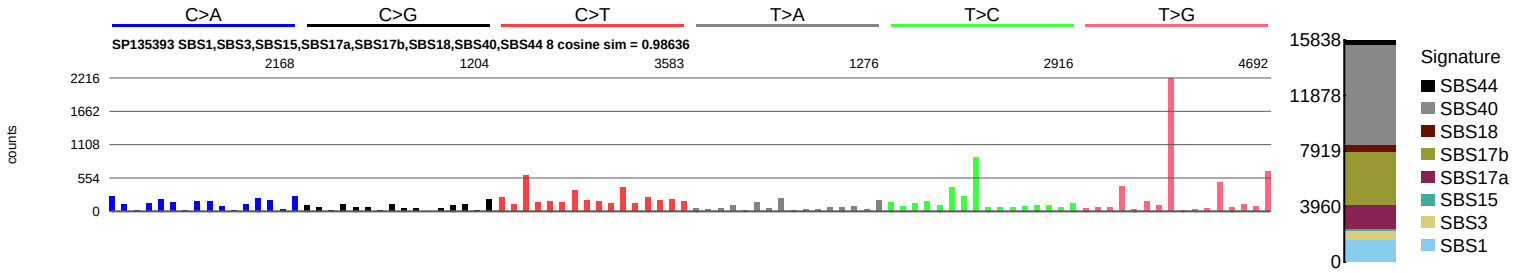
SBS17b
SBS17a
SBS5
SBS3
SBS1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SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98636
2168
1204
3583
1276
2916
4692
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
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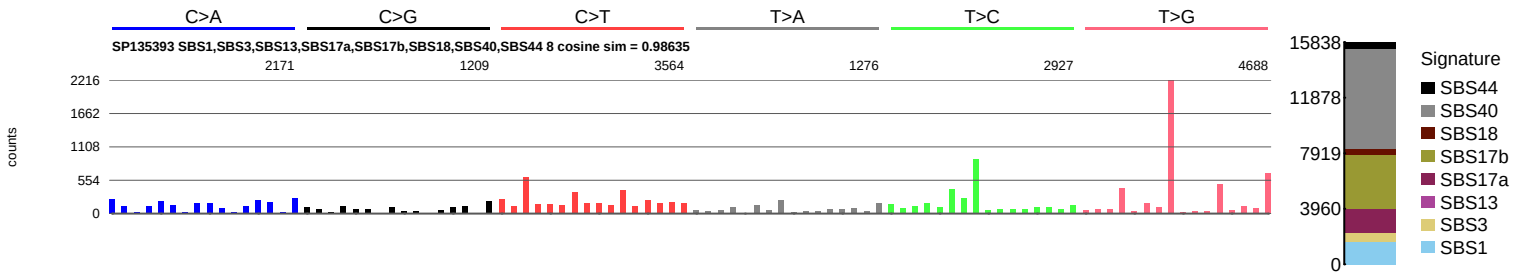
SBS17b
SBS17a
SBS15
SBS3
SBS1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SP135393 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98635
2171
1209
3564
1276
2927
4688
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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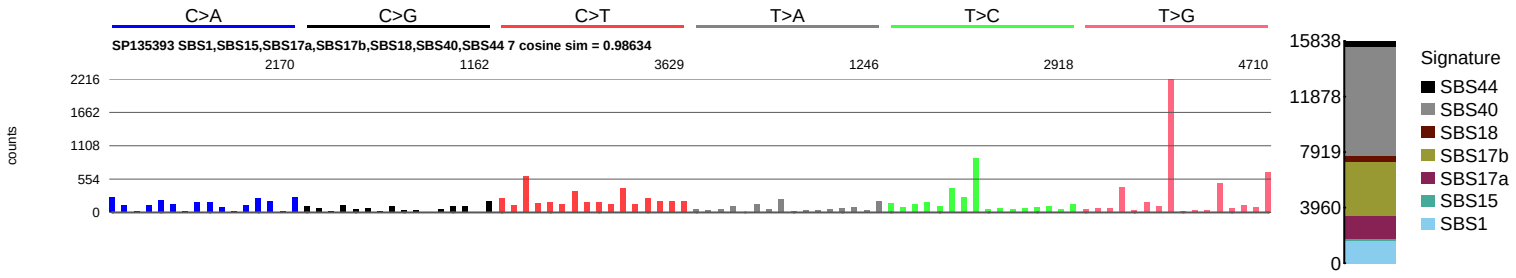
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98634
2170
1162
3629
1246
2918
4710
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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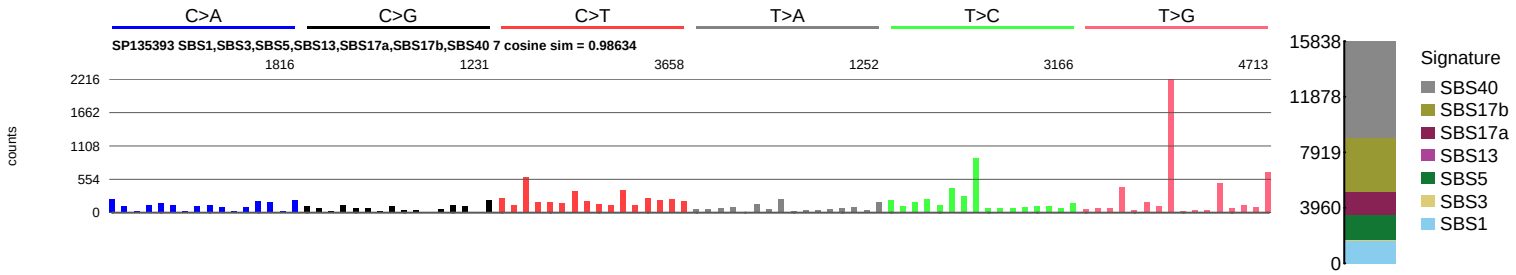
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
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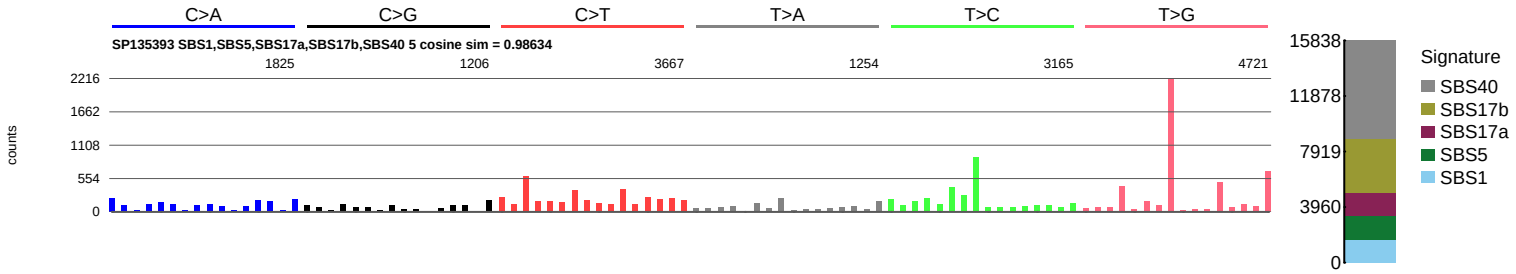
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS40 7 cosine sim = 0.98634
1816
1231
3658
1252
3166
4713
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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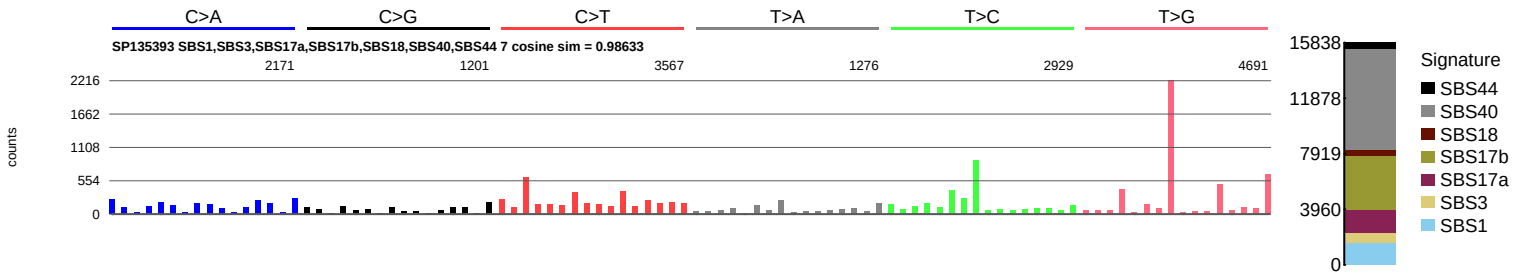
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98634
1825
1206
3667
1254
3165
4721
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS17b
SBS17a
SBS5
SBS1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C>G
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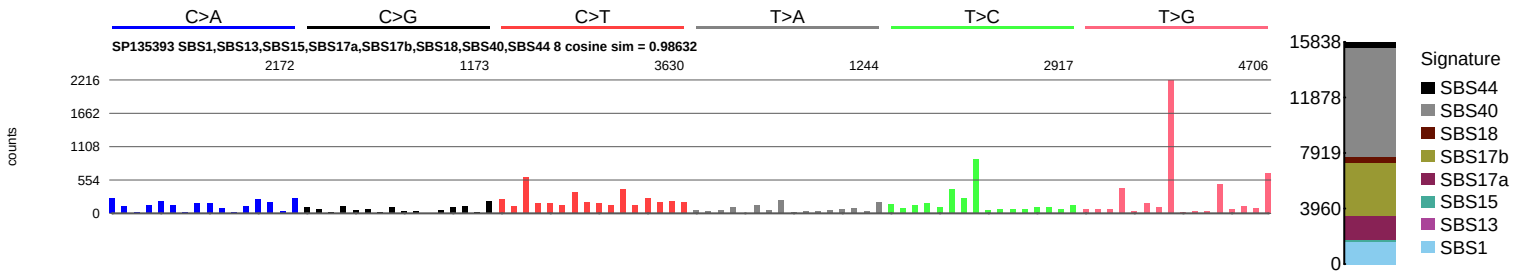
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.98633
2171
1201
3567
1276
2929
4691
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS1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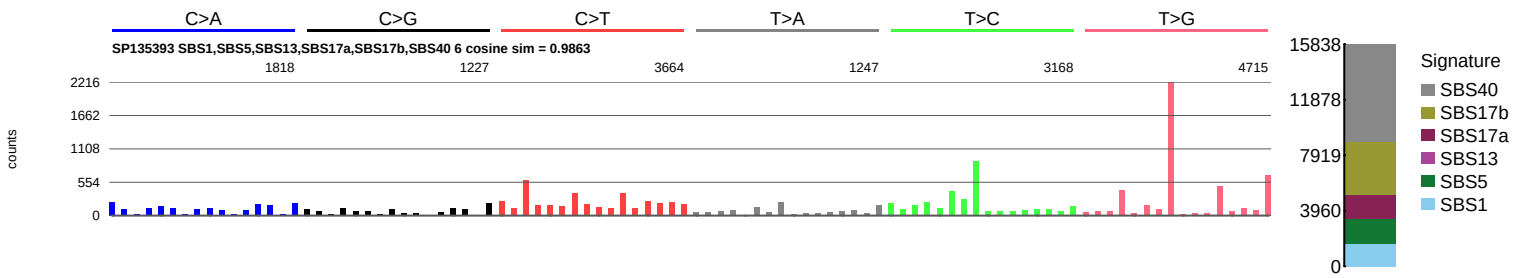
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40,SBS44 8 cosine sim = 0.98632
2172
1173
3630
1244
2917
4706
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
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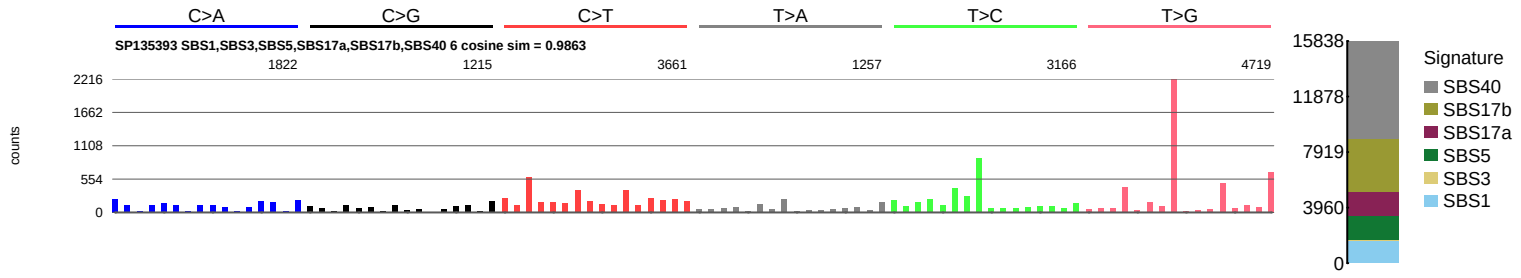
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS40 6 cosine sim = 0.9863
1818
1227
3664
1247
3168
4715
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS40 6 cosine sim = 0.9863
1822
1215
3661
1257
3166
4719
counts
2216
1662
1108
554
0
15838
11878
7919
3960
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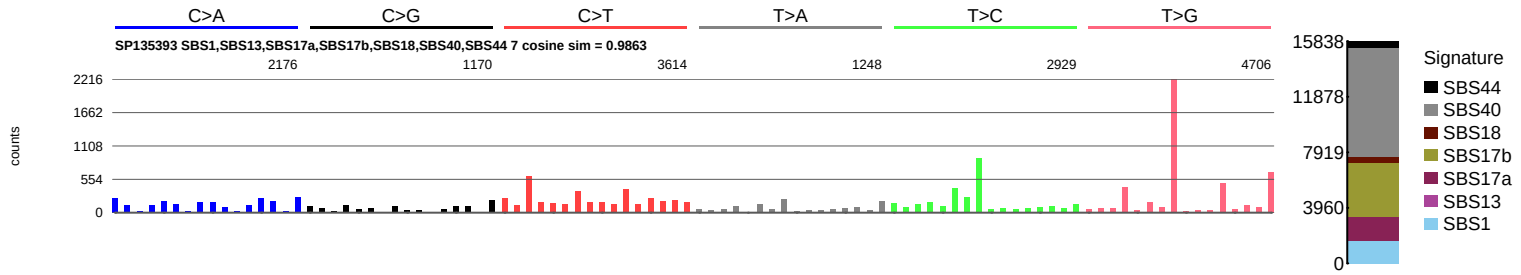
Signature
SBS40
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
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T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS40,SBS44 7 cosine sim = 0.9863
2176
1170
3614
1248
2929
4706
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
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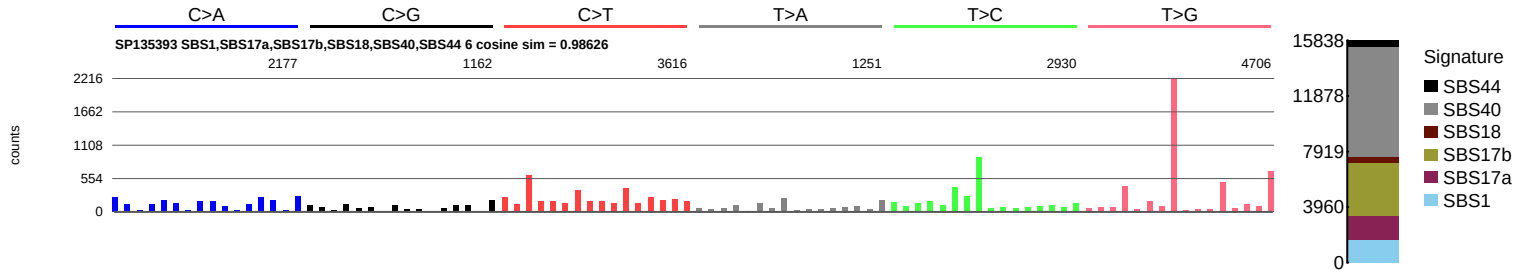
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS13
SBS1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C>G
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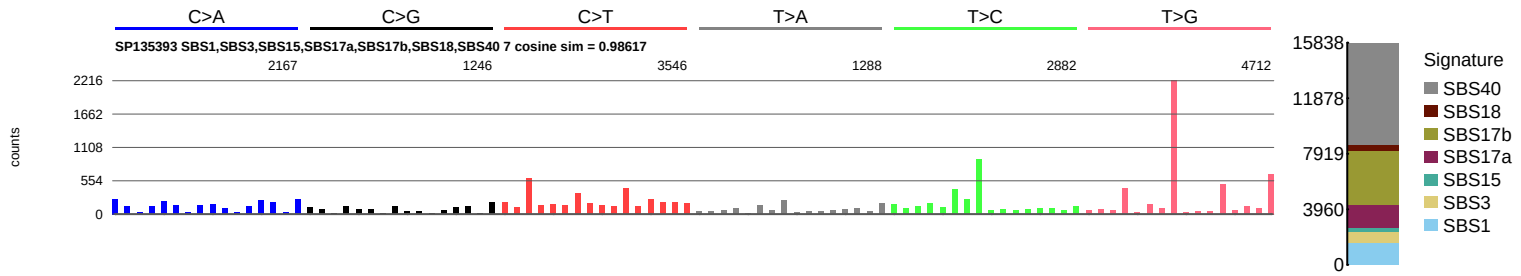
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS40,SBS44 6 cosine sim = 0.98626
2177
1162
3616
1251
2930
4706
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98617
2167
1246
3546
1288
2882
4712
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
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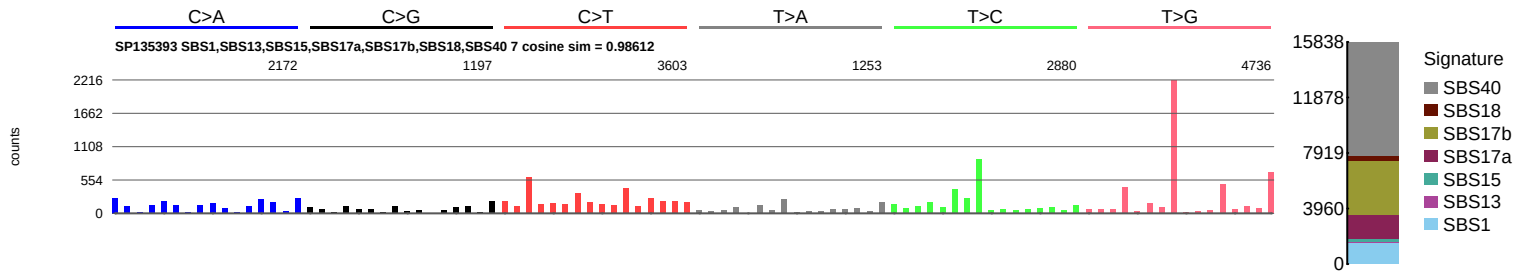
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>G
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T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS40 7 cosine sim = 0.98612
2172
1197
3603
1253
2880
4736
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
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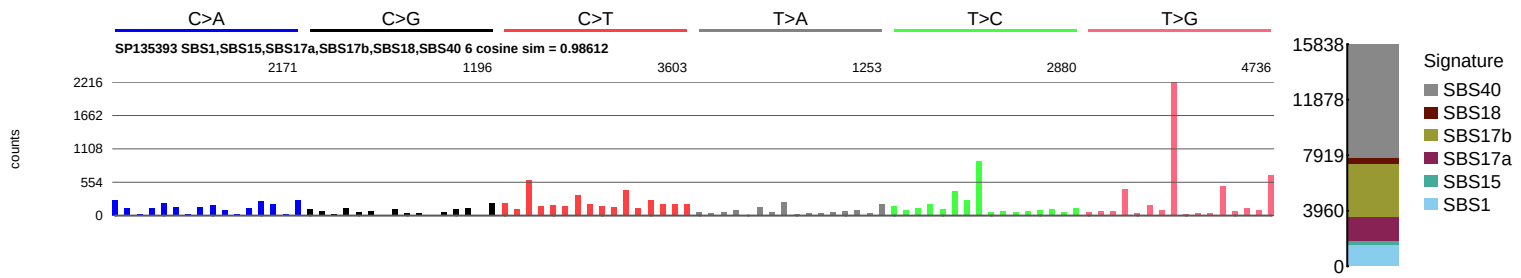
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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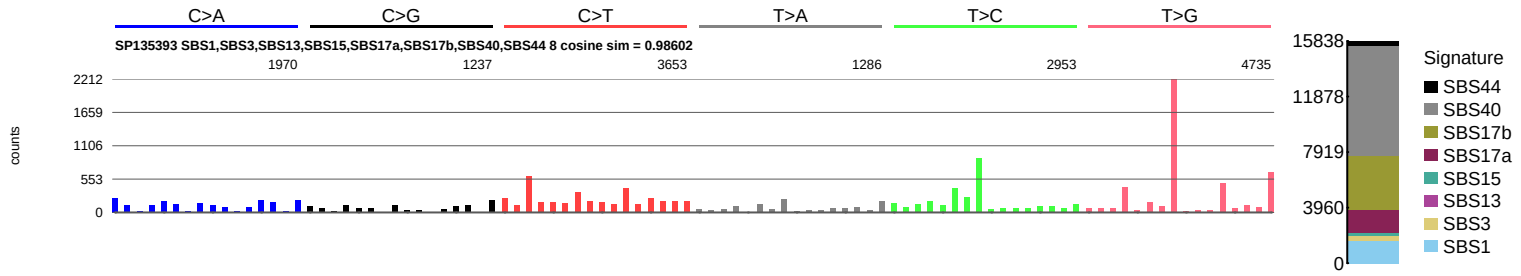
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98612
2171
1196
3603
1253
2880
4736
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS15
SBS1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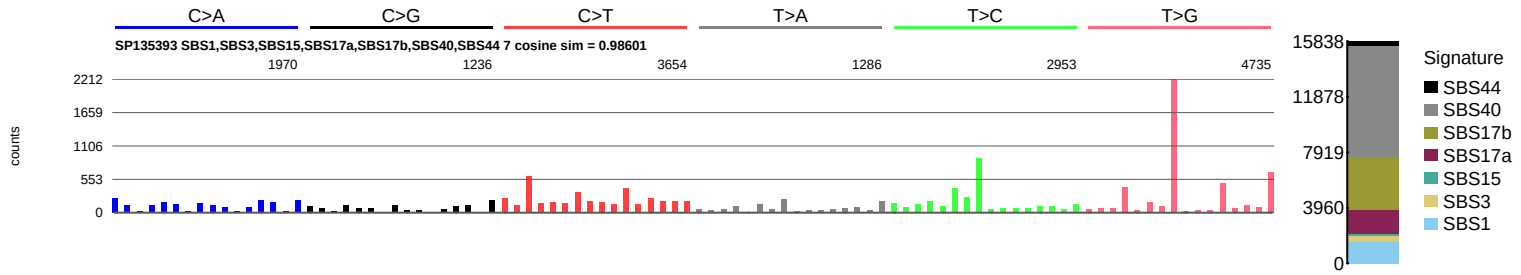
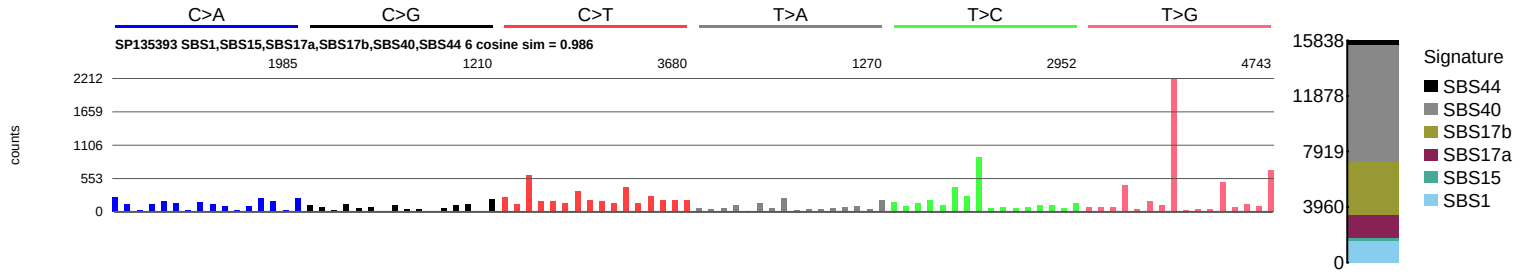
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 8 cosine sim = 0.98602
1970
1237
3653
1286
2953
4735
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98601
1970
1236
3654
1286
2953
4735
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.986
1985
1210
3680
1270
2952
4743
counts
2212
1659
1106
553
0
15838
11878
7919
3960
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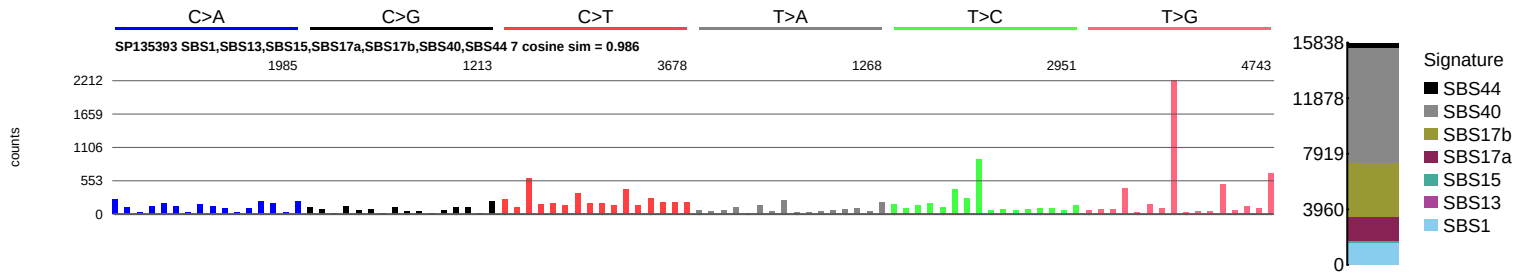
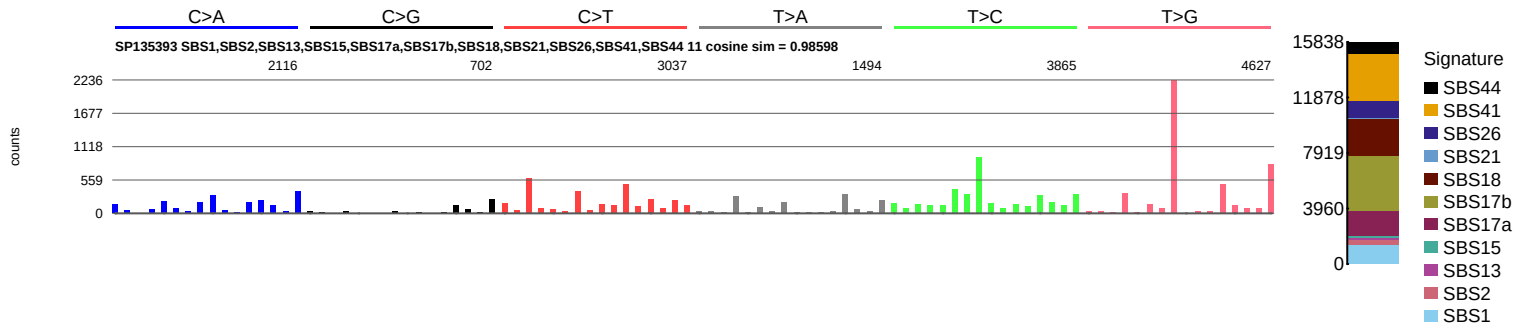
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.986
1985
1213
3678
1268
2951
4743
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.98598
2116
702
3037
1494
3865
4627
counts
2236
1677
1118
559
0
15838
11878
7919
3960
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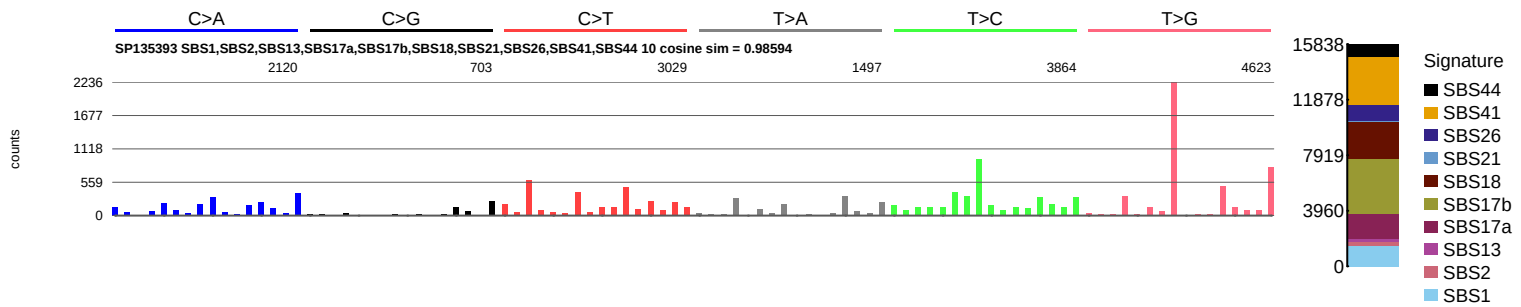
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98594
2120
703
3029
1497
3864
4623
counts
2236
1677
1118
559
0
15838
11878
7919
3960
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SBS41
SBS26
SBS21
SBS18
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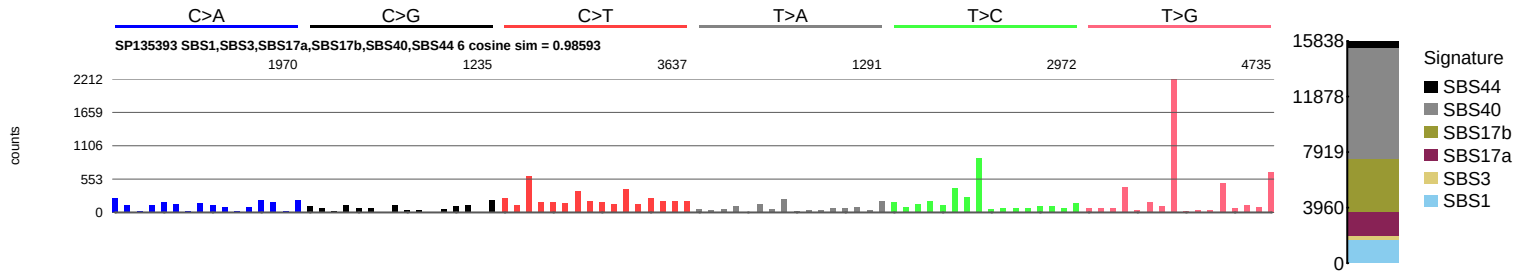
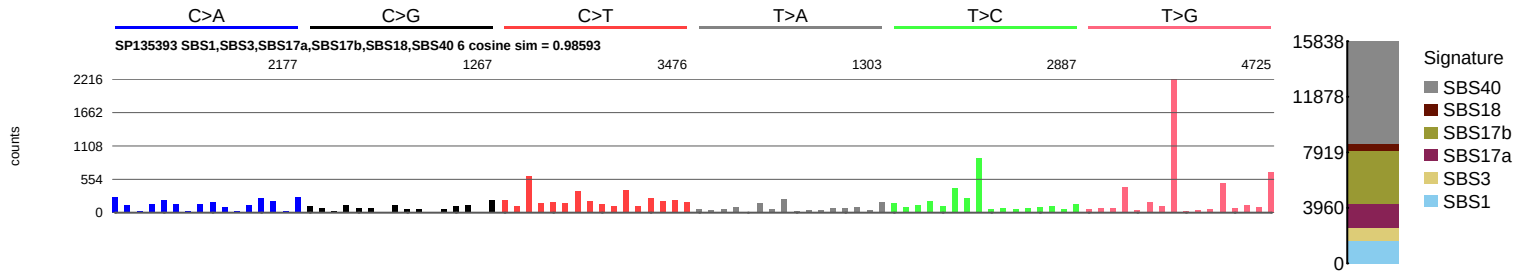
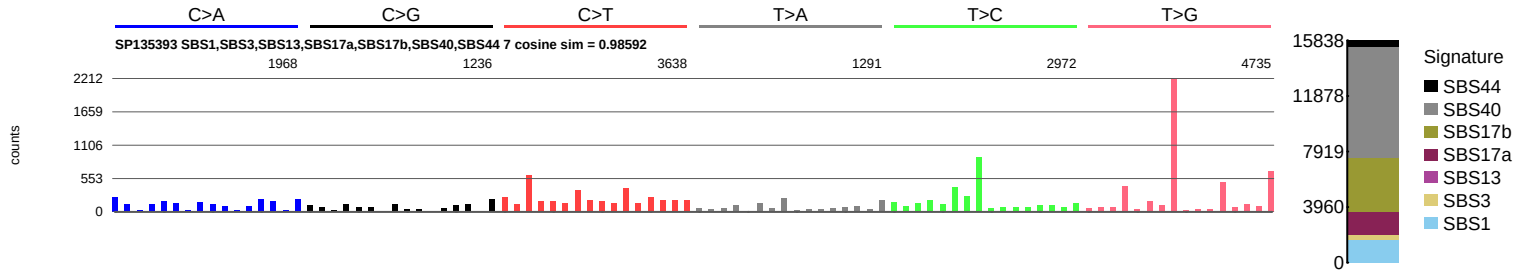
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.98593
1970
1235
3637
1291
2972
4735
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS40 6 cosine sim = 0.98593
2177
1267
3476
1303
2887
4725
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
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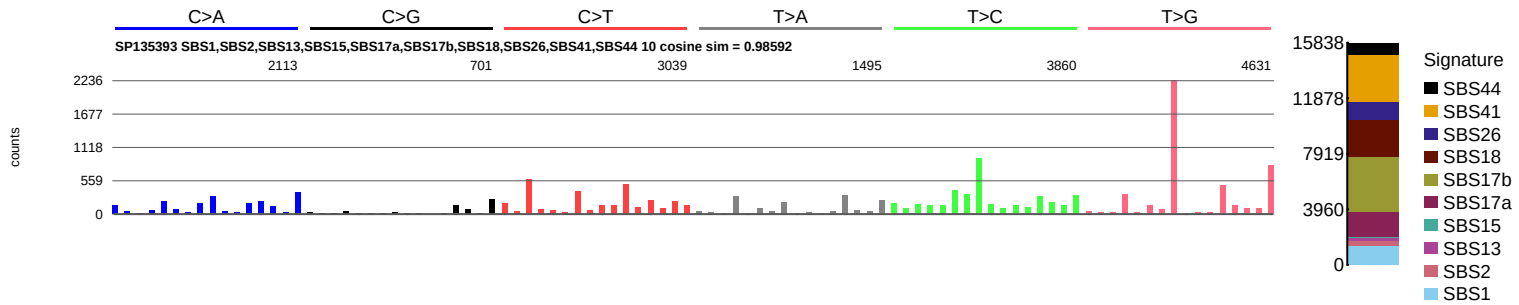
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS40,SBS44 7 cosine sim = 0.98592
1968
1236
3638
1291
2972
4735
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS3
SBS1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C>T
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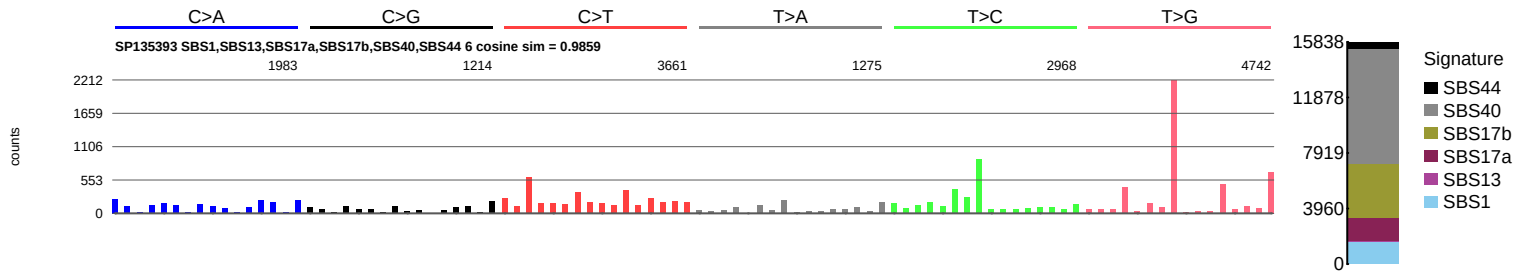
T>C
T>G
SP135393 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 10 cosine sim = 0.98592
2113
701
3039
1495
3860
4631
counts
2236
1677
1118
559
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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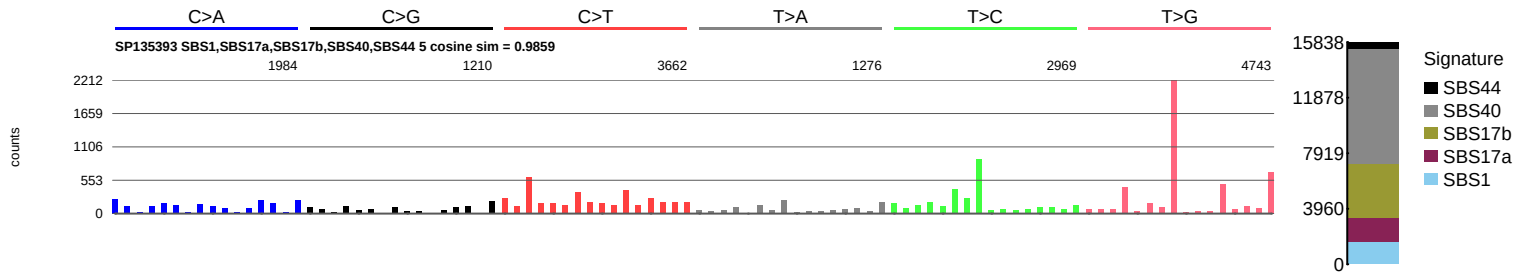
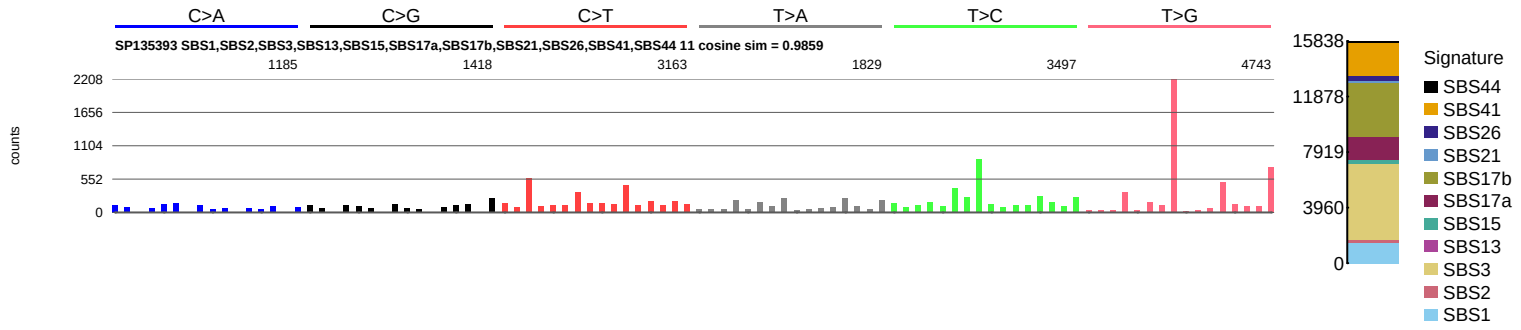
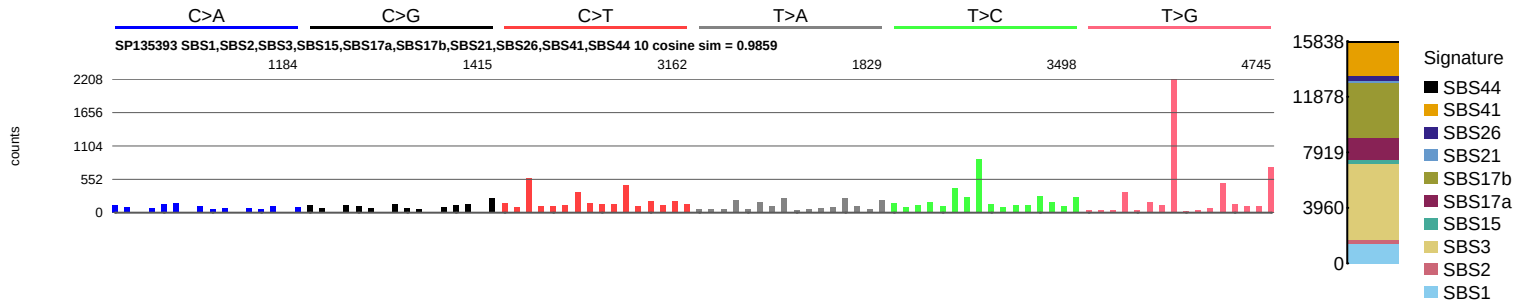
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS40,SBS44 6 cosine sim = 0.9859
1983
1214
3661
1275
2968
4742
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS40,SBS44 5 cosine sim = 0.9859
1984
1210
3662
1276
2969
4743
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS44
SBS40
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 11 cosine sim = 0.9859
1185
1418
3163
1829
3497
4743
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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C>G
C>T
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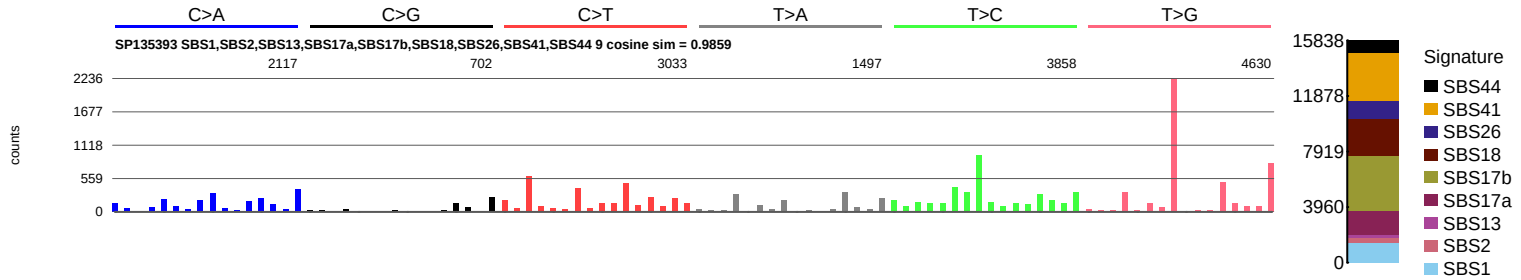
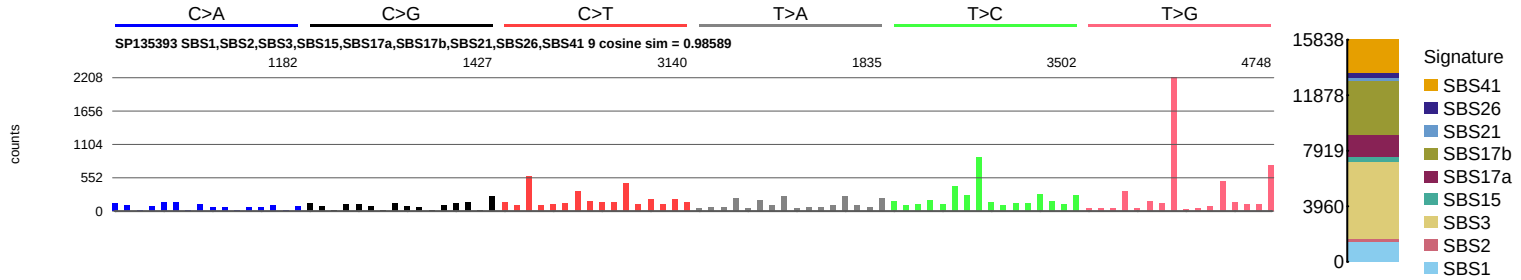
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.9859
1184
1415
3162
1829
3498
4745
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.9859
2117
702
3033
1497
3858
4630
counts
2236
1677
1118
559
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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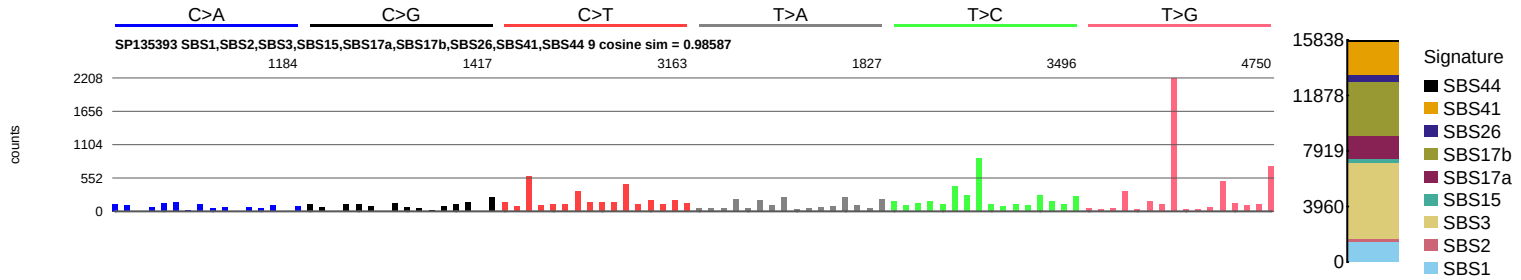
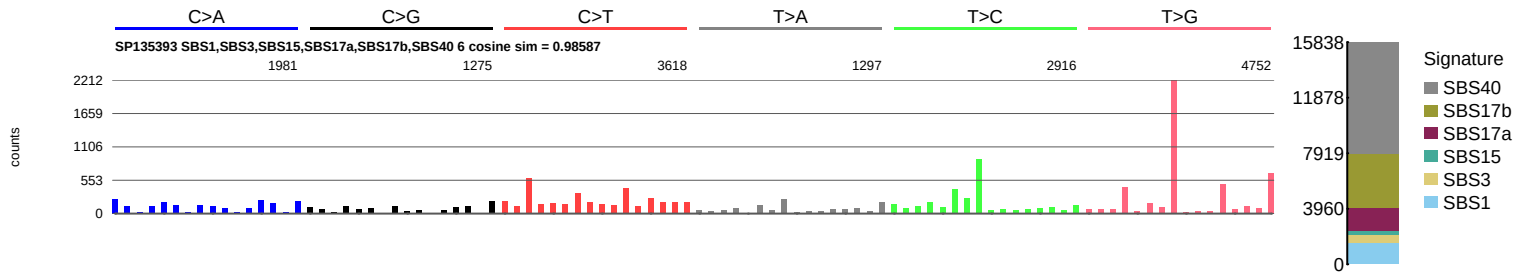
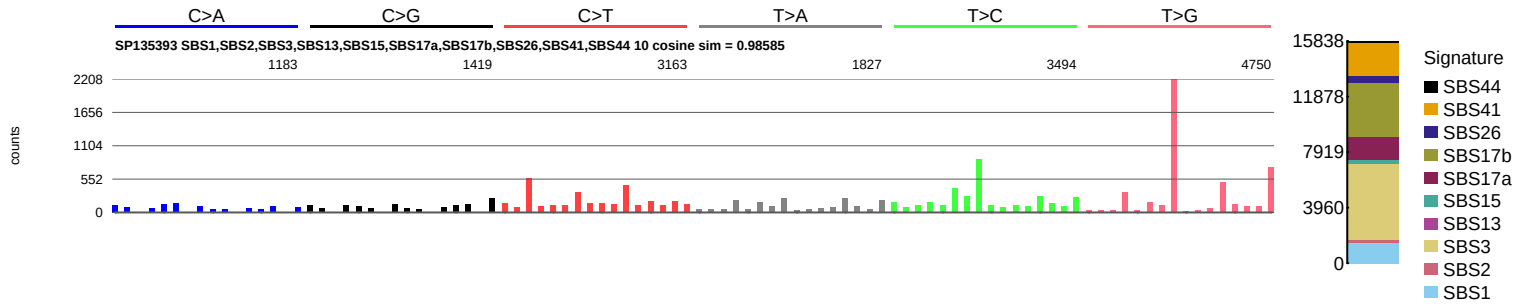
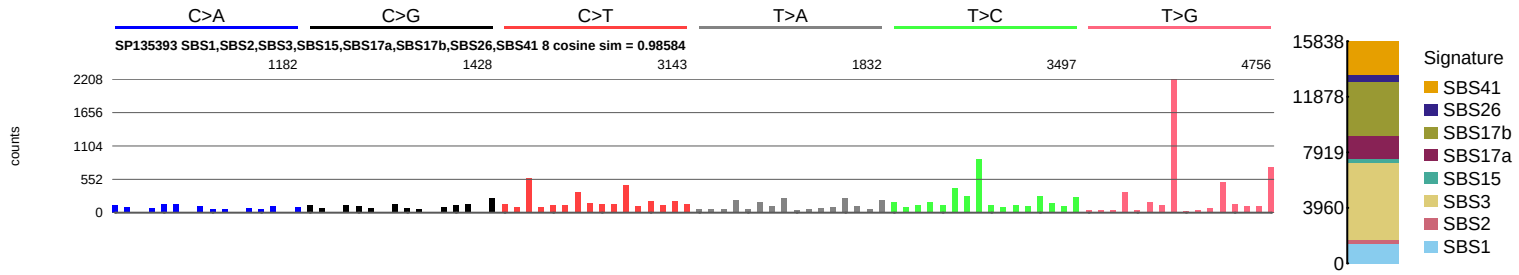
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 9 cosine sim = 0.98589
1182
1427
3140
1835
3502
4748
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98587
1184
1417
3163
1827
3496
4750
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS40 6 cosine sim = 0.98587
1981
1275
3618
1297
2916
4752
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 10 cosine sim = 0.98585
1183
1419
3163
1827
3494
4750
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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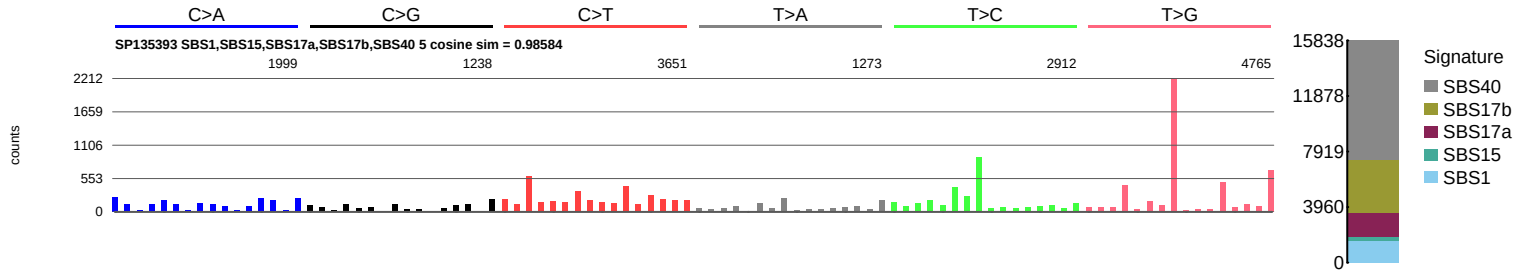
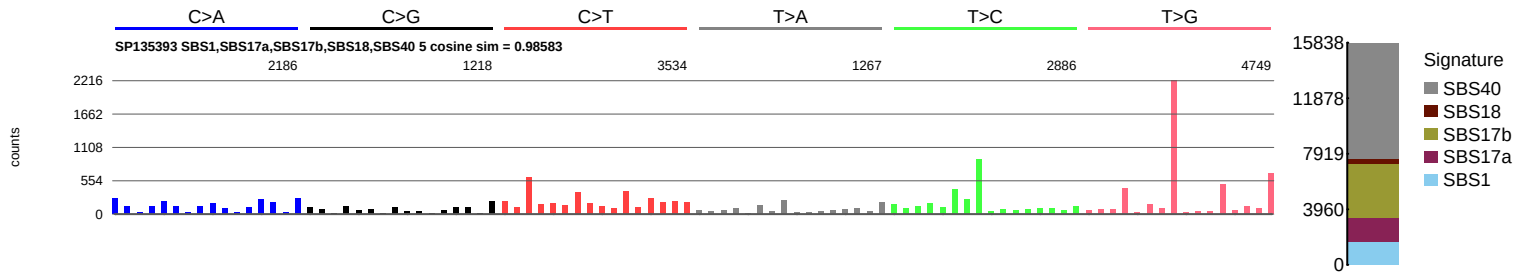
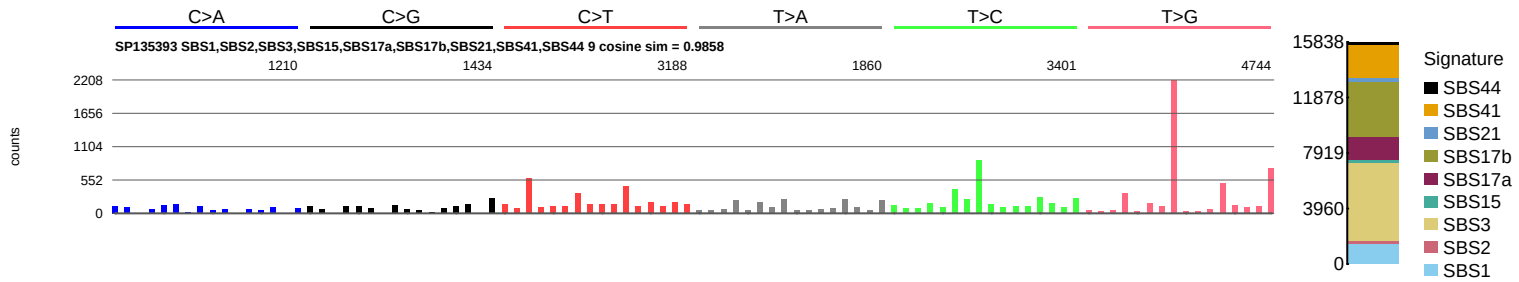
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 8 cosine sim = 0.98584
1182
1428
3143
1832
3497
4756
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98584
1999
1238
3651
1273
2912
4765
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS40 5 cosine sim = 0.98583
2186
1218
3534
1267
2886
4749
counts
2216
1662
1108
554
0
15838
11878
7919
3960
0
Signature
SBS40
SBS18
SBS17b
SBS17a
SBS1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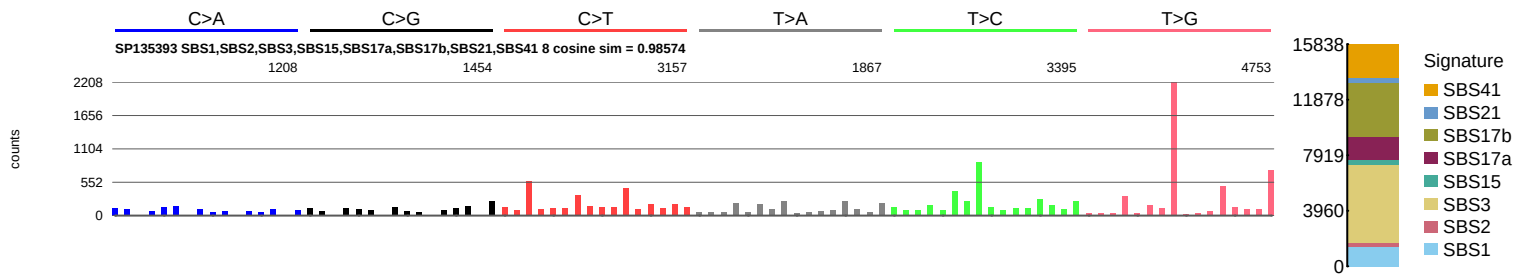
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 9 cosine sim = 0.9858
1210
1434
3188
1860
3401
4744
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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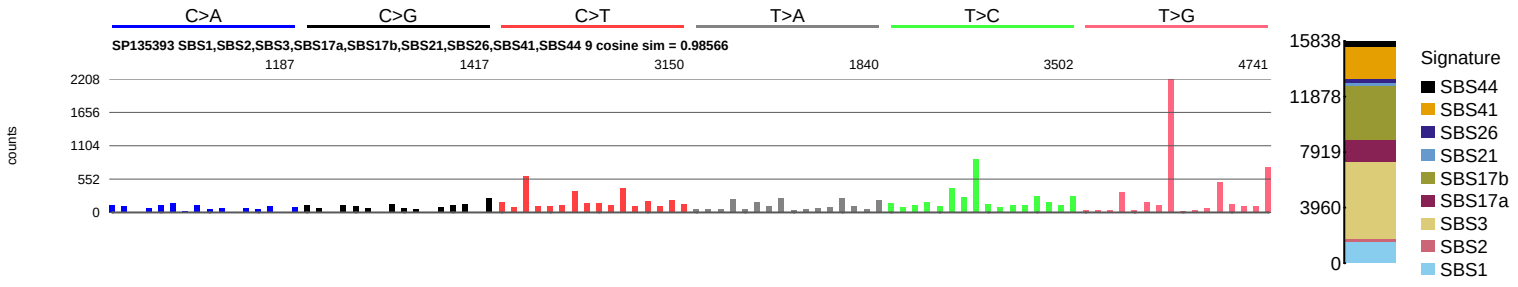
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98574
1208
1454
3157
1867
3395
4753
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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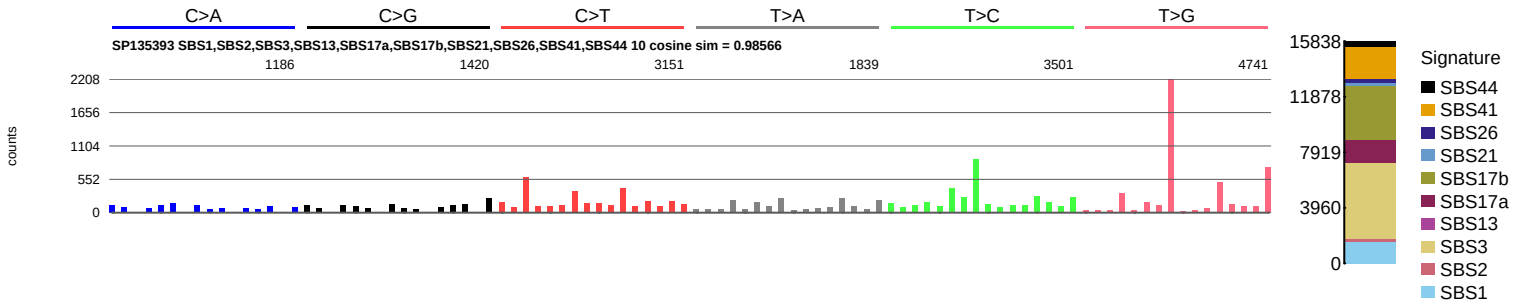
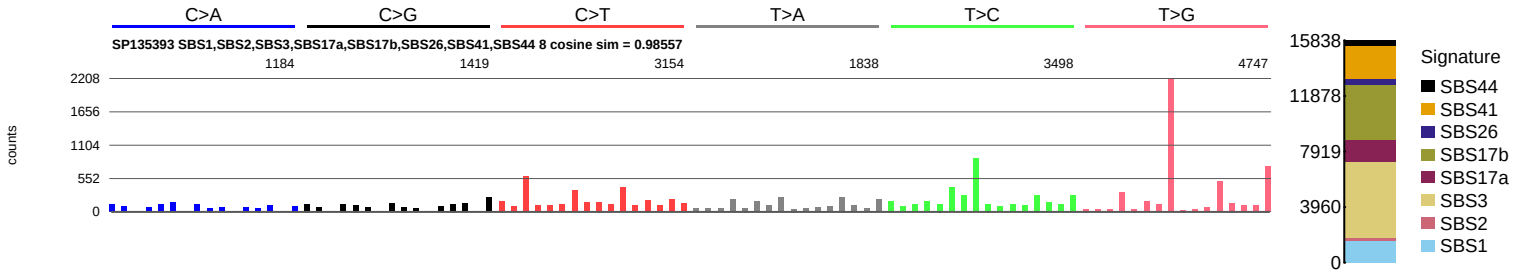
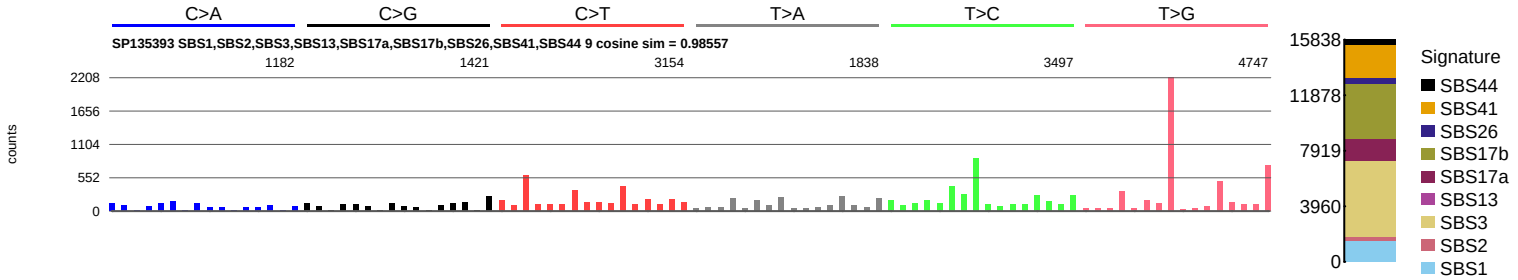
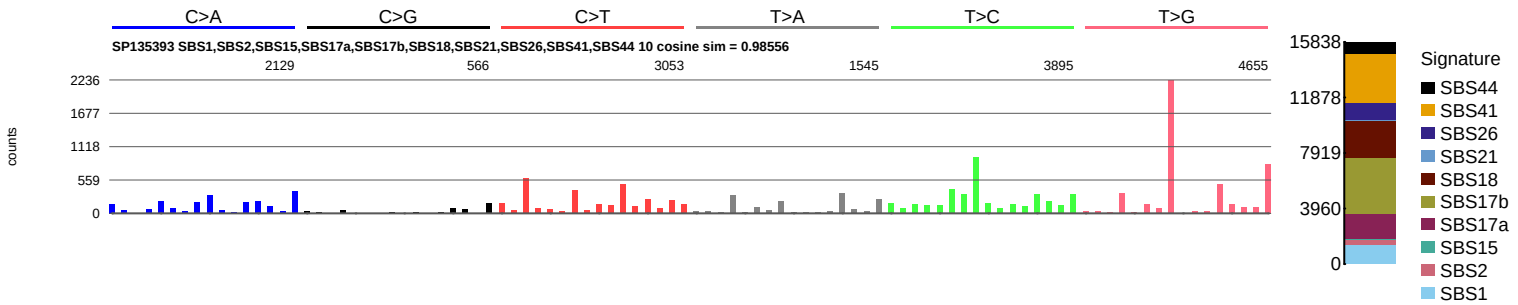
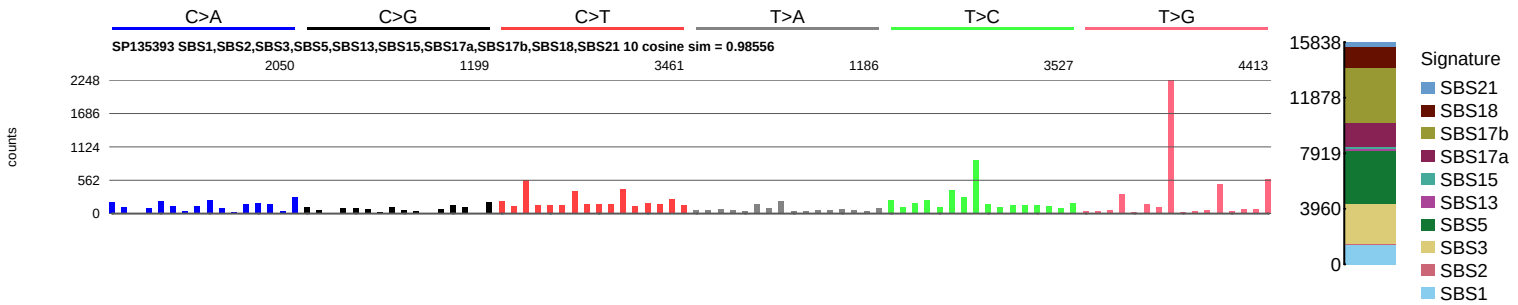
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98566
1187
1417
3150
1840
3502
4741
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

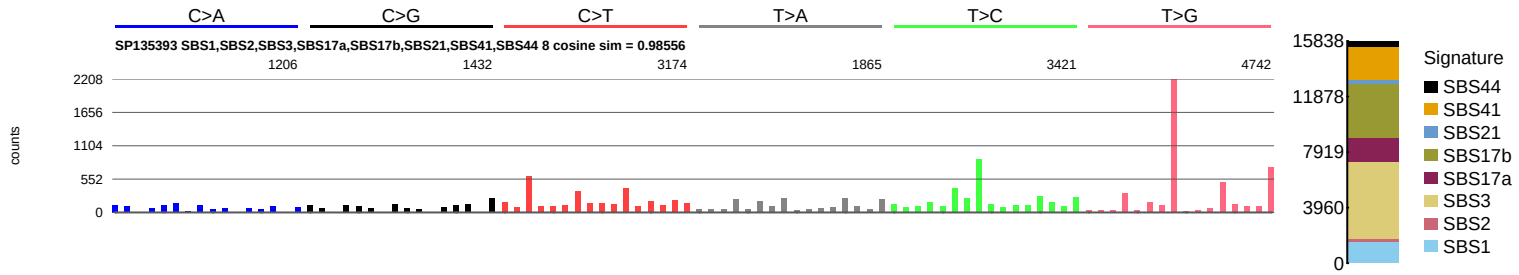

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98566
1186
1420
3151
1839
3501
4741
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

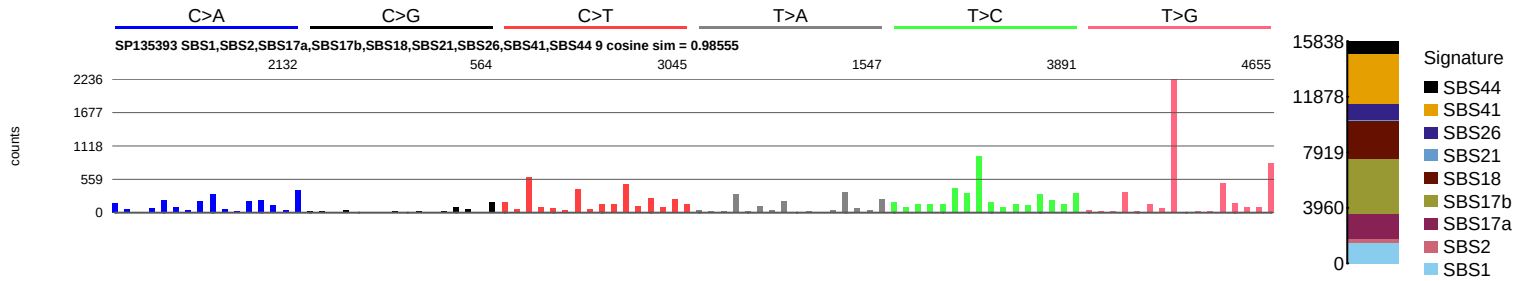

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98557
1184
1419
3154
1838
3498
4747
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

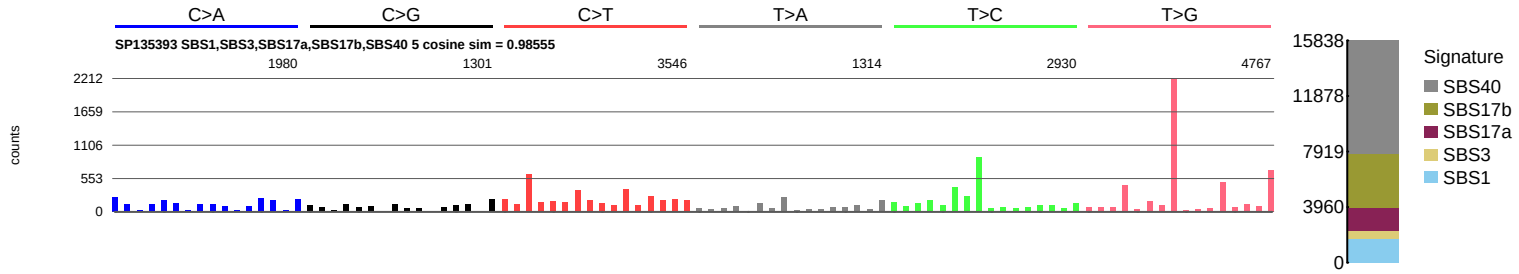

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS26,SBS41,SBS44 9 cosine sim = 0.98557
1182
1421
3154
1838
3497
4747
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

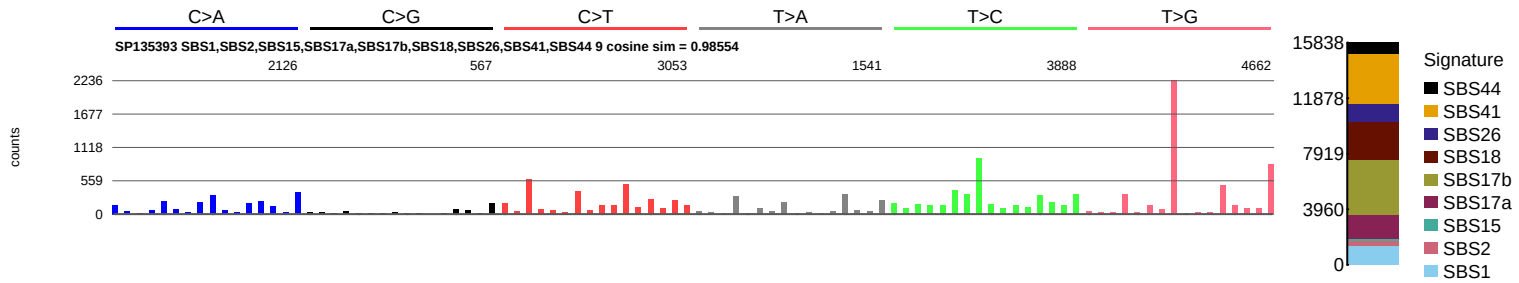

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98556
2129
566
3053
1545
3895
4655
counts
2236
1677
1118
559
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 10 cosine sim = 0.98556
2050
1199
3461
1186
3527
4413
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.98556
1206
1432
3174
1865
3421
4742
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98555
2132
564
3045
1547
3891
4655
counts
2236
1677
1118
559
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS40 5 cosine sim = 0.98555
1980
1301
3546
1314
2930
4767
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS17b
SBS17a
SBS3
SBS1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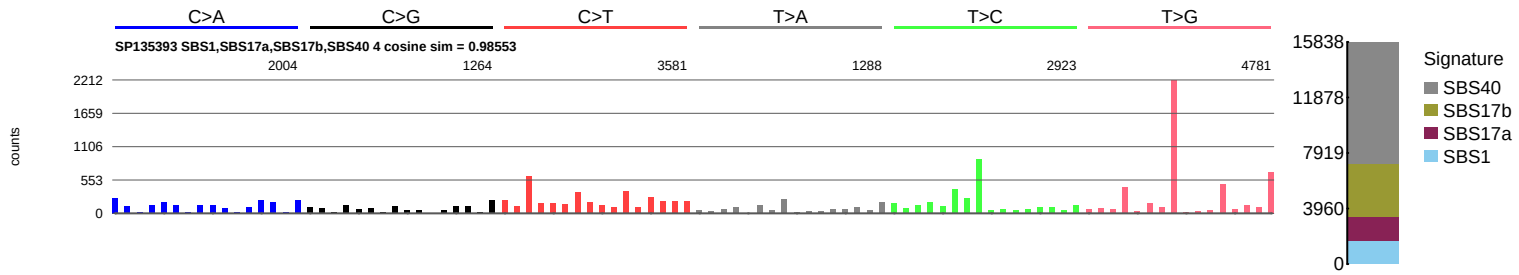
SP135393 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98554
2126
567
3053
1541
3888
4662
counts
2236
1677
1118
559
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
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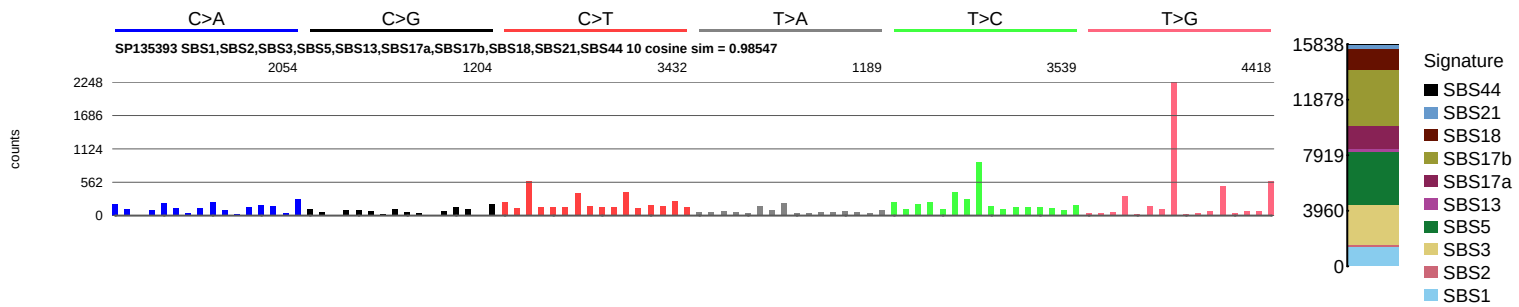
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS40 4 cosine sim = 0.98553
2004
1264
3581
1288
2923
4781
counts
2212
1659
1106
553
0
15838
11878
7919
3960
0
Signature
SBS40
SBS17b
SBS17a
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98547
2054
1204
3432
1189
3539
4418
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
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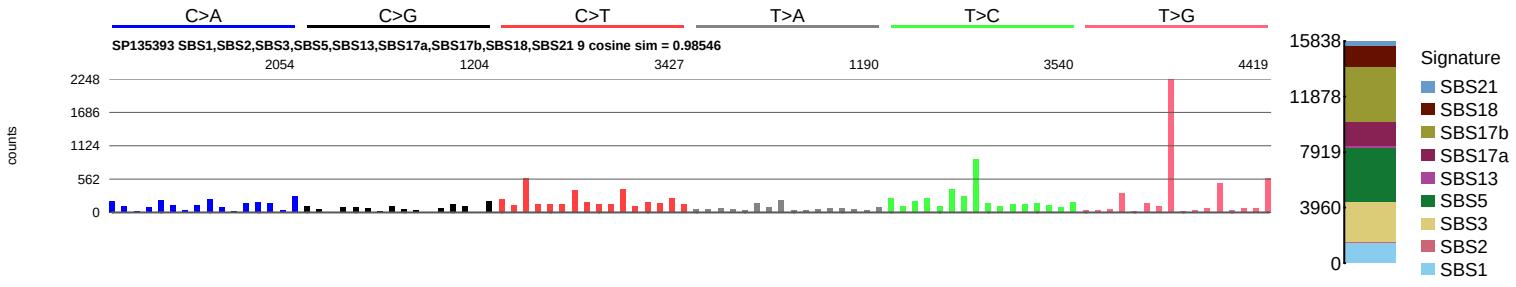
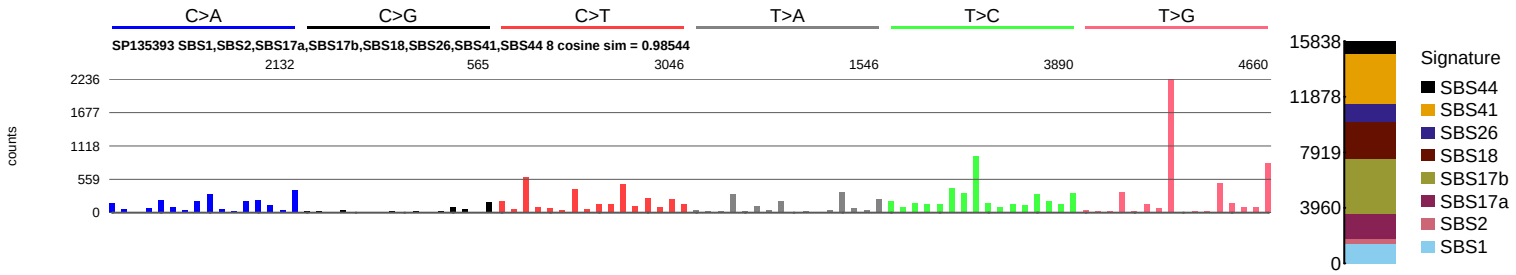
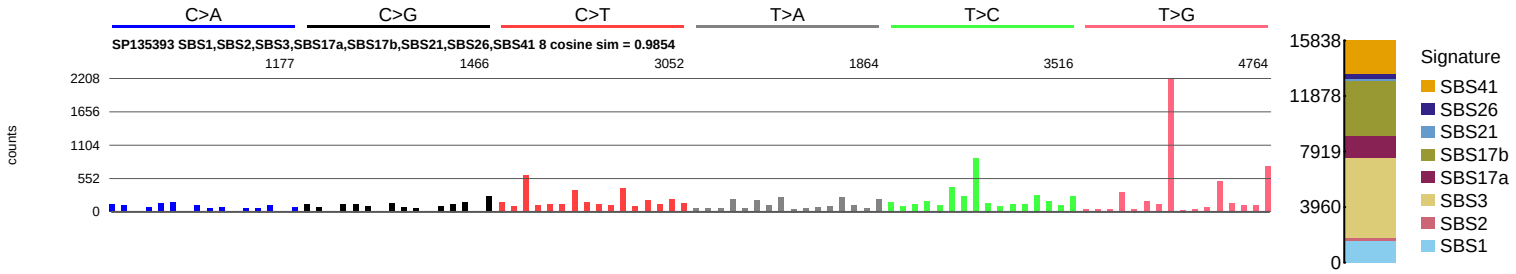
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98546
2054
1204
3427
1190
3540
4419
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98544
2132
565
3046
1546
3890
4660
counts
2236
1677
1118
559
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.9854
1177
1466
3052
1864
3516
4764
counts
2208
1656
1104
552
0
15838
11878
7919
3960
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Signature
SBS41
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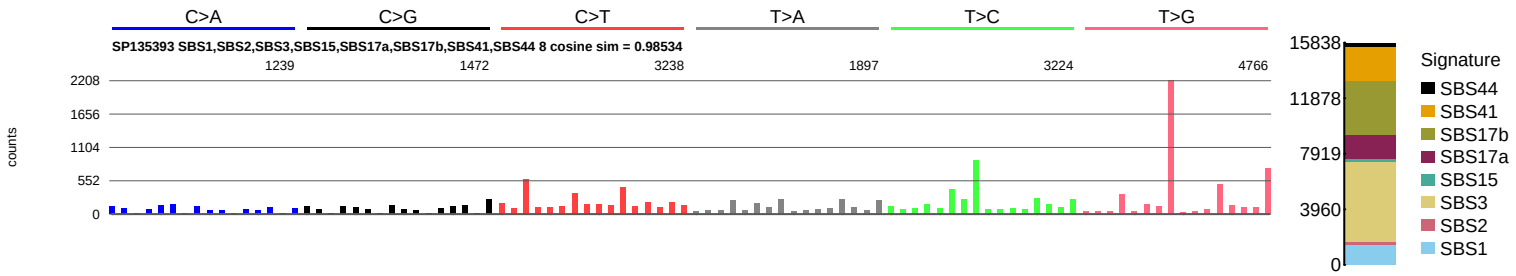
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 8 cosine sim = 0.98534
1239
1472
3238
1897
3224
4766
counts
2208
1656
1104
552
0
15838
11878
7919
3960
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Signature
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SBS17b
SBS17a
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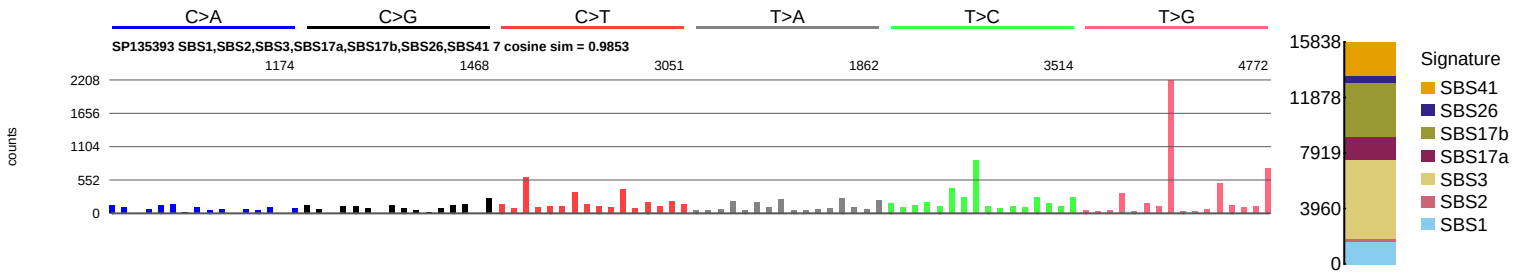
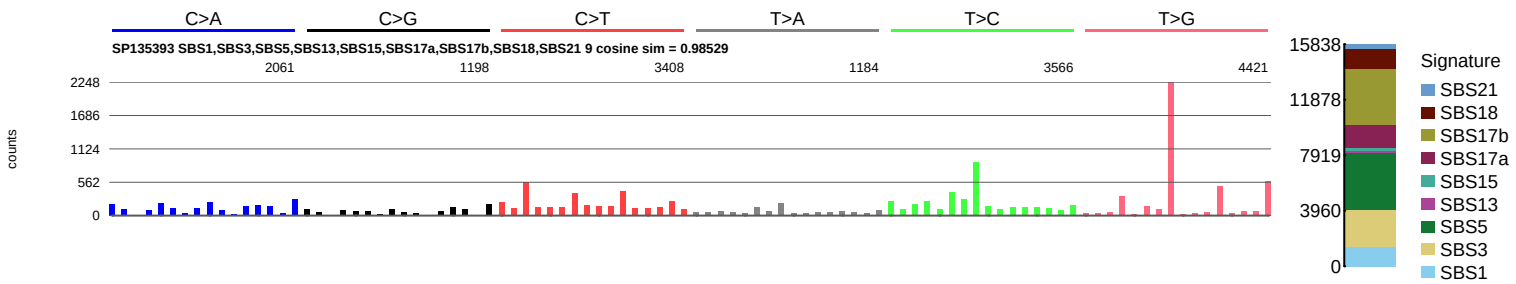
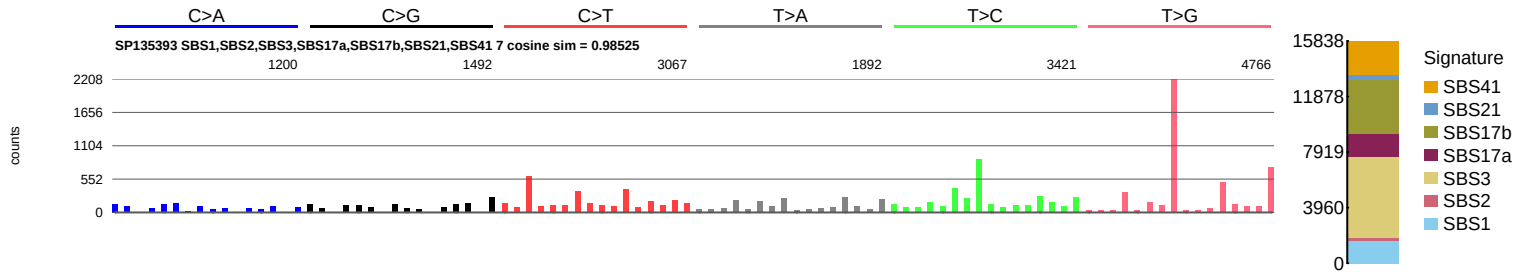
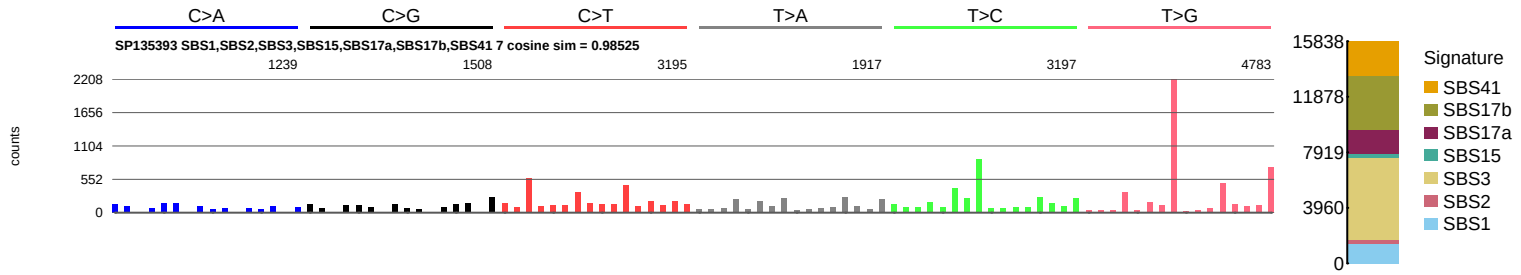
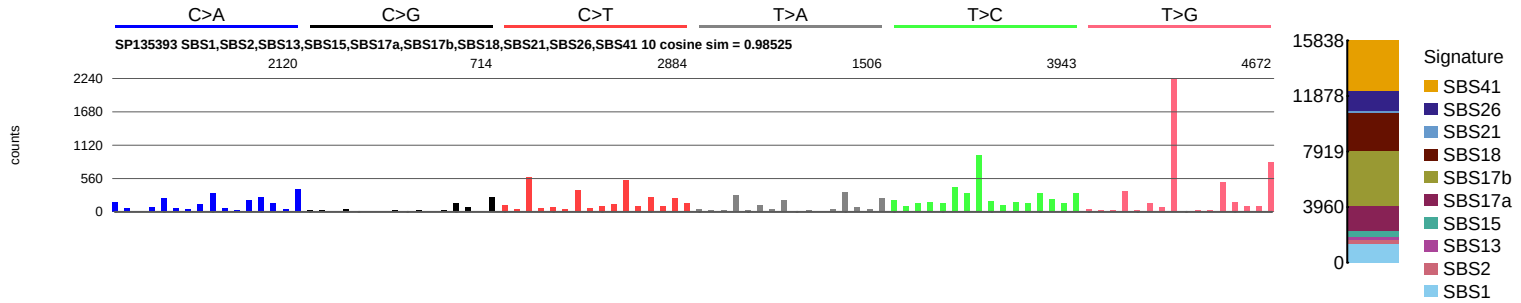
SBS3
SBS2
SBS1

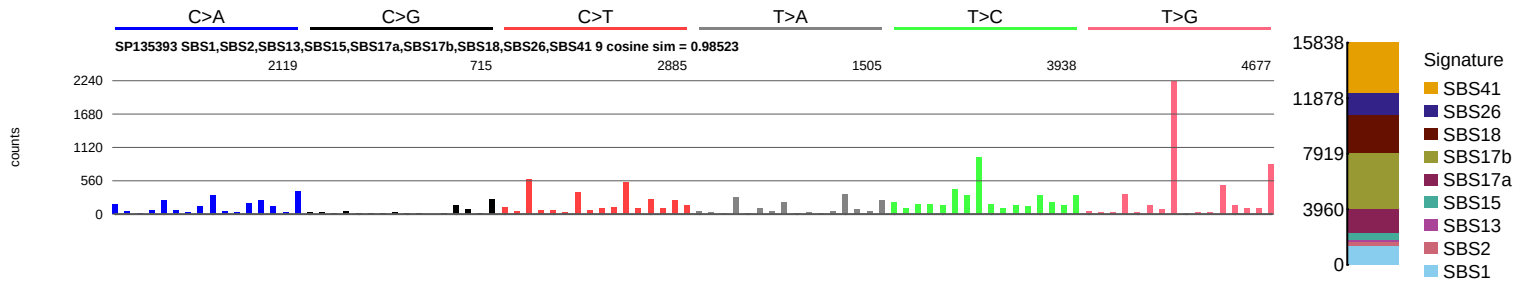

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.9853
1174
1468
3051
1862
3514
4772
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS2
SBS1

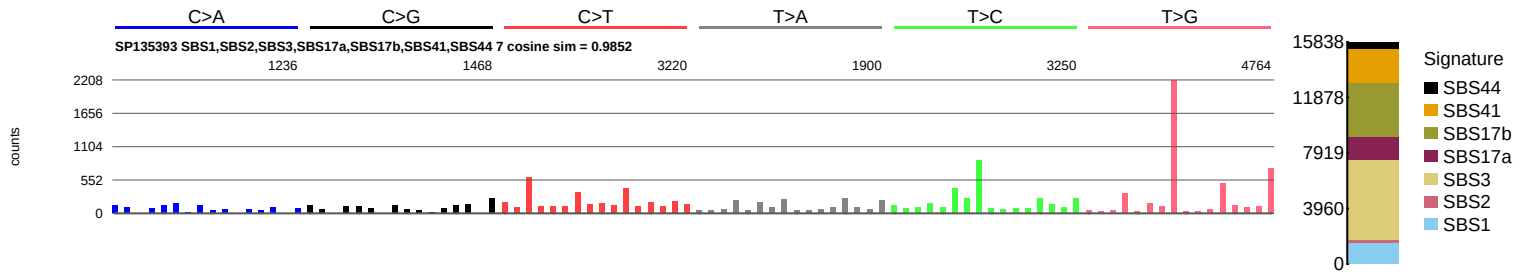

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98529
2061
1198
3408
1184
3566
4421
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1

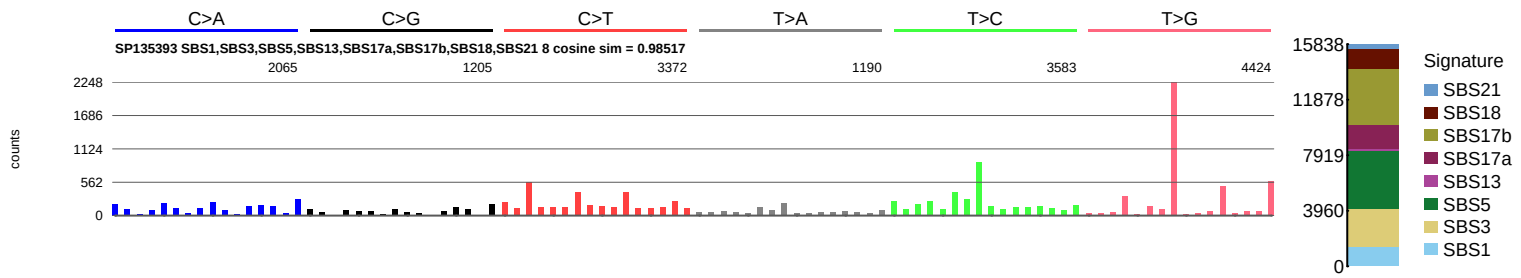

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98525
1200
1492
3067
1892
3421
4766
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS2
SBS1

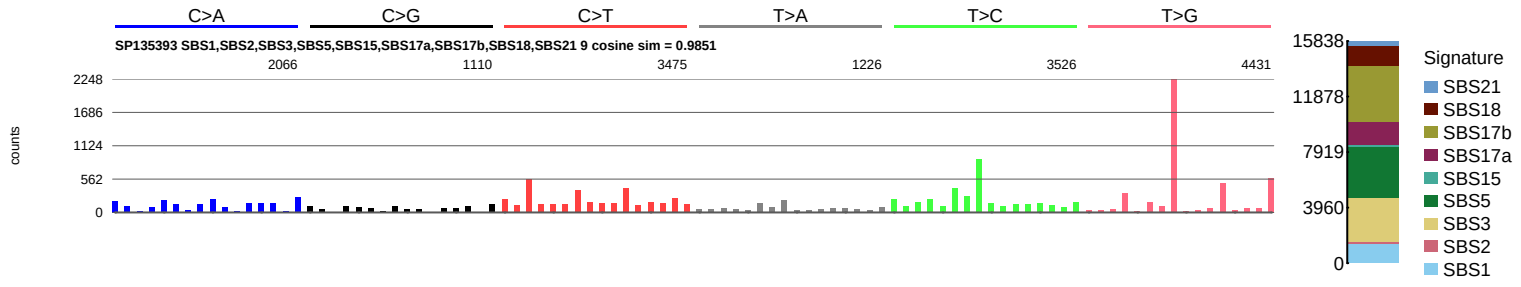

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98525
1239
1508
3195
1917
3197
4783
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

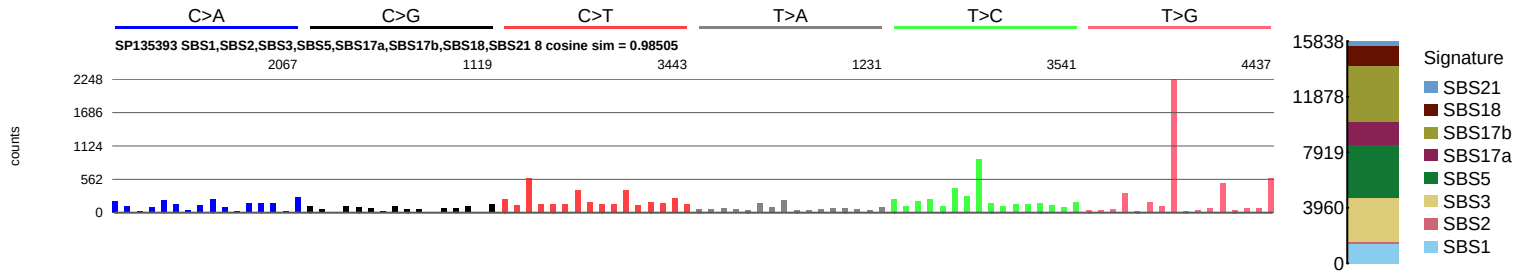

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 10 cosine sim = 0.98525
2120
714
2884
1506
3943
4672
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

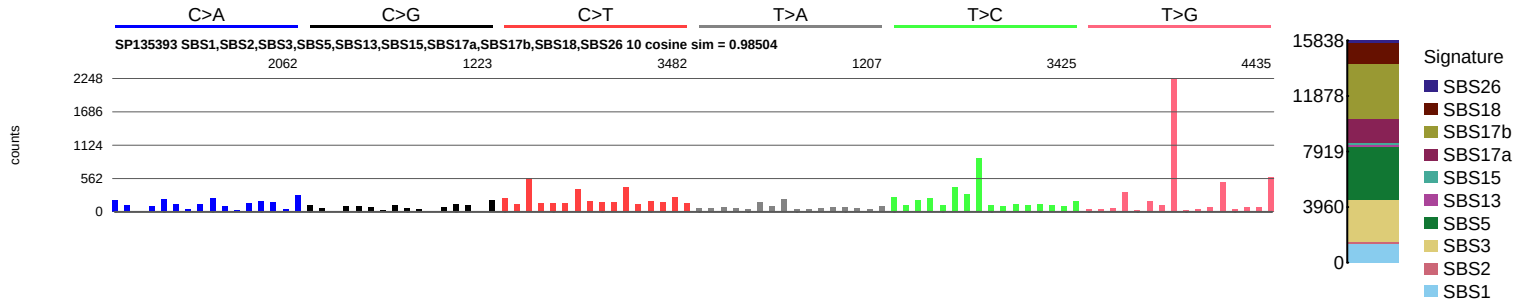

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 9 cosine sim = 0.98523
2119
715
2885
1505
3938
4677
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1

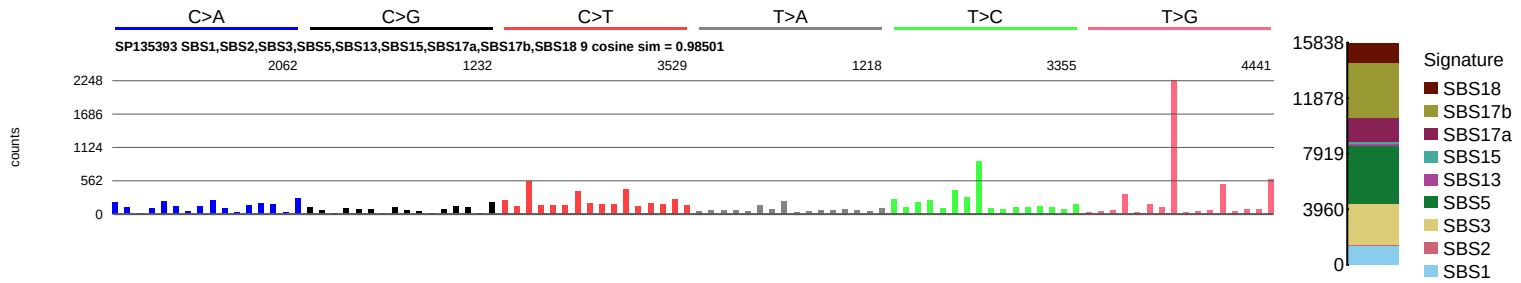

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.9852
1236
1468
3220
1900
3250
4764
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1

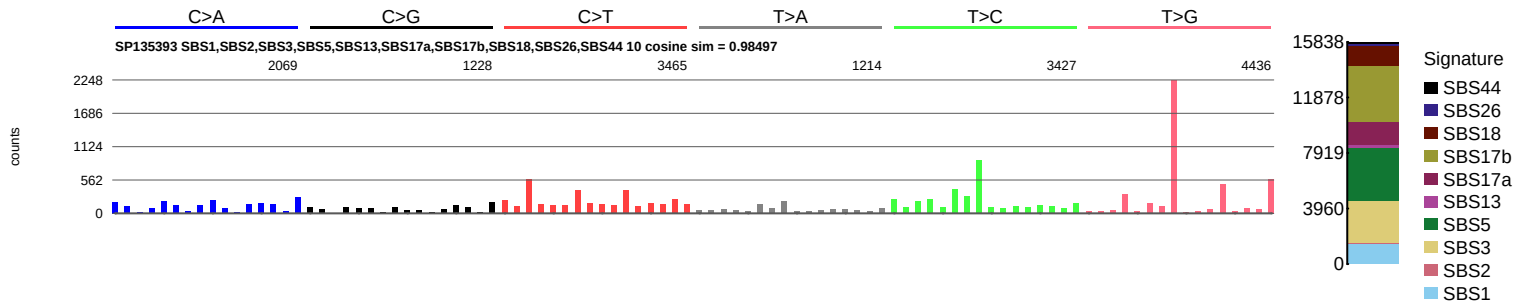

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98517
2065
1205
3372
1190
3583
4424
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

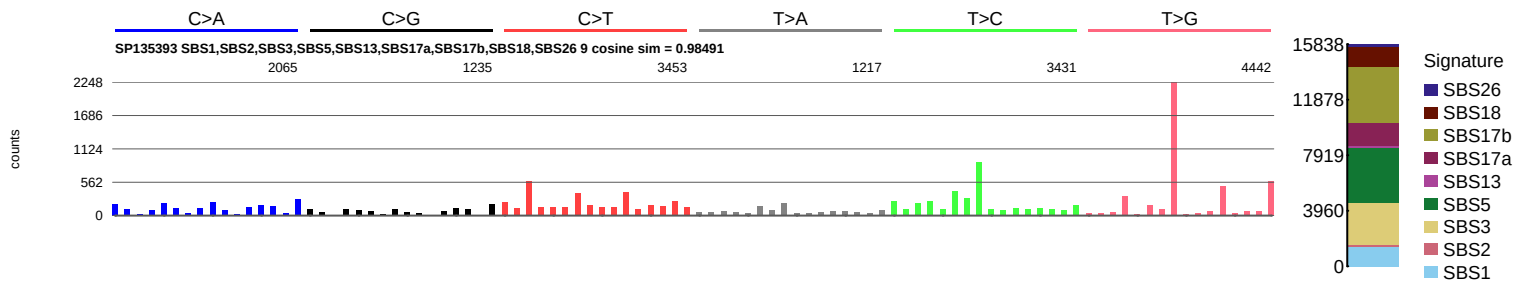

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.9851
2066
1110
3475
1226
3526
4431
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1

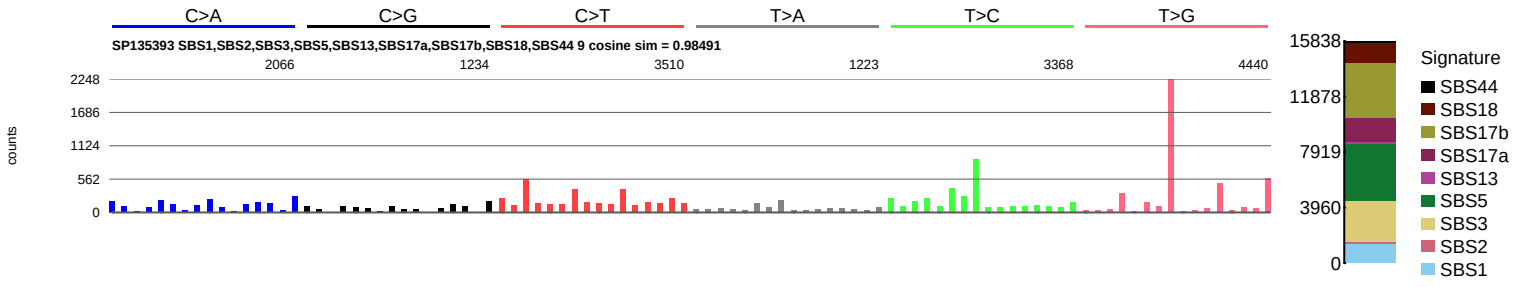

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98505
2067
1119
3443
1231
3541
4437
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1

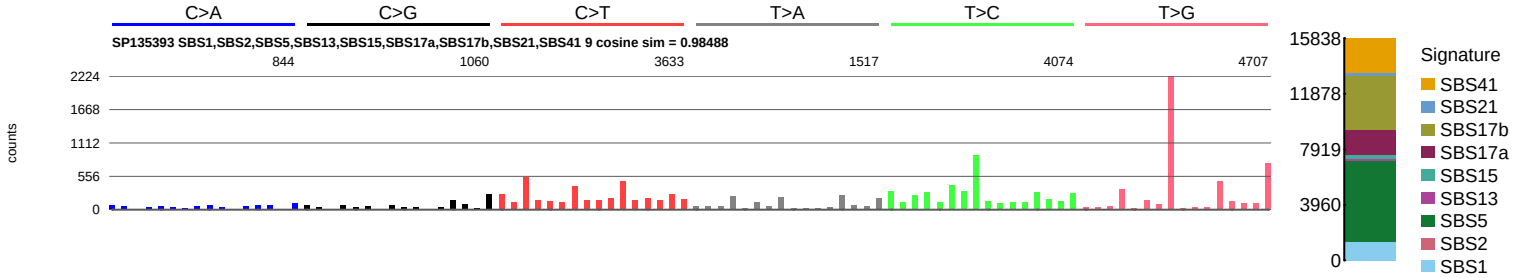

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 10 cosine sim = 0.98504
2062
1223
3482
1207
3425
4435
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

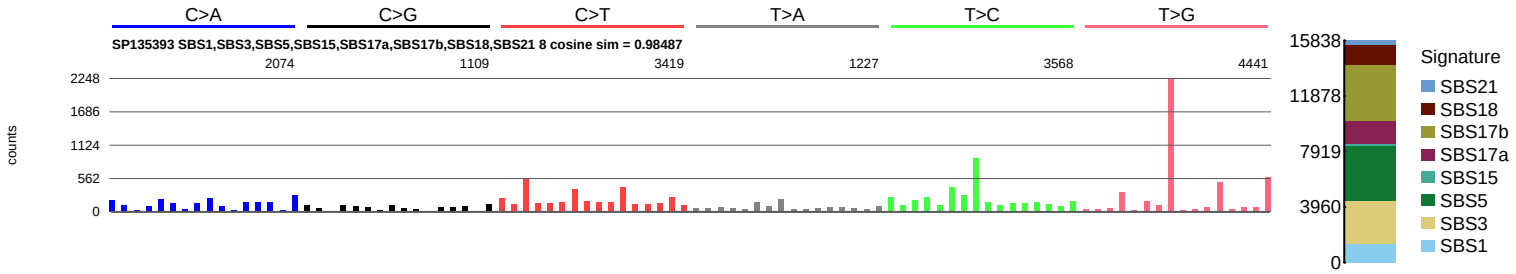

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 9 cosine sim = 0.98501
2062
1232
3529
1218
3355
4441
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS2
SBS1

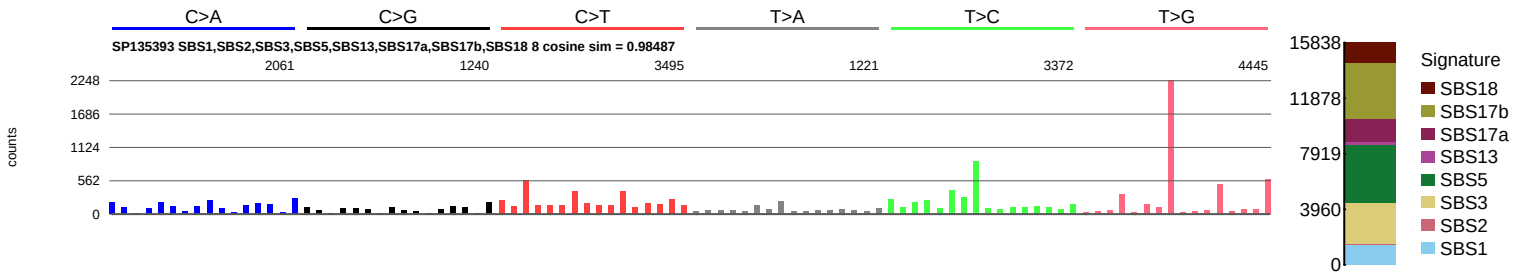

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98497
2069
1228
3465
1214
3427
4436
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98491
2065
1235
3453
1217
3431
4442
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98491
2066
1234
3510
1223
3368
4440
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 9 cosine sim = 0.98488
844
1060
3633
1517
4074
4707
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98487
2074
1109
3419
1227
3568
4441
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1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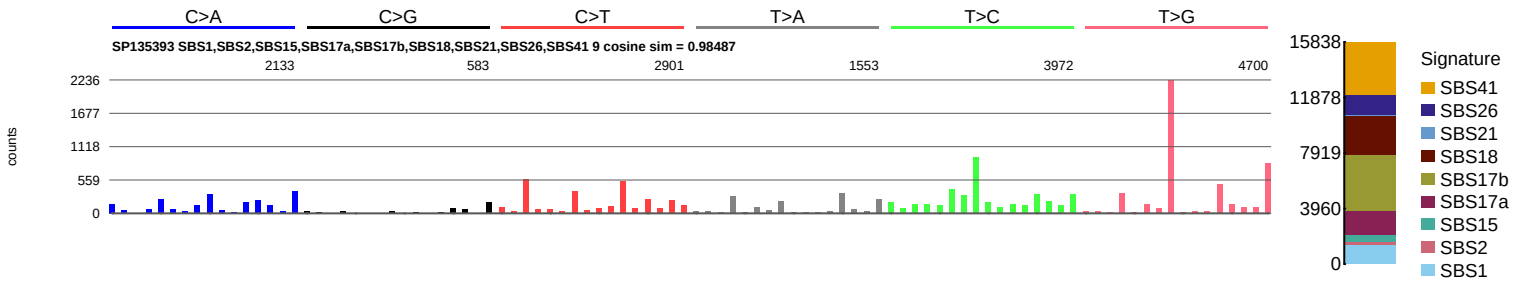
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98487
2061
1240
3495
1221
3372
4445
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1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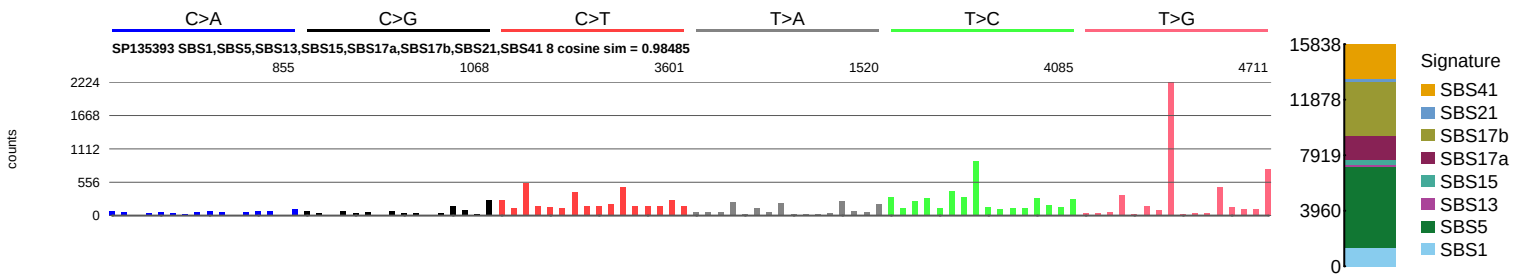
T>G
SP135393 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98487
2133
583
2901
1553
3972
4700
counts
2236
1677
1118
559
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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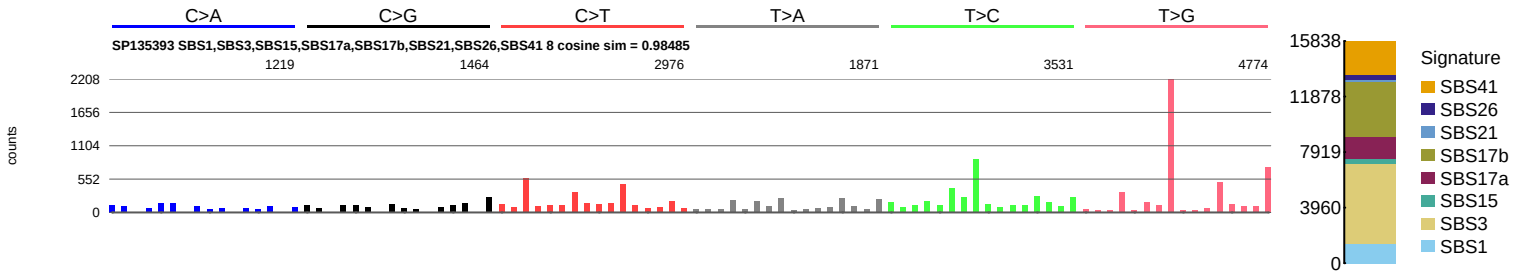
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98485
855
1068
3601
1520
4085
4711
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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C>G
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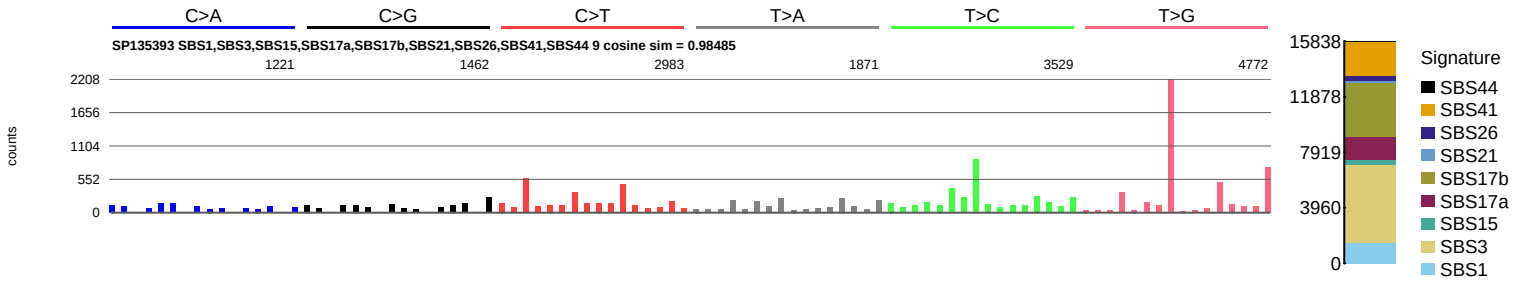
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41 8 cosine sim = 0.98485
1219
1464
2976
1871
3531
4774
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>G
C>T
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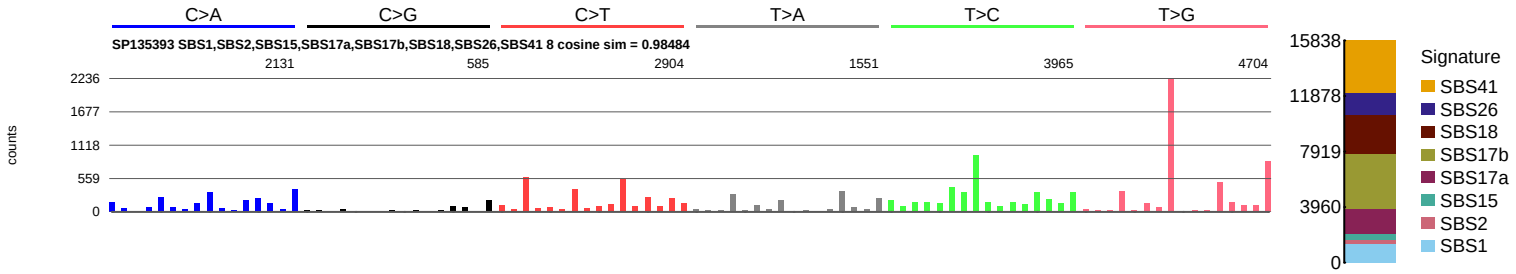
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98485
1221
1462
2983
1871
3529
4772
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1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C>T
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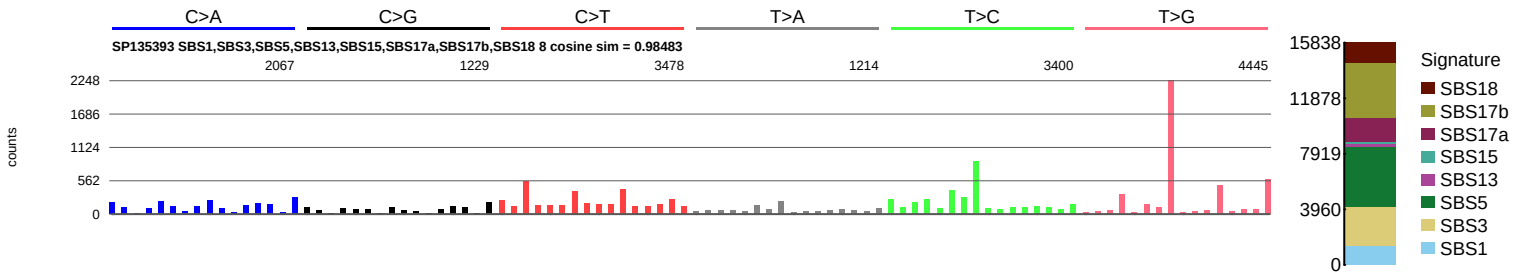
SP135393 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98484
2131
585
2904
1551
3965
4704
counts
2236
1677
1118
559
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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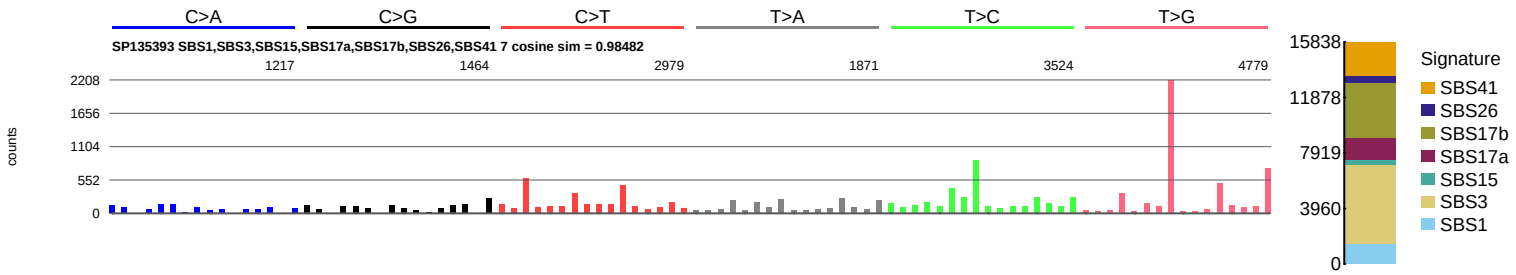
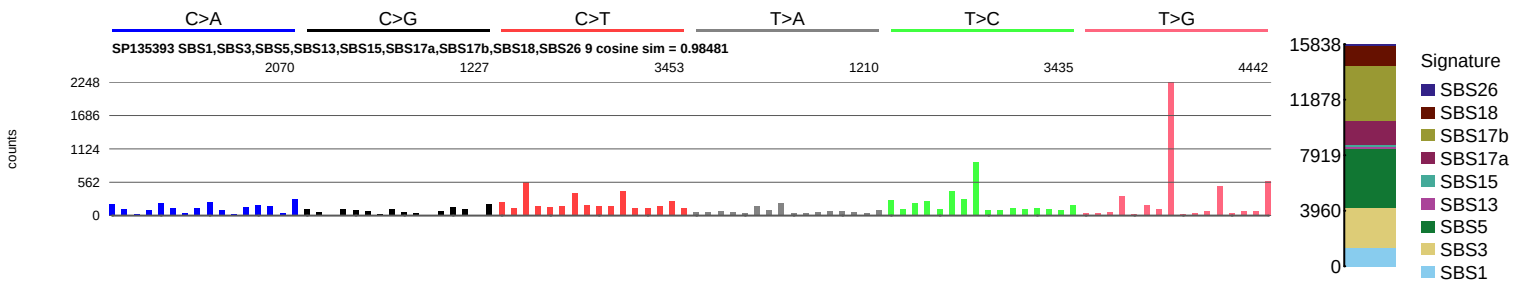
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98483
2067
1229
3478
1214
3400
4445
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41 7 cosine sim = 0.98482
1217
1464
2979
1871
3524
4779
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1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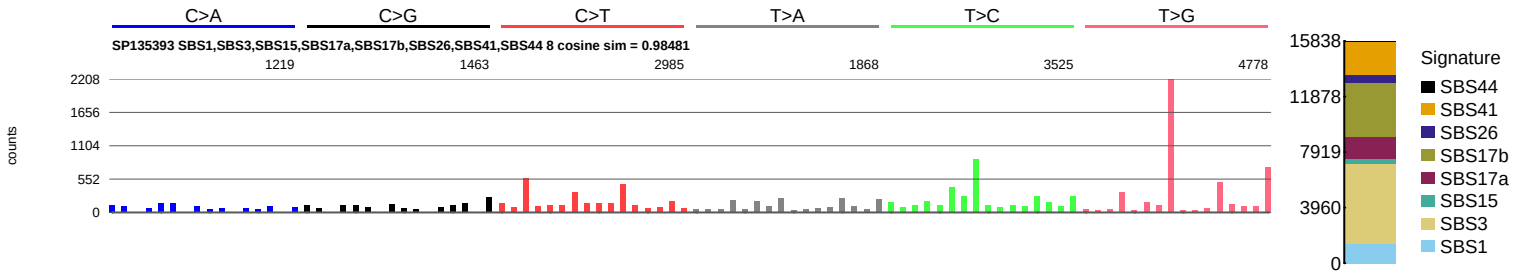
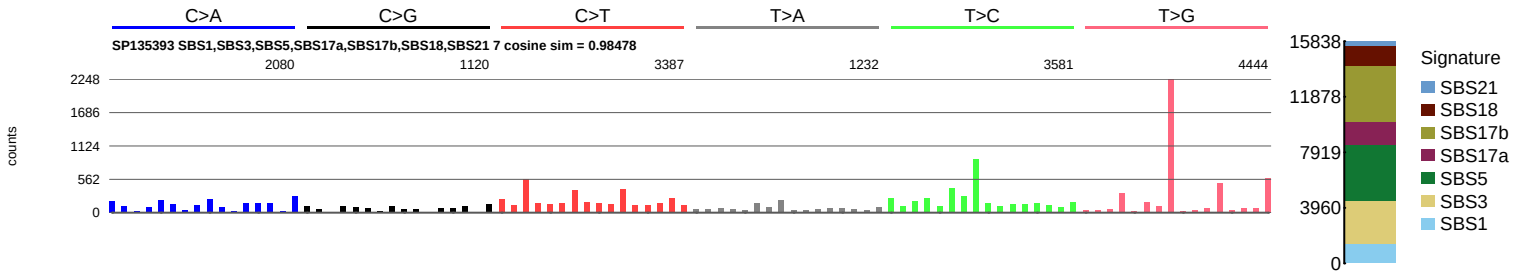
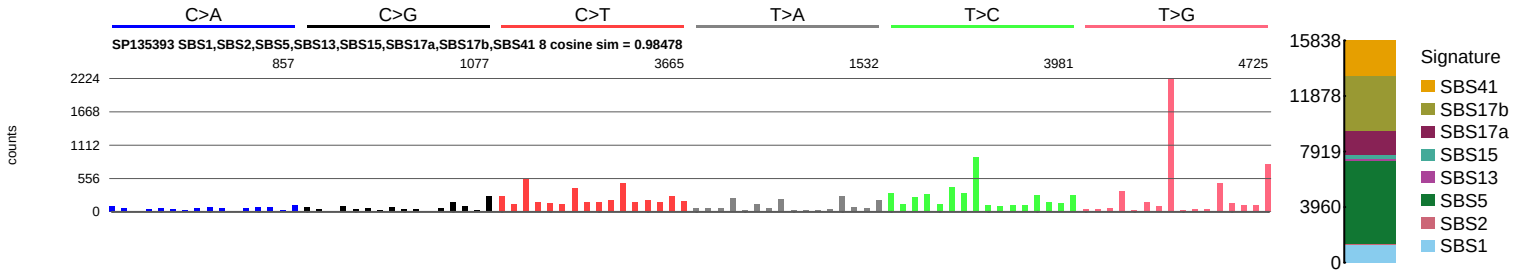
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98481
2070
1227
3453
1210
3435
4442
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS26,SBS41,SBS44 8 cosine sim = 0.98481
1219
1463
2985
1868
3525
4778
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98478
2080
1120
3387
1232
3581
4444
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 8 cosine sim = 0.98478
857
1077
3665
1532
3981
4725
counts
2224
1668
1112
556
0
15838
11878
7919
3960
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Signature
SBS41
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SBS17a
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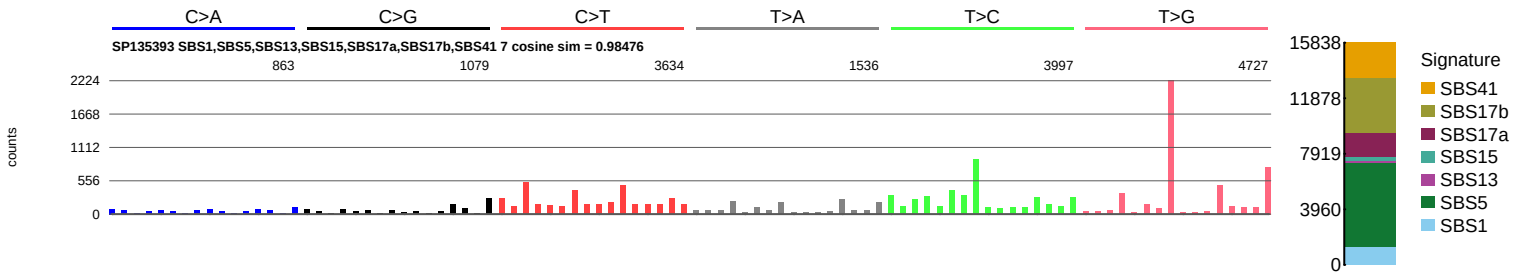
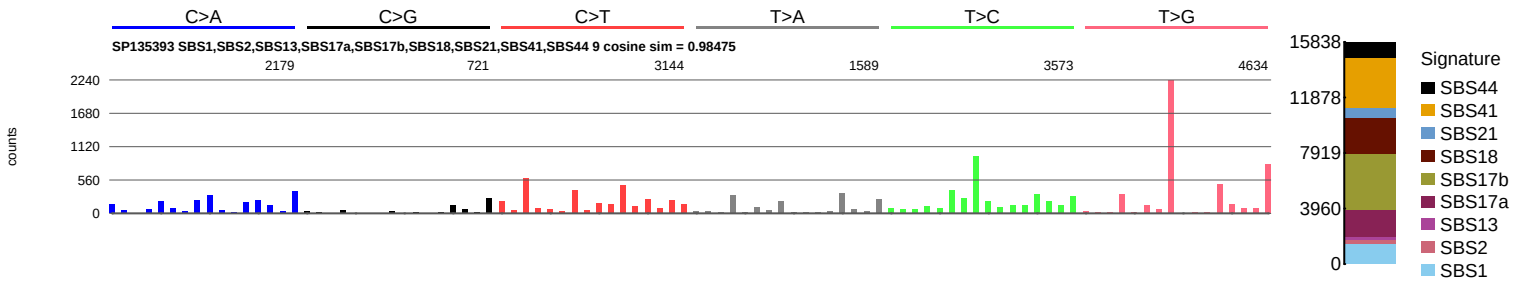
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98476
863
1079
3634
1536
3997
4727
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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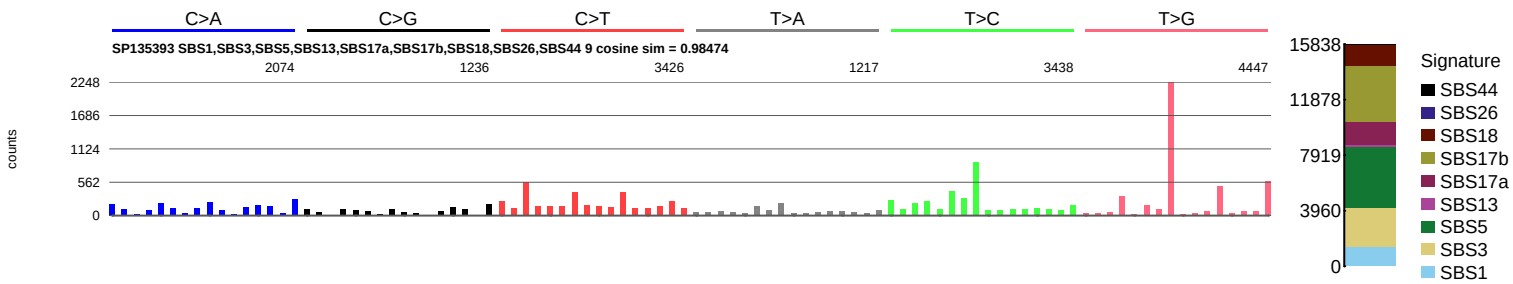
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98475
2179
721
3144
1589
3573
4634
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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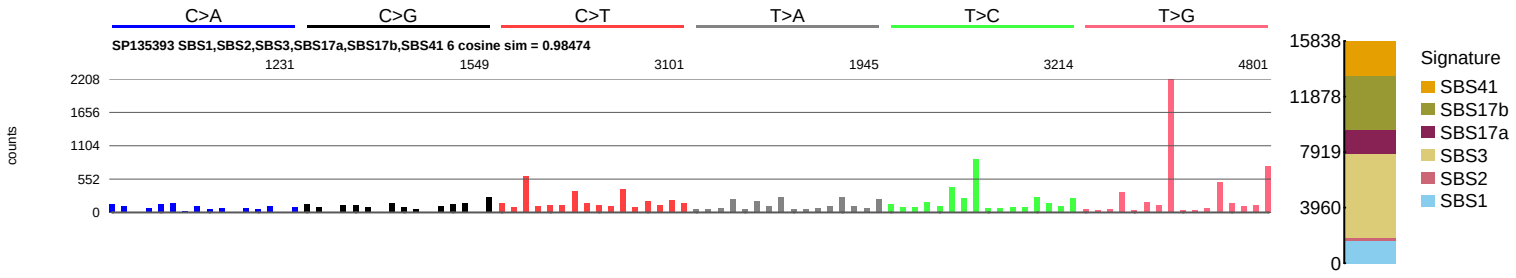
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98474
2074
1236
3426
1217
3438
4447
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1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C>G
C>T
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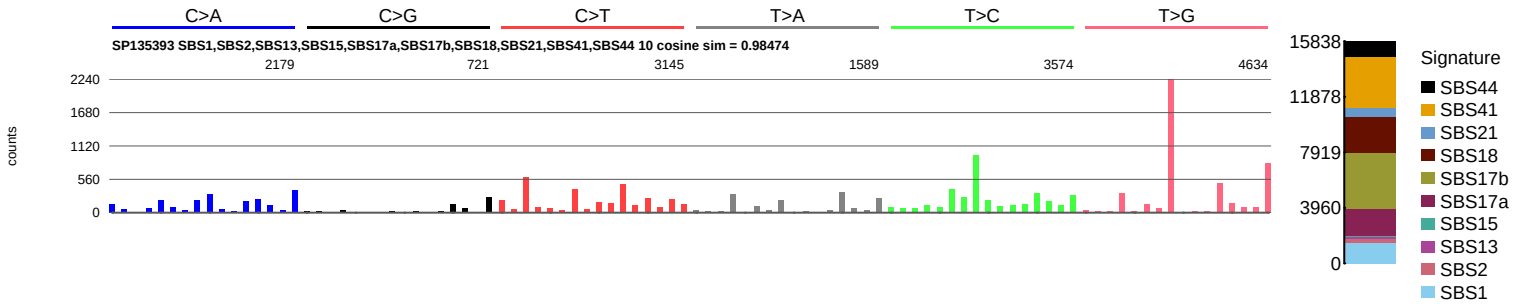
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98474
1231
1549
3101
1945
3214
4801
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS2
SBS1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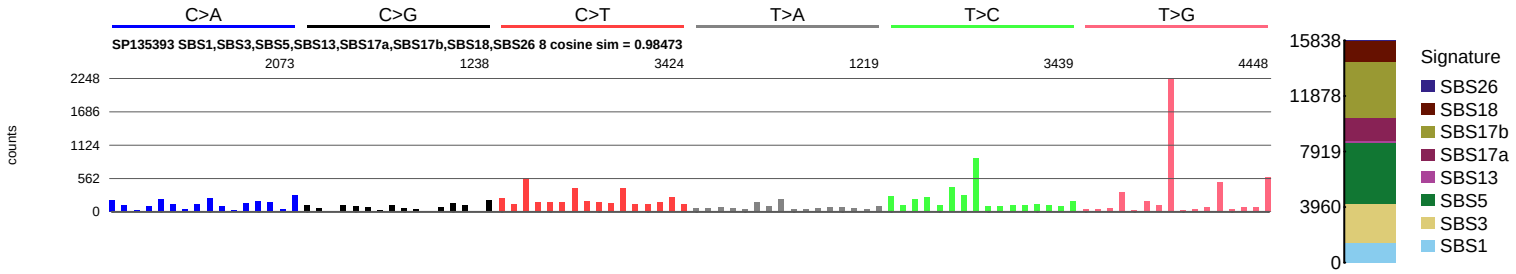
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 10 cosine sim = 0.98474
2179
721
3145
1589
3574
4634
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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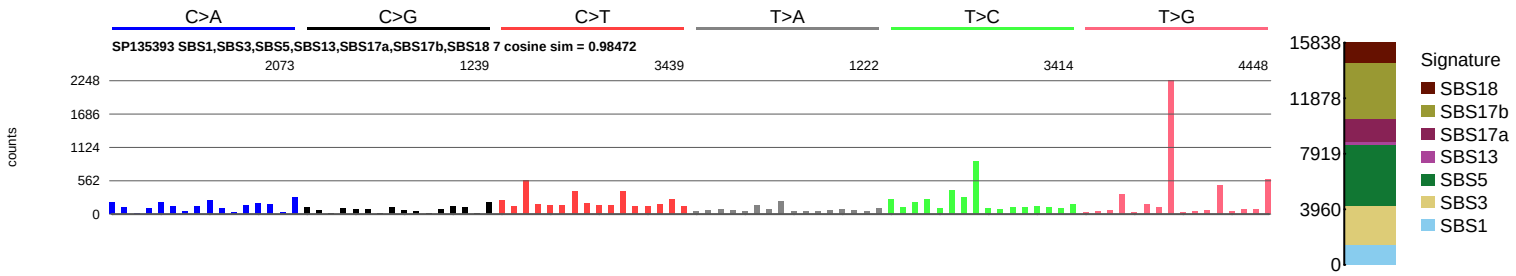
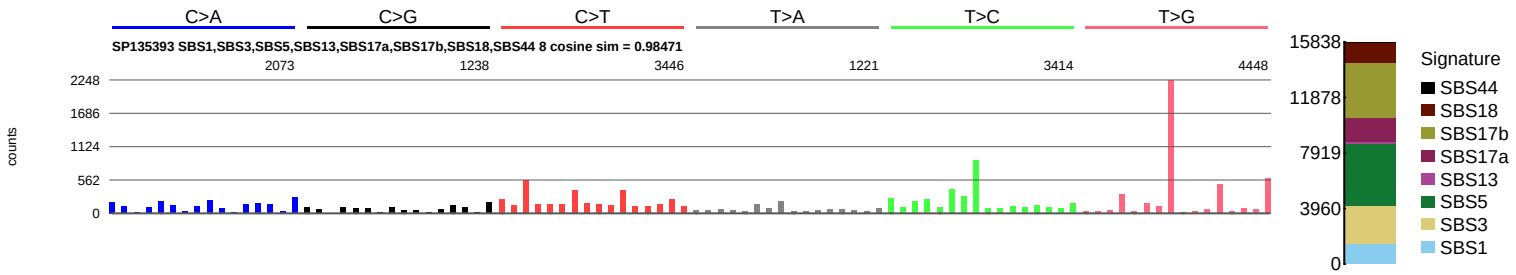
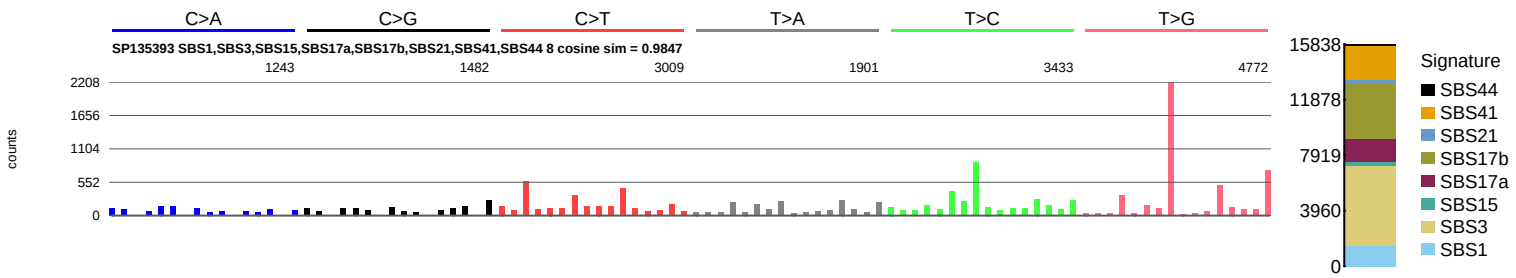
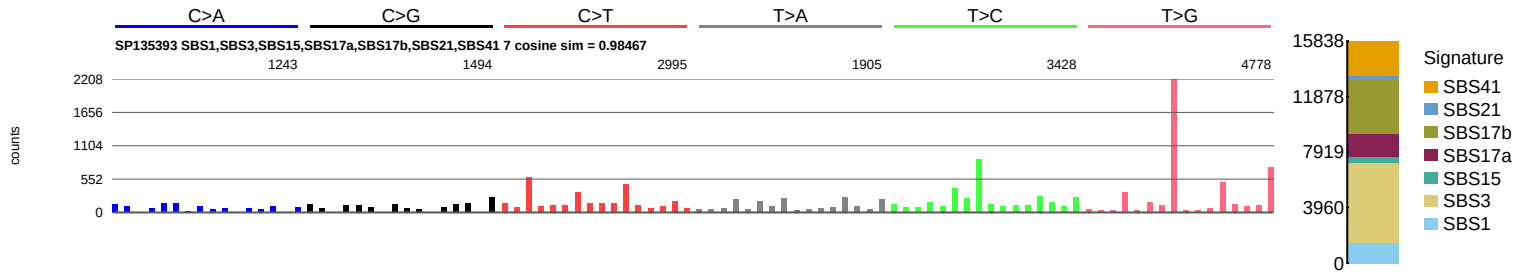
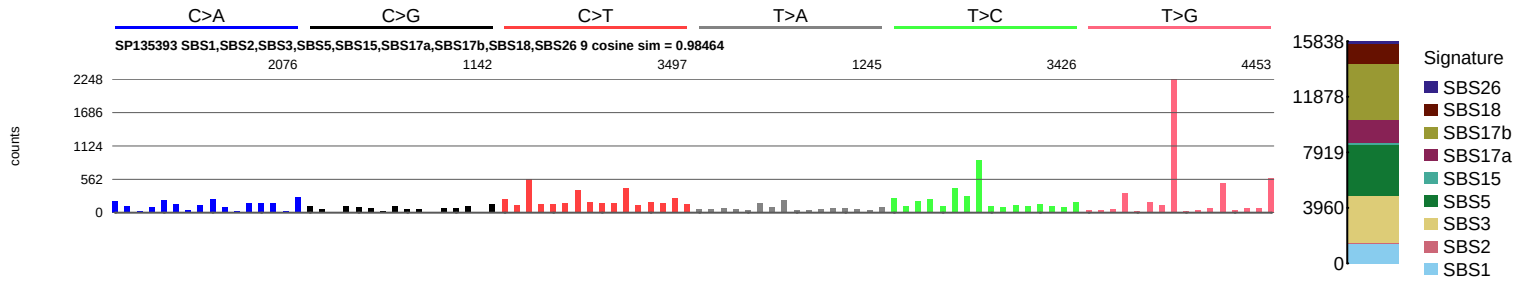
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98473
2073
1238
3424
1219
3439
4448
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

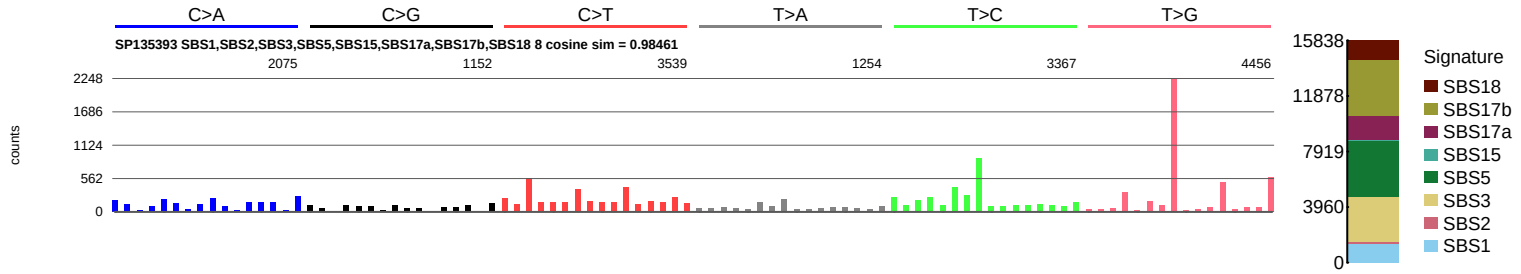

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98472
2073
1239
3439
1222
3414
4448
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1

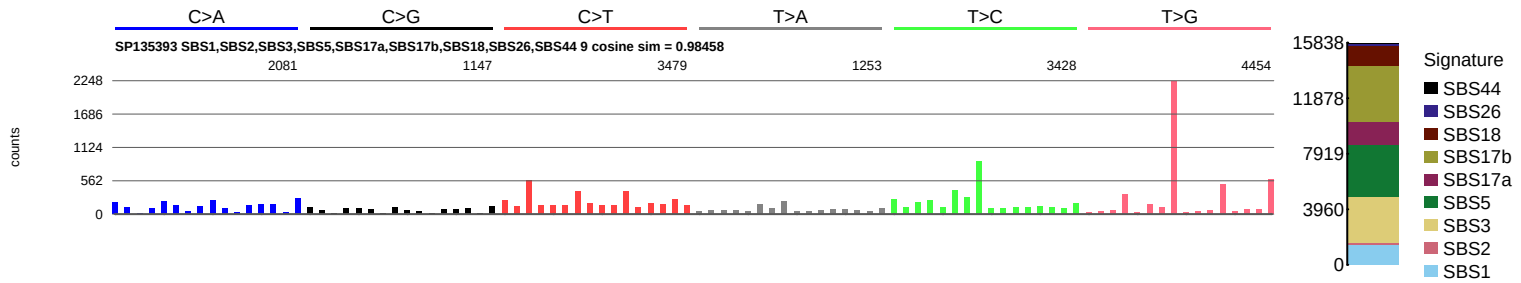

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98471
2073
1238
3446
1221
3414
4448
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41,SBS44 8 cosine sim = 0.9847
1243
1482
3009
1901
3433
4772
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98467
1243
1494
2995
1905
3428
4778
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98464
2076
1142
3497
1245
3426
4453
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98461
2075
1152
3539
1254
3367
4456
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98458
2081
1147
3479
1253
3428
4454
counts
2248
1686
1124
562
0
15838
11878
7919
3960
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SBS44
SBS26
SBS18
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SBS17a
SBS5
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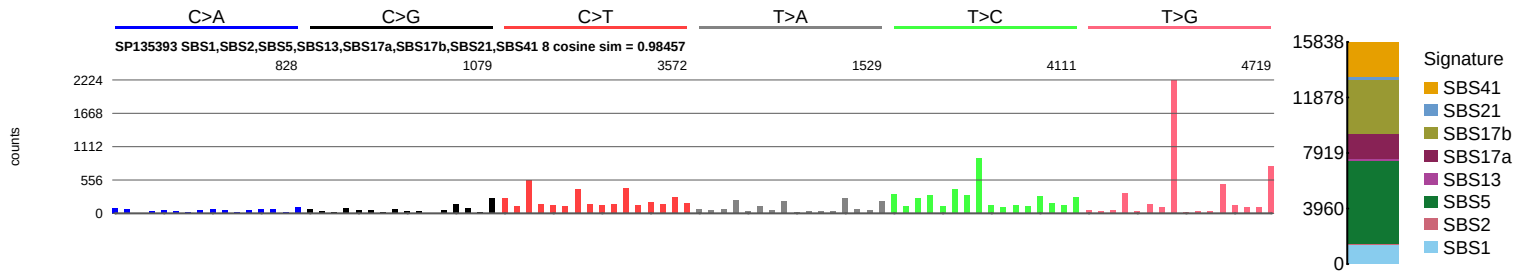
SBS2
SBS1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C>T
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SP135393 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98457
828
1079
3572
1529
4111
4719
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
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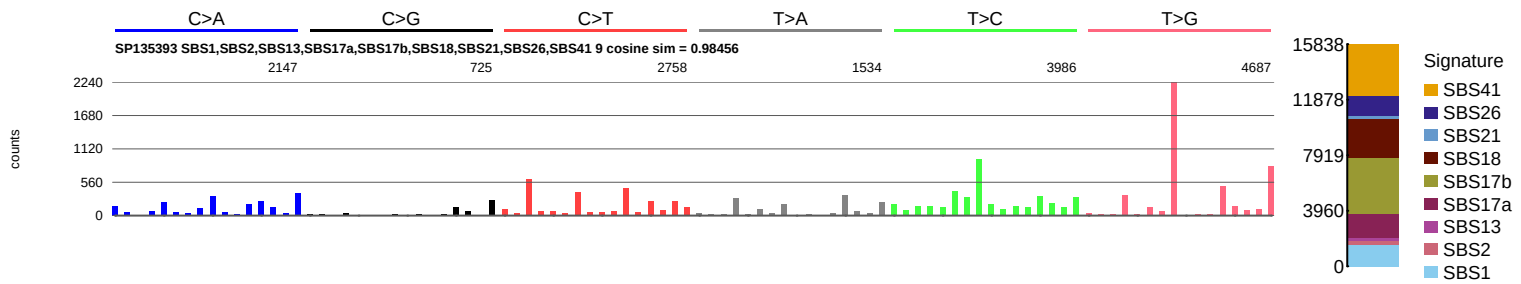
SBS1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SP135393 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.98456
2147
725
2758
1534
3986
4687
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
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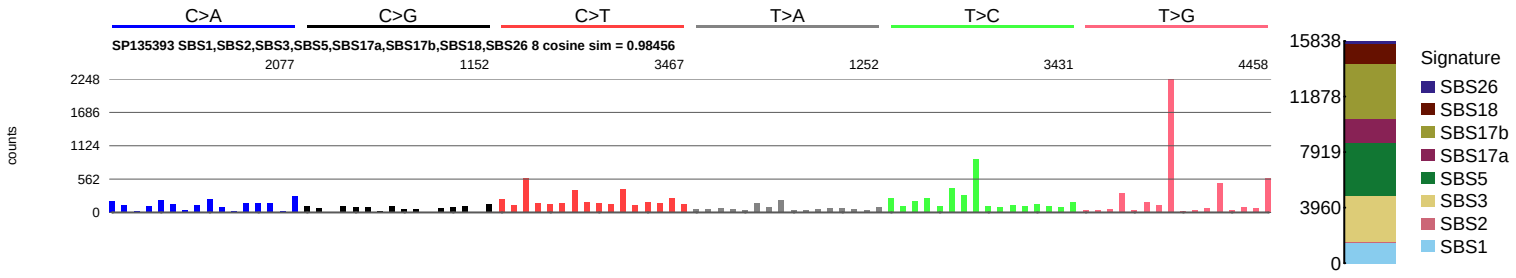
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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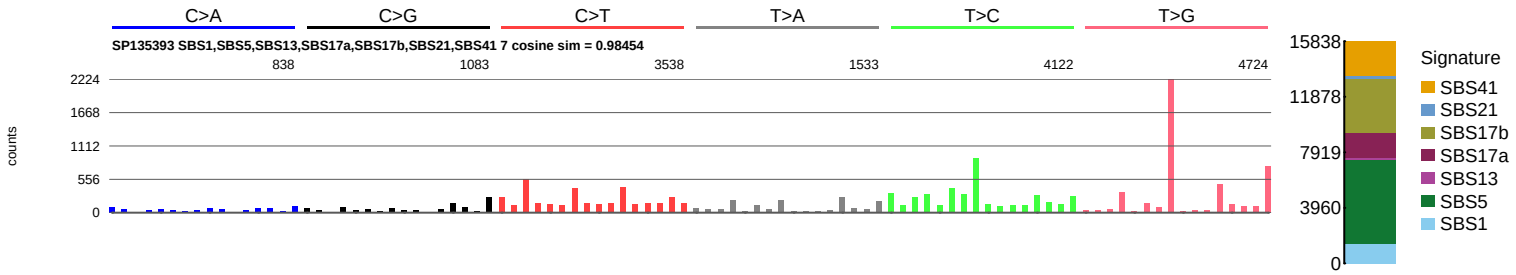
SP135393 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98456
2077
1152
3467
1252
3431
4458
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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C>G
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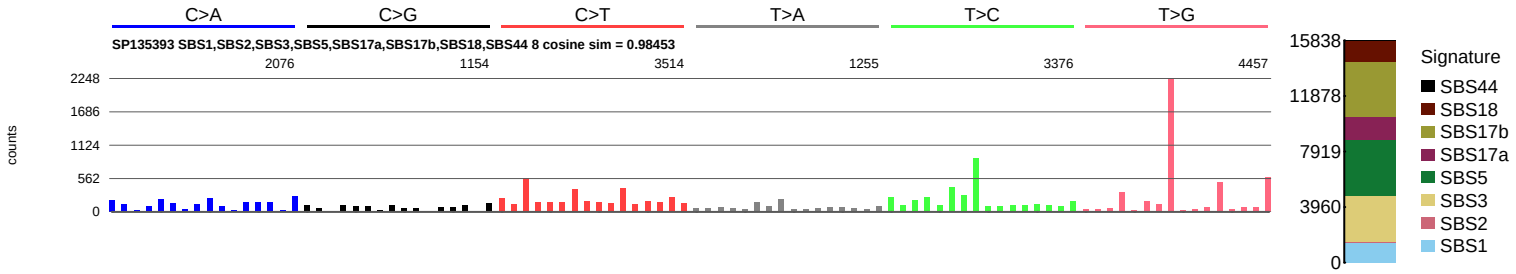
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98454
838
1083
3538
1533
4122
4724
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS1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C>T
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T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98453
2076
1154
3514
1255
3376
4457
counts
2248
1686
1124
562
0
15838
11878
7919
3960
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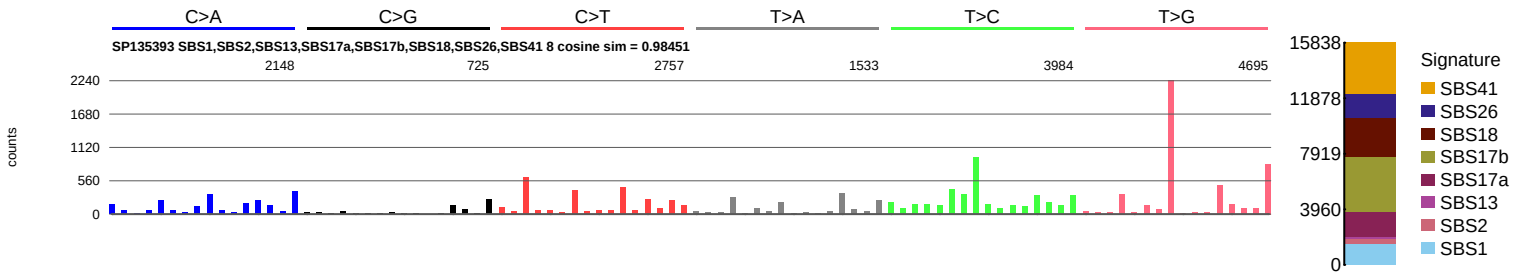
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98451
2148
725
2757
1533
3984
4695
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
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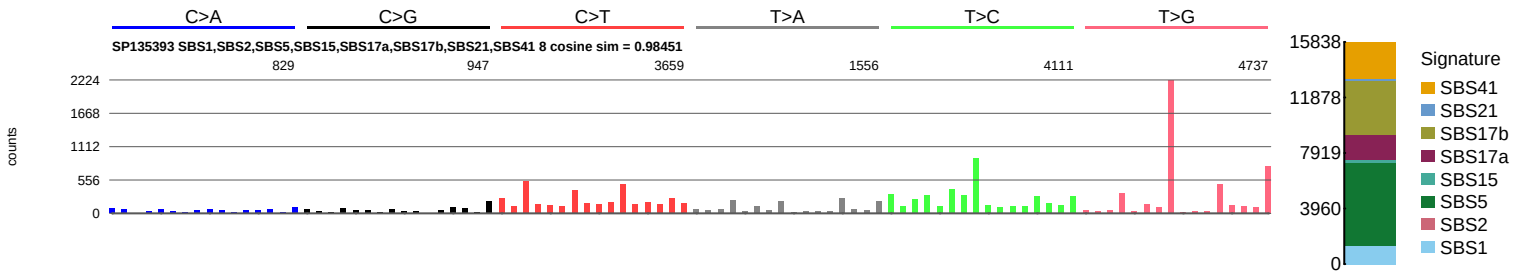
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 8 cosine sim = 0.98451
829
947
3659
1556
4111
4737
counts
2224
1668
1112
556
0
15838
11878
7919
3960
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Signature
SBS41
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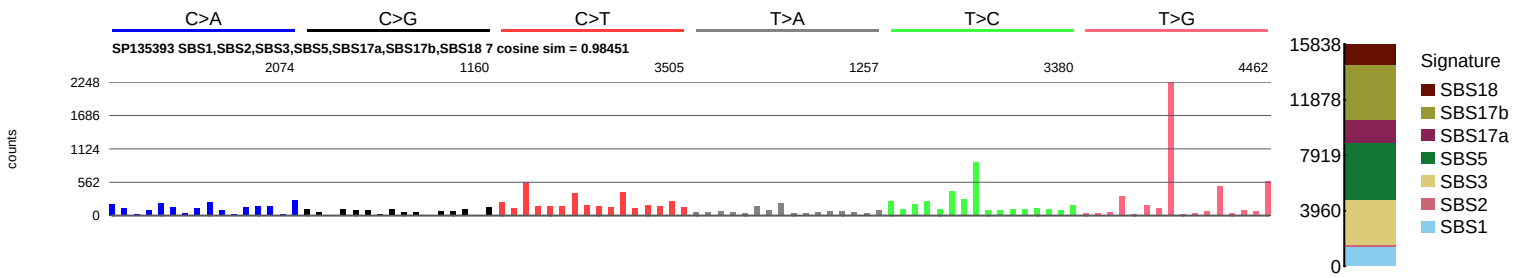
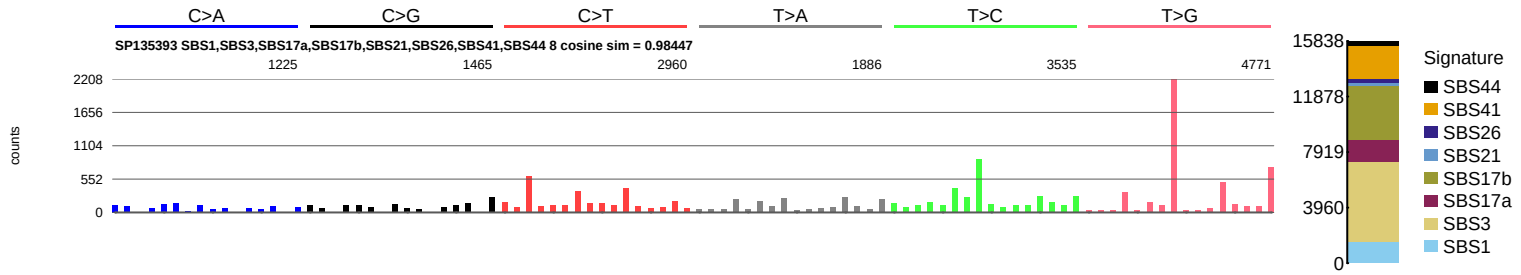
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98451
2074
1160
3505
1257
3380
4462
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1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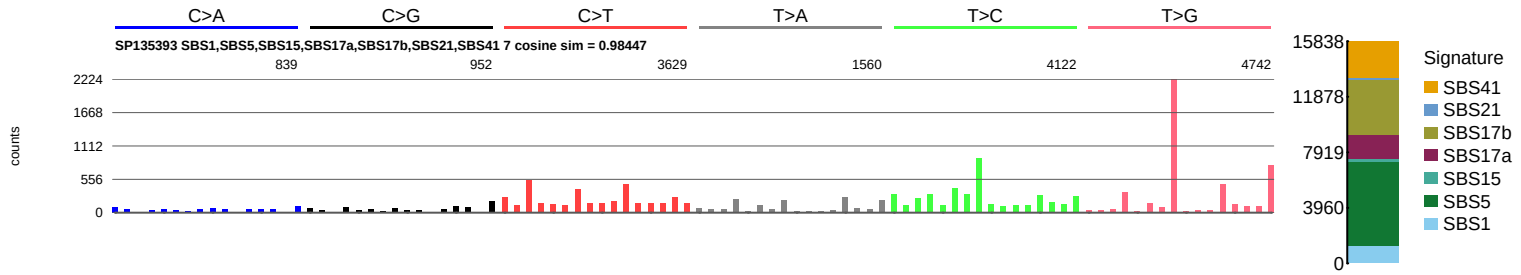
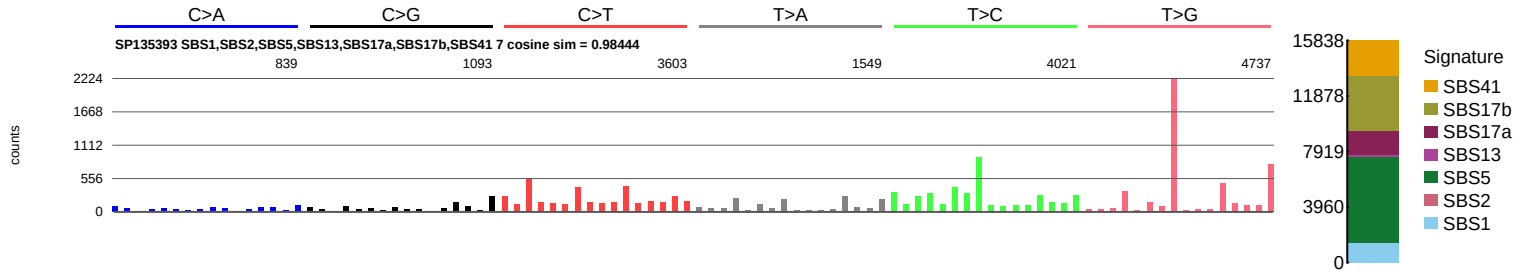
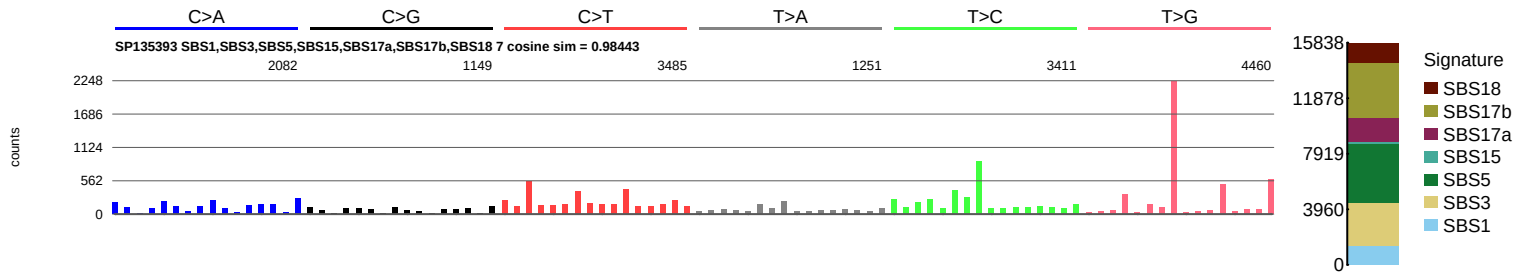
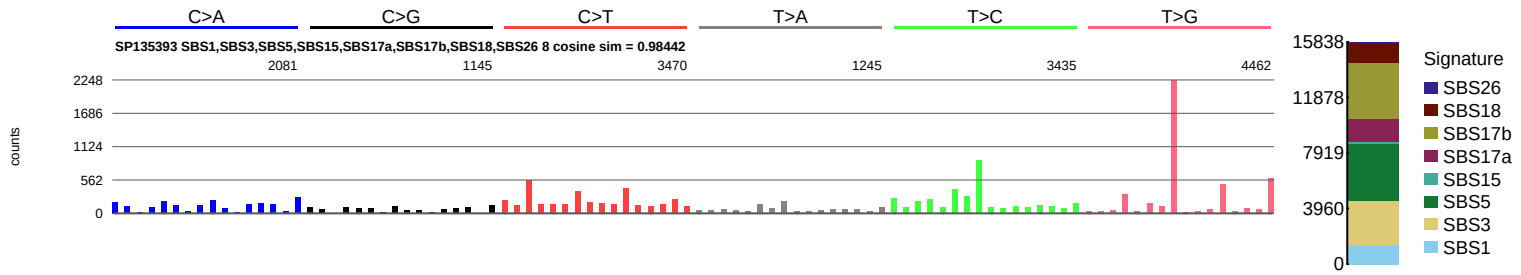
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.98447
1225
1465
2960
1886
3535
4771
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98447
839
952
3629
1560
4122
4742
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98444
839
1093
3603
1549
4021
4737
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98443
2082
1149
3485
1251
3411
4460
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98442
2081
1145
3470
1245
3435
4462
counts
2248
1686
1124
562
0
15838
11878
7919
3960
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Signature
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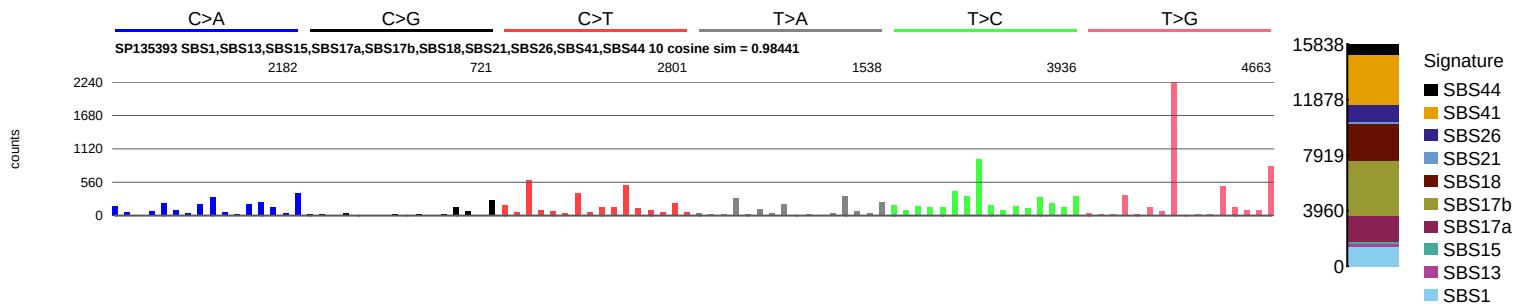
SBS3
SBS1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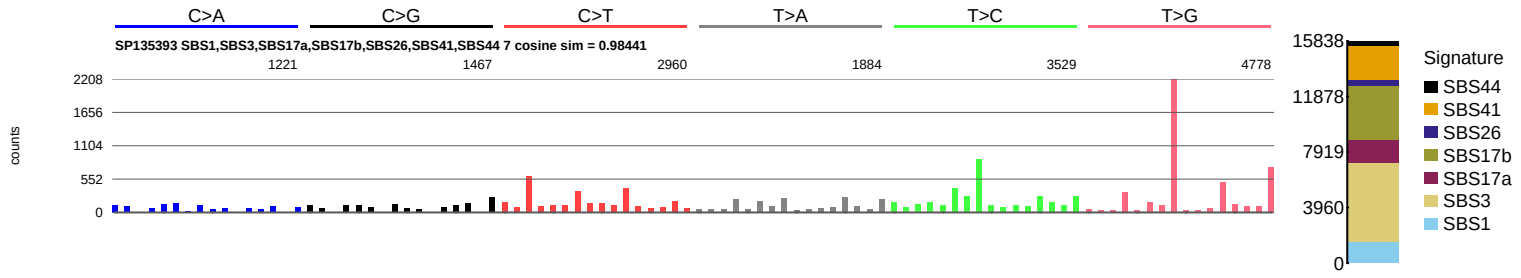
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 10 cosine sim = 0.98441
2182
721
2801
1538
3936
4663
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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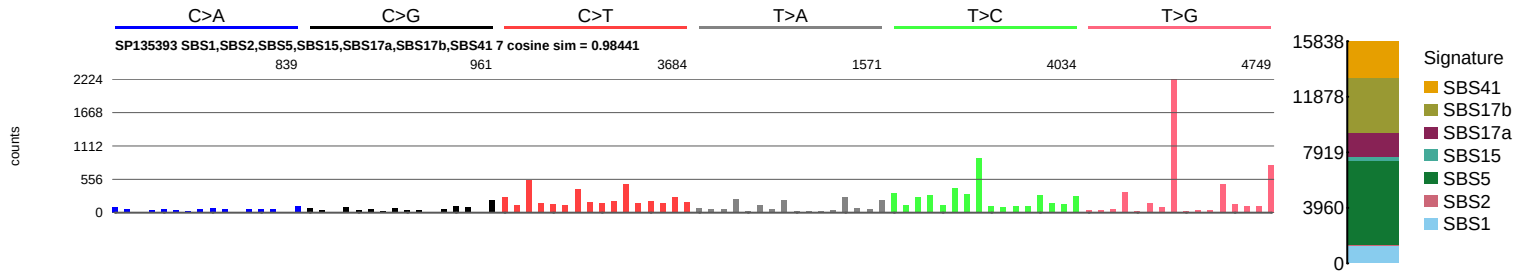
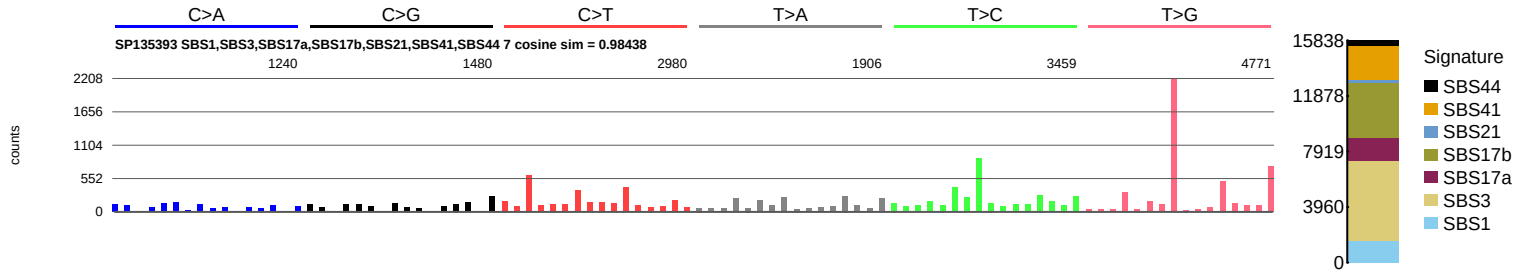
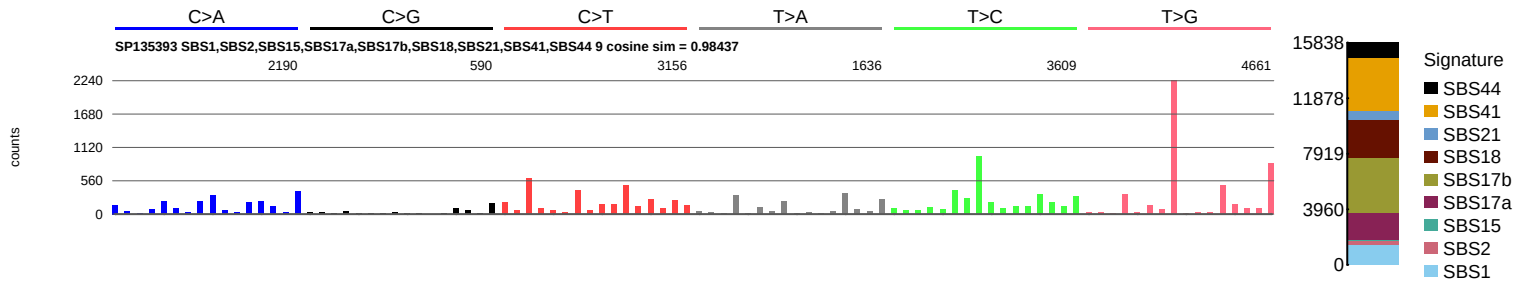
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS41,SBS44 7 cosine sim = 0.98441
1221
1467
2960
1884
3529
4778
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS41 7 cosine sim = 0.98441
839
961
3684
1571
4034
4749
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41,SBS44 7 cosine sim = 0.98438
1240
1480
2980
1906
3459
4771
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS1


C>A
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C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98437
2190
590
3156
1636
3609
4661
counts
2240
1680
1120
560
0
15838
11878
7919
3960
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SBS21
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SBS17b
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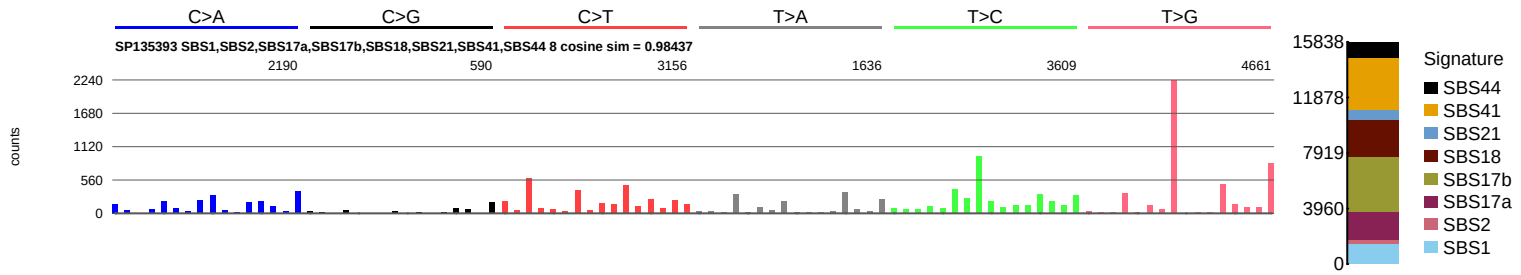
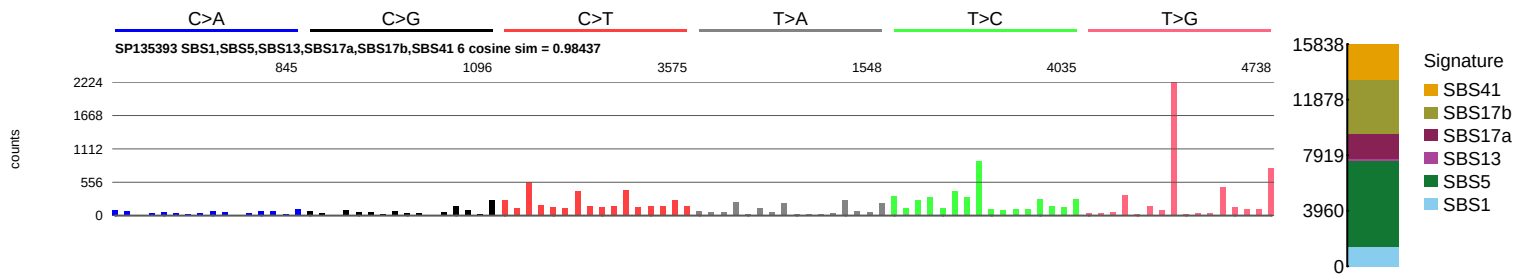
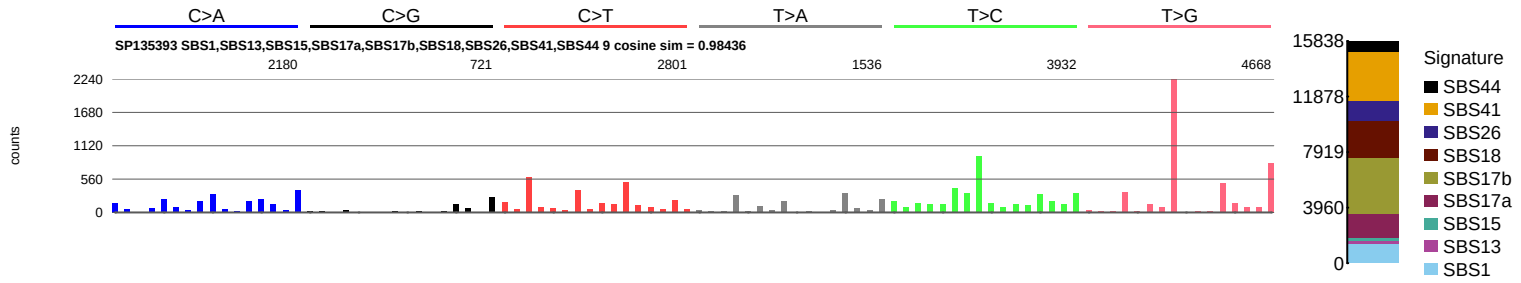
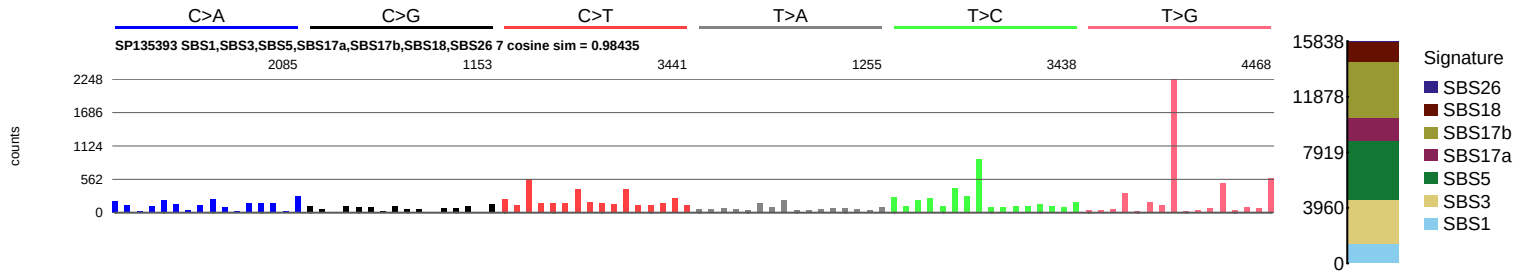
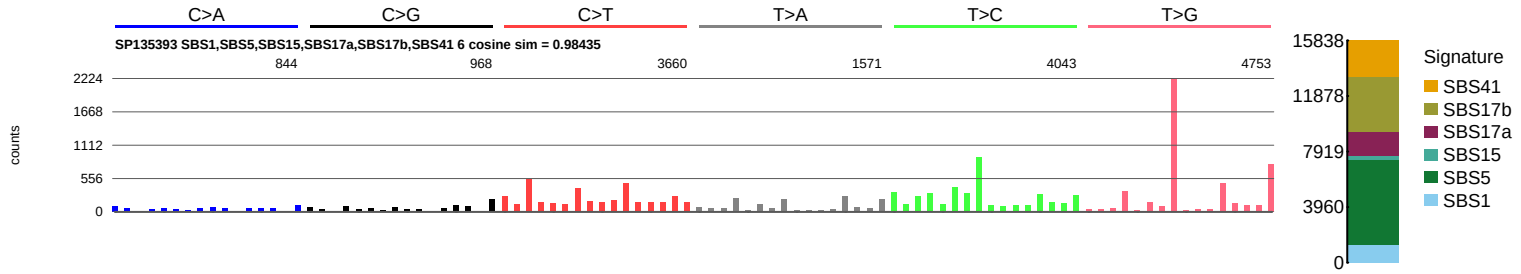
SBS15
SBS2
SBS1

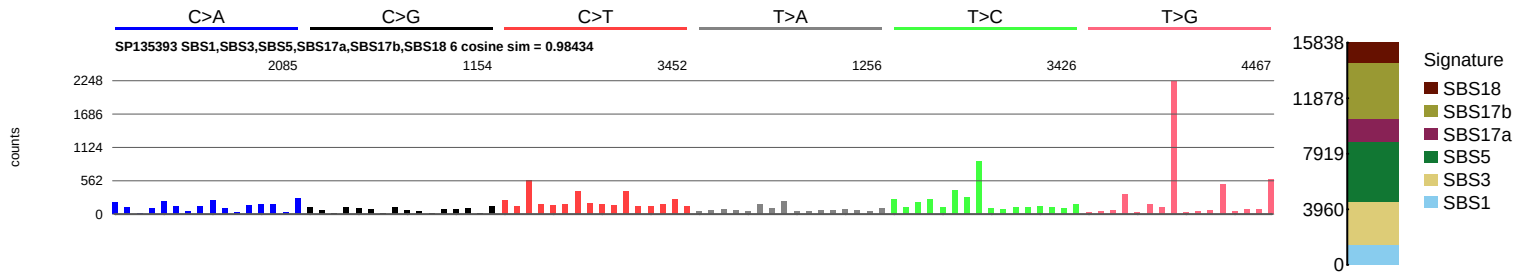

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98437
2190
590
3156
1636
3609
4661
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1

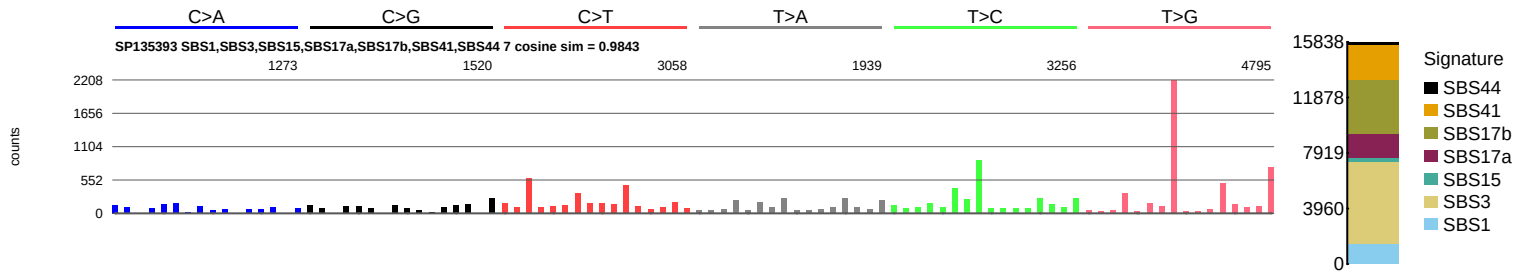

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98437
845
1096
3575
1548
4035
4738
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 9 cosine sim = 0.98436
2180
721
2801
1536
3932
4668
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98435
2085
1153
3441
1255
3438
4468
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98435
844
968
3660
1571
4043
4753
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98434
2085
1154
3452
1256
3426
4467
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS41,SBS44 7 cosine sim = 0.9843
1273
1520
3058
1939
3256
4795
counts
2208
1656
1104
552
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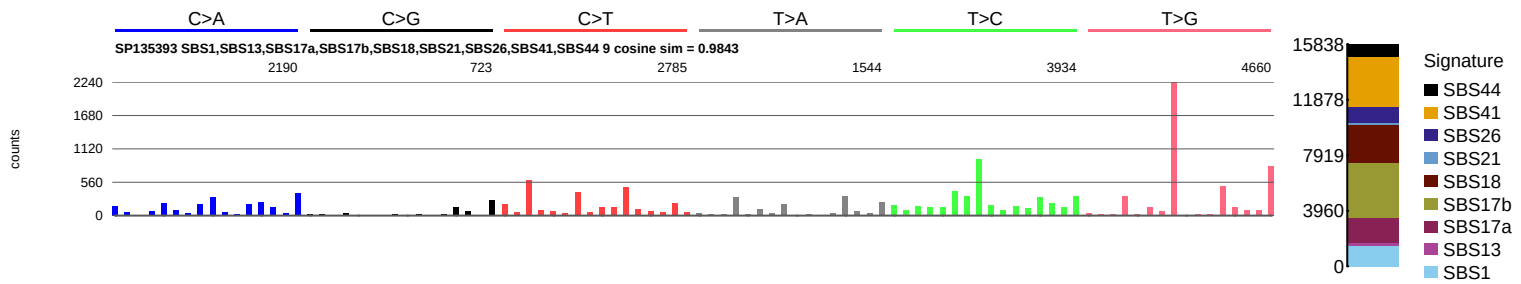
SBS1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SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.9843
2190
723
2785
1544
3934
4660
counts
2240
1680
1120
560
0
15838
11878
7919
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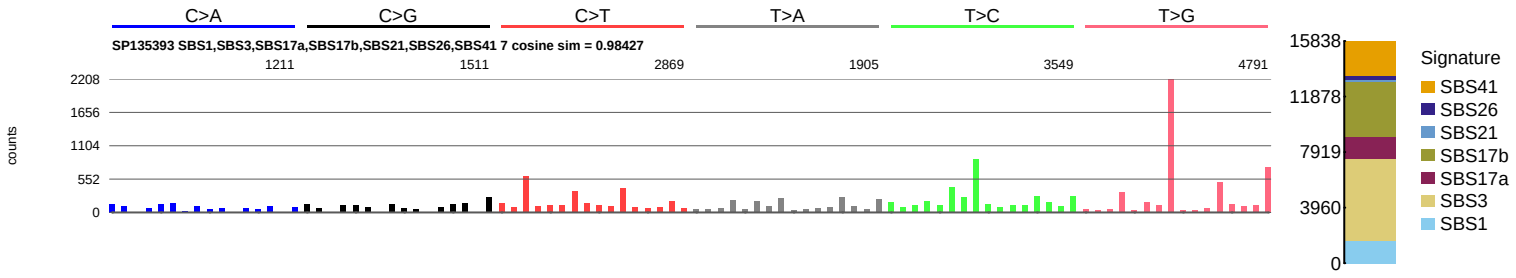
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SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS26,SBS41 7 cosine sim = 0.98427
1211
1511
2869
1905
3549
4791
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
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SBS17b
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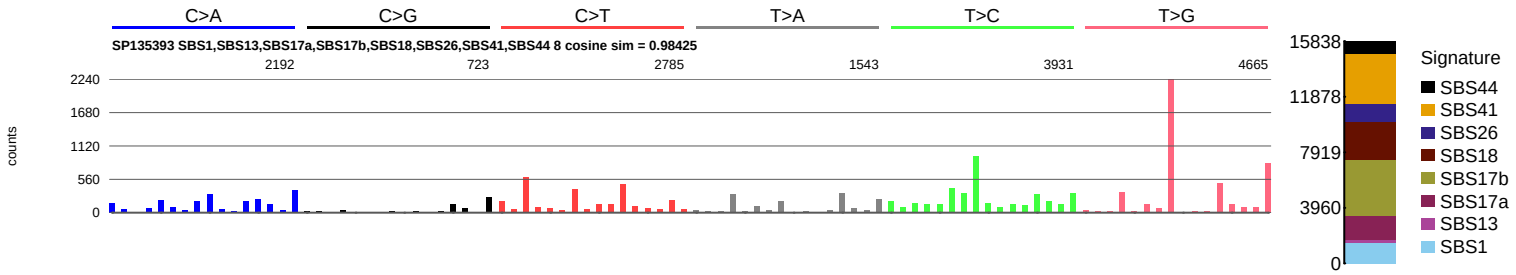
SBS3
SBS1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SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98425
2192
723
2785
1543
3931
4665
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS44
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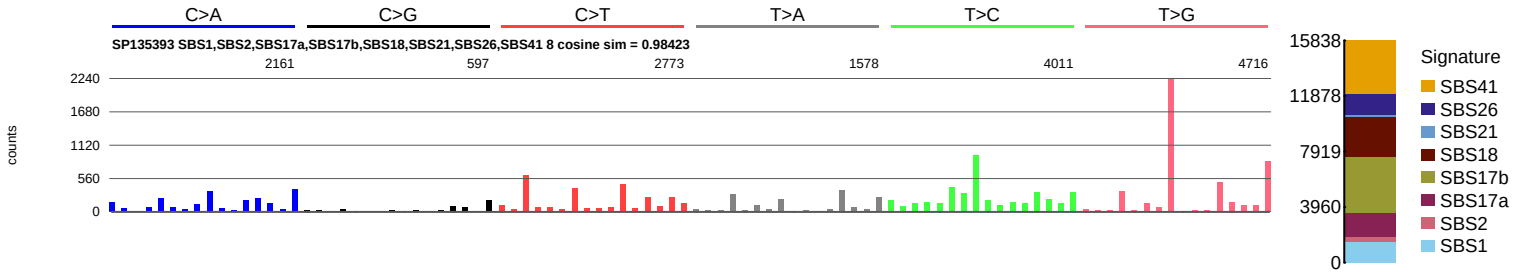
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1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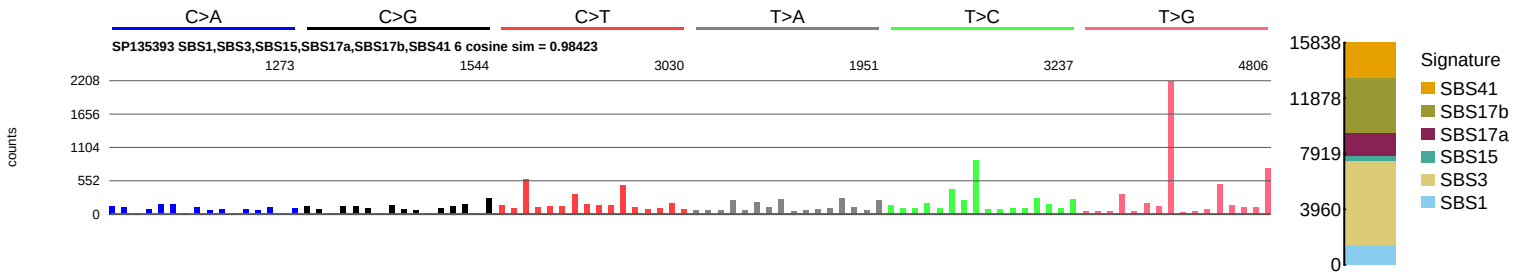
SP135393 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98423
2161
597
2773
1578
4011
4716
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1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SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98423
1273
1544
3030
1951
3237
4806
counts
2208
1656
1104
552
0
15838
11878
7919
3960
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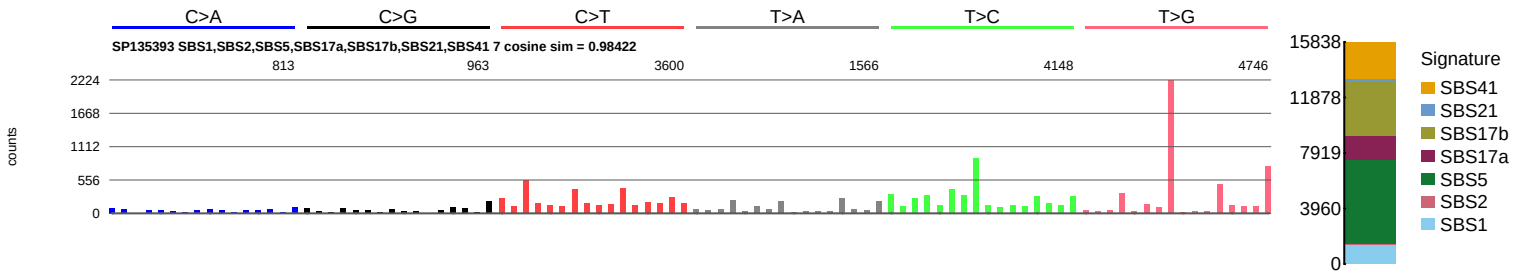
SBS41
SBS17b
SBS17a
SBS15
SBS3
SBS1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SP135393 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS21,SBS41 7 cosine sim = 0.98422
813
963
3600
1566
4148
4746
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
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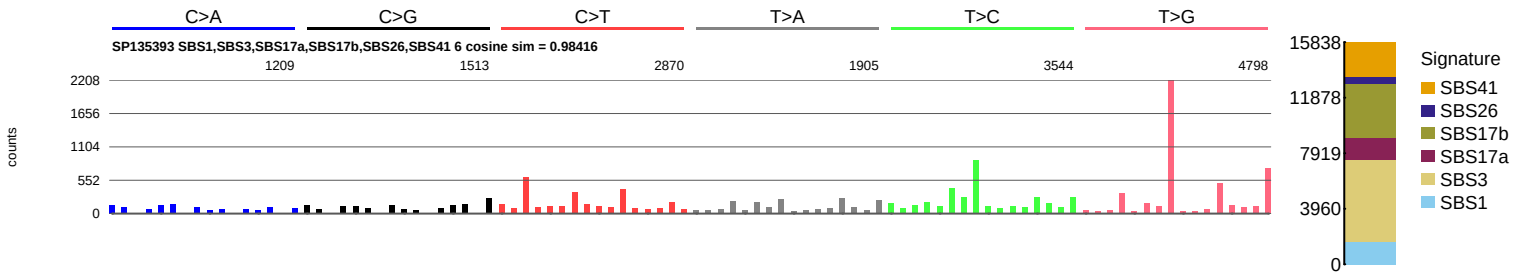
SBS21
SBS17b
SBS17a
SBS5
SBS2
SBS1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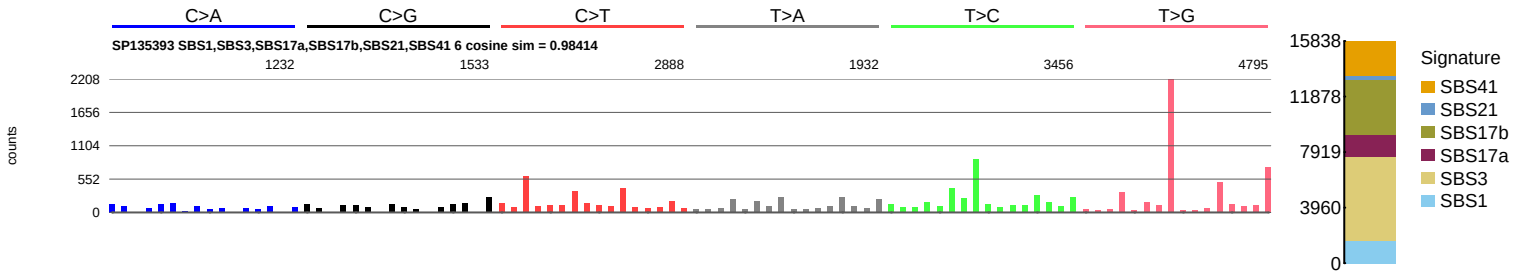
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS26,SBS41 6 cosine sim = 0.98416
1209
1513
2870
1905
3544
4798
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS17b
SBS17a
SBS3
SBS1


C>A
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C>T
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T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98414
1232
1533
2888
1932
3456
4795
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
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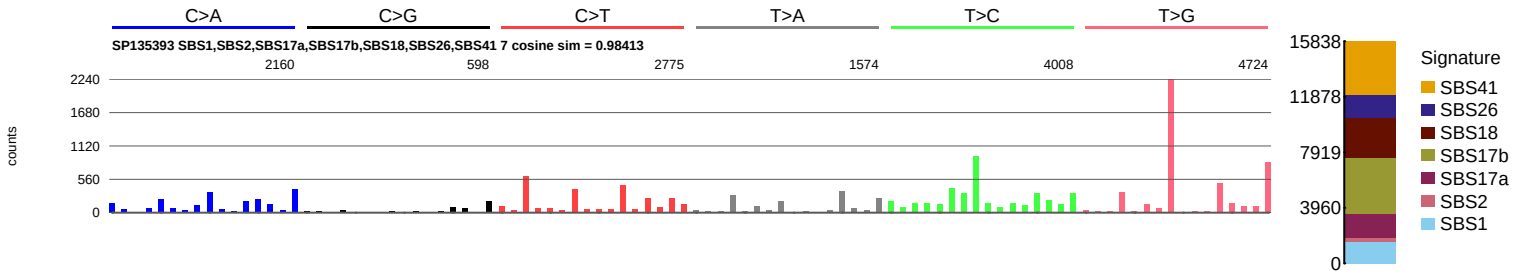
SBS41
SBS21
SBS17b
SBS17a
SBS3
SBS1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C>G
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T>G
SP135393 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98413
2160
598
2775
1574
4008
4724
counts
2240
1680
1120
560
0
15838
11878
7919
3960
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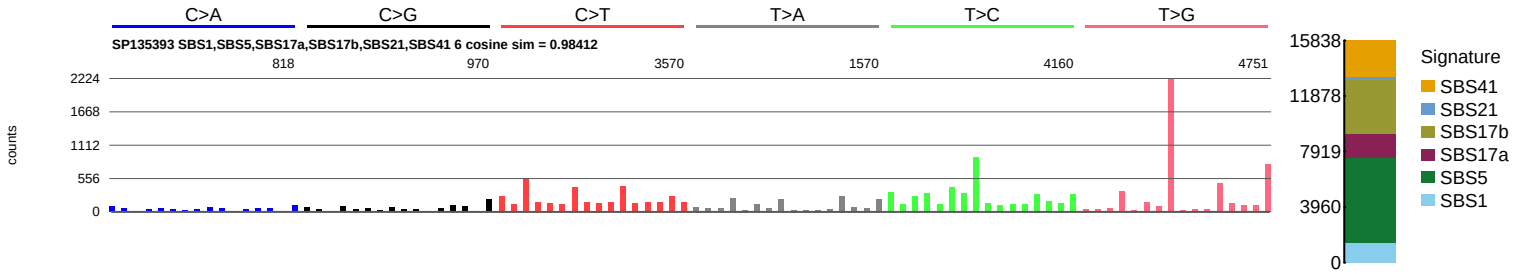
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS2
SBS1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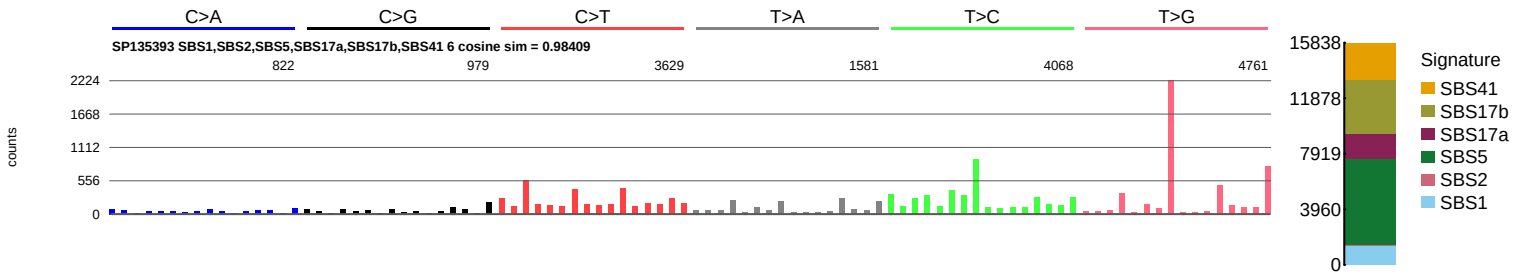
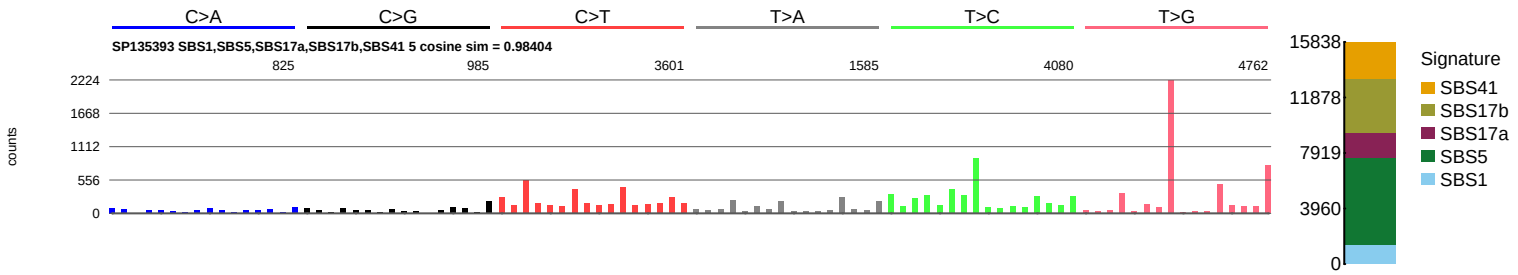
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS21,SBS41 6 cosine sim = 0.98412
818
970
3570
1570
4160
4751
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS41 6 cosine sim = 0.98409
822
979
3629
1581
4068
4761
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS2
SBS1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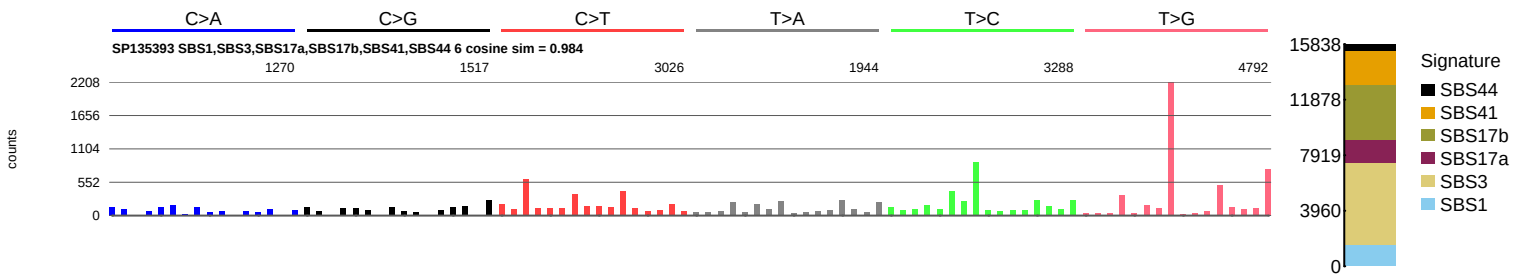
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98404
825
985
3601
1585
4080
4762
counts
2224
1668
1112
556
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS41,SBS44 6 cosine sim = 0.984
1270
1517
3026
1944
3288
4792
counts
2208
1656
1104
552
0
15838
11878
7919
3960
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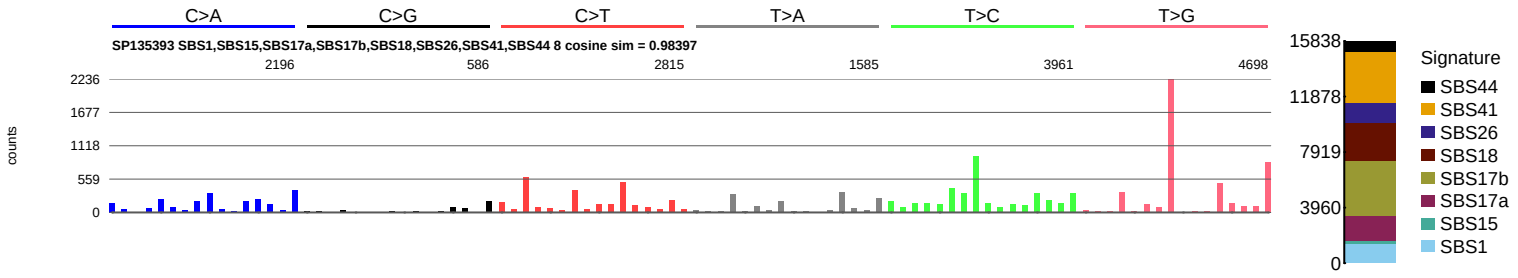
SBS44
SBS41
SBS17b
SBS17a
SBS3
SBS1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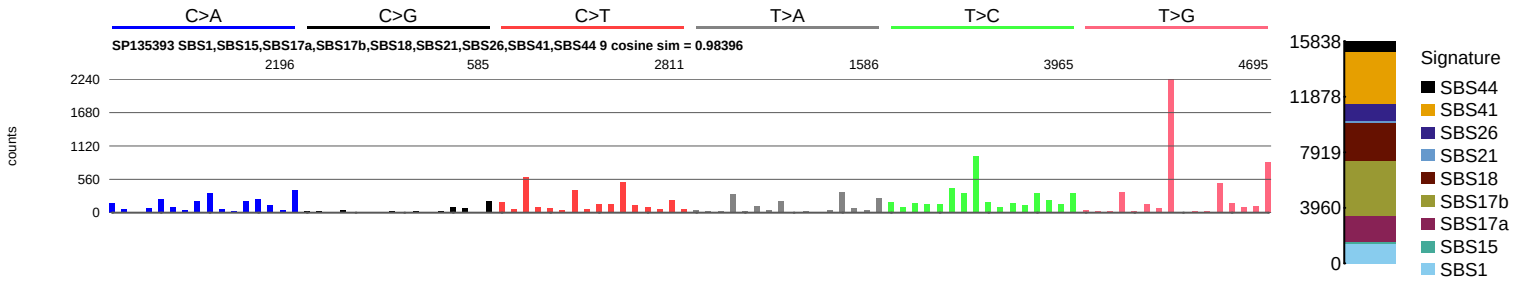
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 8 cosine sim = 0.98397
2196
586
2815
1585
3961
4698
counts
2236
1677
1118
559
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 9 cosine sim = 0.98396
2196
585
2811
1586
3965
4695
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
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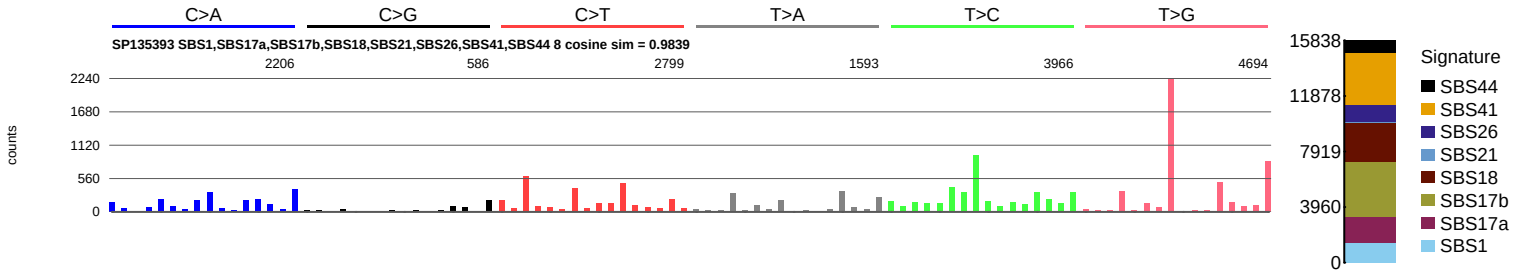
SBS44
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41,SBS44 8 cosine sim = 0.9839
2206
586
2799
1593
3966
4694
counts
2240
1680
1120
560
0
15838
11878
7919
3960
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SBS26
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SBS18
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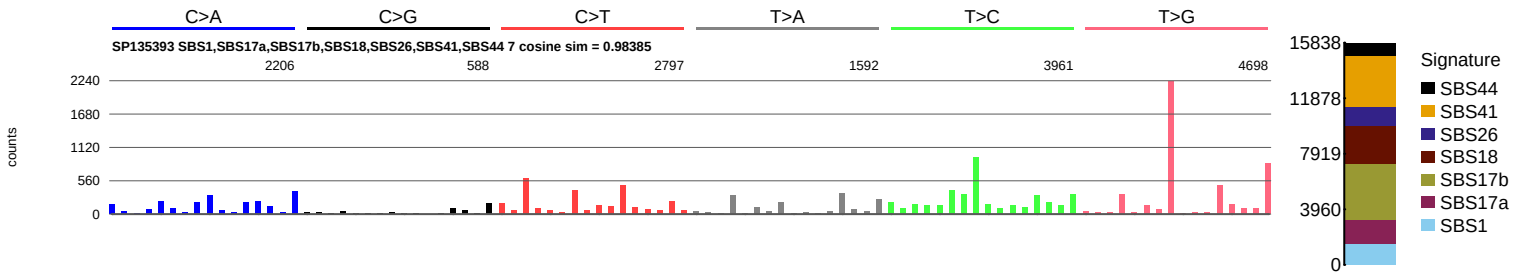
SBS17a
SBS1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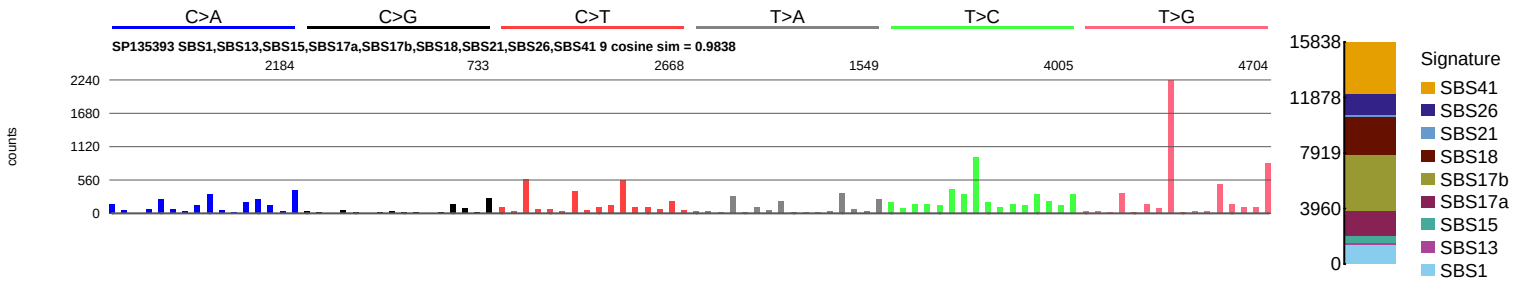
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41,SBS44 7 cosine sim = 0.98385
2206
588
2797
1592
3961
4698
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1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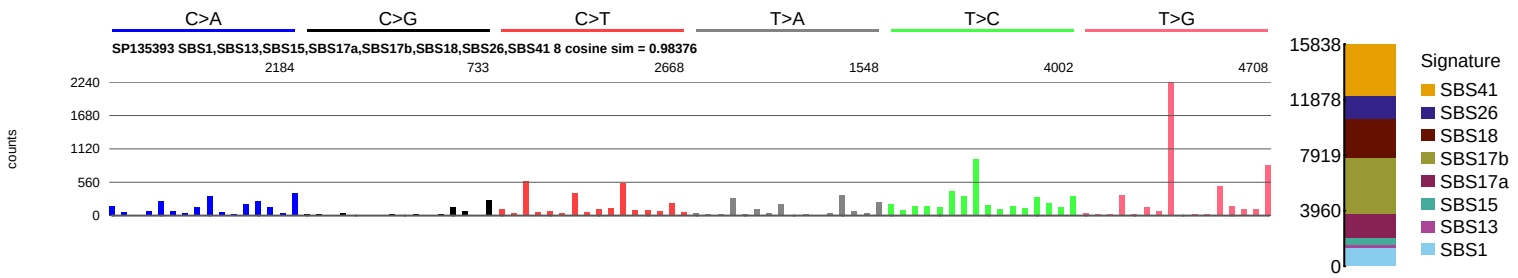
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 9 cosine sim = 0.9838
2184
733
2668
1549
4005
4704
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
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T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 8 cosine sim = 0.98376
2184
733
2668
1548
4002
4708
counts
2240
1680
1120
560
0
15838
11878
7919
3960
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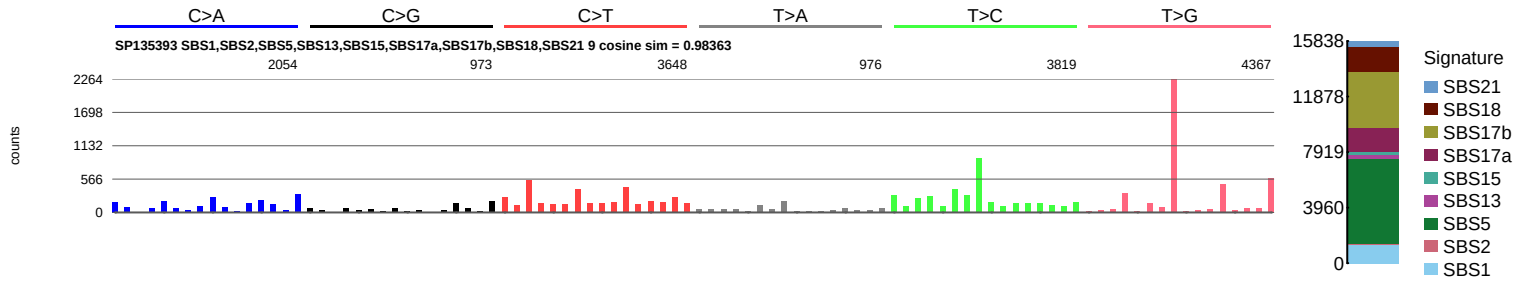
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98363
2054
973
3648
976
3819
4367
counts
2264
1698
1132
566
0
15838
11878
7919
3960
0
Signature
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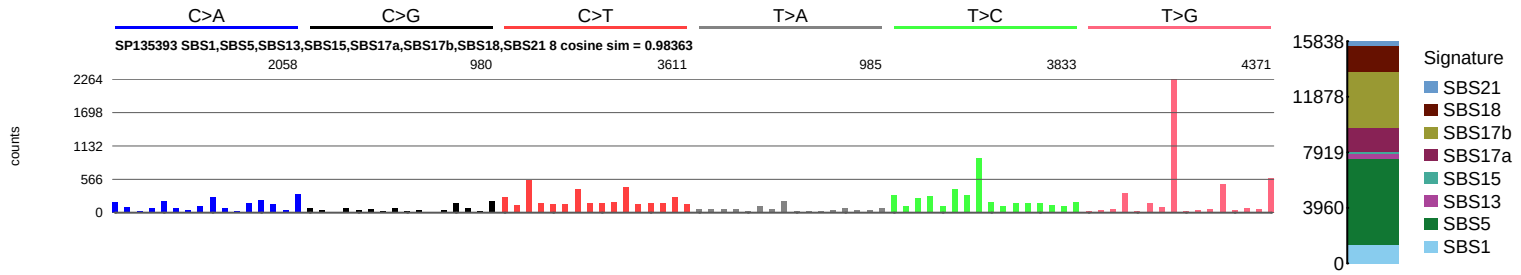
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98363
2058
980
3611
985
3833
4371
counts
2264
1698
1132
566
0
15838
11878
7919
3960
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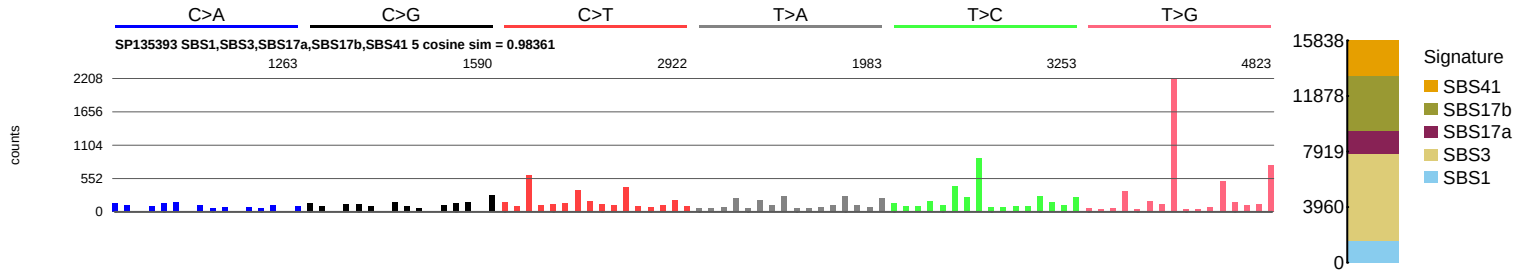
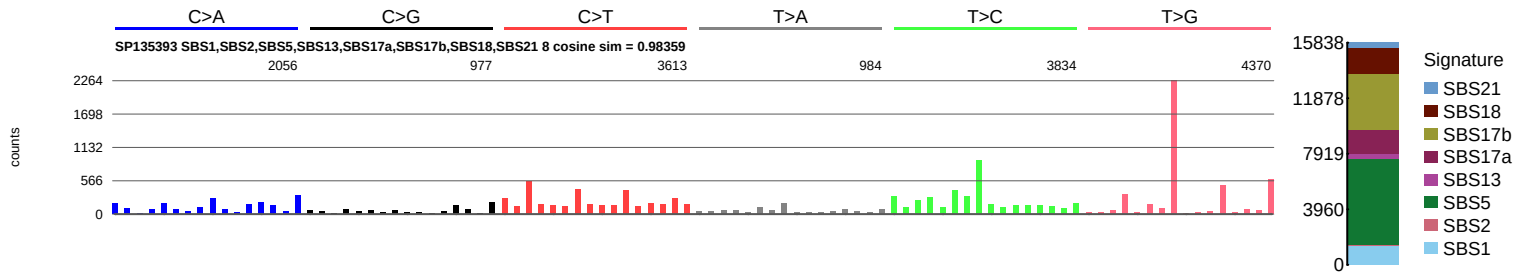
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS41 5 cosine sim = 0.98361
1263
1590
2922
1983
3253
4823
counts
2208
1656
1104
552
0
15838
11878
7919
3960
0
Signature
SBS41
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98359
2056
977
3613
984
3834
4370
counts
2264
1698
1132
566
0
15838
11878
7919
3960
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Signature
SBS21
SBS18
SBS17b
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SBS13
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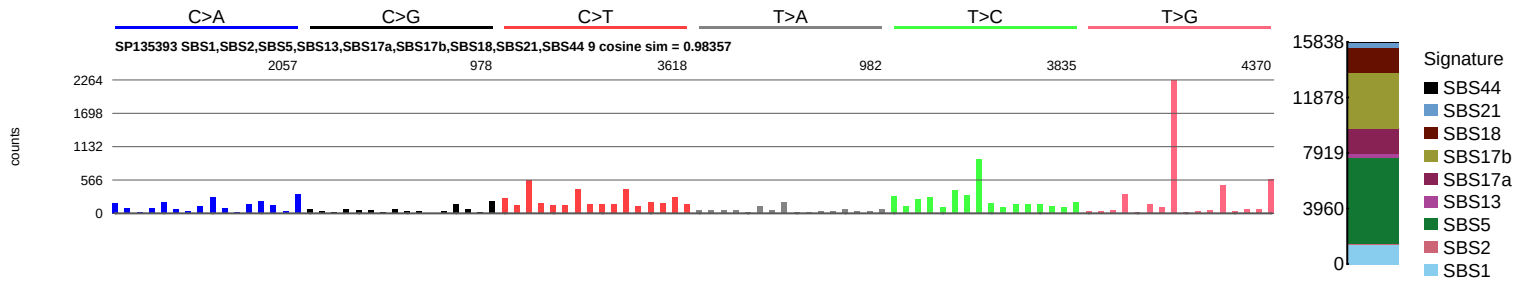
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98357
2057
978
3618
982
3835
4370
counts
2264
1698
1132
566
0
15838
11878
7919
3960
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
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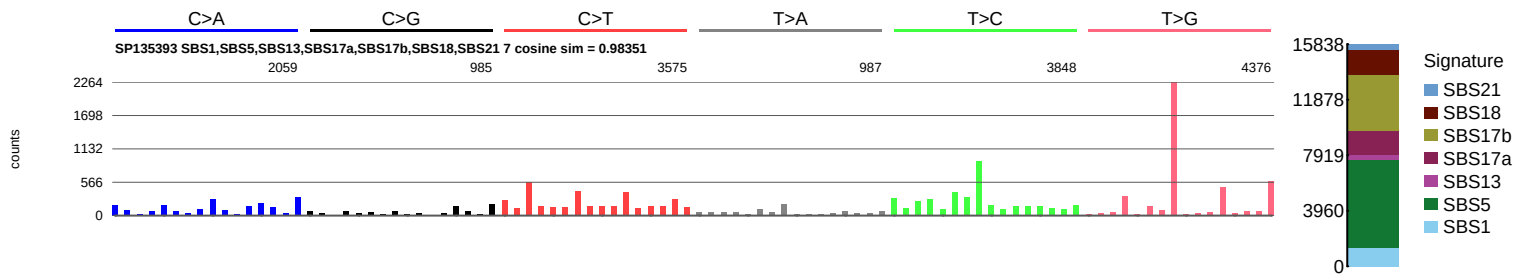
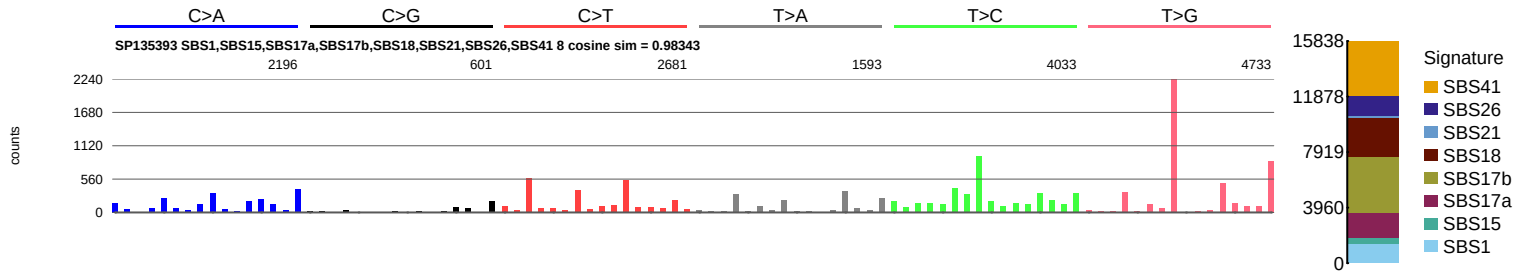
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98351
2059
985
3575
987
3848
4376
counts
2264
1698
1132
566
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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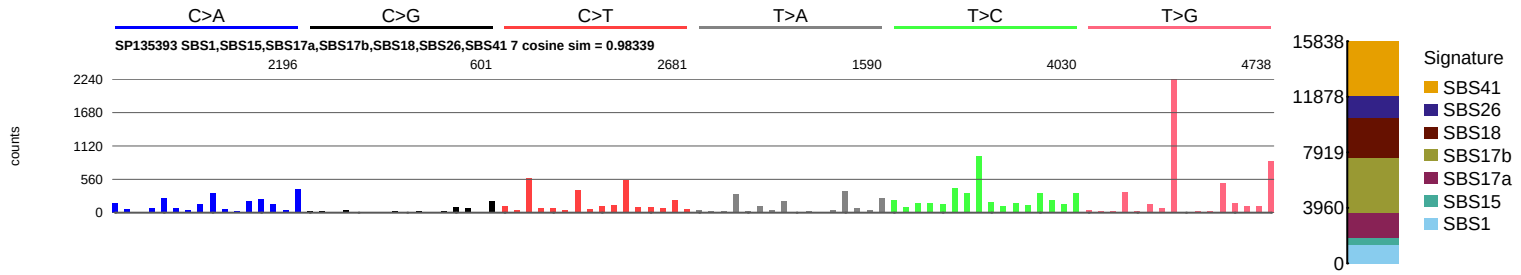
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98343
2196
601
2681
1593
4033
4733
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
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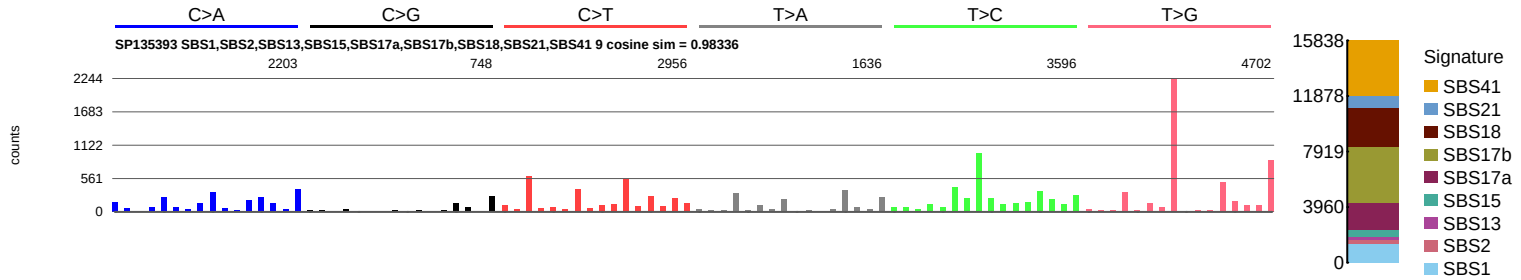
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98339
2196
601
2681
1590
4030
4738
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>T
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T>C
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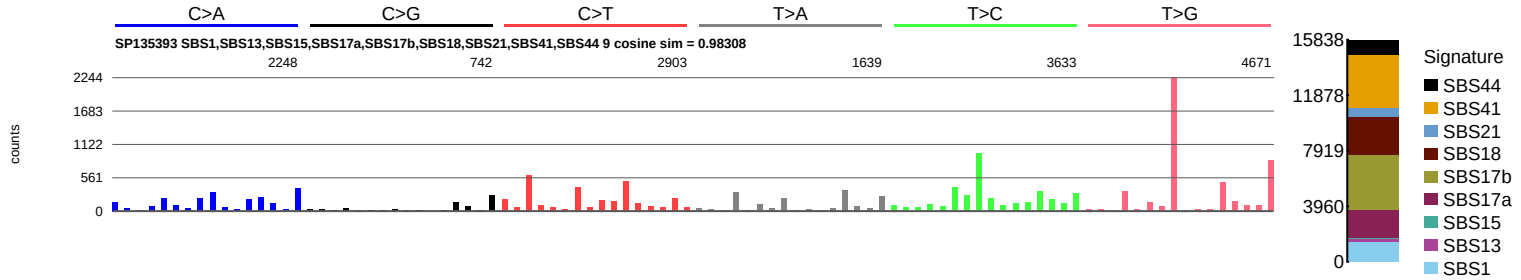
SP135393 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 9 cosine sim = 0.98336
2203
748
2956
1636
3596
4702
counts
2244
1683
1122
561
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS2
SBS1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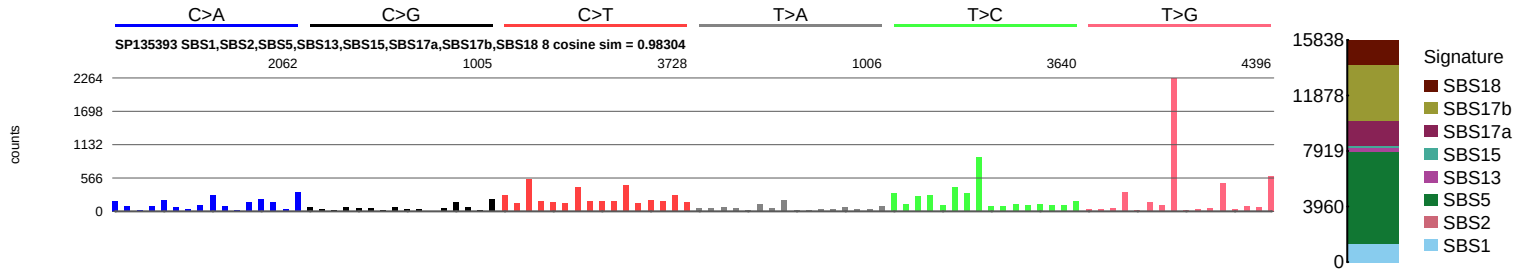
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 9 cosine sim = 0.98308
2248
742
2903
1639
3633
4671
counts
2244
1683
1122
561
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1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C>G
C>T
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T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 8 cosine sim = 0.98304
2062
1005
3728
1006
3640
4396
counts
2264
1698
1132
566
0
15838
11878
7919
3960
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Signature
SBS18
SBS17b
SBS17a
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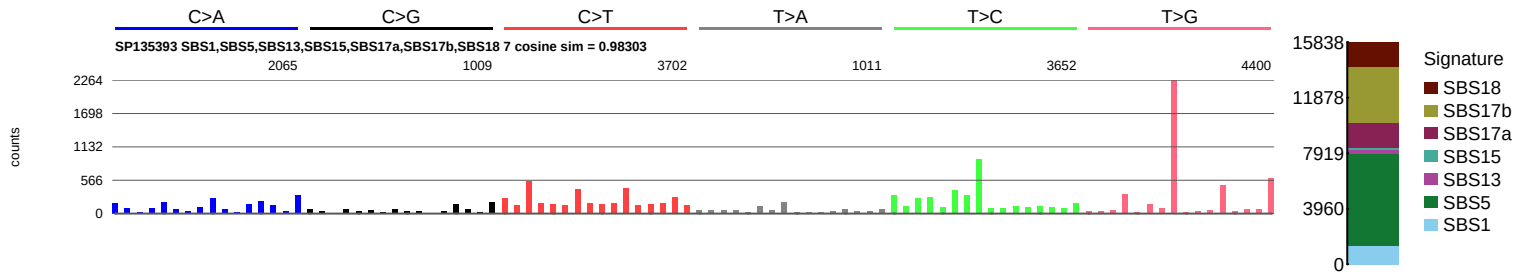
SBS2
SBS1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C>G
C>T
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T>C
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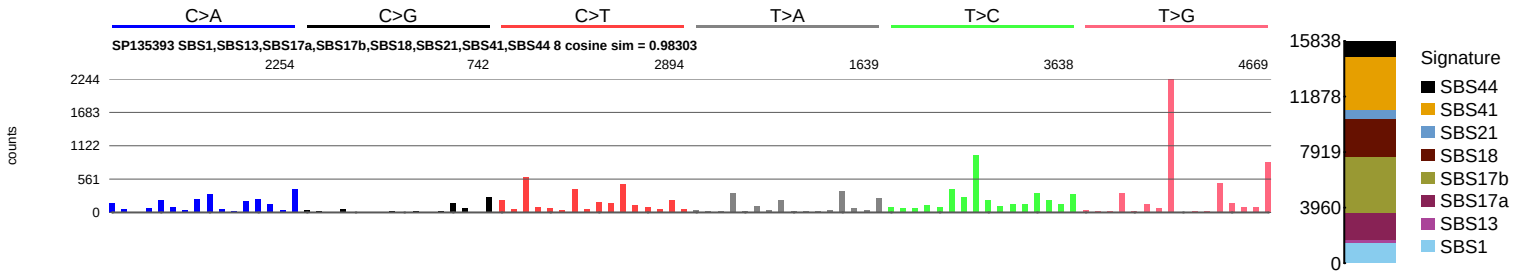
SP135393 SBS1,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98303
2065
1009
3702
1011
3652
4400
counts
2264
1698
1132
566
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS1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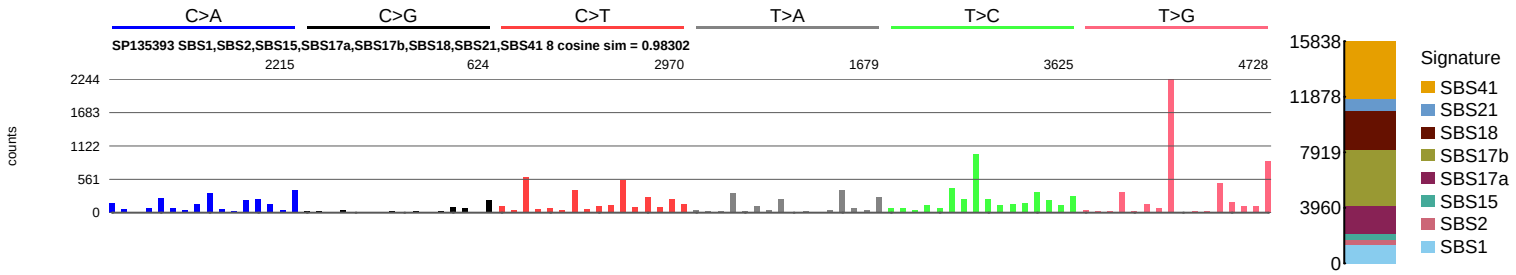
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98303
2254
742
2894
1639
3638
4669
counts
2244
1683
1122
561
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1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C>G
C>T
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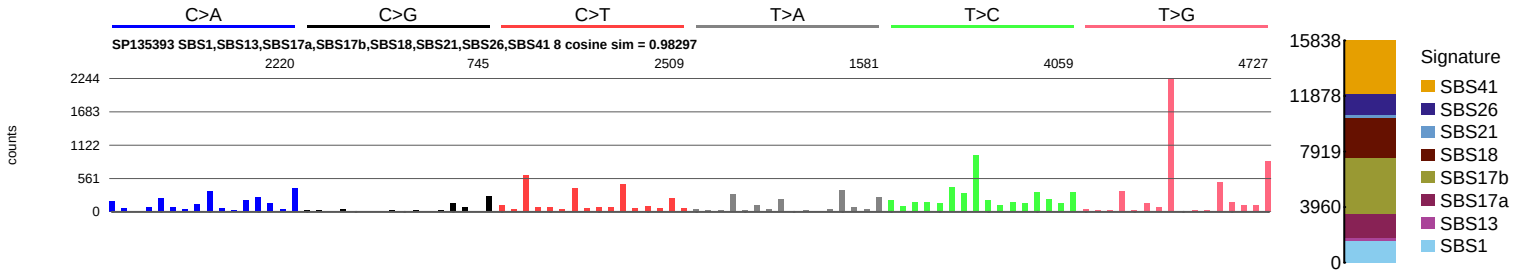
SP135393 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98302
2215
624
2970
1679
3625
4728
counts
2244
1683
1122
561
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1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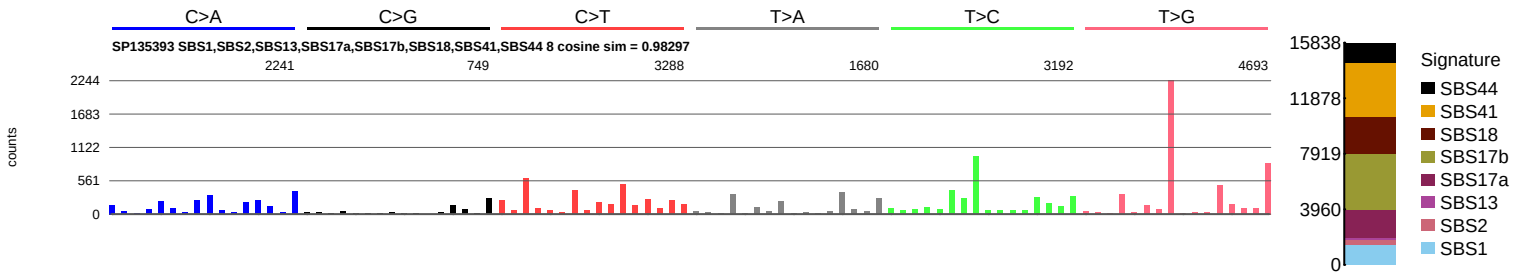
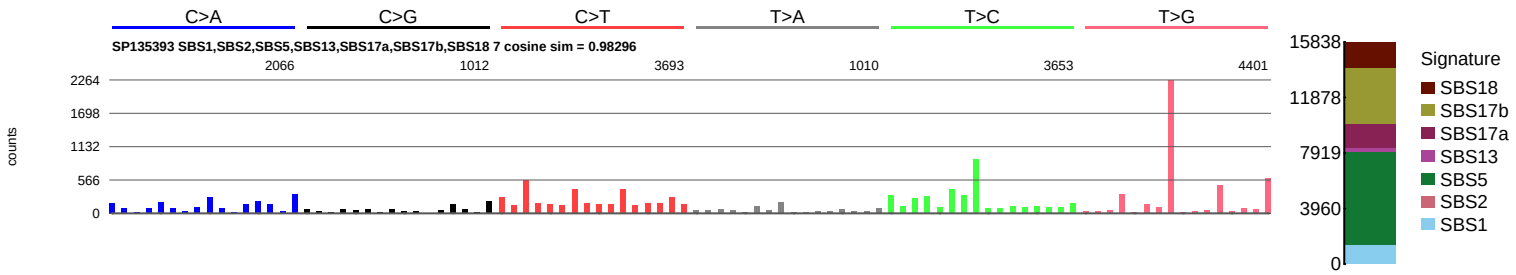
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 8 cosine sim = 0.98297
2220
745
2509
1581
4059
4727
counts
2244
1683
1122
561
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98297
2241
749
3288
1680
3192
4693
counts
2244
1683
1122
561
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1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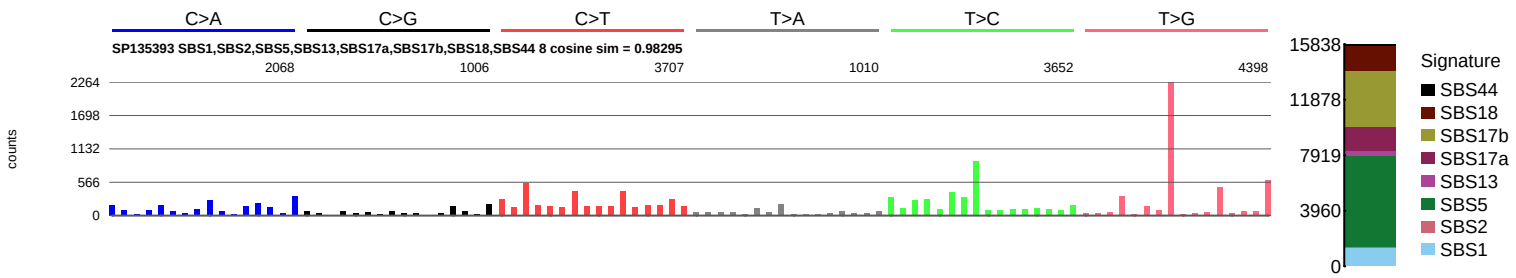
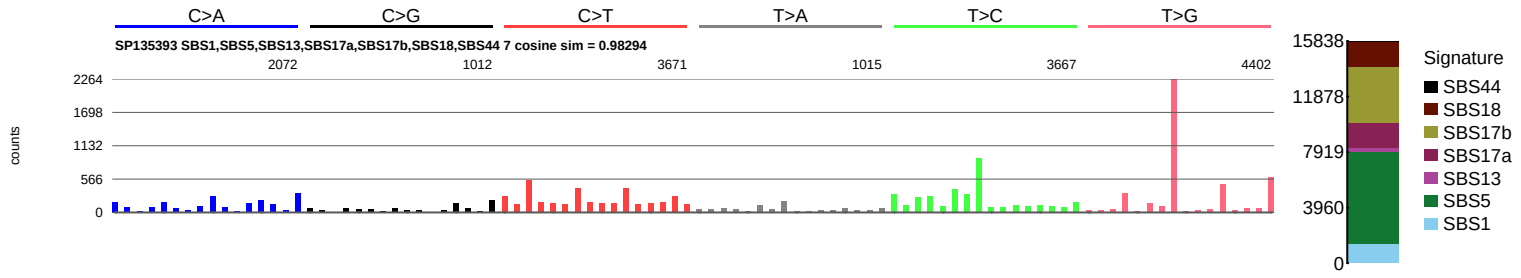
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98296
2066
1012
3693
1010
3653
4401
counts
2264
1698
1132
566
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98295
2068
1006
3707
1010
3652
4398
counts
2264
1698
1132
566
0
15838
11878
7919
3960
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS2
SBS1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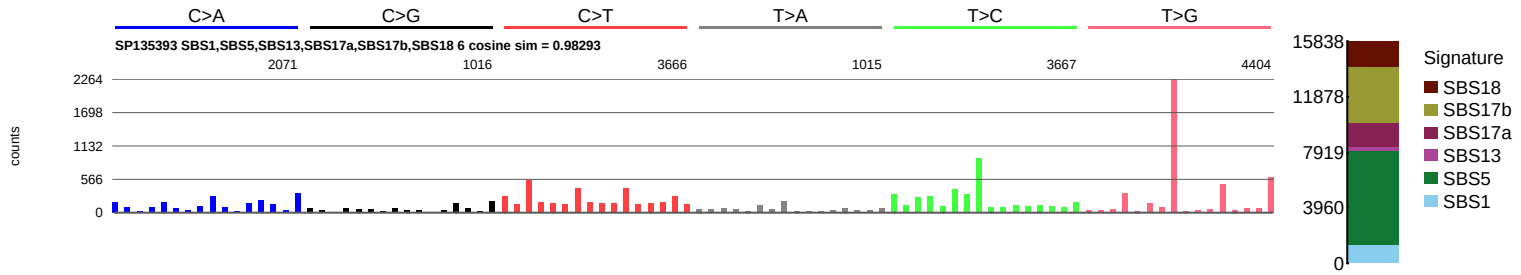
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98294
2072
1012
3671
1015
3667
4402
counts
2264
1698
1132
566
0
15838
11878
7919
3960
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1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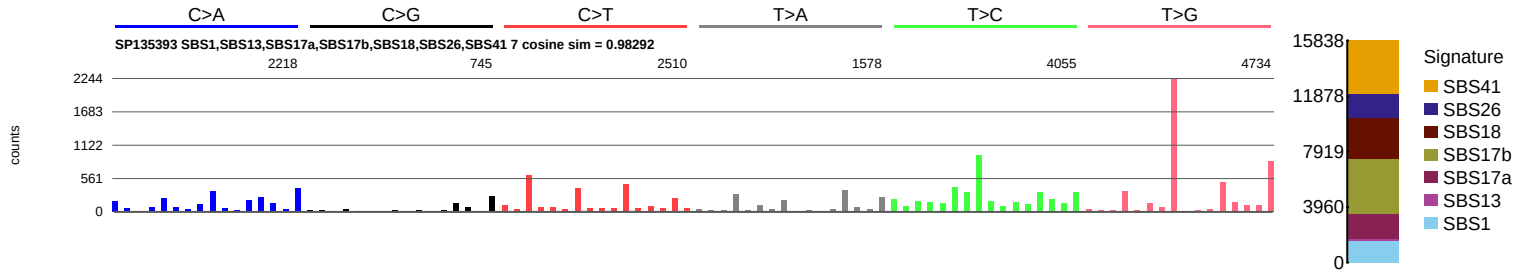
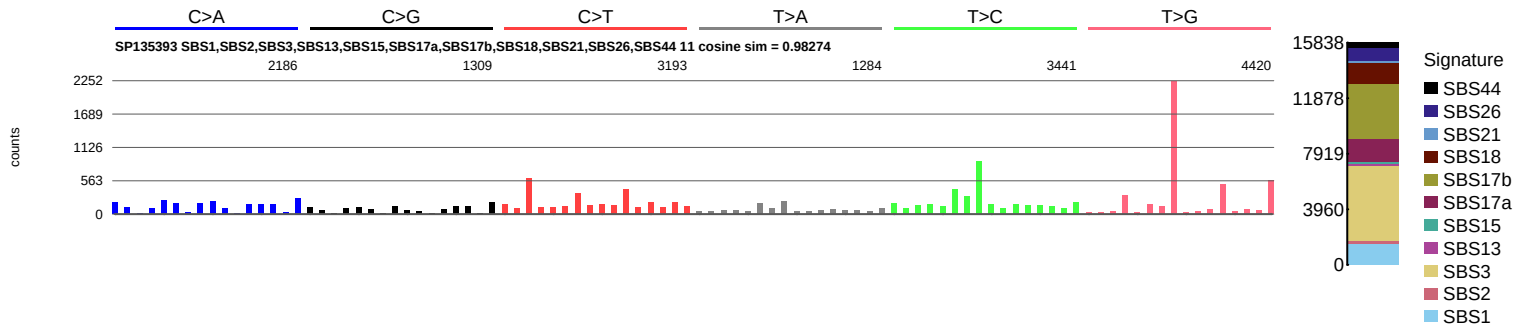
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS13,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98293
2071
1016
3666
1015
3667
4404
counts
2264
1698
1132
566
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS13
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS41 7 cosine sim = 0.98292
2218
745
2510
1578
4055
4734
counts
2244
1683
1122
561
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS1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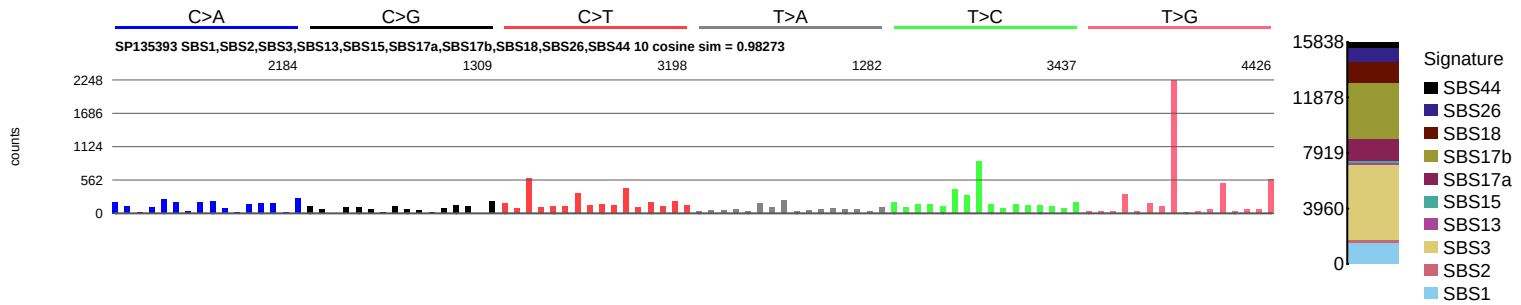
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 11 cosine sim = 0.98274
2186
1309
3193
1284
3441
4420
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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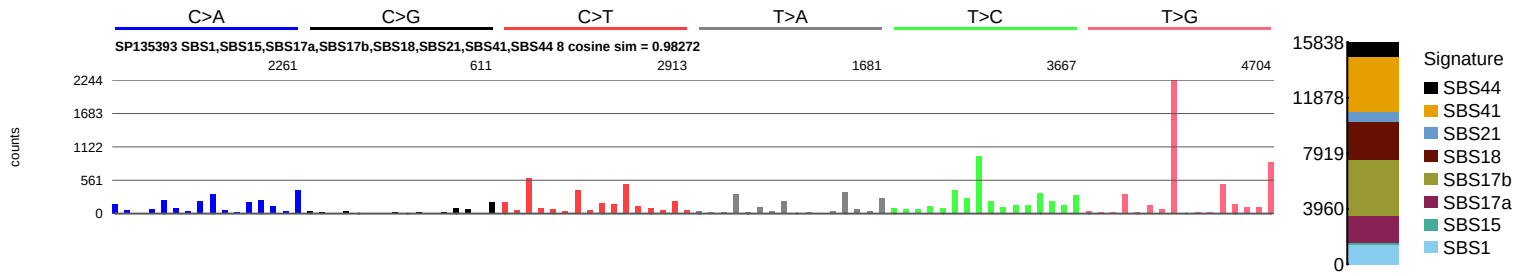
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 10 cosine sim = 0.98273
2184
1309
3198
1282
3437
4426
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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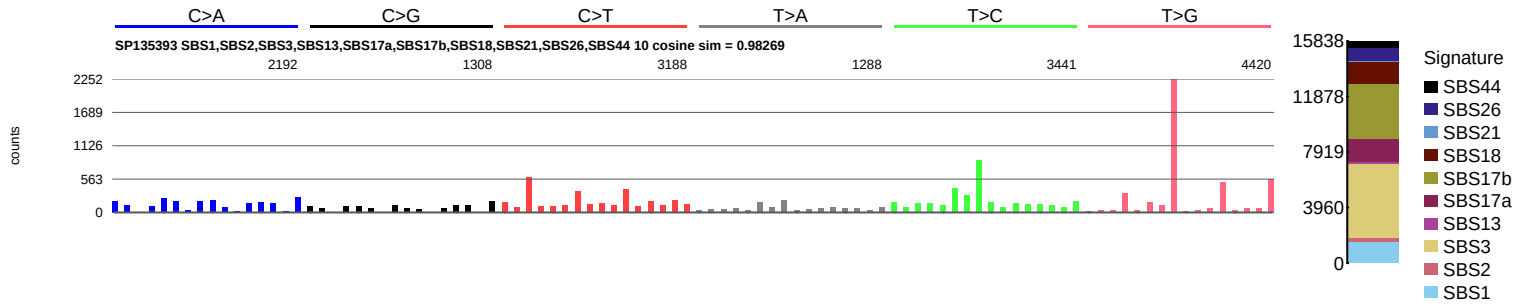
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 8 cosine sim = 0.98272
2261
611
2913
1681
3667
4704
counts
2244
1683
1122
561
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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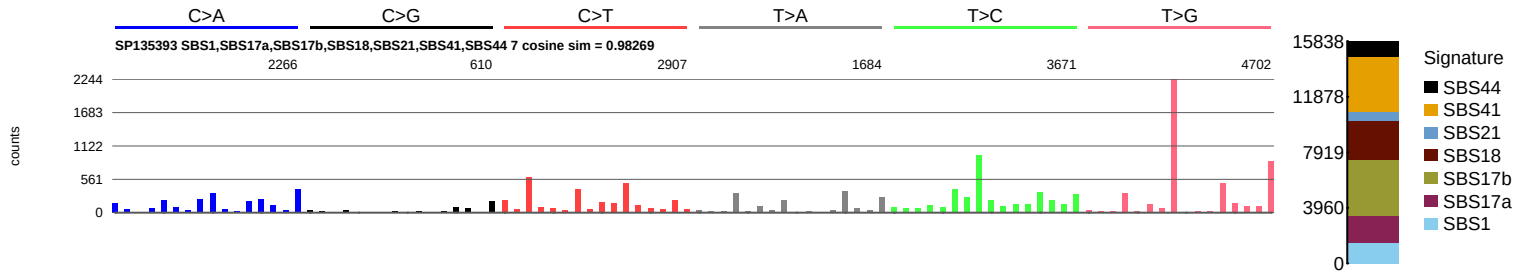
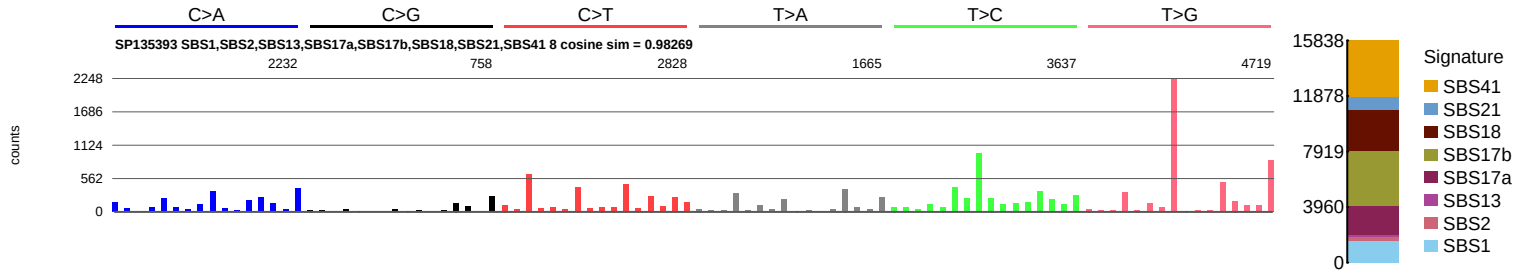
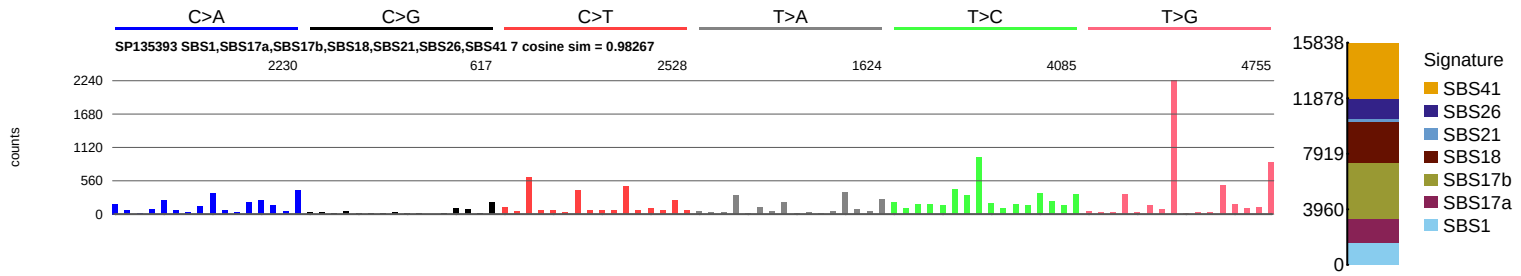
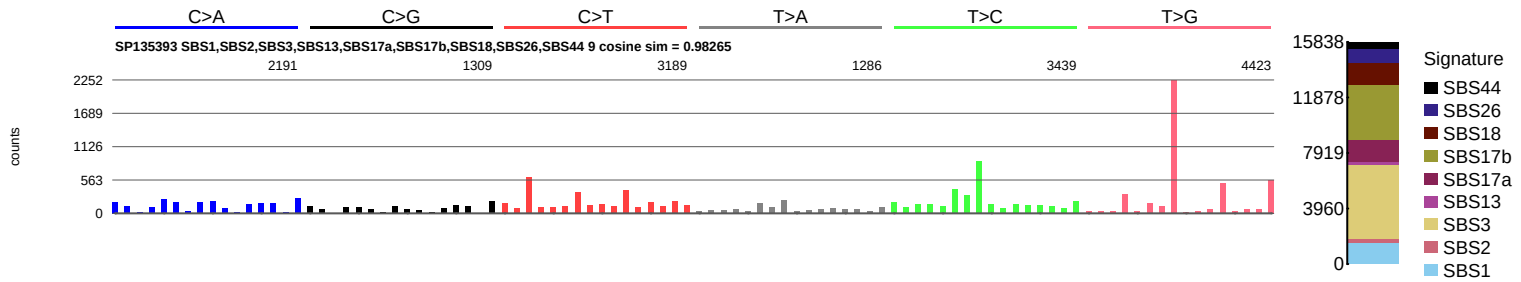
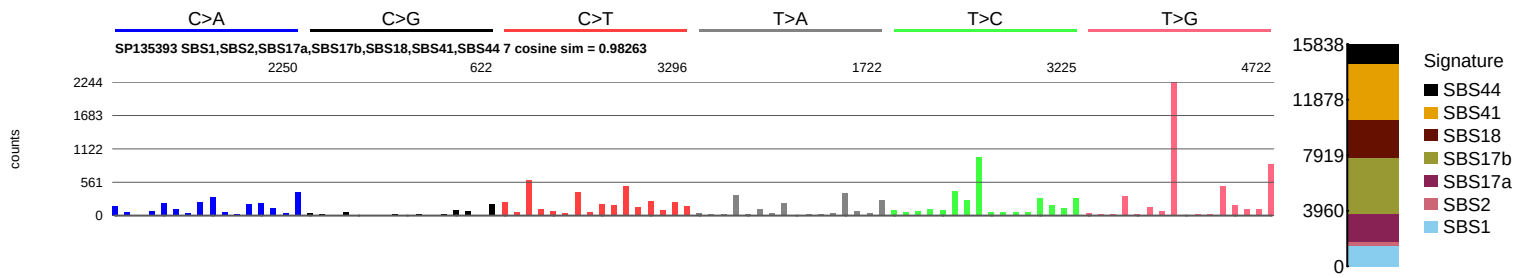
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98269
2192
1308
3188
1288
3441
4420
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1

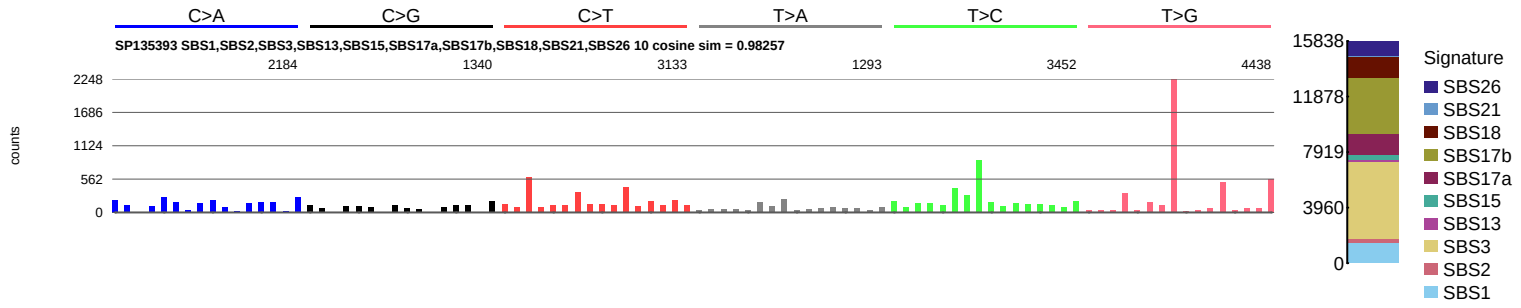

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41,SBS44 7 cosine sim = 0.98269
2266
610
2907
1684
3671
4702
counts
2244
1683
1122
561
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS1

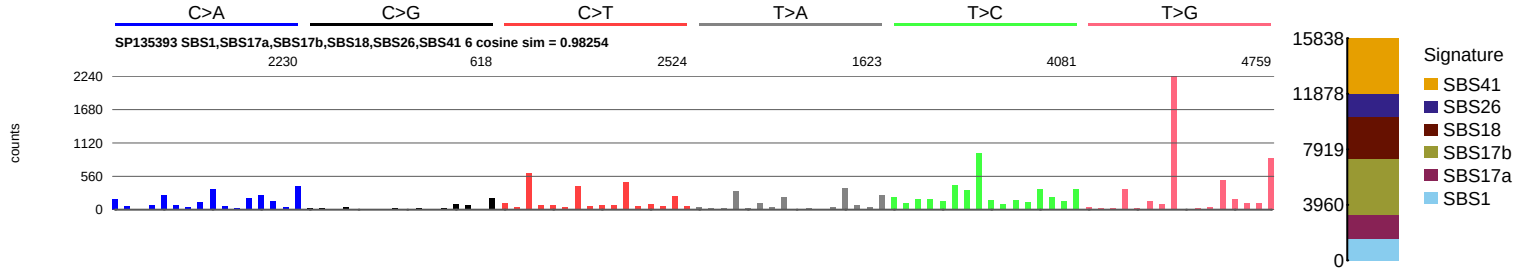

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98269
2232
758
2828
1665
3637
4719
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS41 7 cosine sim = 0.98267
2230
617
2528
1624
4085
4755
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98265
2191
1309
3189
1286
3439
4423
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98263
2250
622
3296
1722
3225
4722
counts
2244
1683
1122
561
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 10 cosine sim = 0.98257
2184
1340
3133
1293
3452
4438
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
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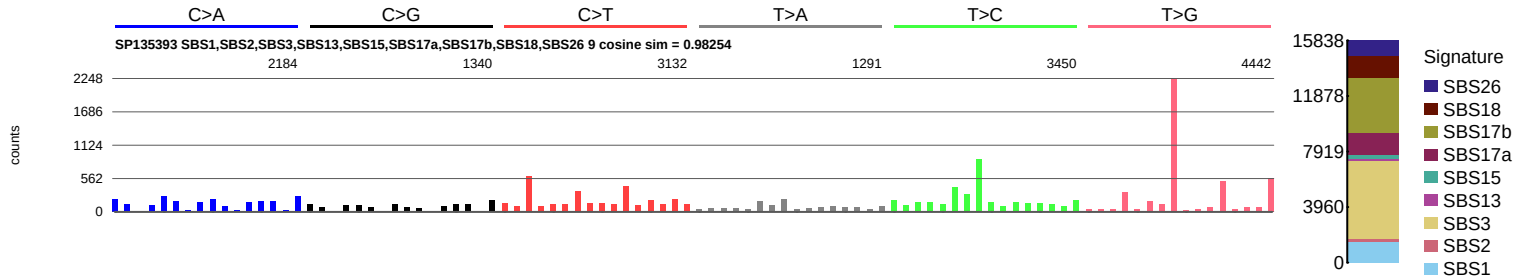
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS26,SBS41 6 cosine sim = 0.98254
2230
618
2524
1623
4081
4759
counts
2240
1680
1120
560
0
15838
11878
7919
3960
0
Signature
SBS41
SBS26
SBS18
SBS17b
SBS17a
SBS1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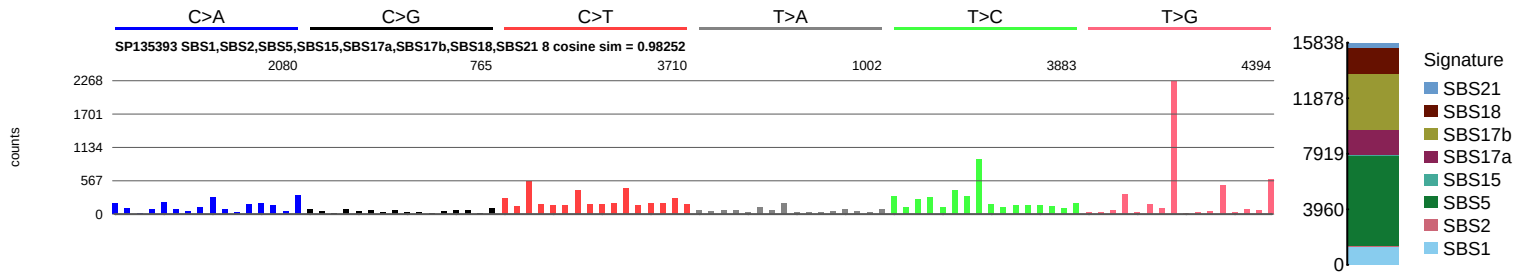
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 9 cosine sim = 0.98254
2184
1340
3132
1291
3450
4442
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98252
2080
765
3710
1002
3883
4394
counts
2268
1701
1134
567
0
15838
11878
7919
3960
0
Signature
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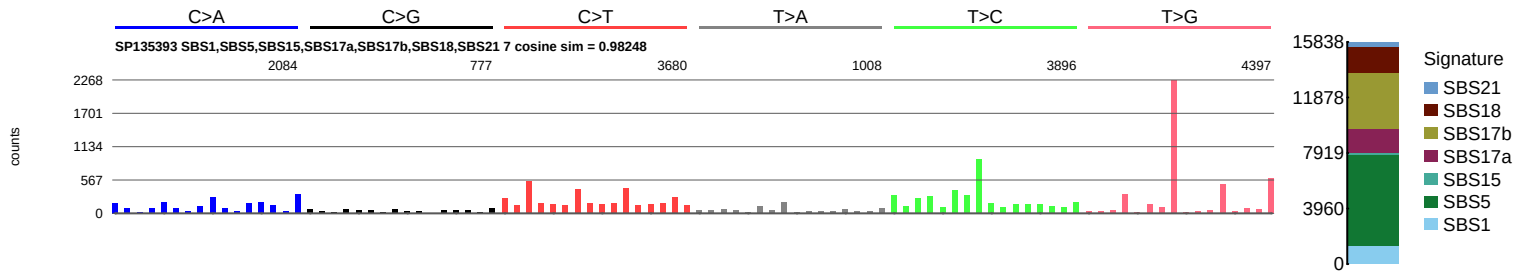
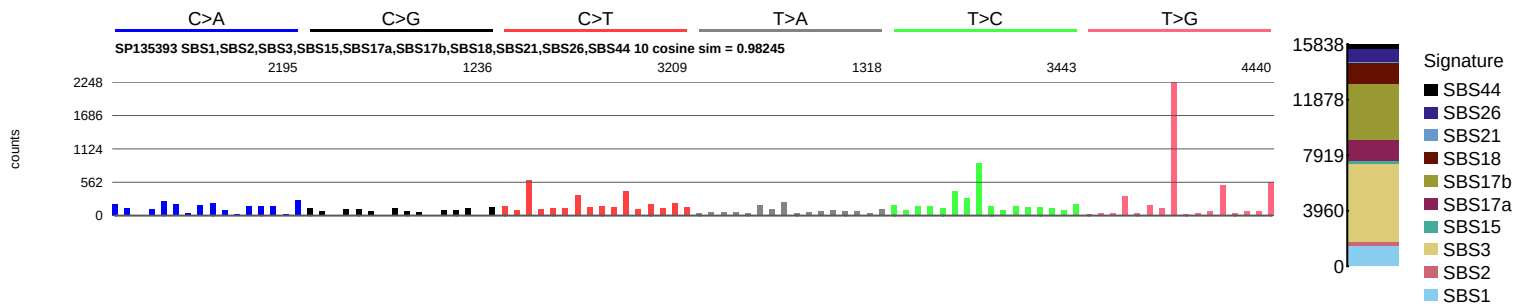
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98248
2084
777
3680
1008
3896
4397
counts
2268
1701
1134
567
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1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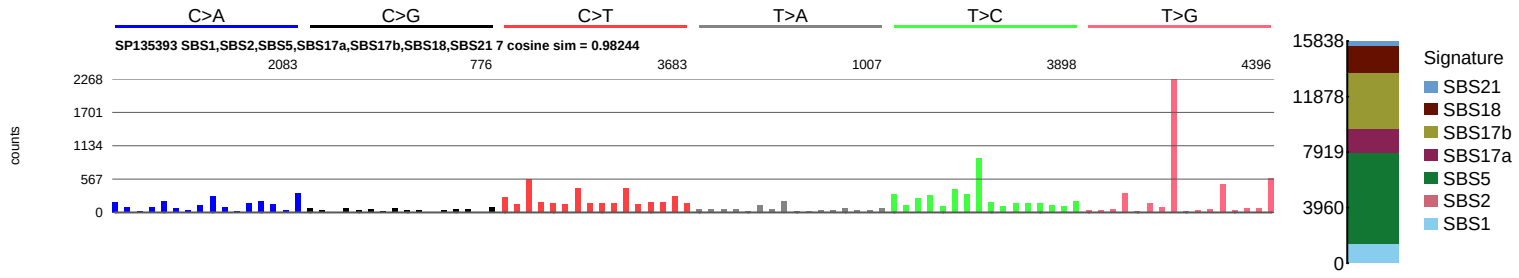
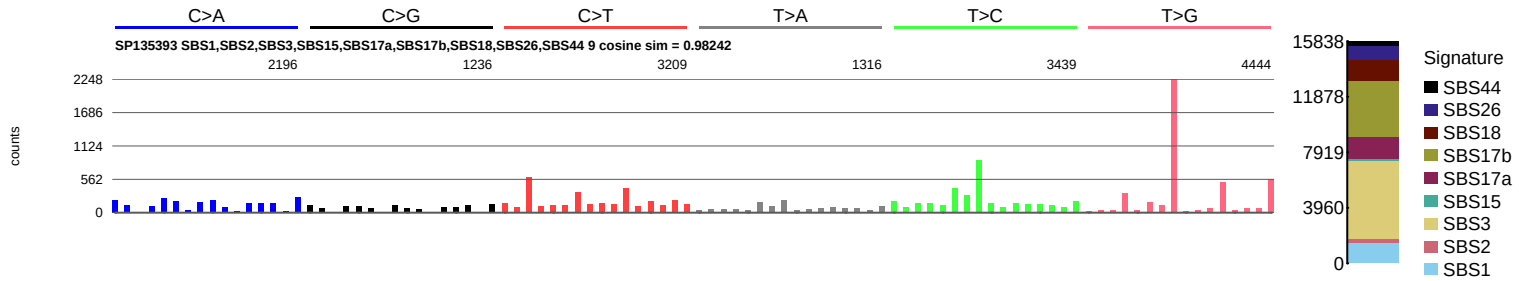
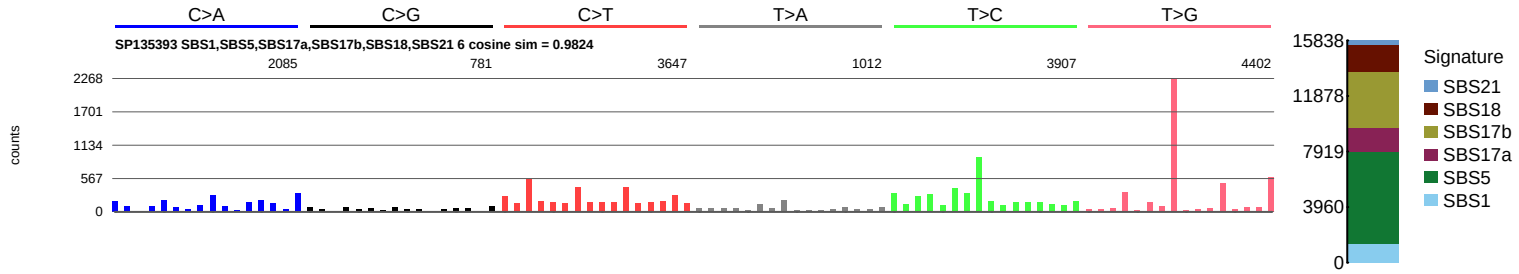
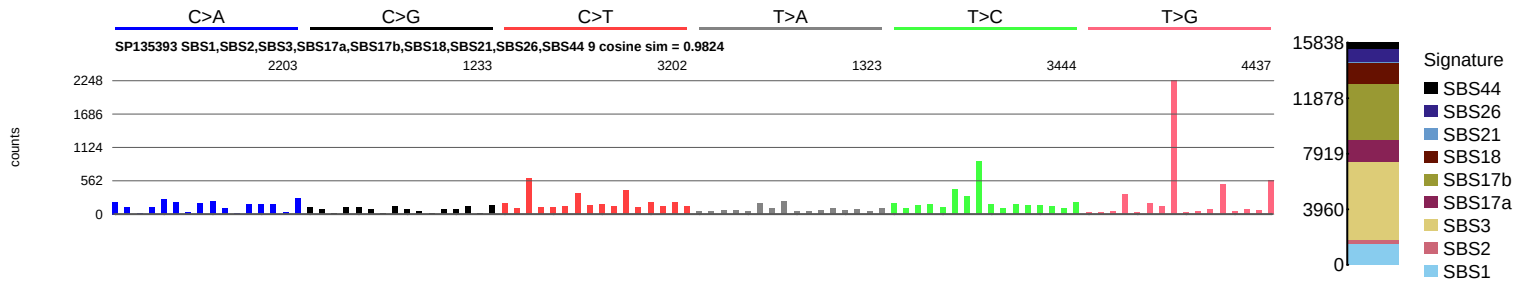
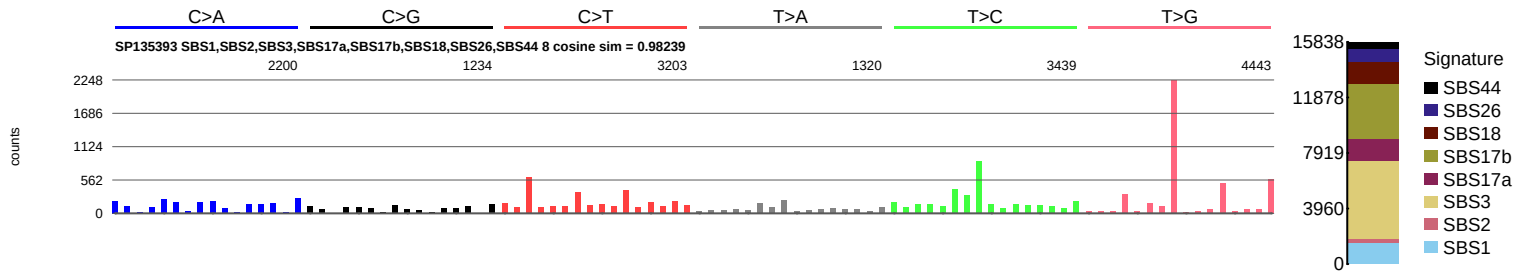
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98245
2195
1236
3209
1318
3443
4440
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

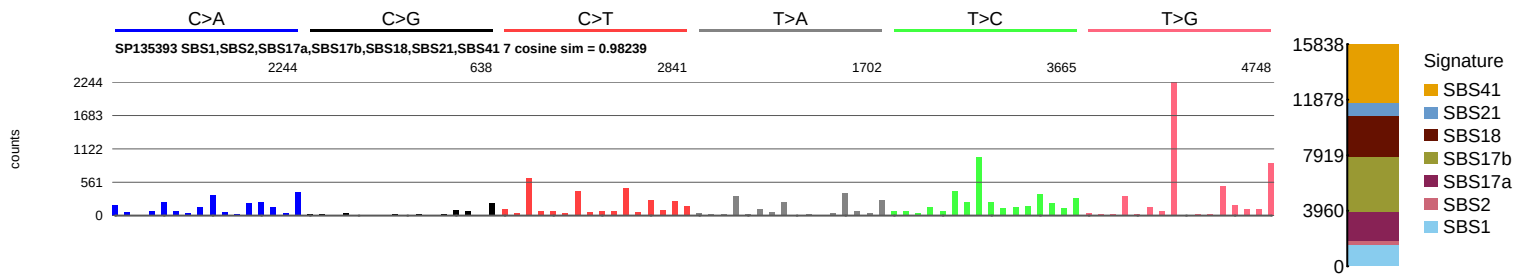

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98244
2083
776
3683
1007
3898
4396
counts
2268
1701
1134
567
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1

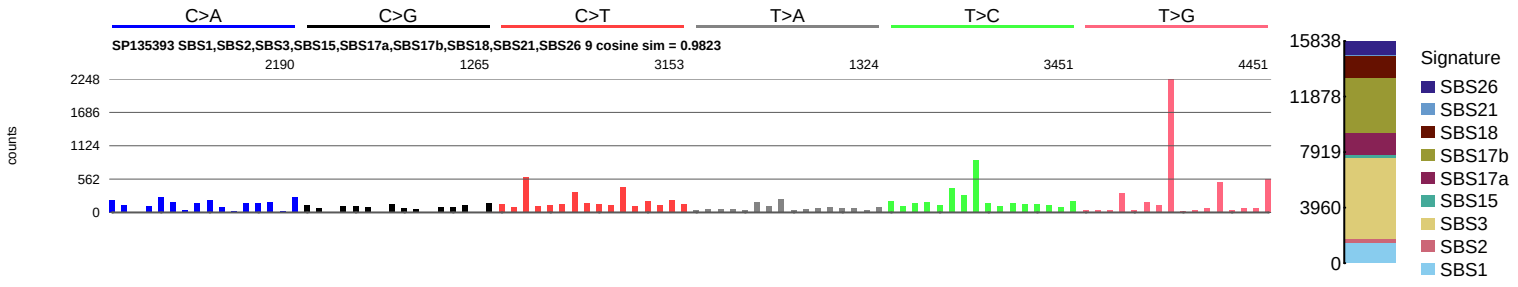

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98242
2196
1236
3209
1316
3439
4444
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

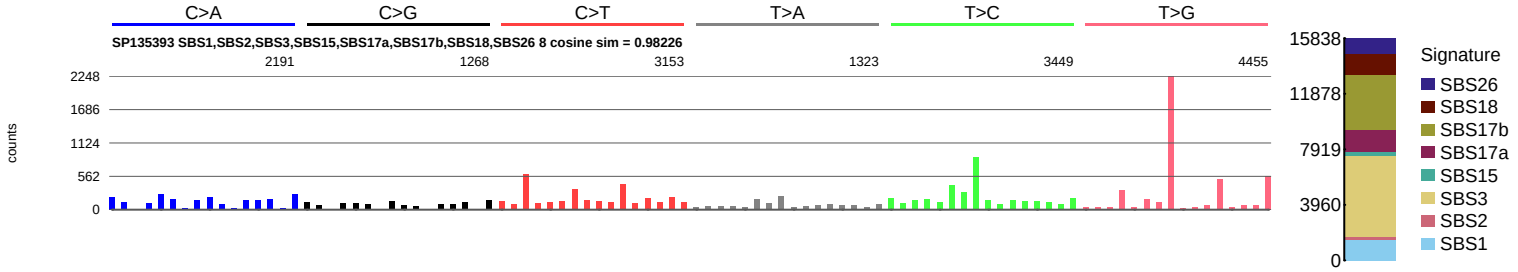

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS18,SBS21 6 cosine sim = 0.9824
2085
781
3647
1012
3907
4402
counts
2268
1701
1134
567
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS5
SBS1

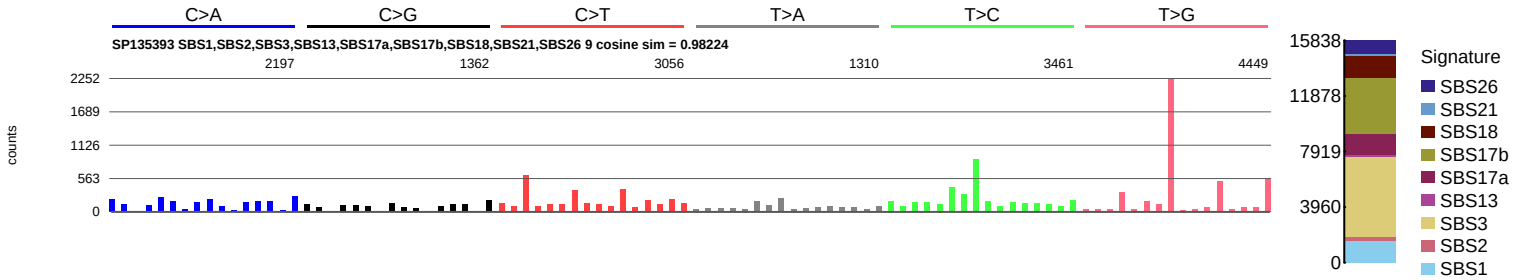

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.9824
2203
1233
3202
1323
3444
4437
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

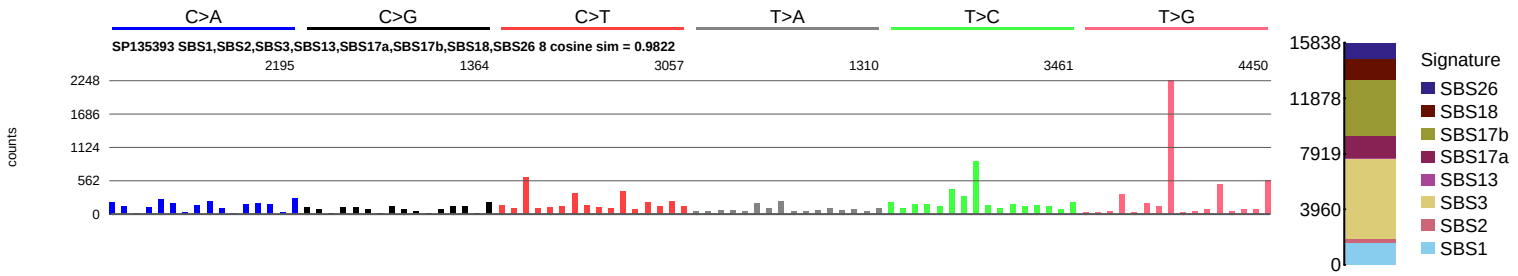

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98239
2200
1234
3203
1320
3439
4443
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1

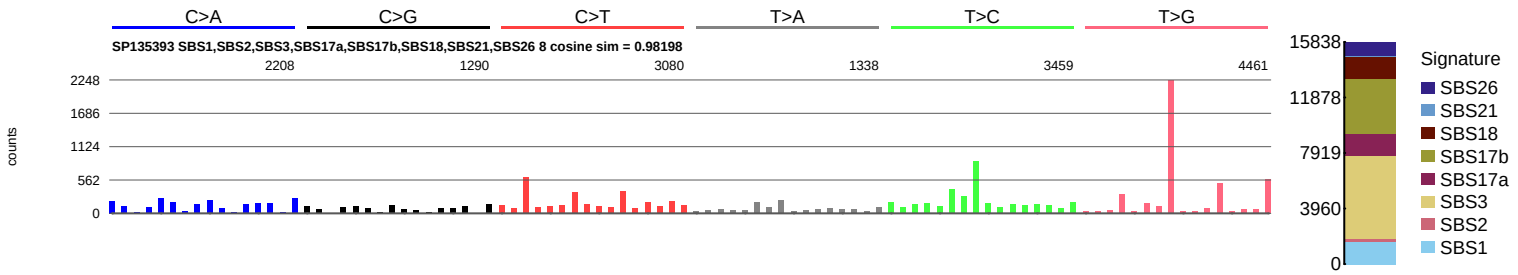

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98239
2244
638
2841
1702
3665
4748
counts
2244
1683
1122
561
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.9823
2190
1265
3153
1324
3451
4451
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98226
2191
1268
3153
1323
3449
4455
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98224
2197
1362
3056
1310
3461
4449
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.9822
2195
1364
3057
1310
3461
4450
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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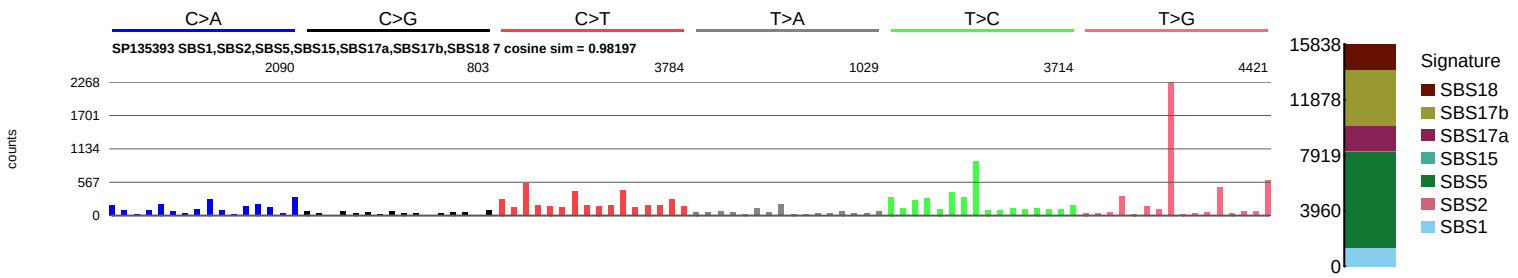
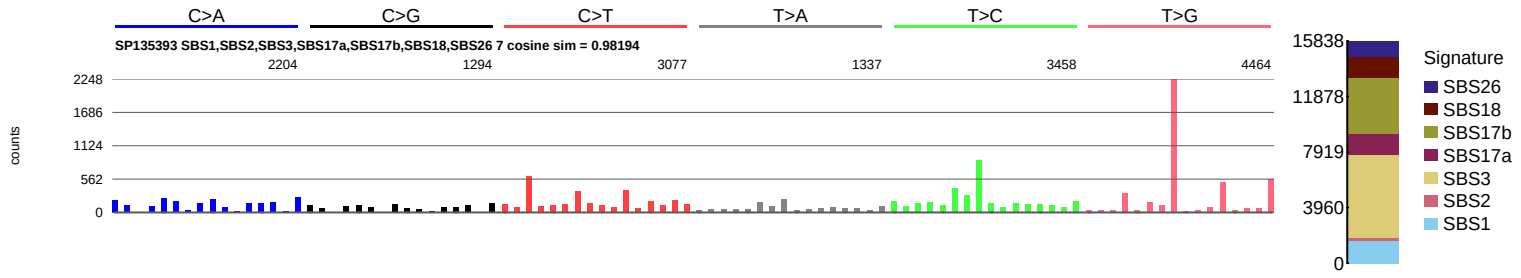
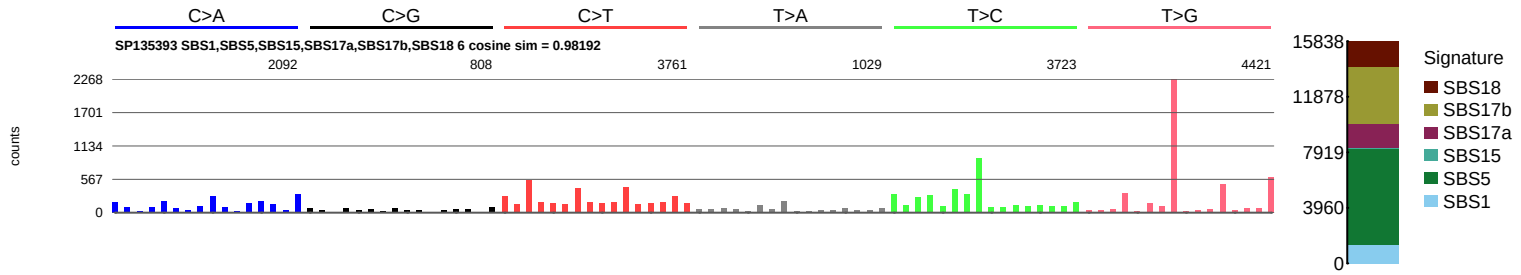
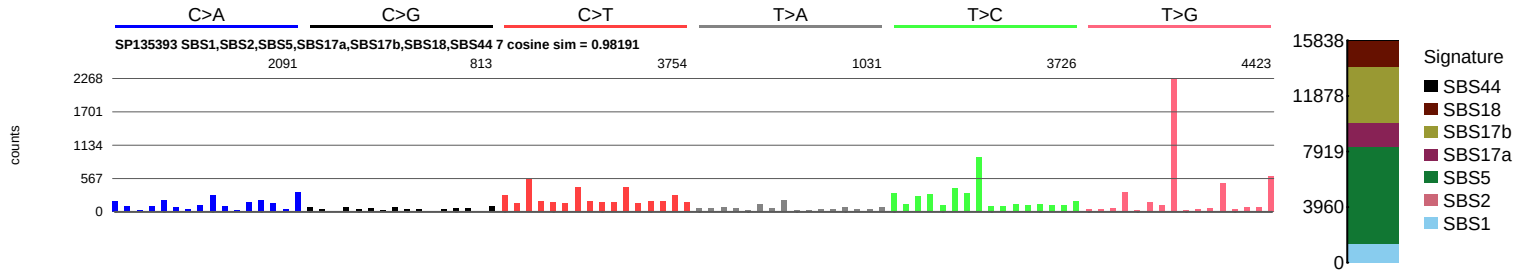
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98198
2208
1290
3080
1338
3459
4461
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS15,SBS17a,SBS17b,SBS18 7 cosine sim = 0.98197
2090
803
3784
1029
3714
4421
counts
2268
1701
1134
567
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98194
2204
1294
3077
1337
3458
4464
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS5,SBS15,SBS17a,SBS17b,SBS18 6 cosine sim = 0.98192
2092
808
3761
1029
3723
4421
counts
2268
1701
1134
567
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS15
SBS5
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98191
2091
813
3754
1031
3726
4423
counts
2268
1701
1134
567
0
15838
11878
7919
3960
0
Signature
SBS44
SBS18
SBS17b
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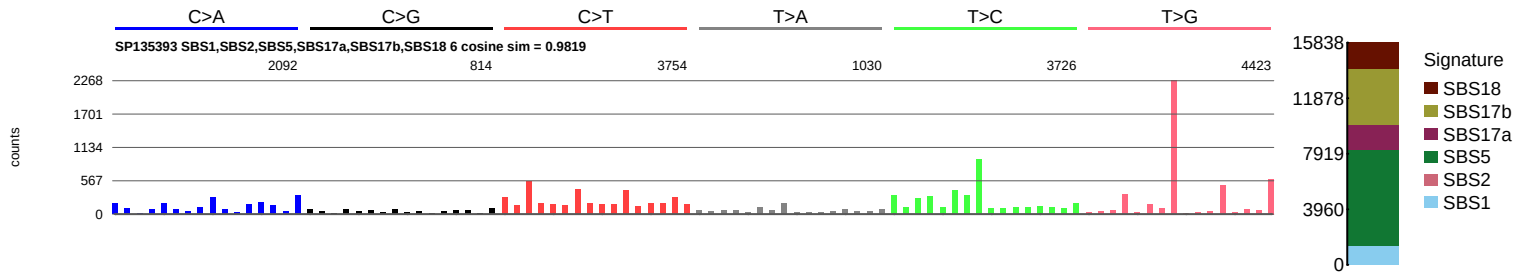
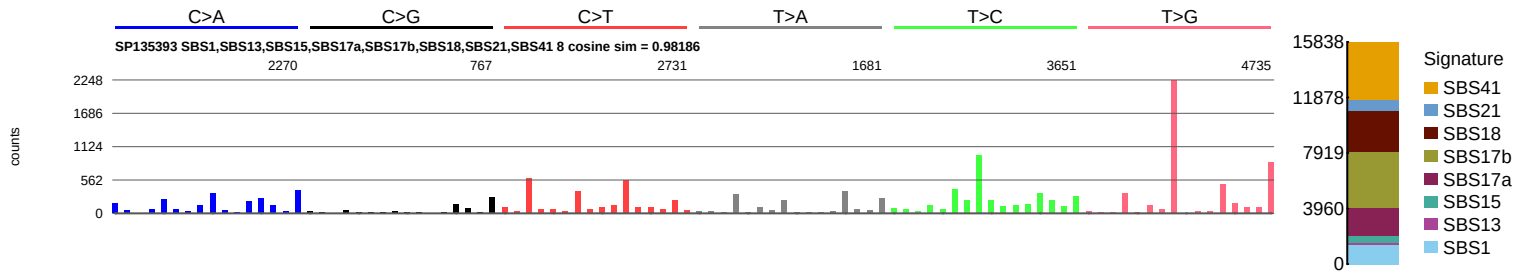
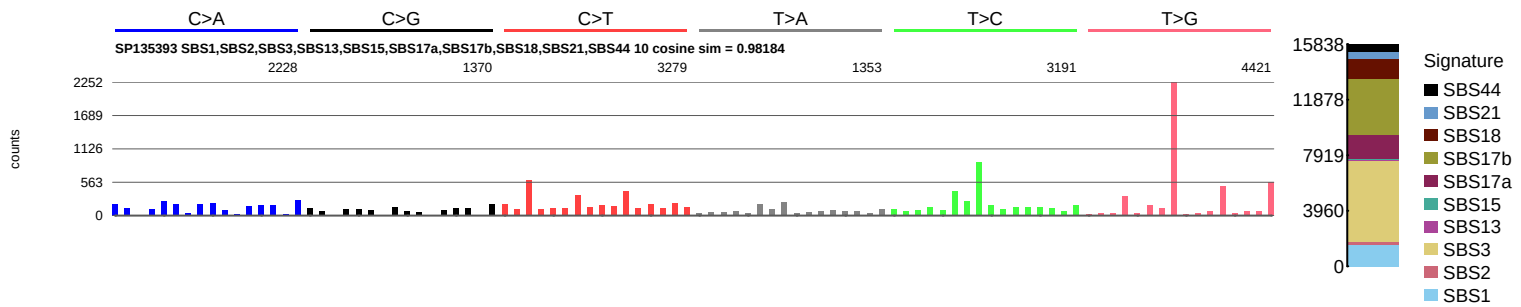
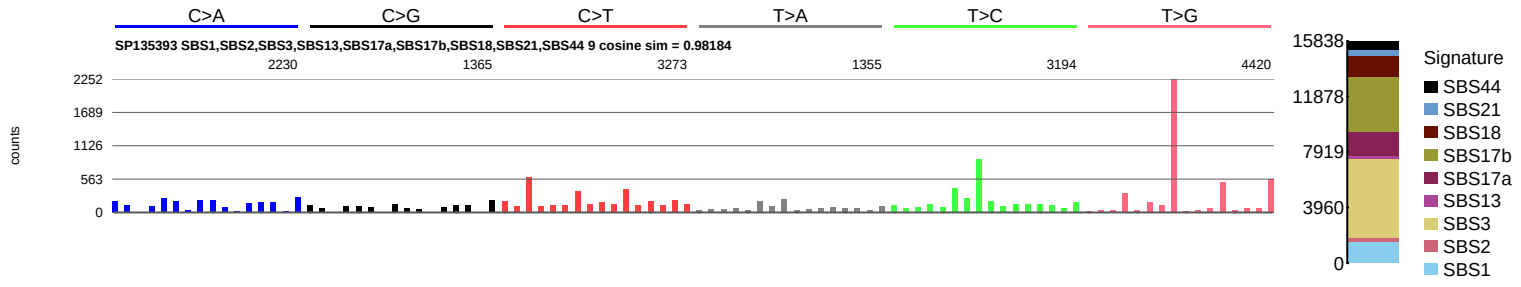
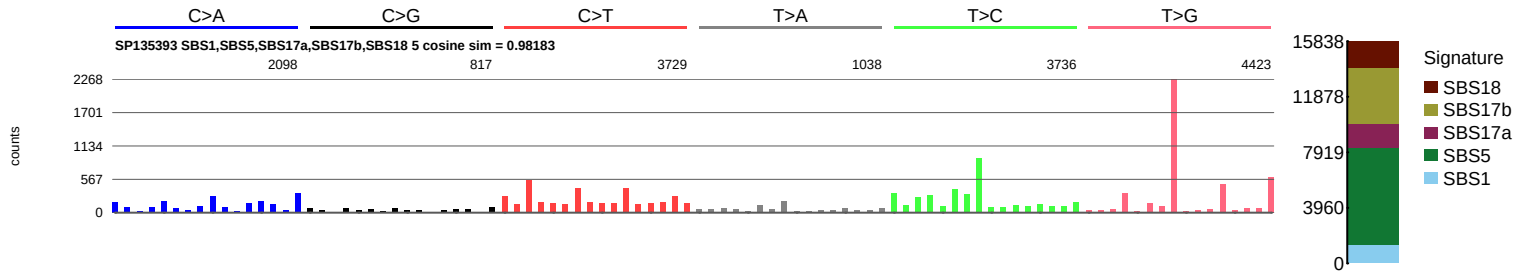
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS5,SBS17a,SBS17b,SBS18 6 cosine sim = 0.9819
2092
814
3754
1030
3726
4423
counts
2268
1701
1134
567
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 8 cosine sim = 0.98186
2270
767
2731
1681
3651
4735
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 10 cosine sim = 0.98184
2228
1370
3279
1353
3191
4421
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98184
2230
1365
3273
1355
3194
4420
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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C>G
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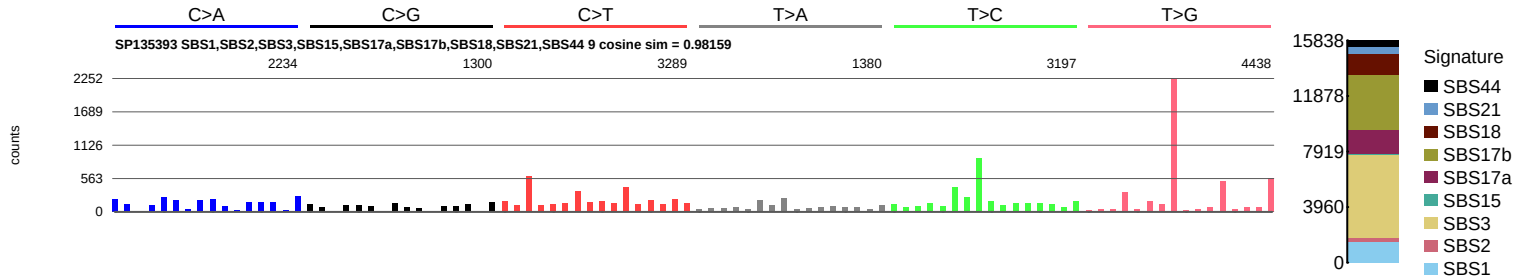
SP135393 SBS1,SBS5,SBS17a,SBS17b,SBS18 5 cosine sim = 0.98183
2098
817
3729
1038
3736
4423
counts
2268
1701
1134
567
0
15838
11878
7919
3960
0
Signature
SBS18
SBS17b
SBS17a
SBS5
SBS1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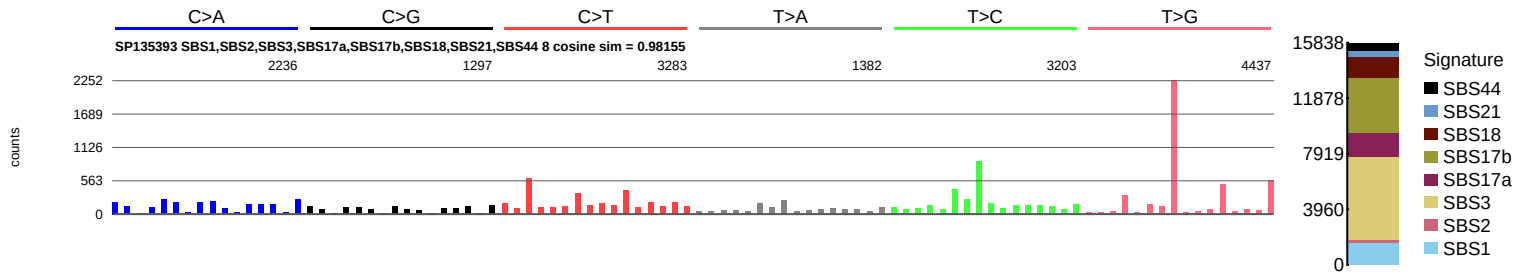
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98159
2234
1300
3289
1380
3197
4438
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1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C>T
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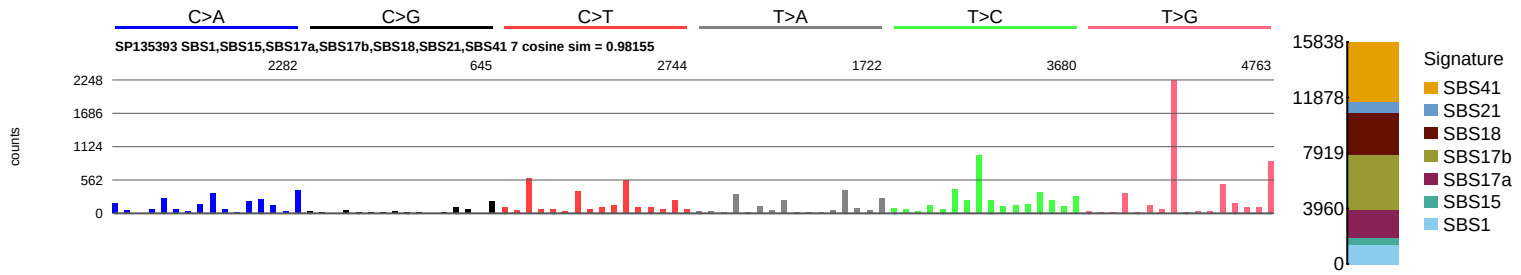
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98155
2236
1297
3283
1382
3203
4437
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1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C>G
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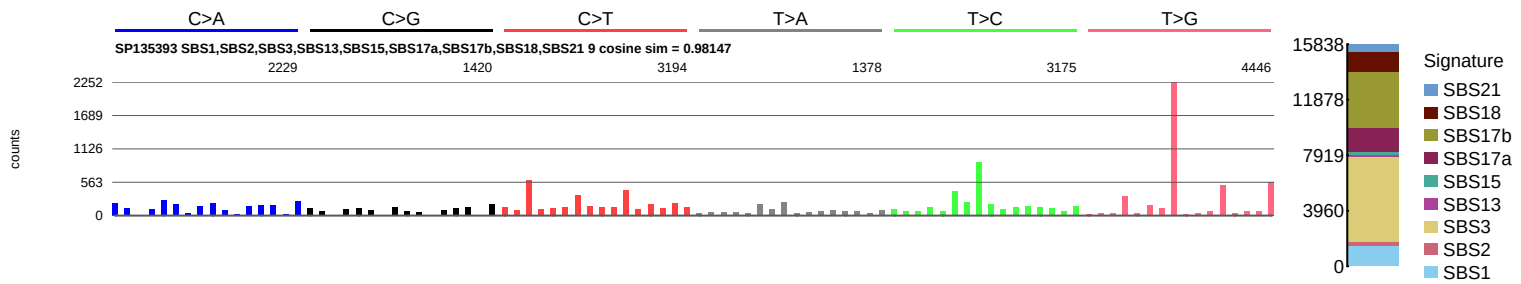
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98155
2282
645
2744
1722
3680
4763
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS1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C>G
C>T
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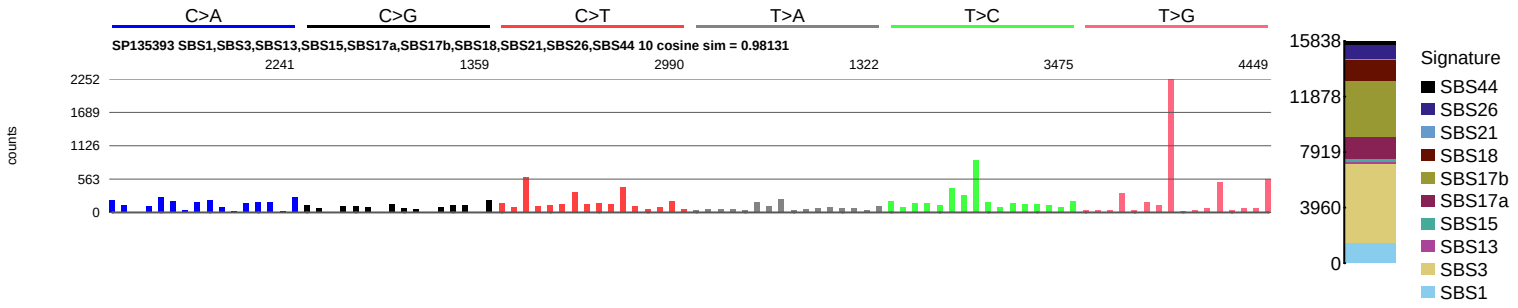
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 9 cosine sim = 0.98147
2229
1420
3194
1378
3175
4446
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS2
SBS1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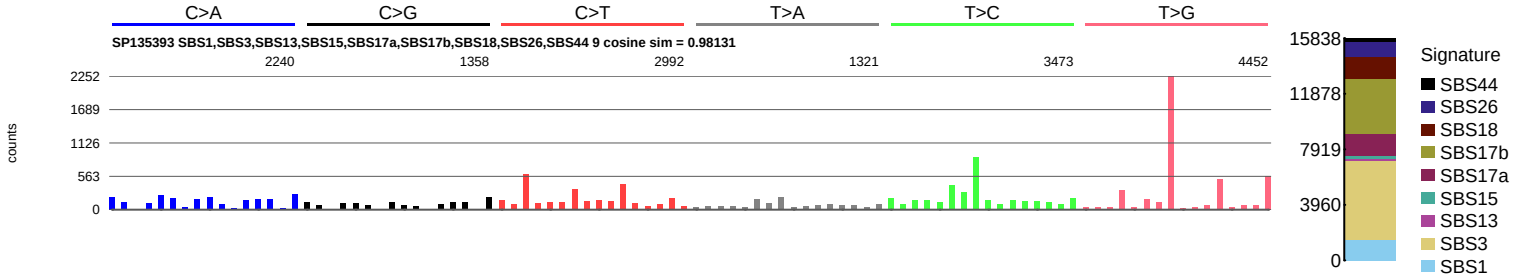
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 10 cosine sim = 0.98131
2241
1359
2990
1322
3475
4449
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1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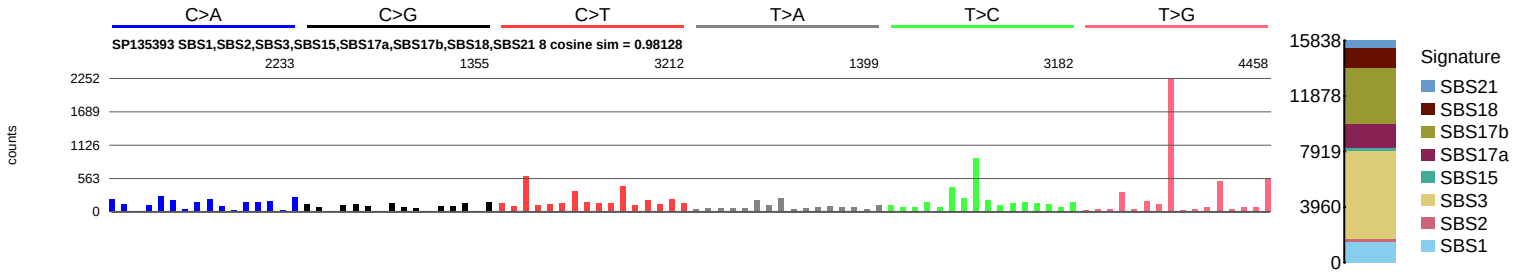
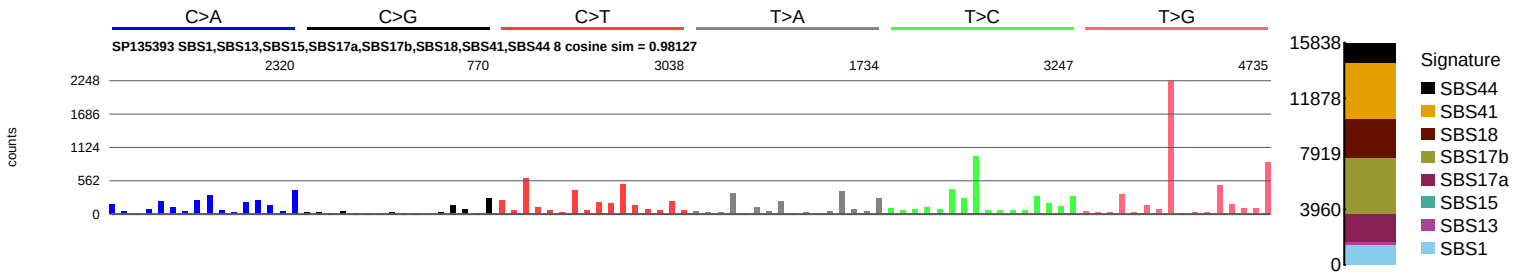
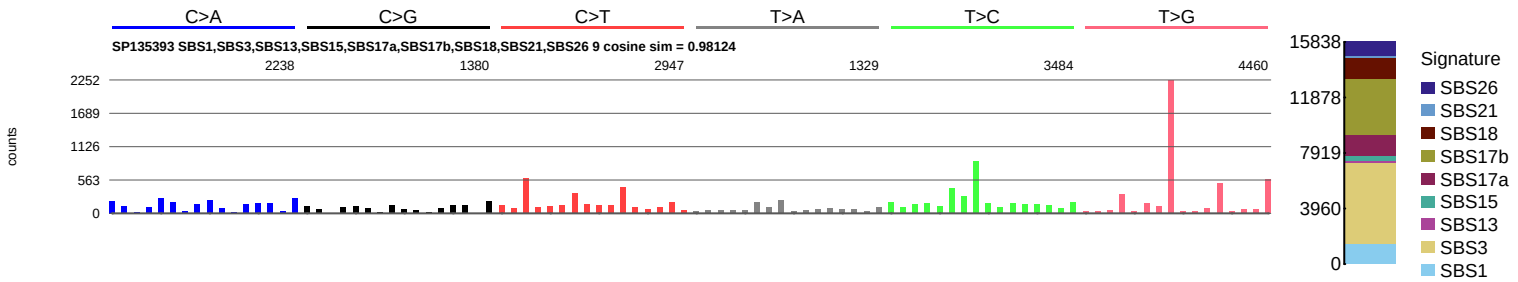
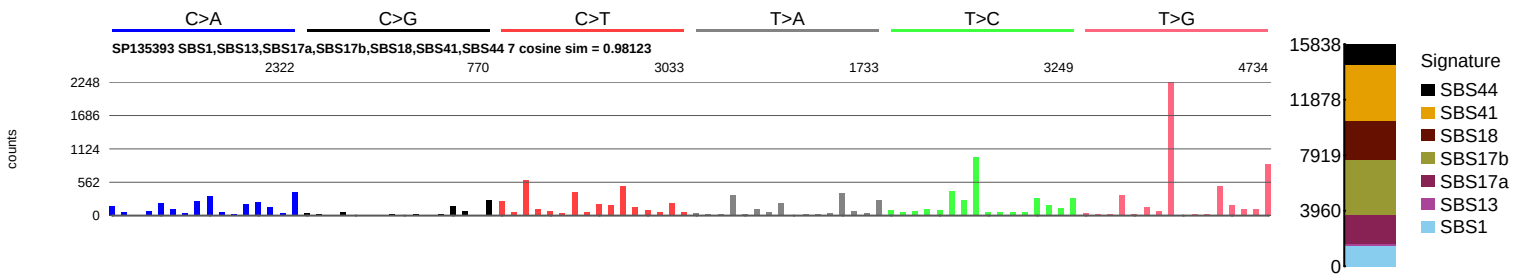
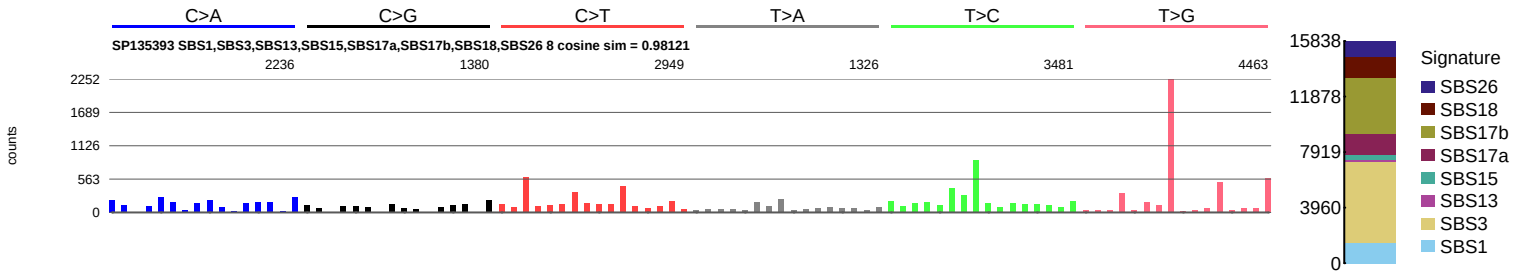
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 9 cosine sim = 0.98131
2240
1358
2992
1321
3473
4452
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1

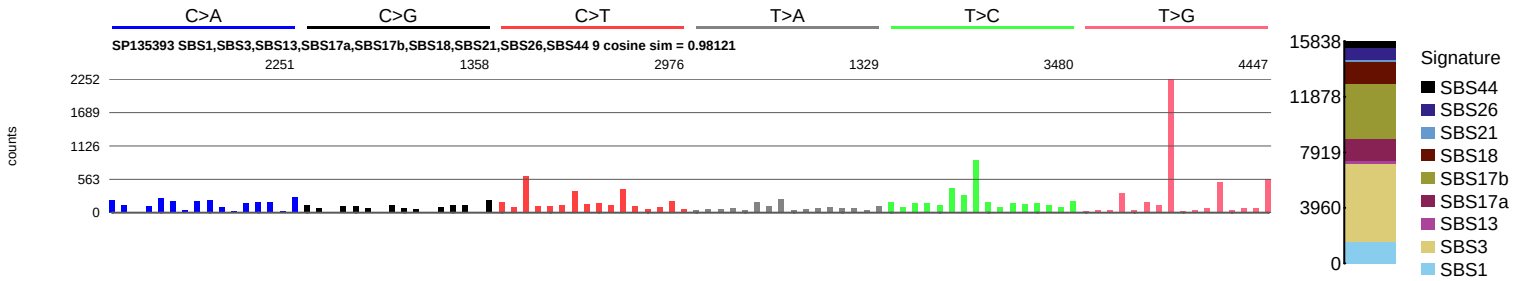

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98128
2233
1355
3212
1399
3182
4458
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS2
SBS1

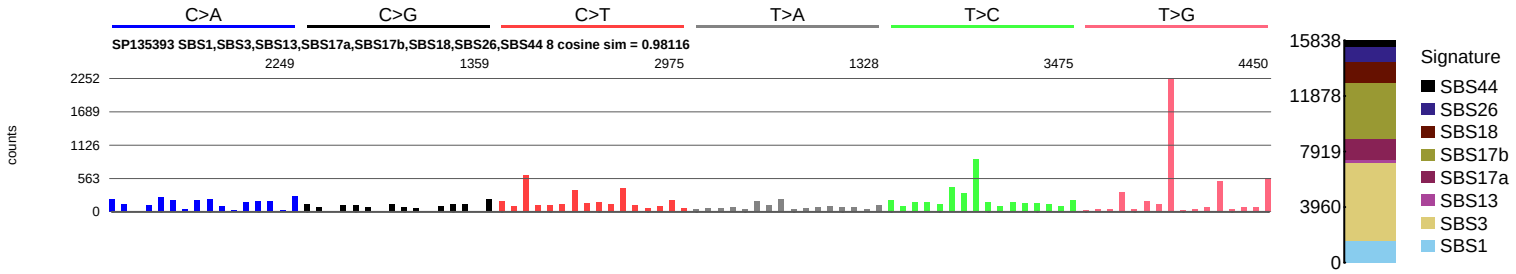

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 8 cosine sim = 0.98127
2320
770
3038
1734
3247
4735
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS1

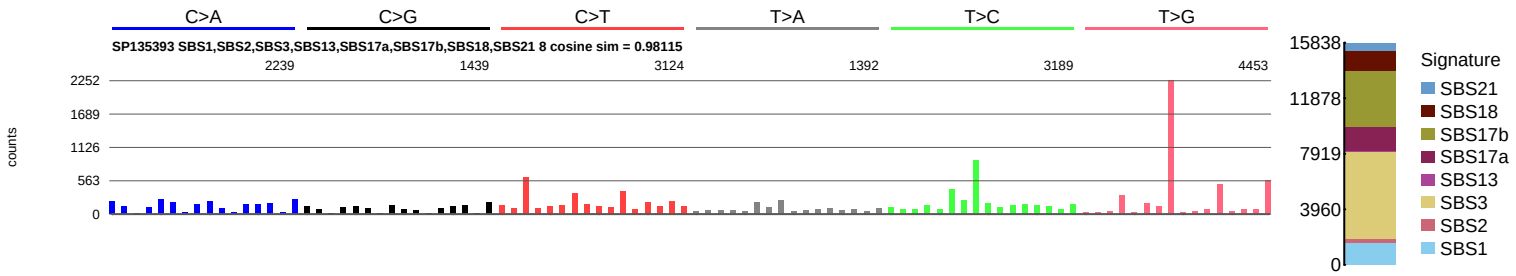

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 9 cosine sim = 0.98124
2238
1380
2947
1329
3484
4460
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98123
2322
770
3033
1733
3249
4734
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS13
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS26 8 cosine sim = 0.98121
2236
1380
2949
1326
3481
4463
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98121
2251
1358
2976
1329
3480
4447
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98116
2249
1359
2975
1328
3475
4450
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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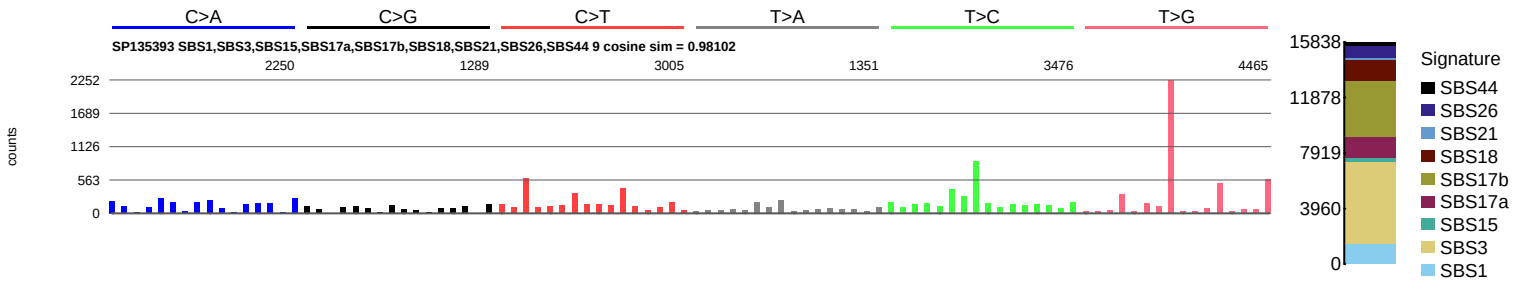
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.98115
2239
1439
3124
1392
3189
4453
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS2
SBS1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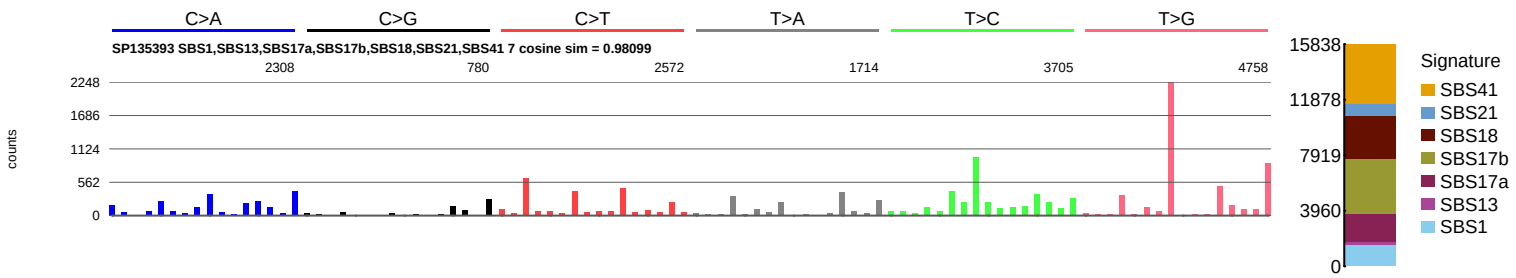
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 9 cosine sim = 0.98102
2250
1289
3005
1351
3476
4465
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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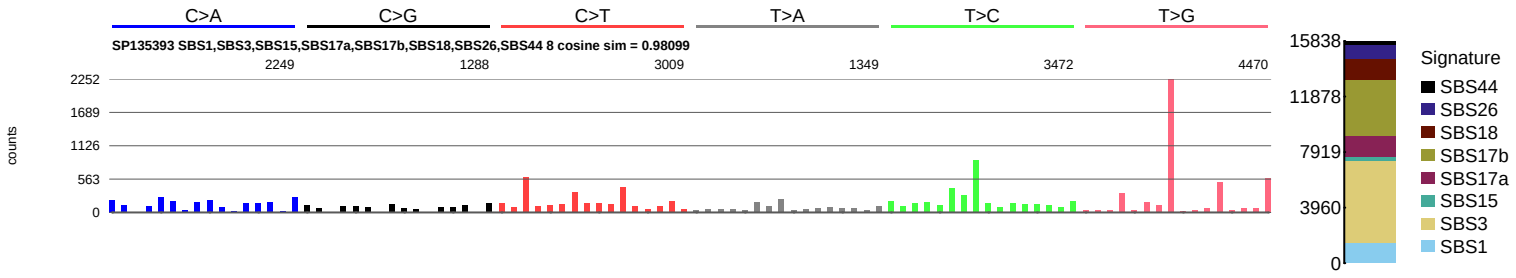
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS41 7 cosine sim = 0.98099
2308
780
2572
1714
3705
4758
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS1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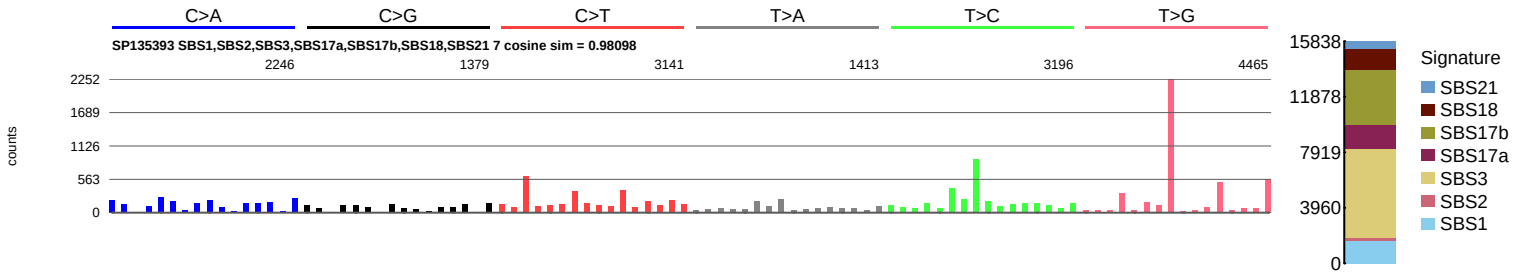
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26,SBS44 8 cosine sim = 0.98099
2249
1288
3009
1349
3472
4470
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1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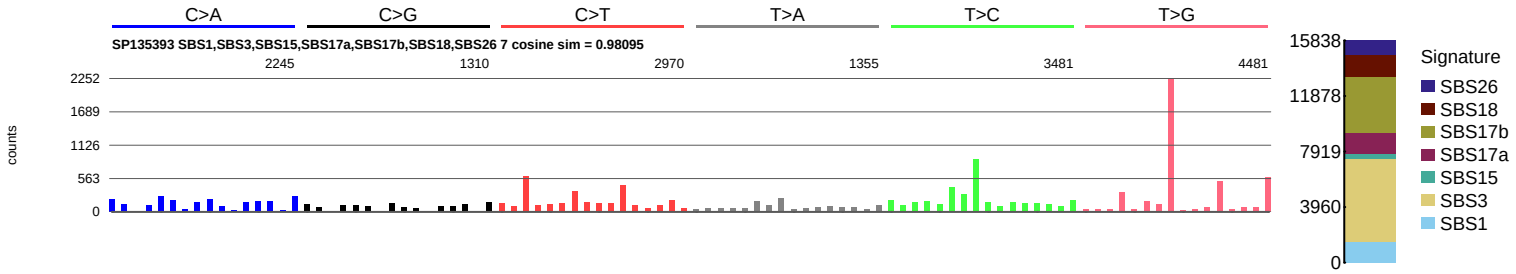
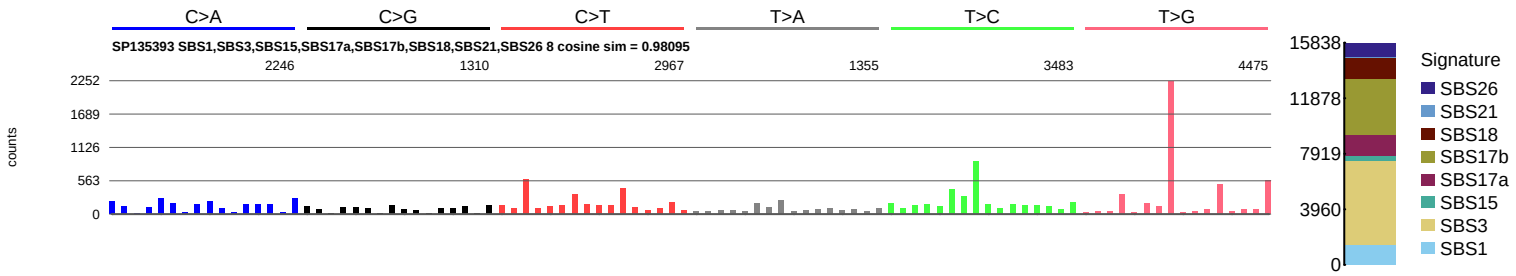
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS21 7 cosine sim = 0.98098
2246
1379
3141
1413
3196
4465
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98095
2245
1310
2970
1355
3481
4481
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS26 8 cosine sim = 0.98095
2246
1310
2967
1355
3483
4475
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
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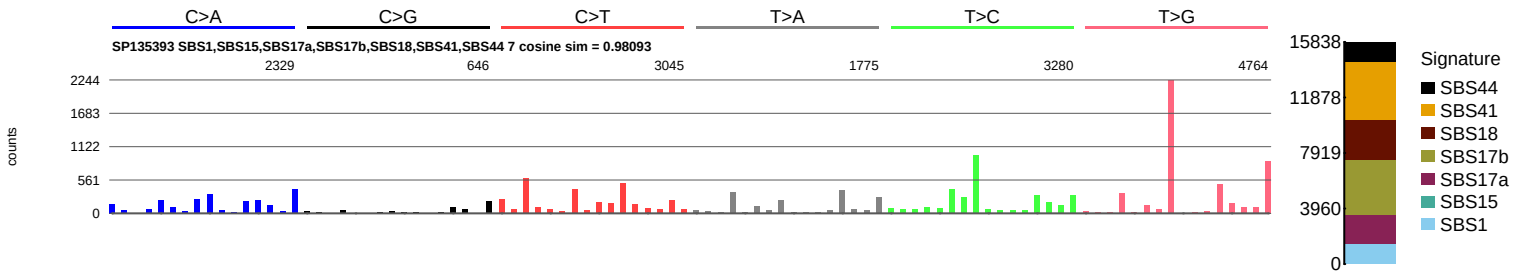
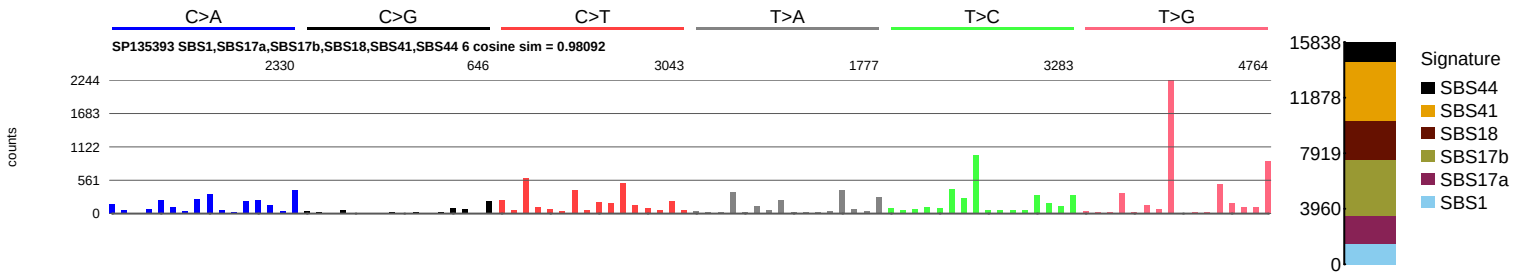
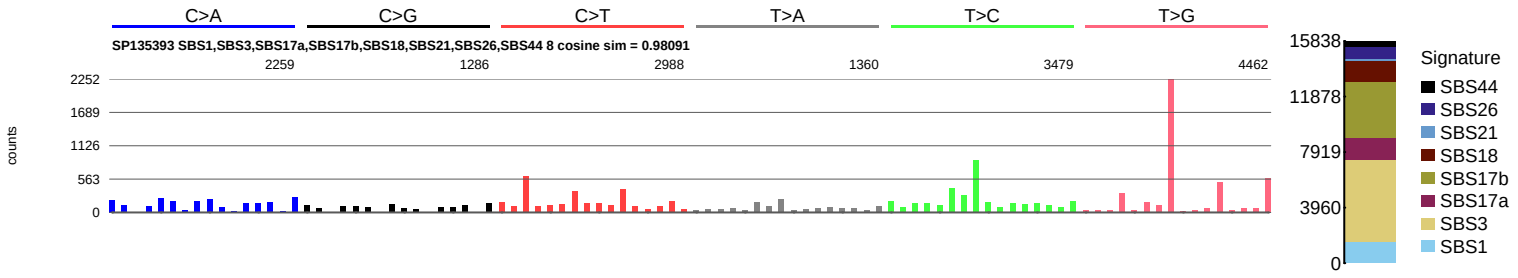
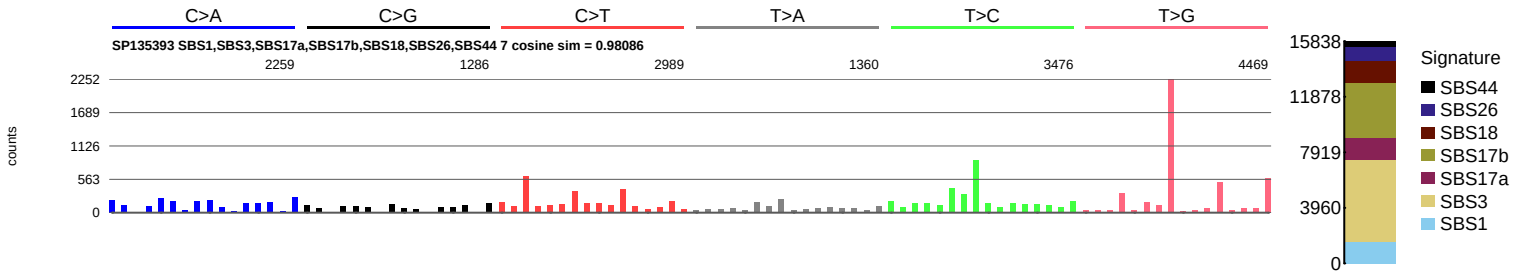
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS15,SBS17a,SBS17b,SBS18,SBS41,SBS44 7 cosine sim = 0.98093
2329
646
3045
1775
3280
4764
counts
2244
1683
1122
561
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS41,SBS44 6 cosine sim = 0.98092
2330
646
3043
1777
3283
4764
counts
2244
1683
1122
561
0
15838
11878
7919
3960
0
Signature
SBS44
SBS41
SBS18
SBS17b
SBS17a
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26,SBS44 8 cosine sim = 0.98091
2259
1286
2988
1360
3479
4462
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26,SBS44 7 cosine sim = 0.98086
2259
1286
2989
1360
3476
4469
counts
2252
1689
1126
563
0
15838
11878
7919
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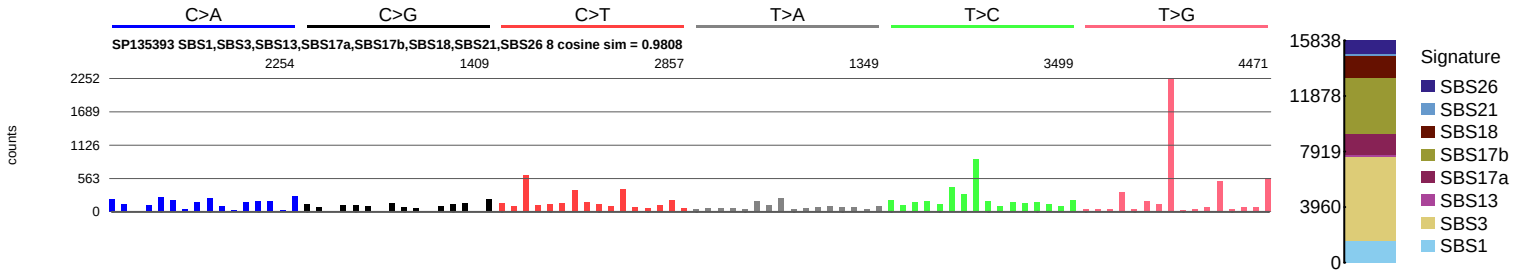
SBS1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2254
1409
2857
1349
3499
4471
counts
2252
1689
1126
563
0
15838
11878
7919
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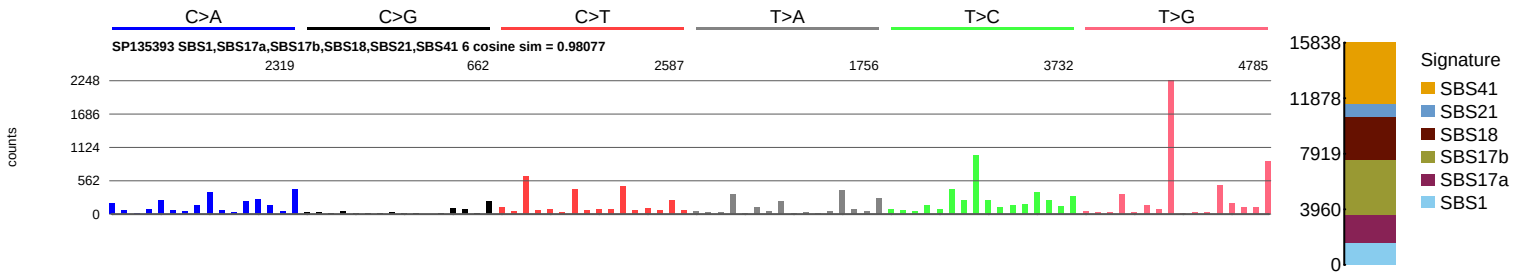
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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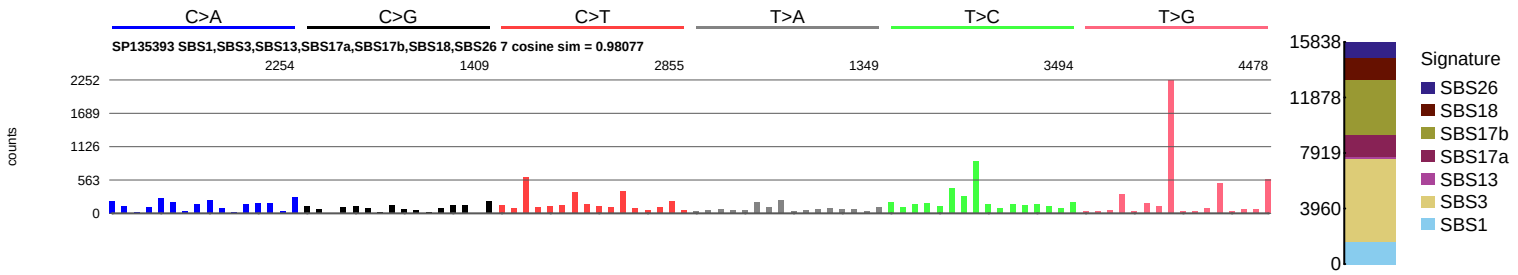
SP135393 SBS1,SBS17a,SBS17b,SBS18,SBS21,SBS41 6 cosine sim = 0.98077
2319
662
2587
1756
3732
4785
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS41
SBS21
SBS18
SBS17b
SBS17a
SBS1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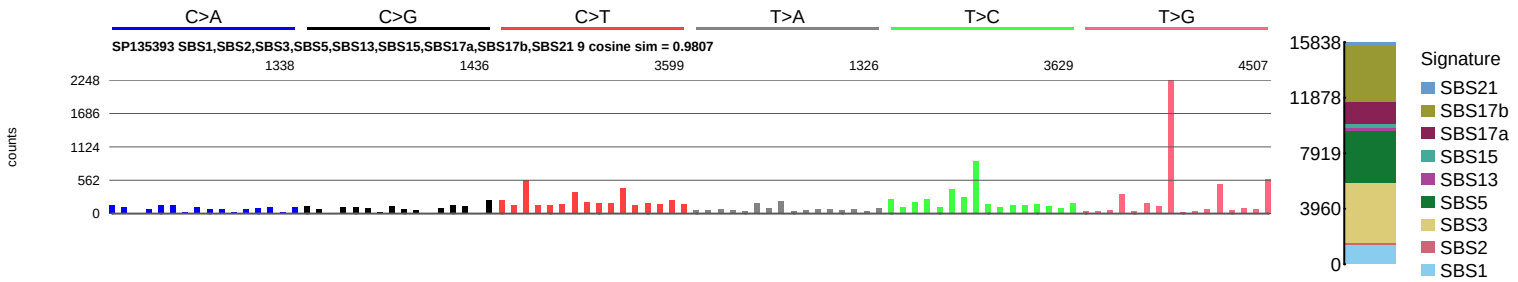
SP135393 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS26 7 cosine sim = 0.98077
2254
1409
2855
1349
3494
4478
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1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C>T
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T>C
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SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 9 cosine sim = 0.9807
1338
1436
3599
1326
3629
4507
counts
2248
1686
1124
562
0
15838
11878
7919
3960
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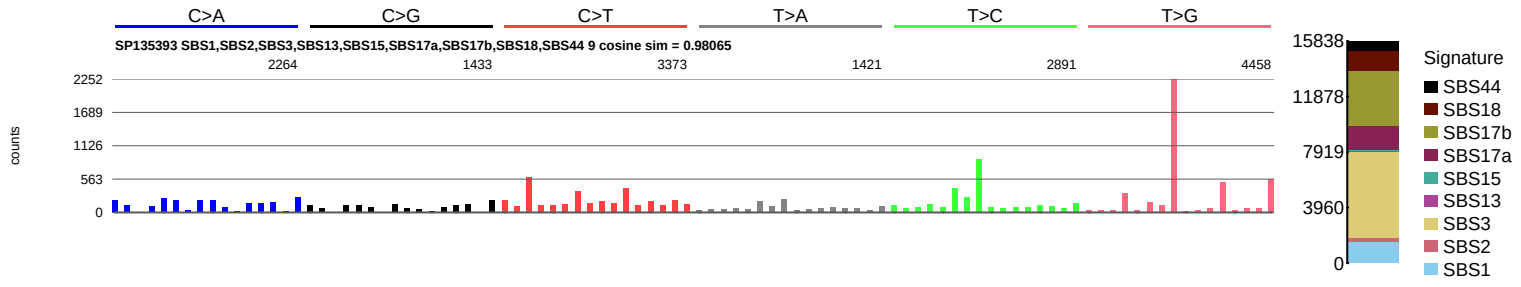
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS44 9 cosine sim = 0.98065
2264
1433
3373
1421
2891
4458
counts
2252
1689
1126
563
0
15838
11878
7919
3960
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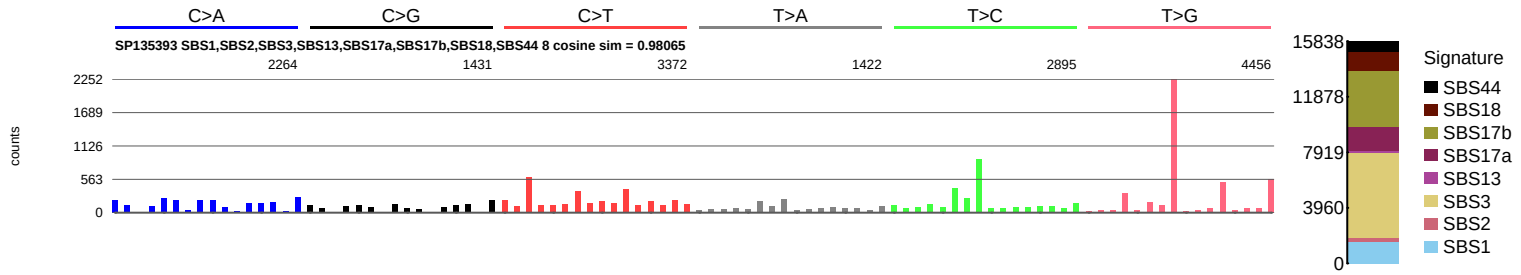
SBS2
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98065
2264
1431
3372
1422
2895
4456
counts
2252
1689
1126
563
0
15838
11878
7919
3960
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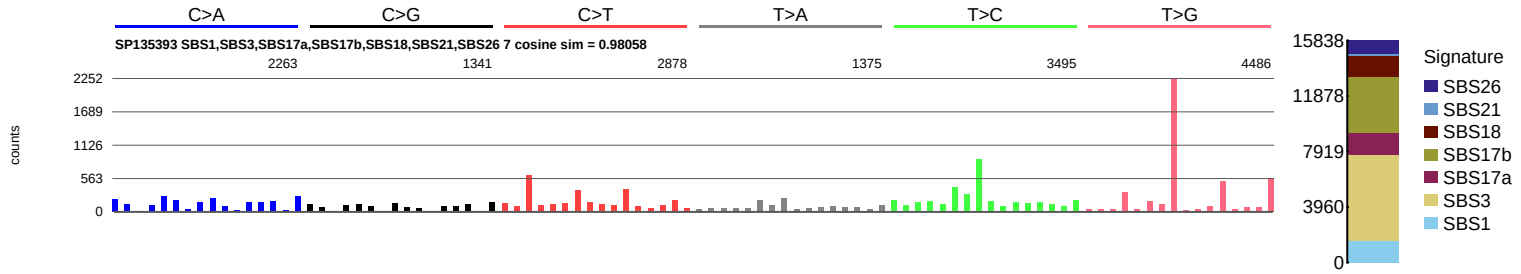
SBS2
SBS1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C>G
C>T
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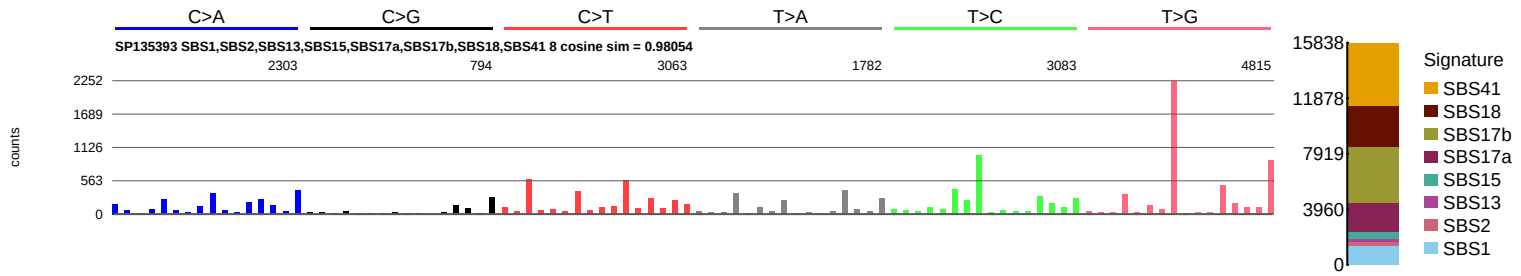
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS26 7 cosine sim = 0.98058
2263
1341
2878
1375
3495
4486
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS26
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1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C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS41 8 cosine sim = 0.98054
2303
794
3063
1782
3083
4815
counts
2252
1689
1126
563
0
15838
11878
7919
3960
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Signature
SBS41
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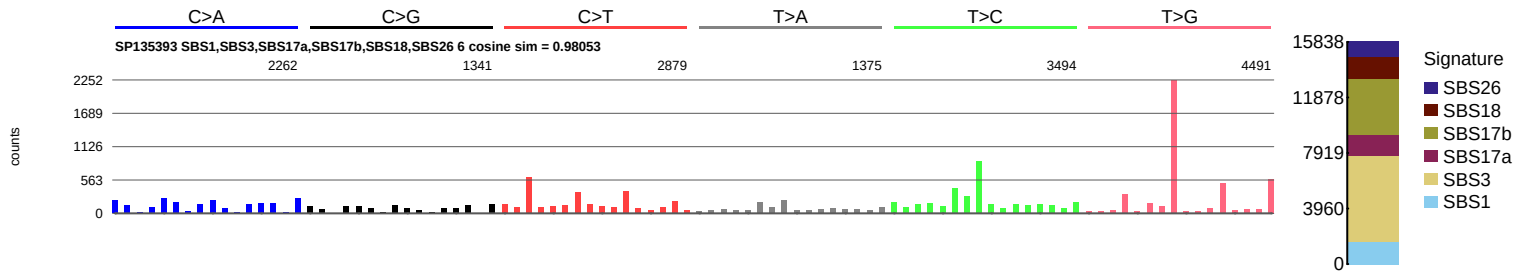
SBS13
SBS2
SBS1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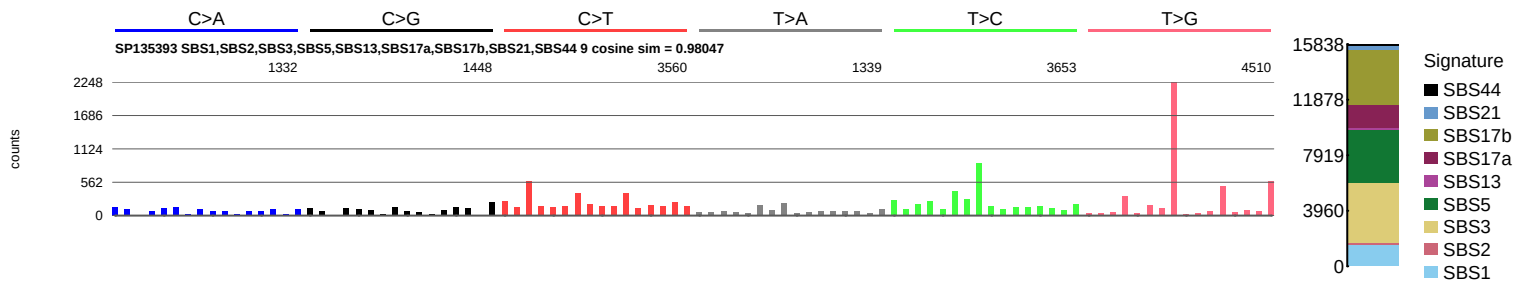
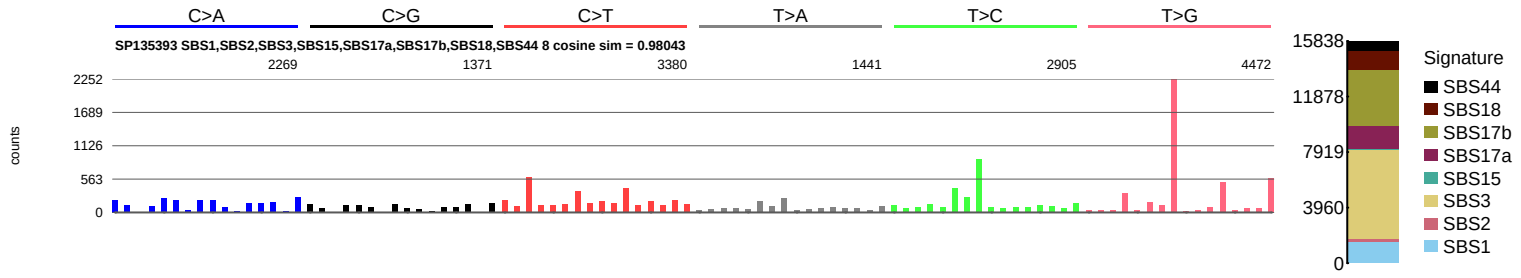
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS26 6 cosine sim = 0.98053
2262
1341
2879
1375
3494
4491
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS26
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 9 cosine sim = 0.98047
1332
1448
3560
1339
3653
4510
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS44 8 cosine sim = 0.98043
2269
1371
3380
1441
2905
4472
counts
2252
1689
1126
563
0
15838
11878
7919
3960
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Signature
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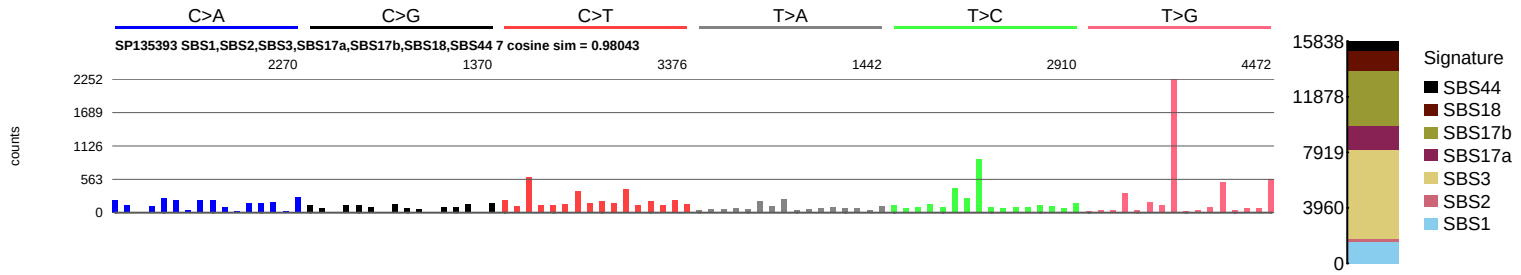
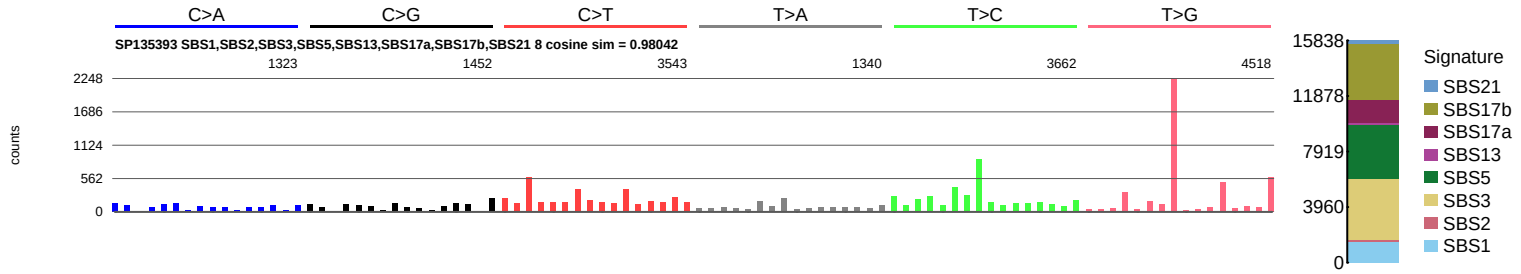
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS17a,SBS17b,SBS18,SBS44 7 cosine sim = 0.98043
2270
1370
3376
1442
2910
4472
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS18
SBS17b
SBS17a
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98042
1323
1452
3543
1340
3662
4518
counts
2248
1686
1124
562
0
15838
11878
7919
3960
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Signature
SBS21
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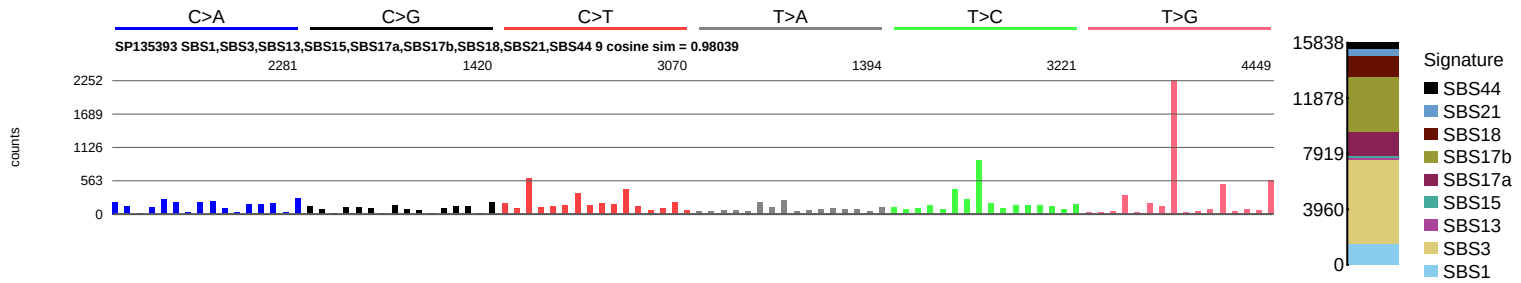
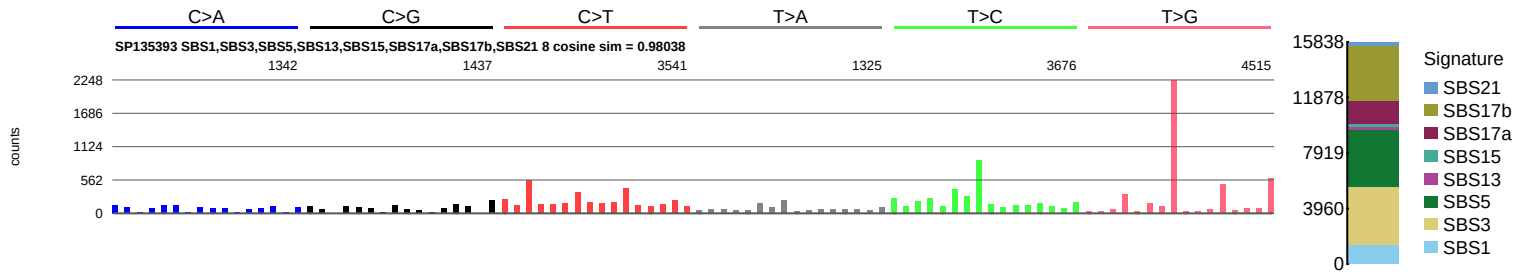
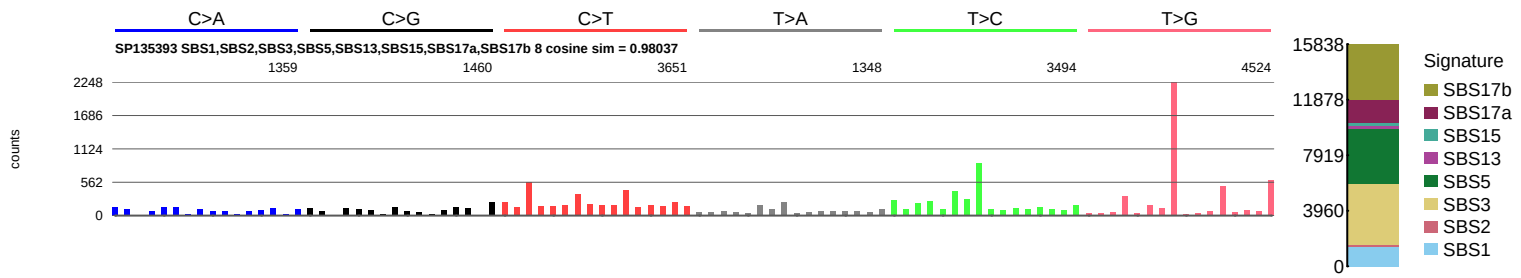
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 9 cosine sim = 0.98039
2281
1420
3070
1394
3221
4449
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98038
1342
1437
3541
1325
3676
4515
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 8 cosine sim = 0.98037
1359
1460
3651
1348
3494
4524
counts
2248
1686
1124
562
0
15838
11878
7919
3960
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Signature
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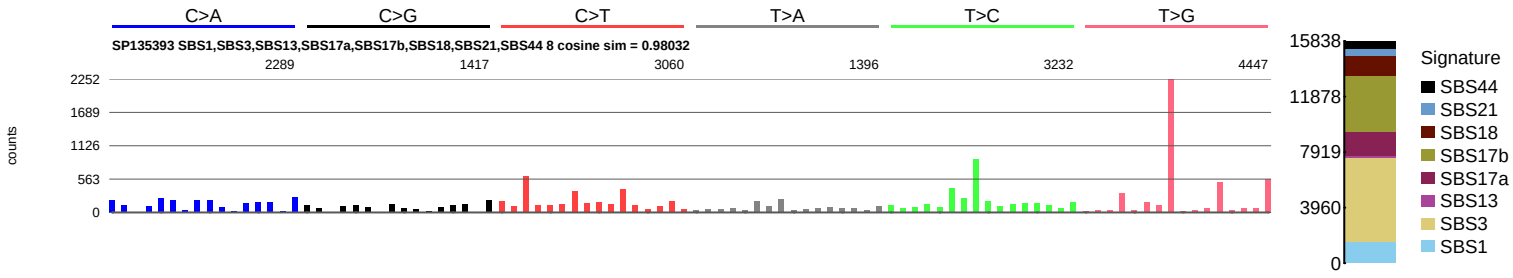
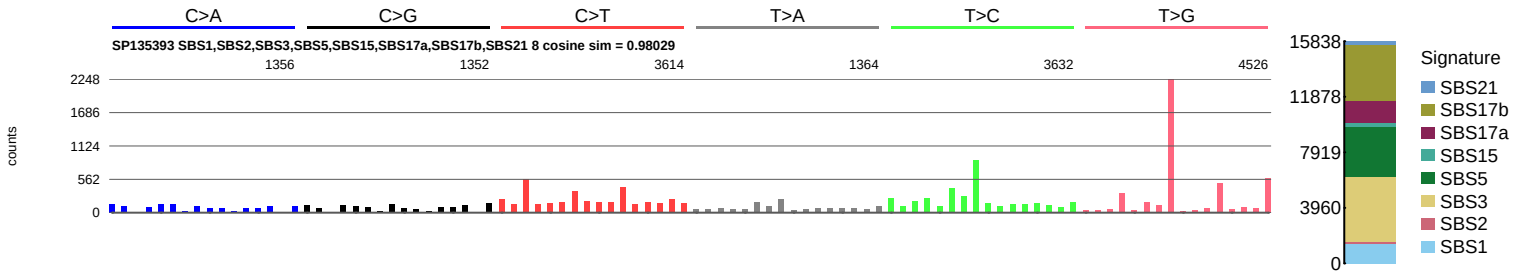
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98032
2289
1417
3060
1396
3232
4447
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS13
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 8 cosine sim = 0.98029
1356
1352
3614
1364
3632
4526
counts
2248
1686
1124
562
0
15838
11878
7919
3960
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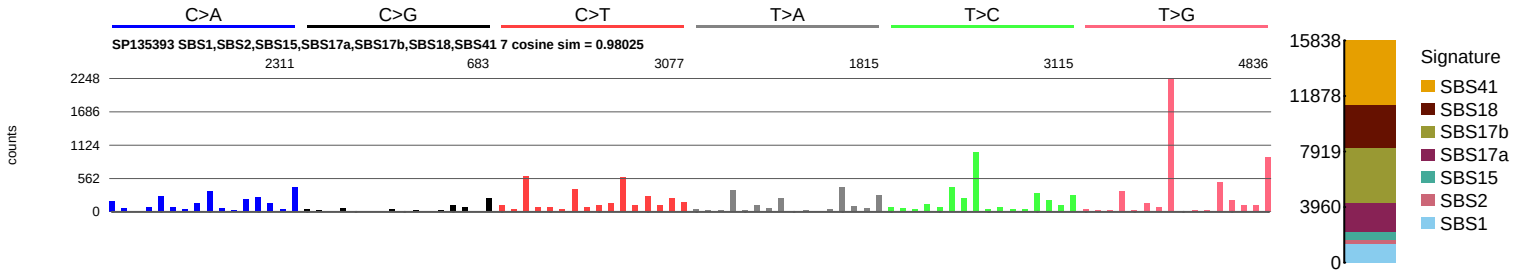
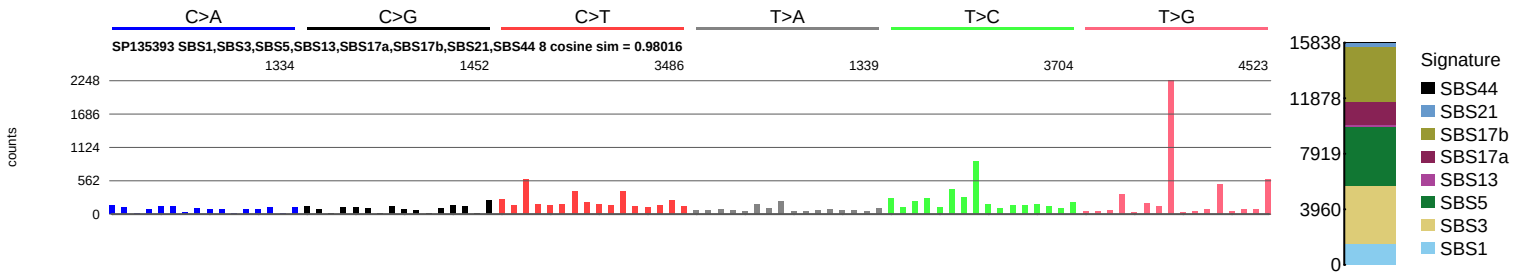
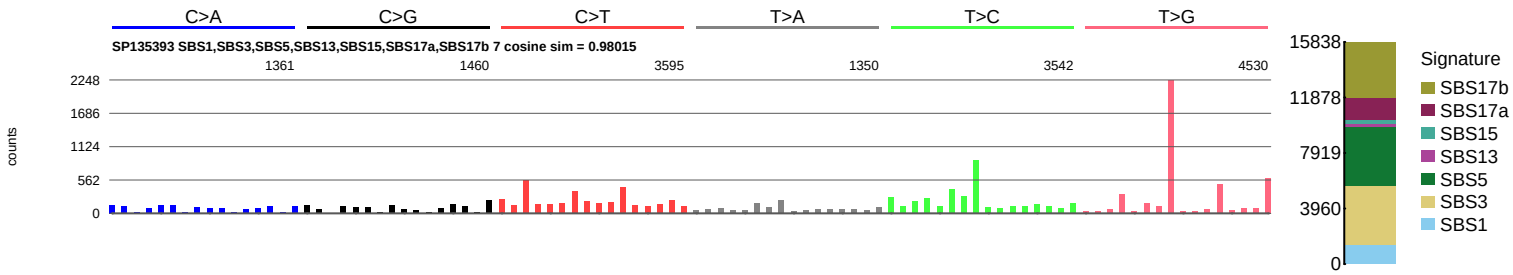
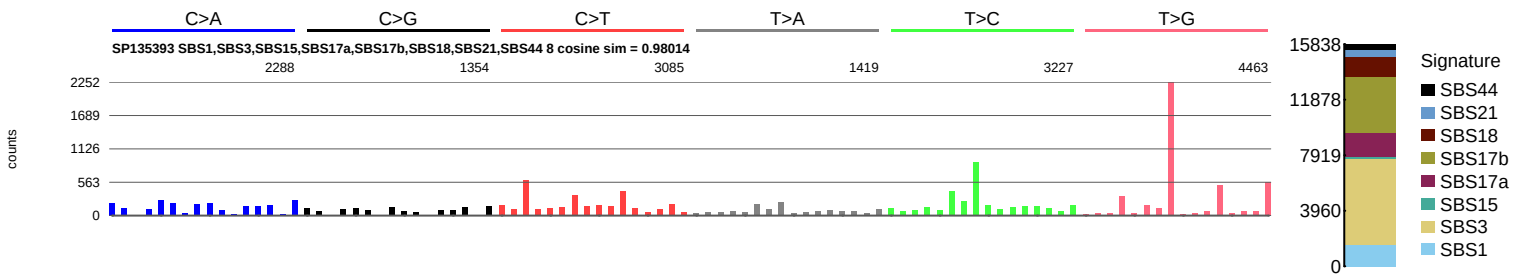
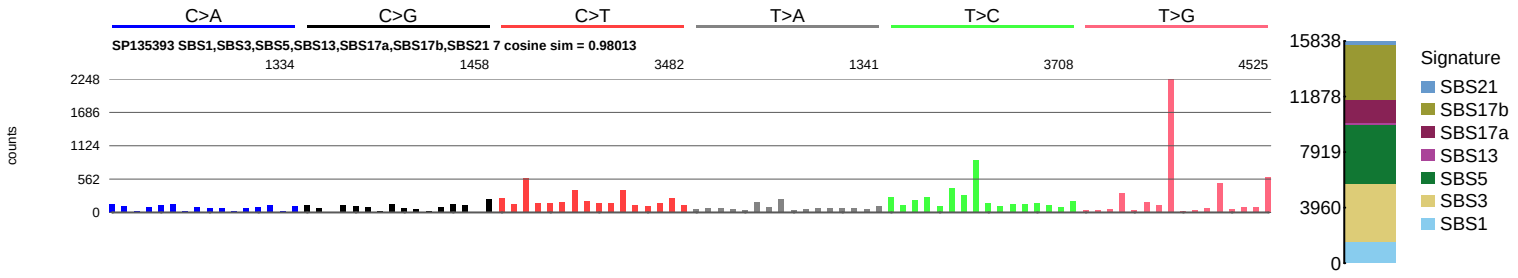
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS15,SBS17a,SBS17b,SBS18,SBS41 7 cosine sim = 0.98025
2311
683
3077
1815
3115
4836
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS41
SBS18
SBS17b
SBS17a
SBS15
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98016
1334
1452
3486
1339
3704
4523
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98015
1361
1460
3595
1350
3542
4530
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS17b
SBS17a
SBS15
SBS13
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS15,SBS17a,SBS17b,SBS18,SBS21,SBS44 8 cosine sim = 0.98014
2288
1354
3085
1419
3227
4463
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS15
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98013
1334
1458
3482
1341
3708
4525
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS21
SBS17b
SBS17a
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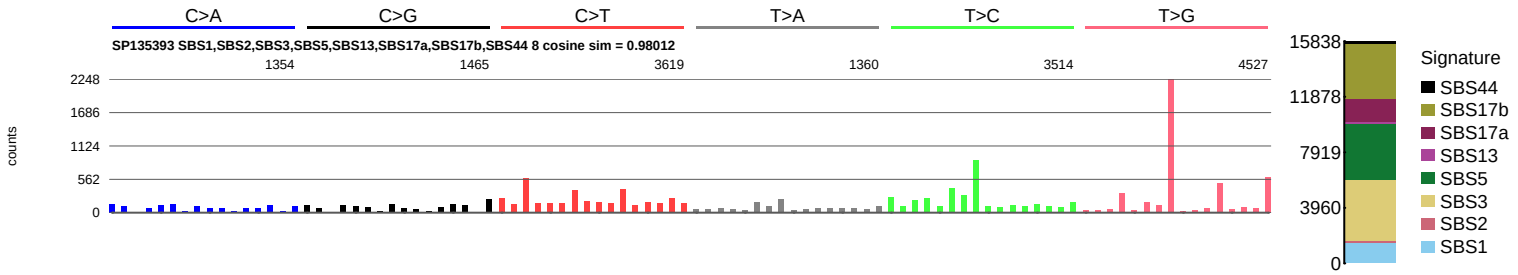
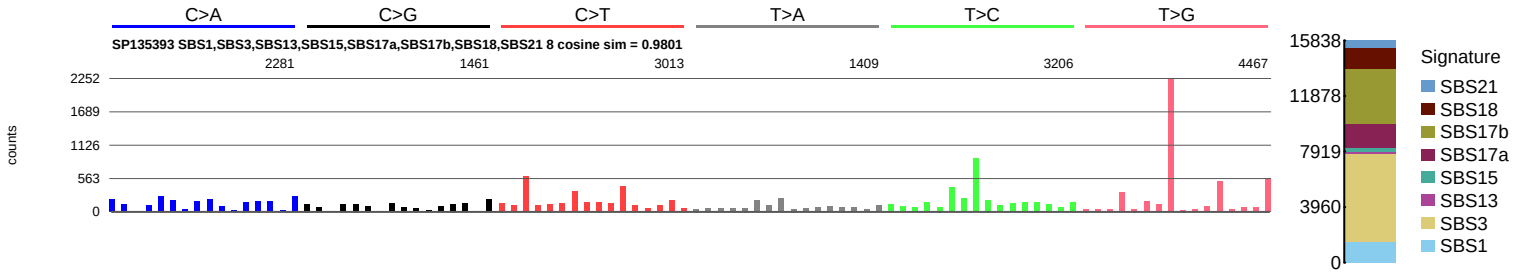
SBS5
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b,SBS44 8 cosine sim = 0.98012
1354
1465
3619
1360
3514
4527
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS13,SBS15,SBS17a,SBS17b,SBS18,SBS21 8 cosine sim = 0.9801
2281
1461
3013
1409
3206
4467
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS21
SBS18
SBS17b
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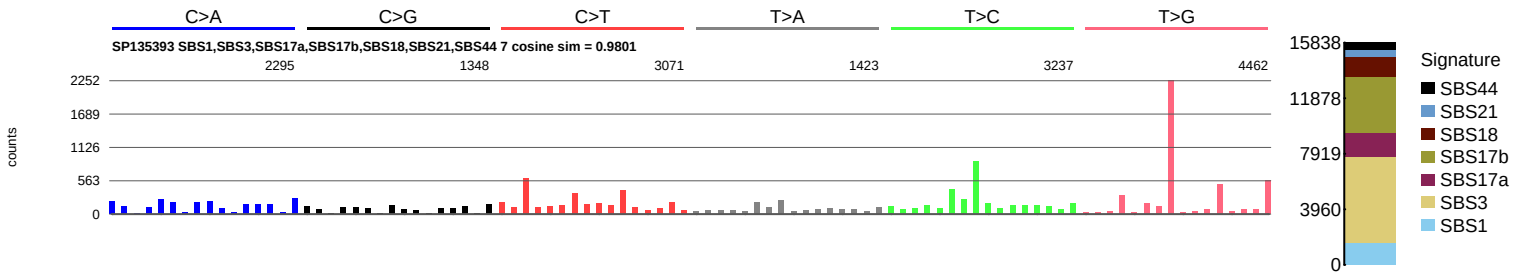
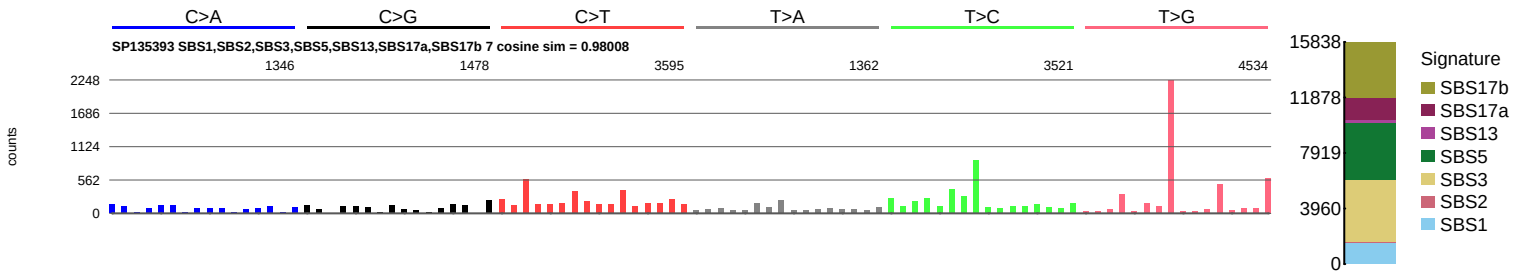
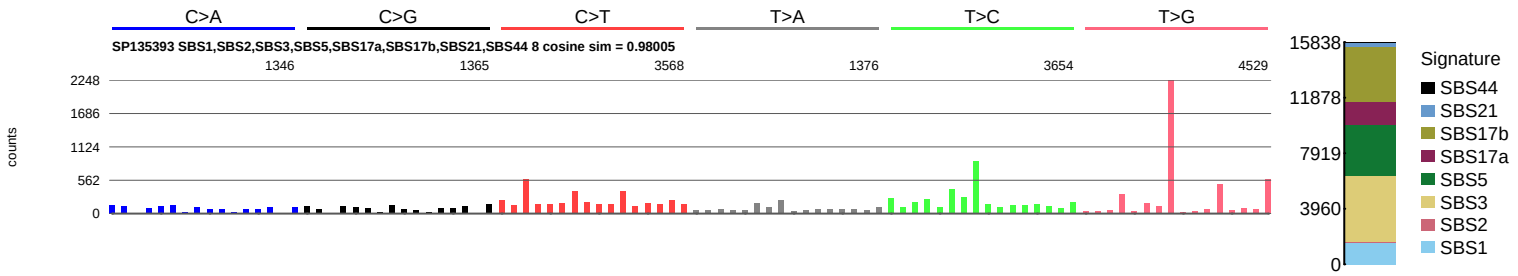
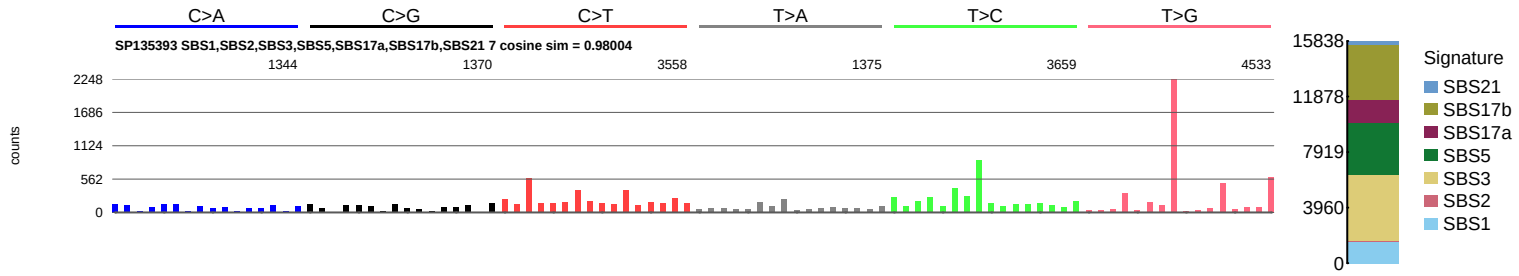
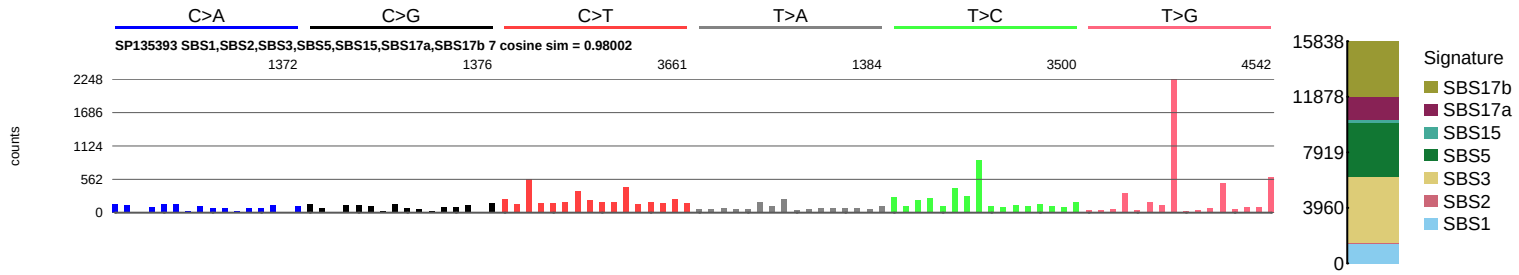
SBS13
SBS3
SBS1

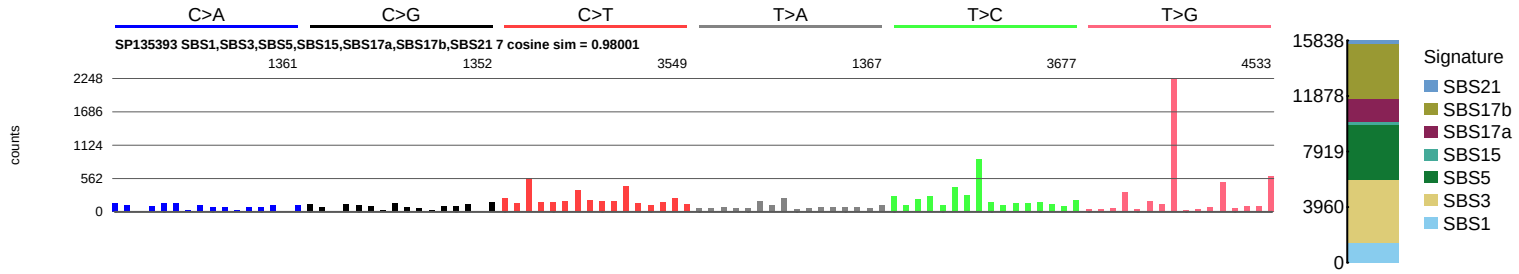

C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS17a,SBS17b,SBS18,SBS21,SBS44 7 cosine sim = 0.9801
2295
1348
3071
1423
3237
4462
counts
2252
1689
1126
563
0
15838
11878
7919
3960
0
Signature
SBS44
SBS21
SBS18
SBS17b
SBS17a
SBS3
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS13,SBS17a,SBS17b 7 cosine sim = 0.98008
1346
1478
3595
1362
3521
4534
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS17b
SBS17a
SBS13
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21,SBS44 8 cosine sim = 0.98005
1346
1365
3568
1376
3654
4529
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS44
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98004
1344
1370
3558
1375
3659
4533
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS21
SBS17b
SBS17a
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS2,SBS3,SBS5,SBS15,SBS17a,SBS17b 7 cosine sim = 0.98002
1372
1376
3661
1384
3500
4542
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS2
SBS1


C>A
C>G
C>T
T>A
T>C
T>G
SP135393 SBS1,SBS3,SBS5,SBS15,SBS17a,SBS17b,SBS21 7 cosine sim = 0.98001
1361
1352
3549
1367
3677
4533
counts
2248
1686
1124
562
0
15838
11878
7919
3960
0
Signature
SBS21
SBS17b
SBS17a
SBS15
SBS5
SBS3
SBS1